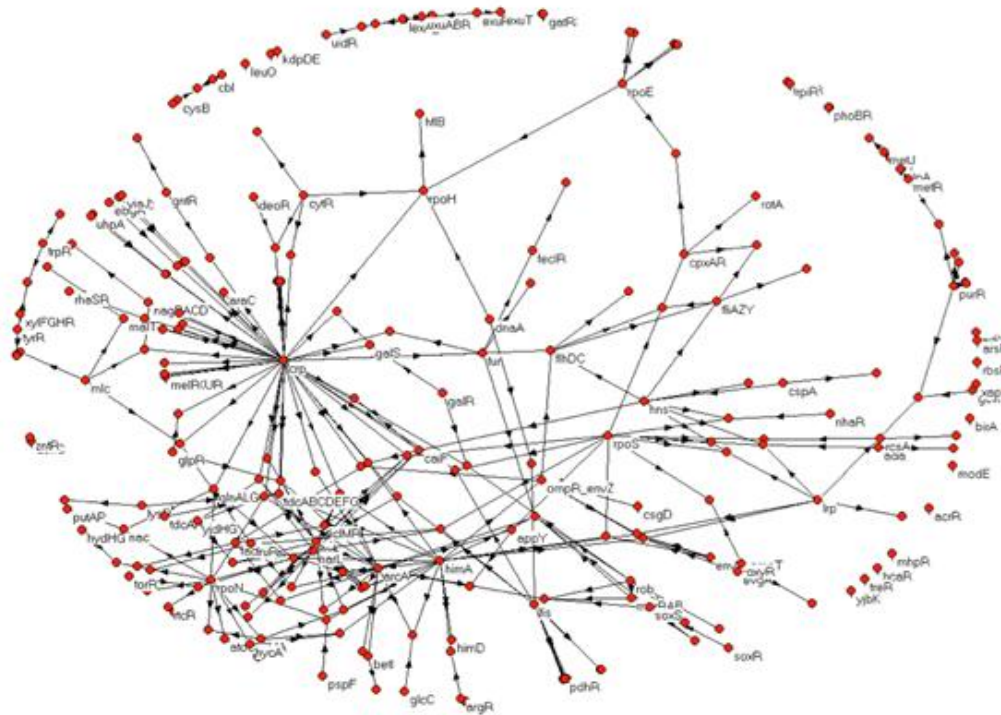


Computational Biology

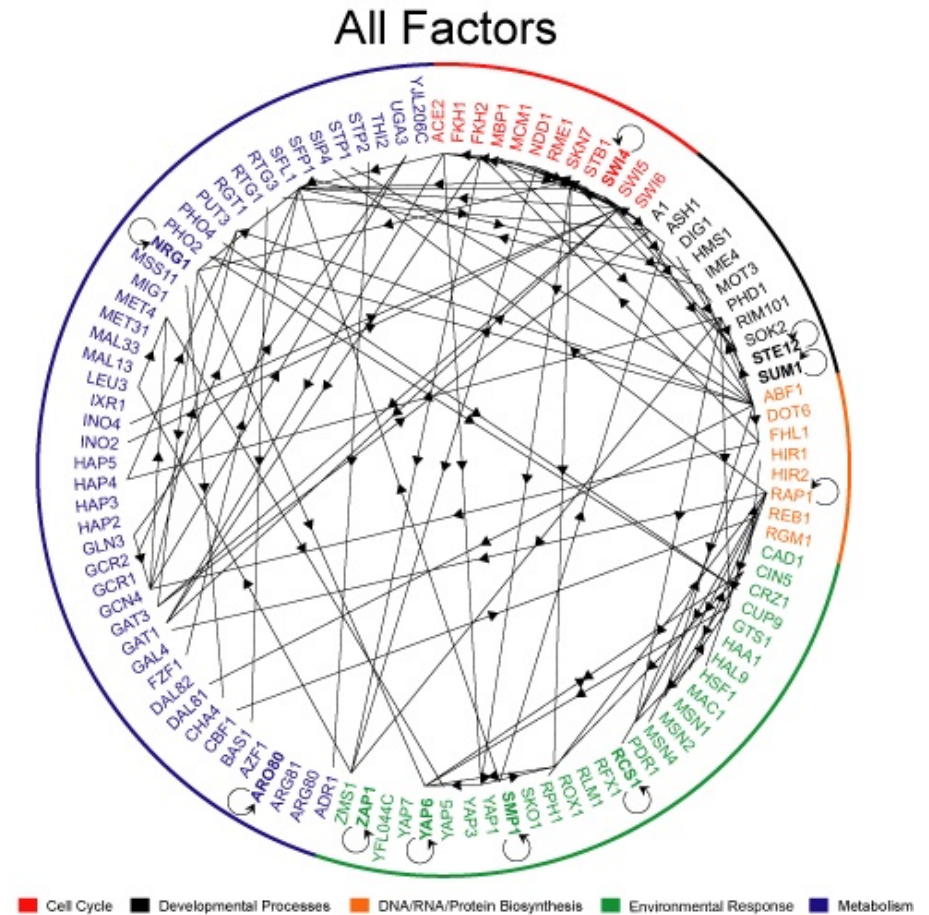
Lecture 3: *Gene Regulation Networks*



Gene regulation, chemical equations, motifs

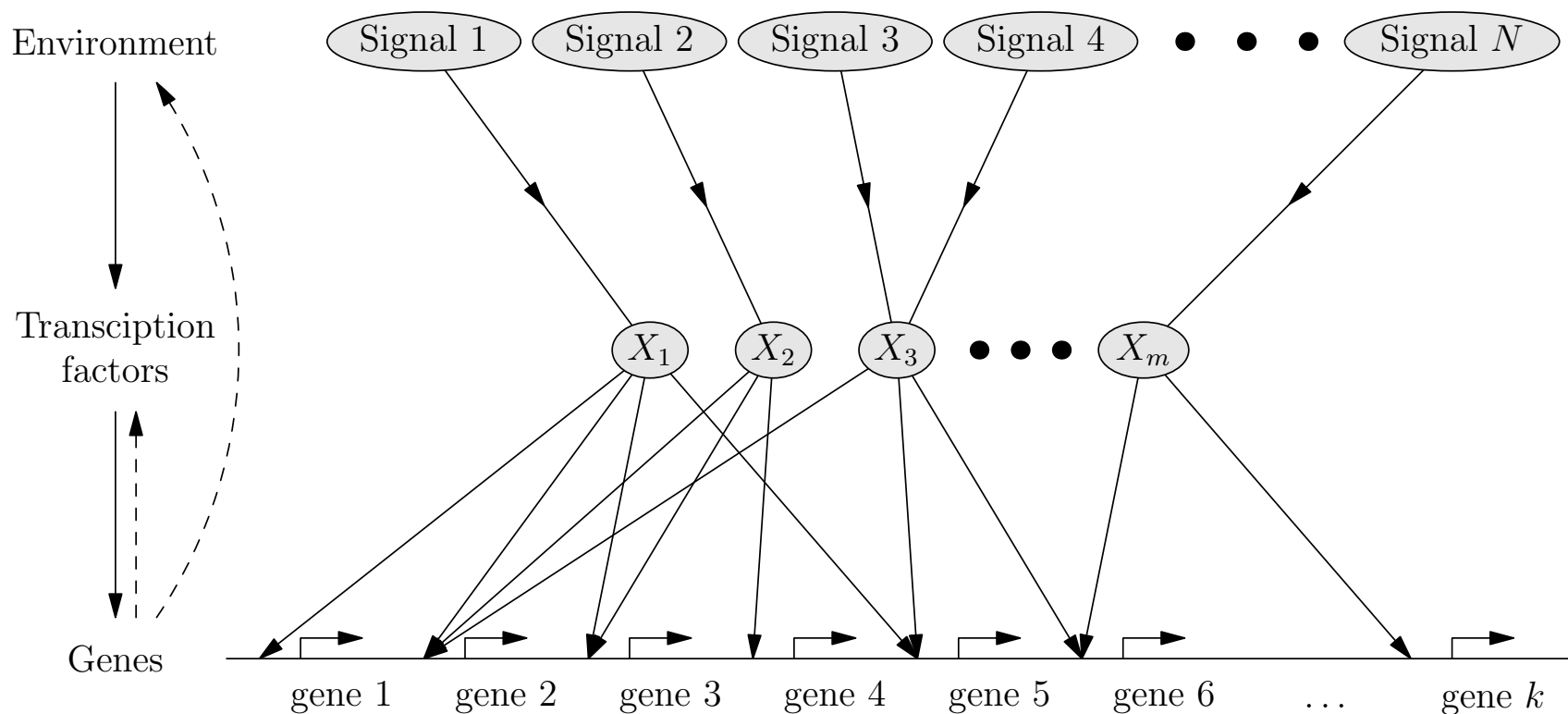
Outline

1. **Gene Regulation**
2. Chemical Equations
3. Motifs

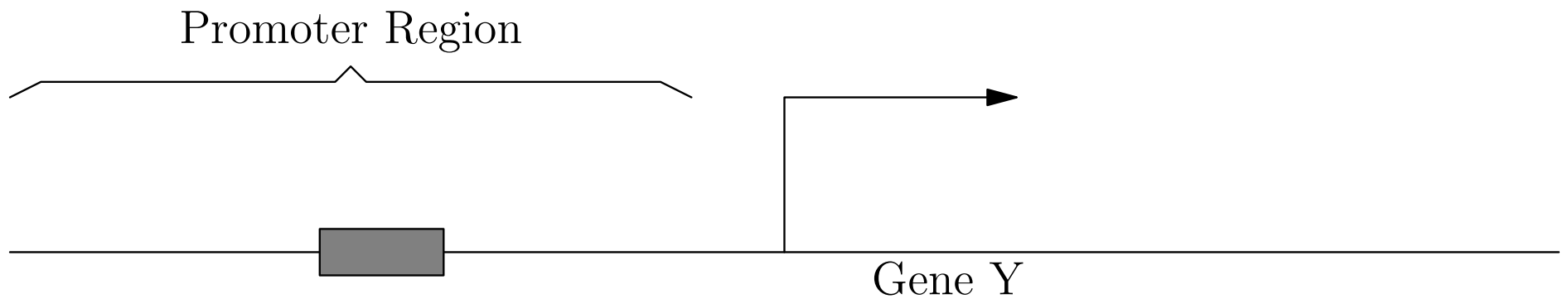


Transcription Networks

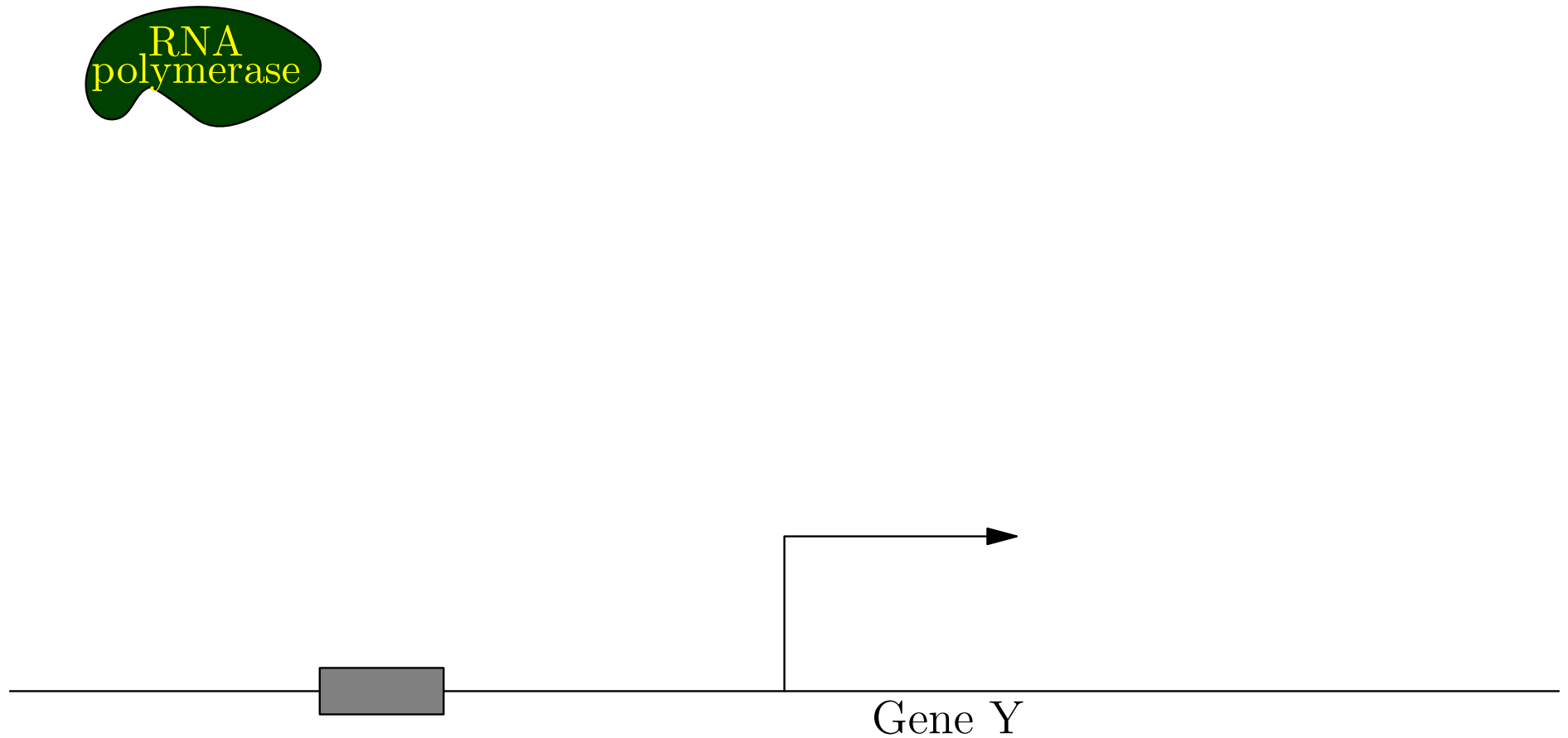
- Signalling molecules combine with transcription factors (proteins) which excite or inhibit the transcription of other proteins



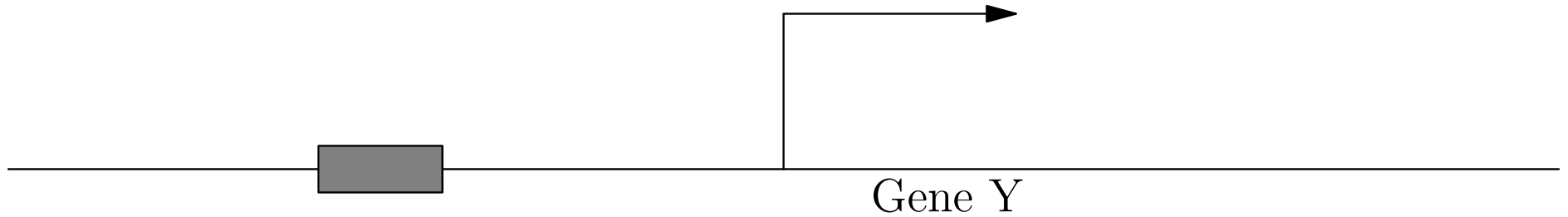
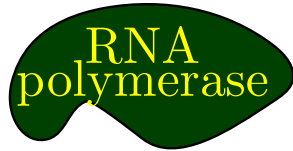
Gene Transcription Regulation



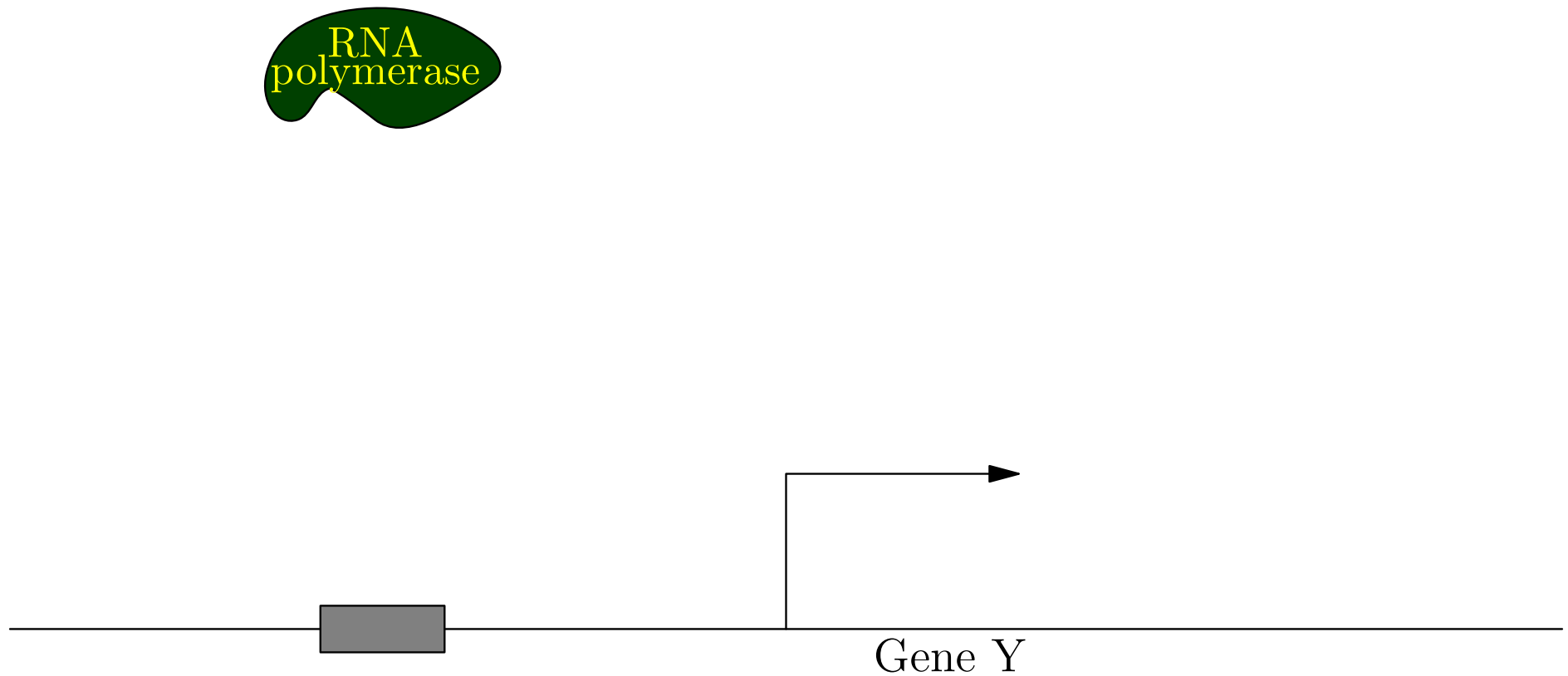
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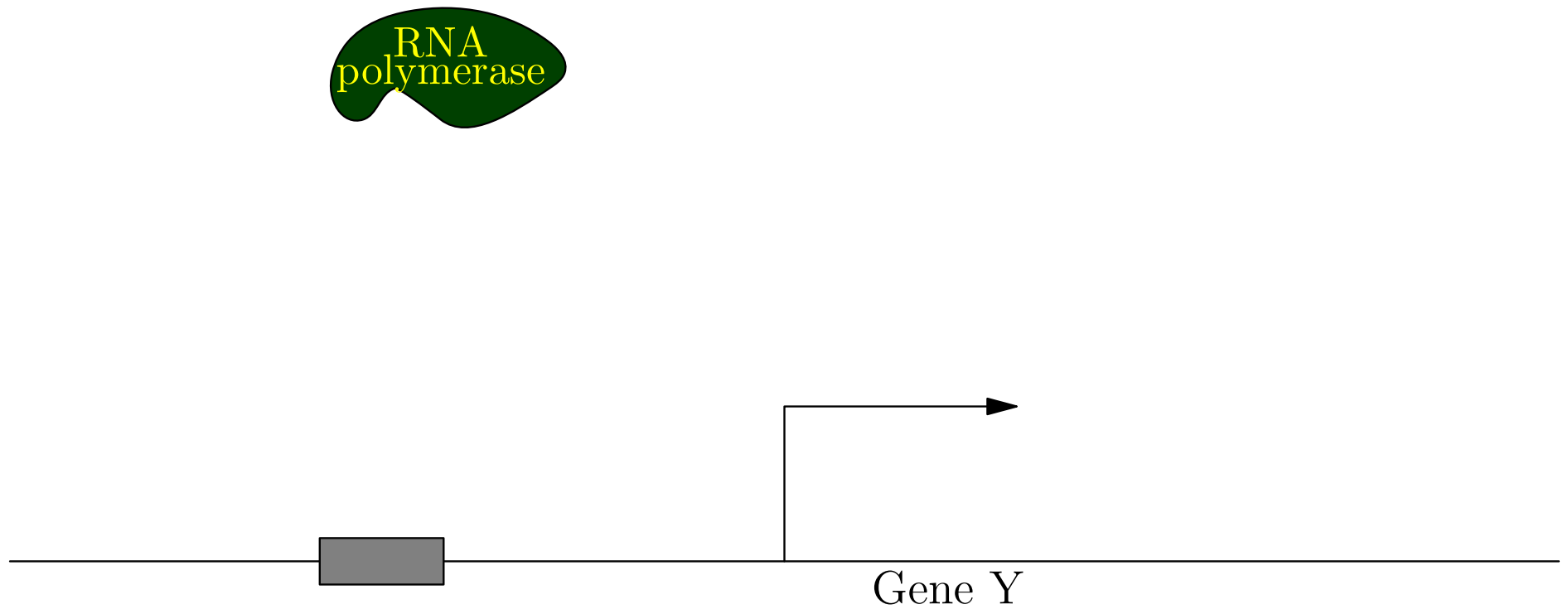
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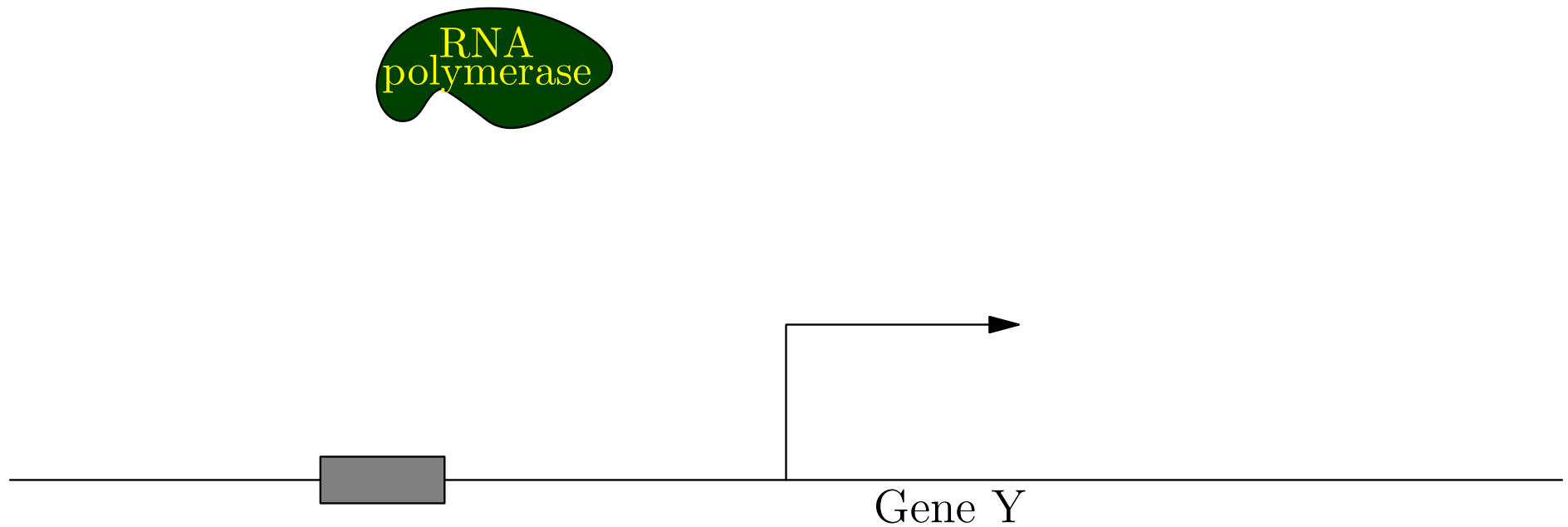
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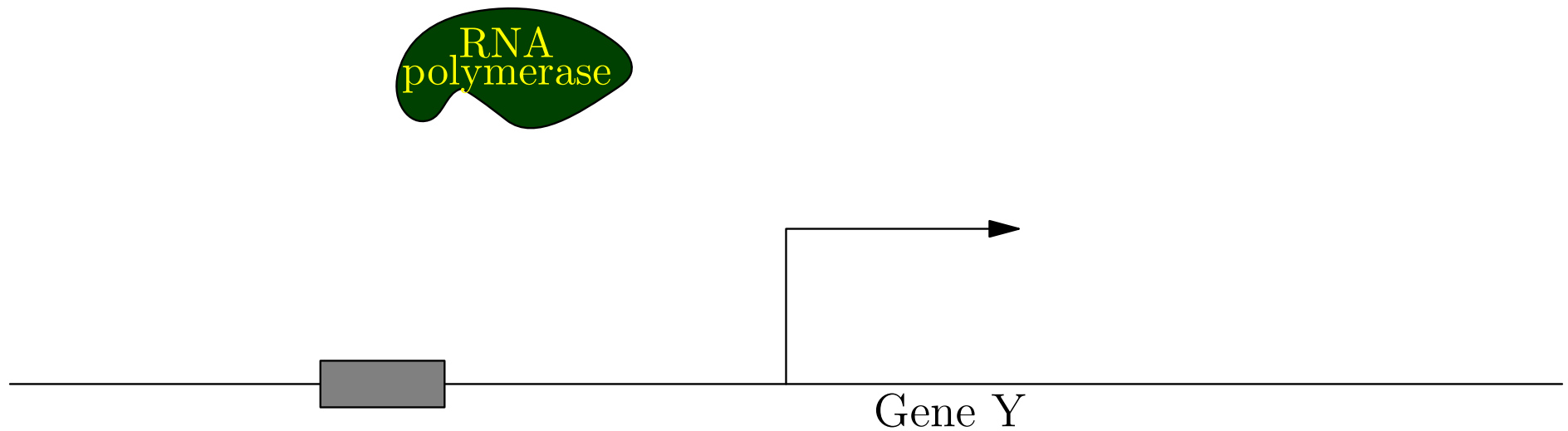
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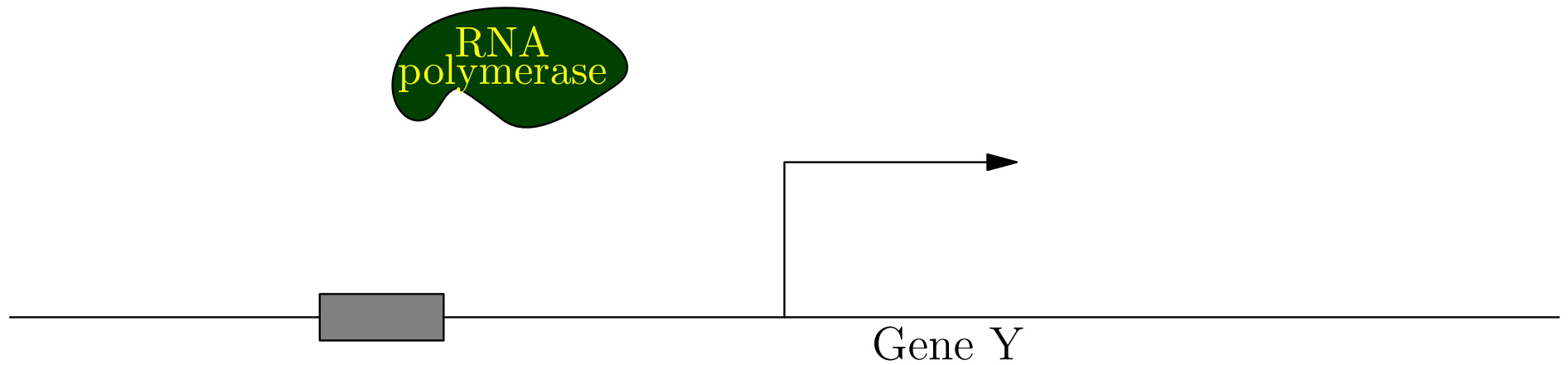
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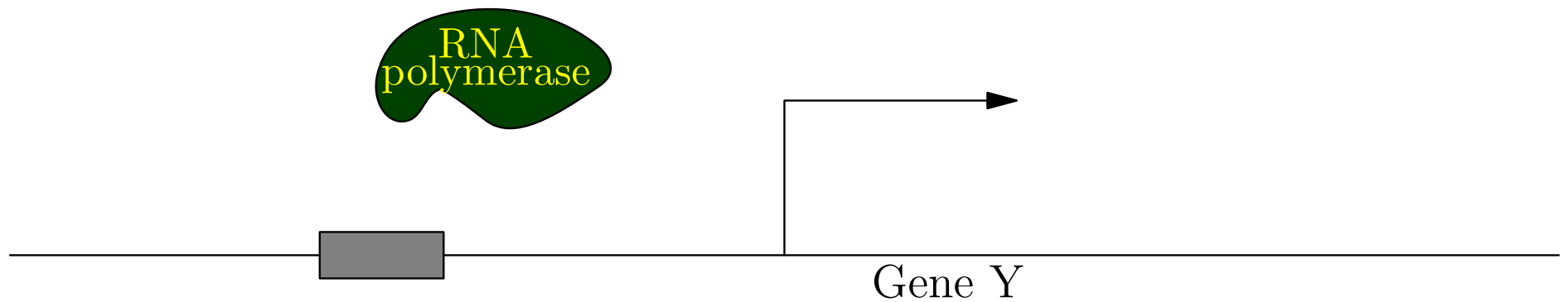
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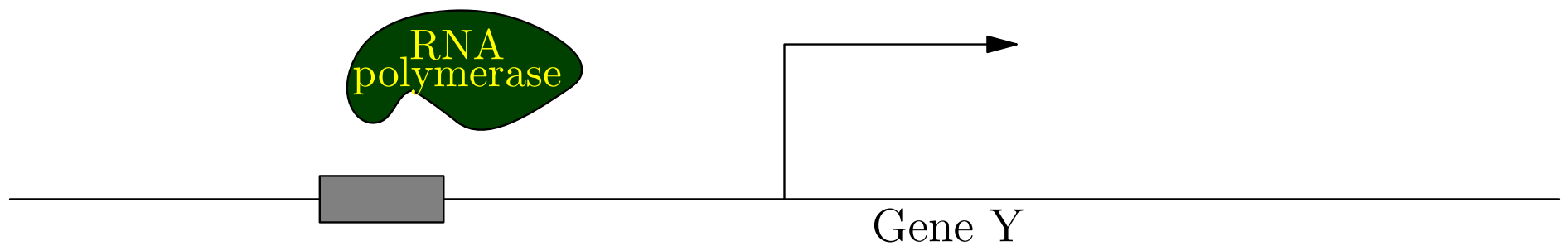
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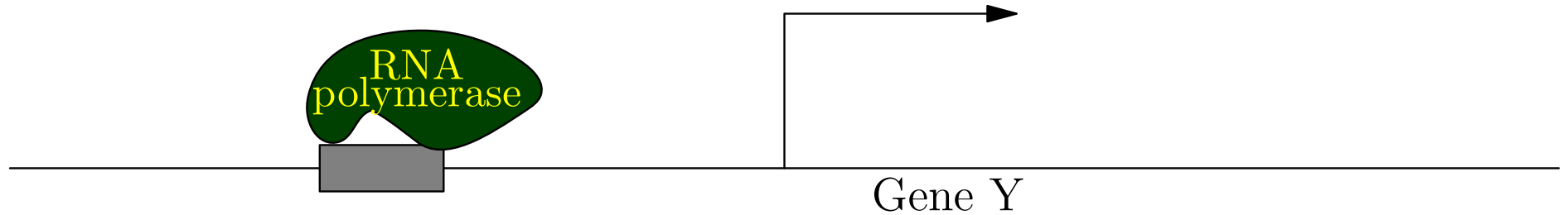
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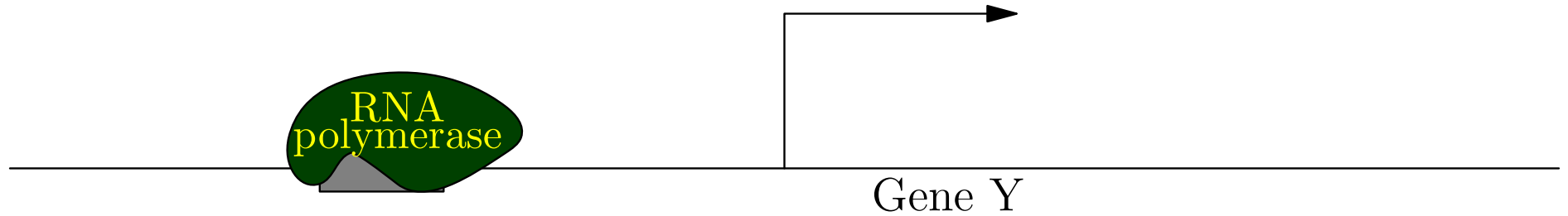
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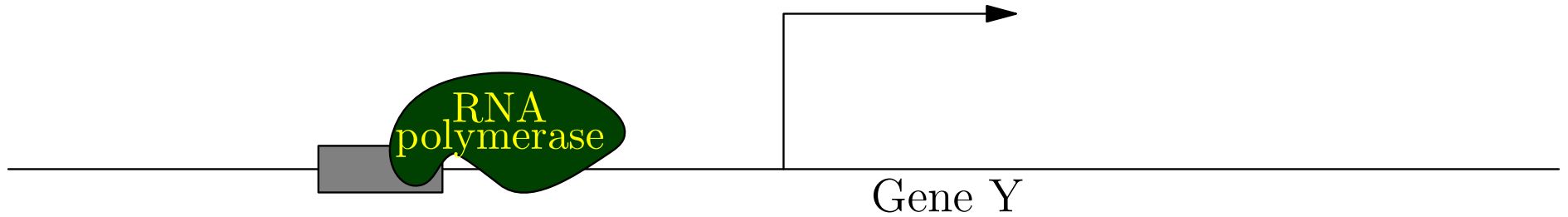
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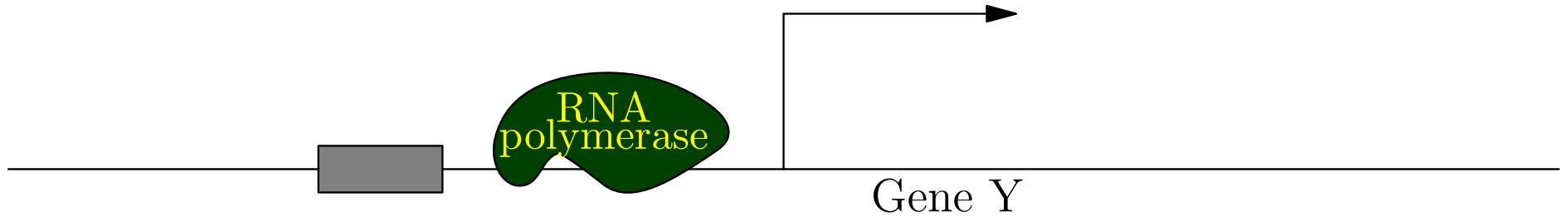
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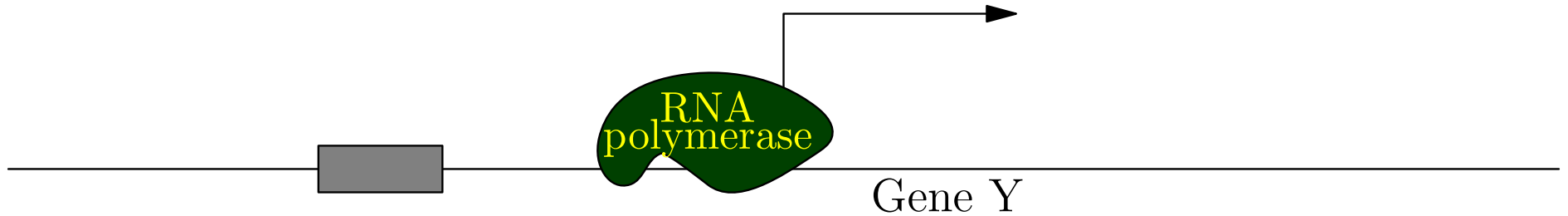
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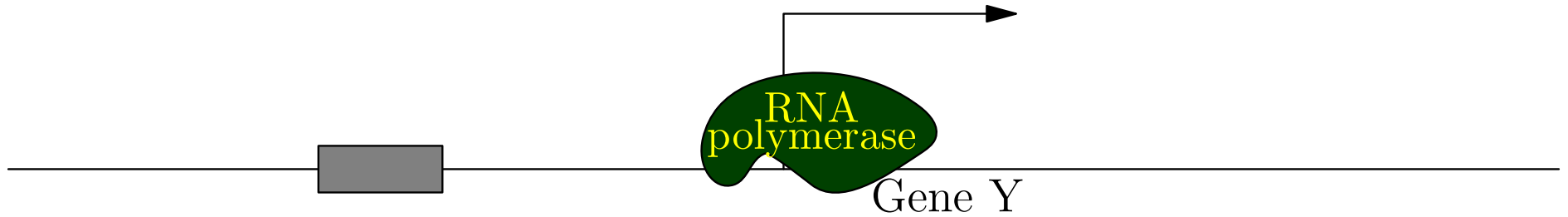
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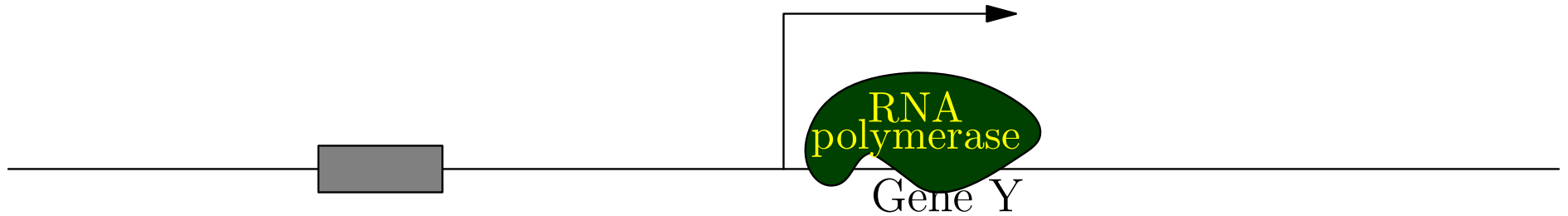
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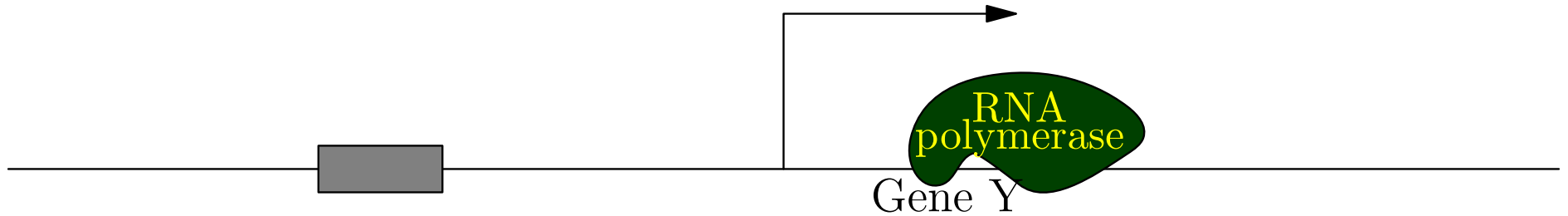
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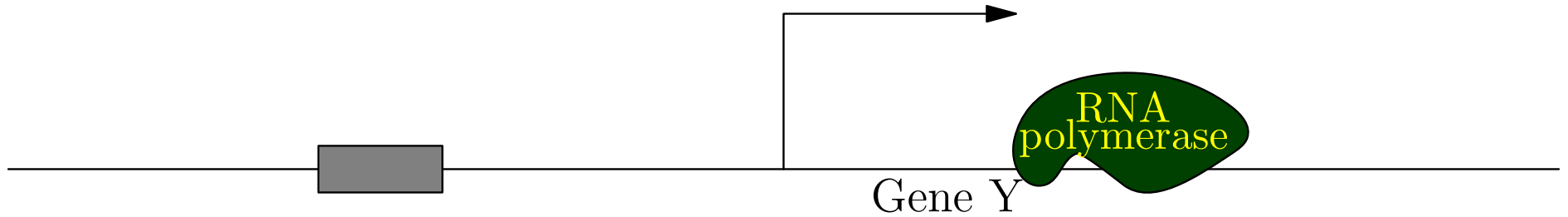
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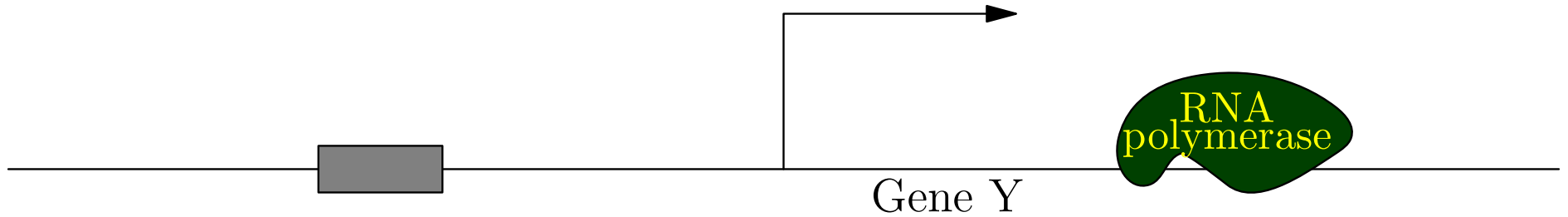
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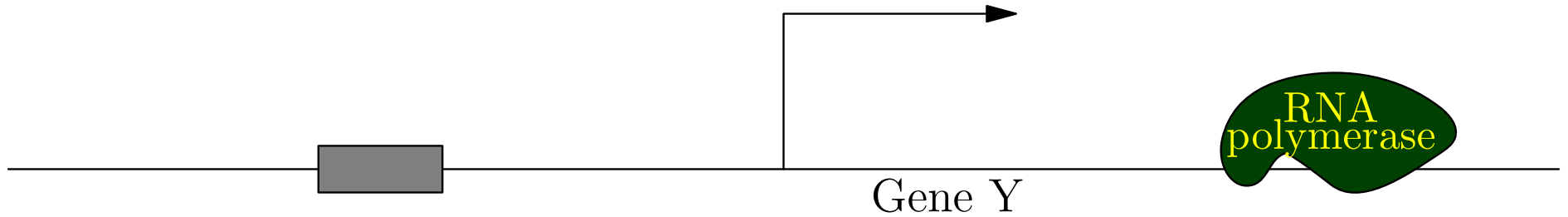
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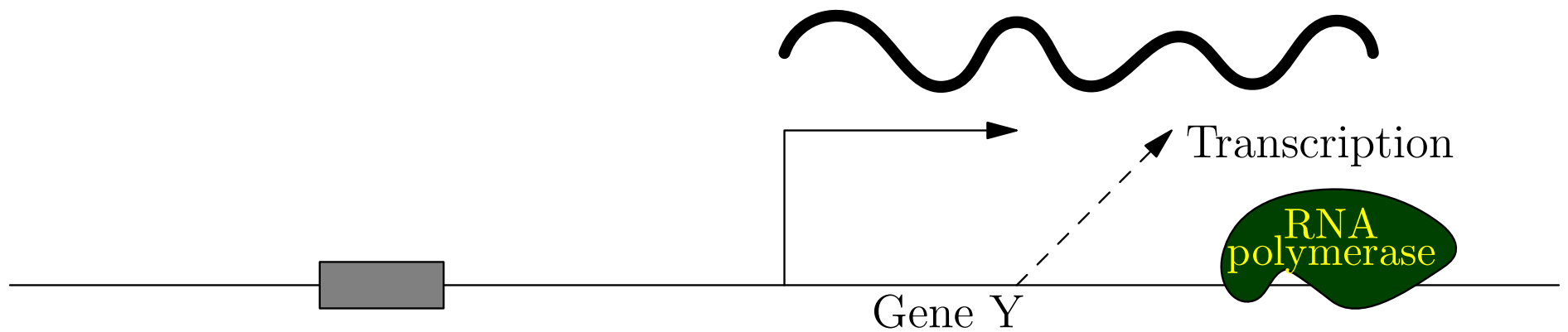
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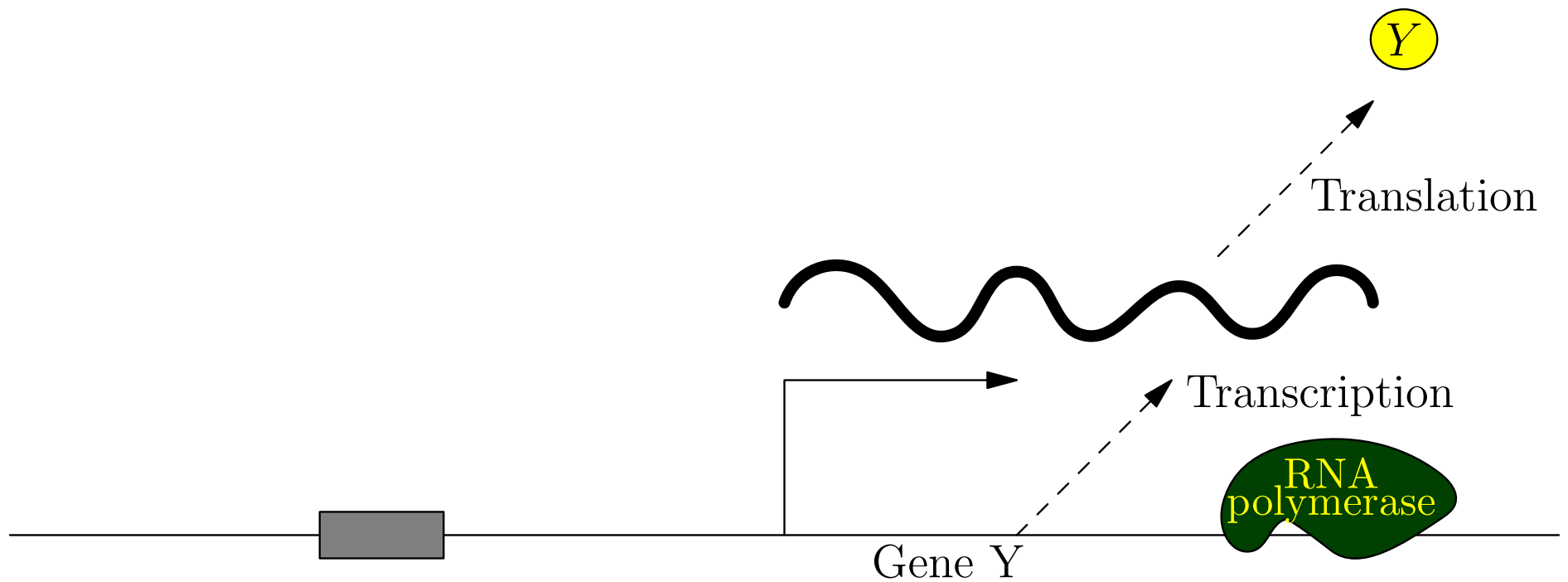
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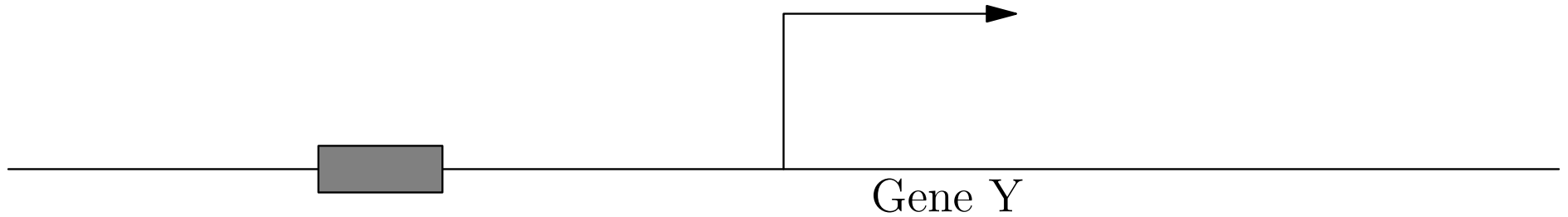
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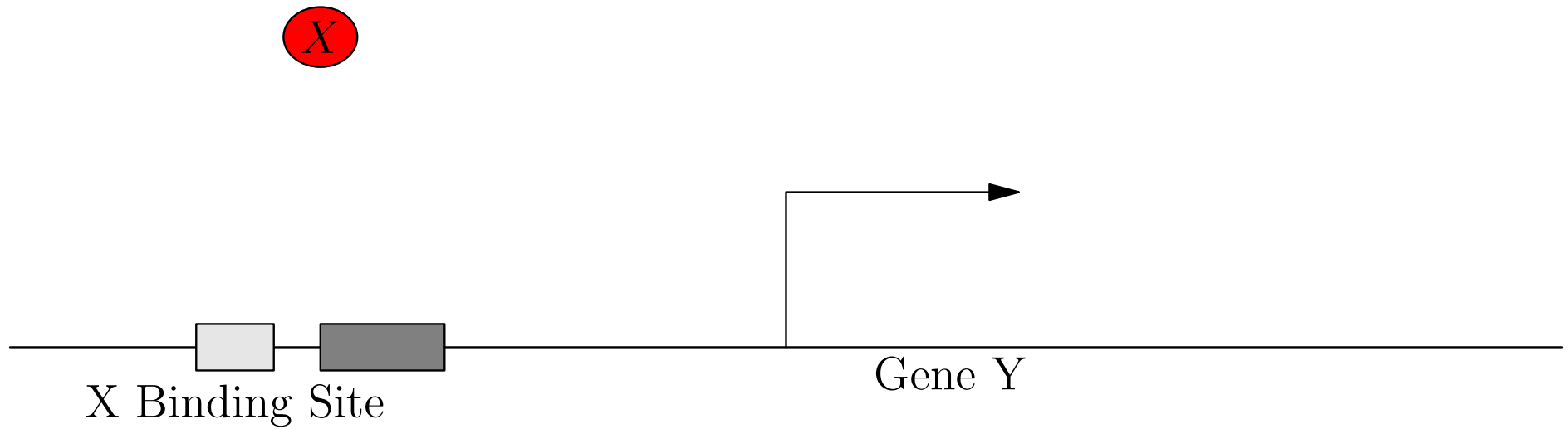
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Gene Transcription Regulation



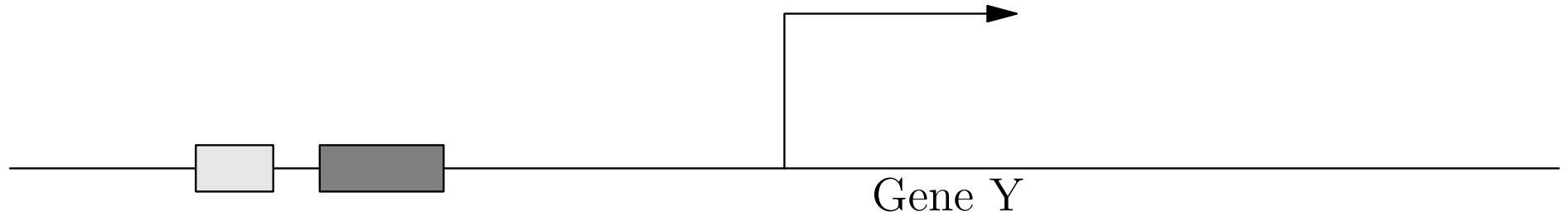
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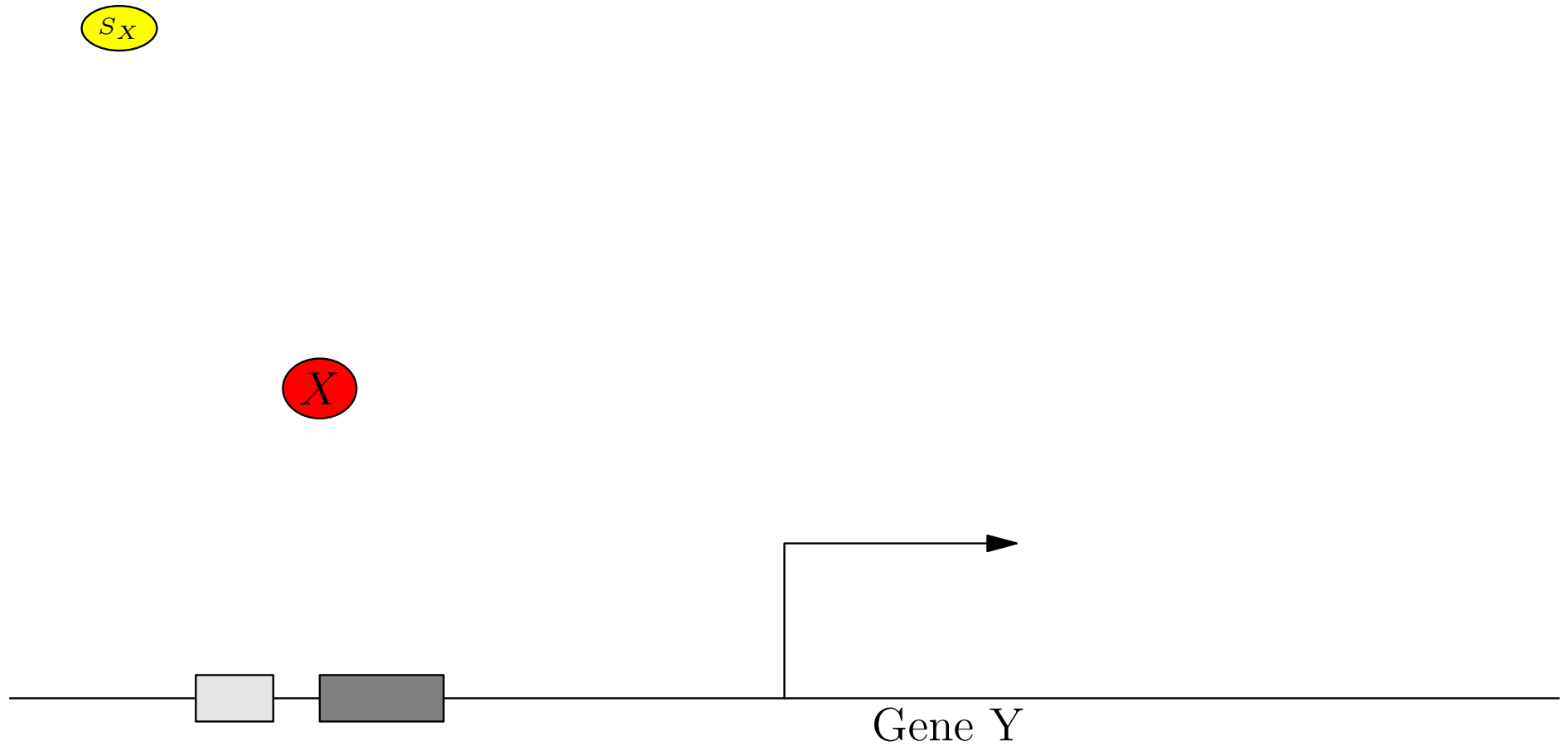
Gene Transcription Regulation

S_X

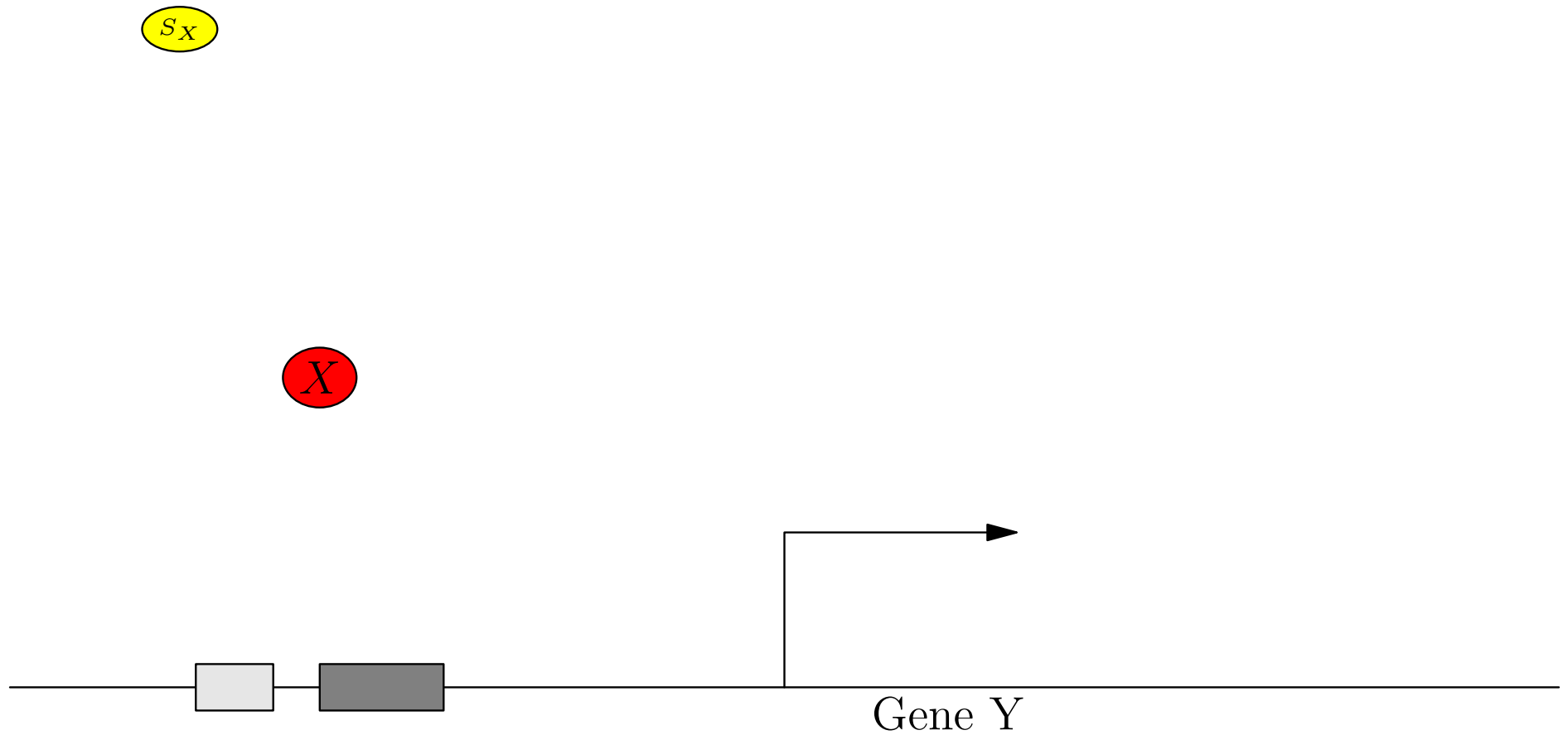
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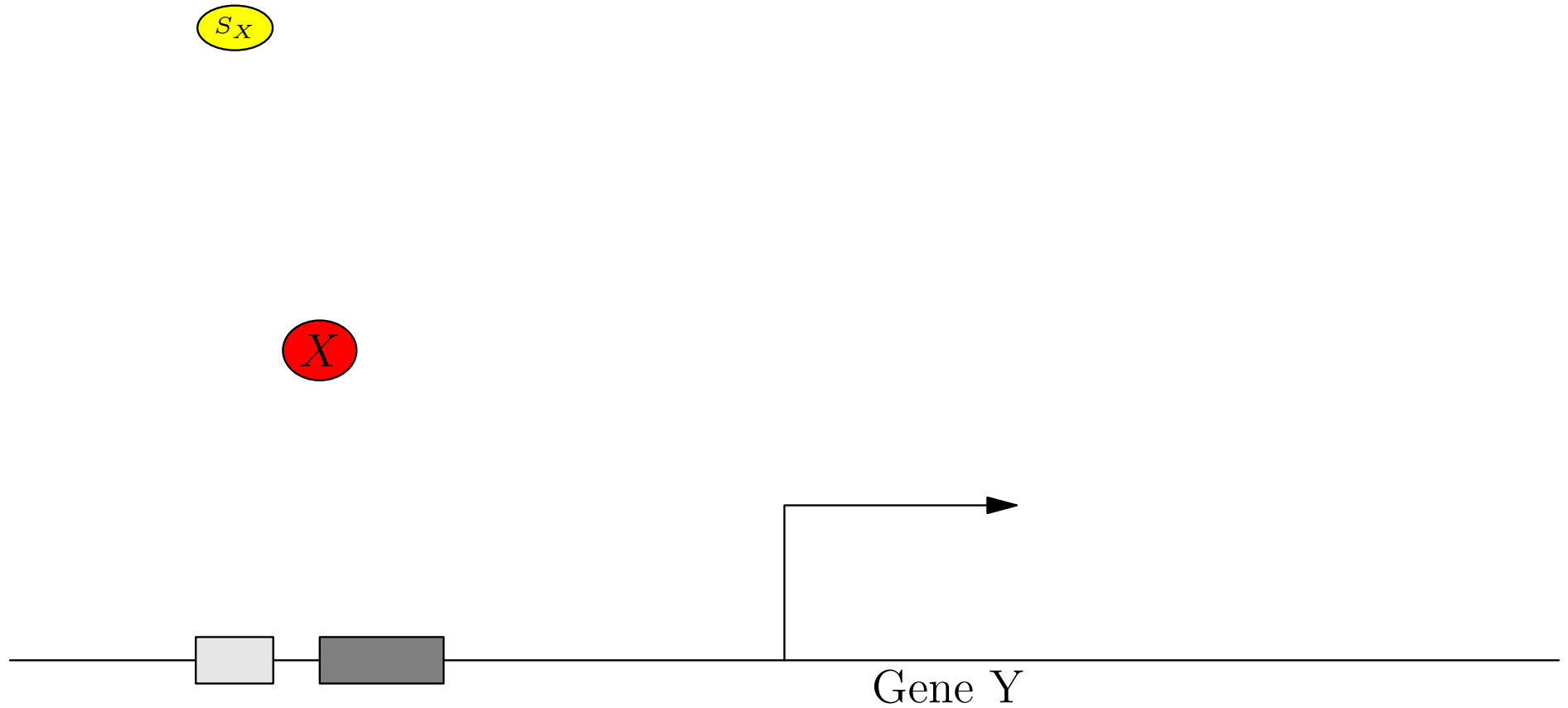
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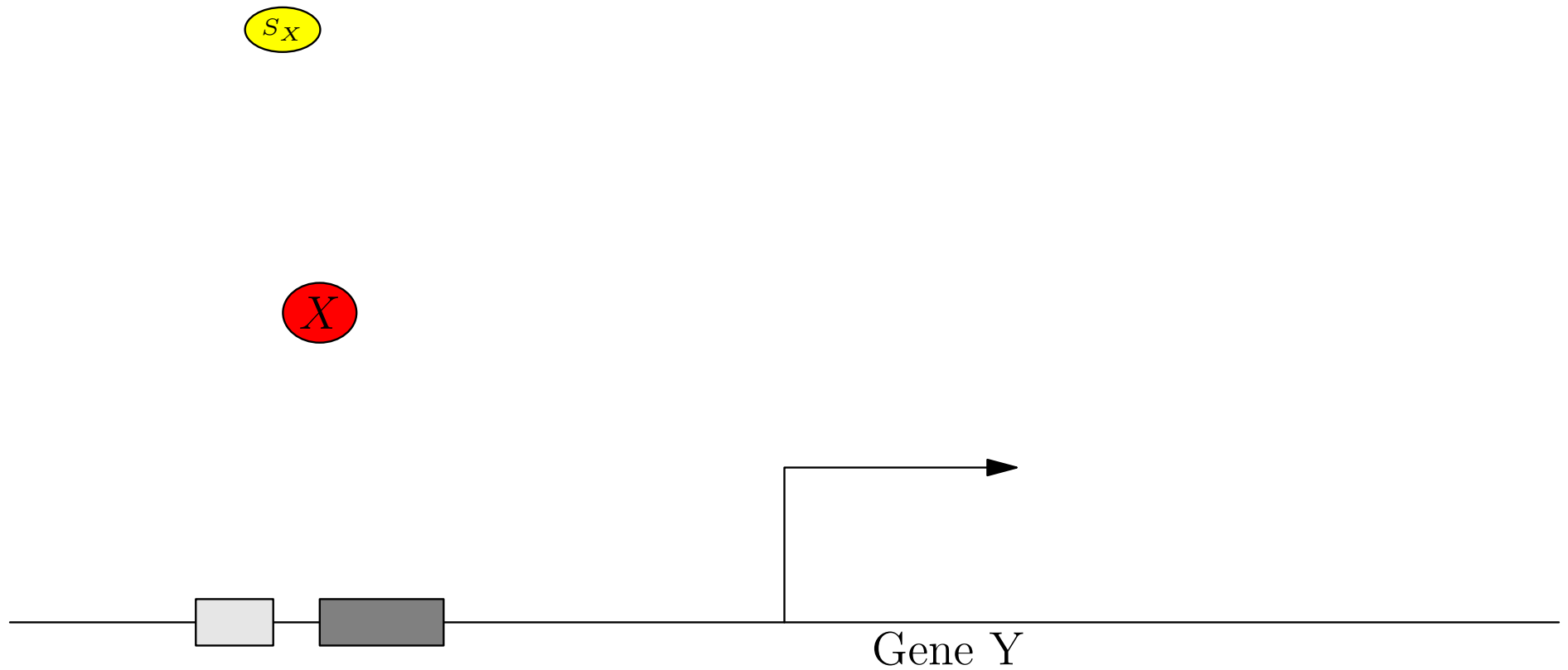
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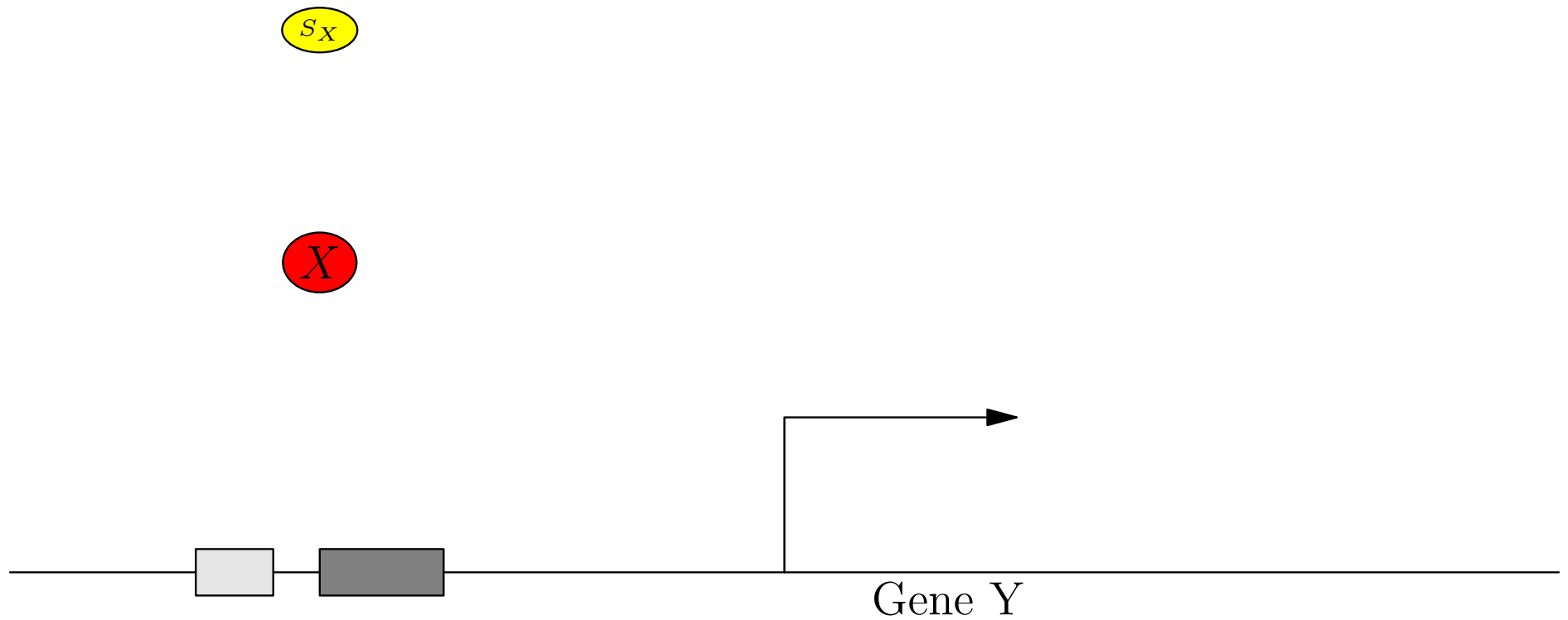
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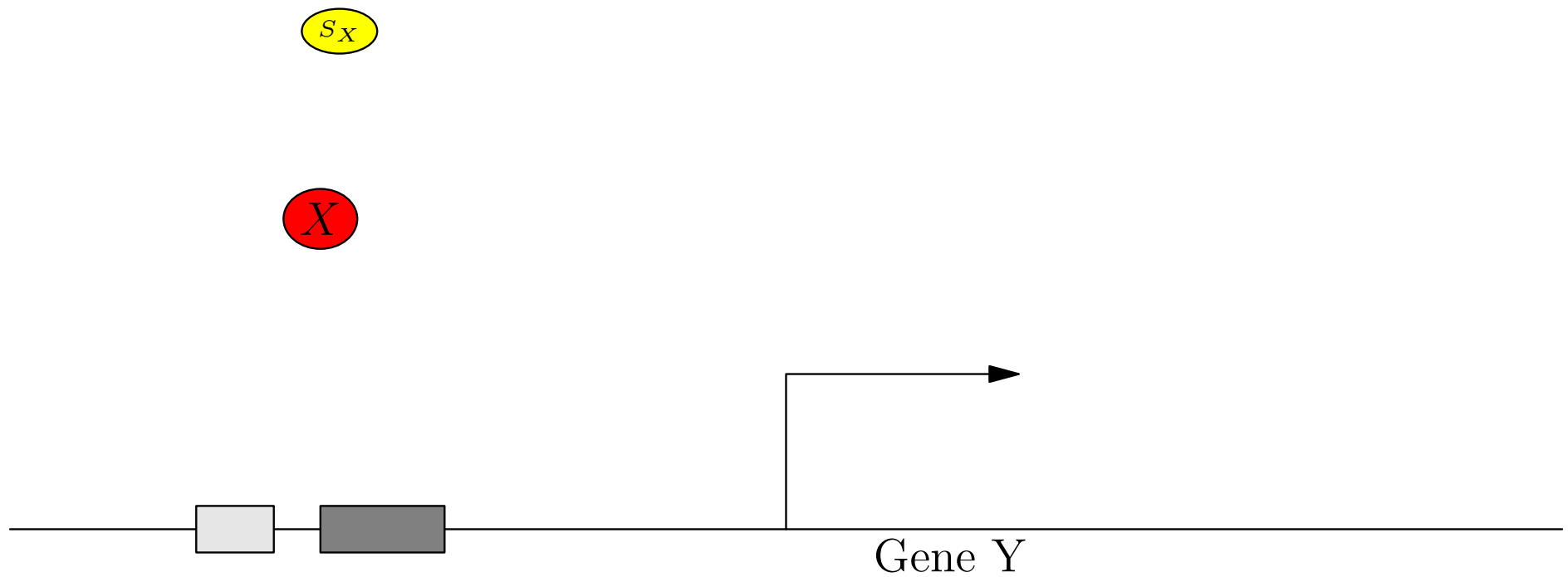
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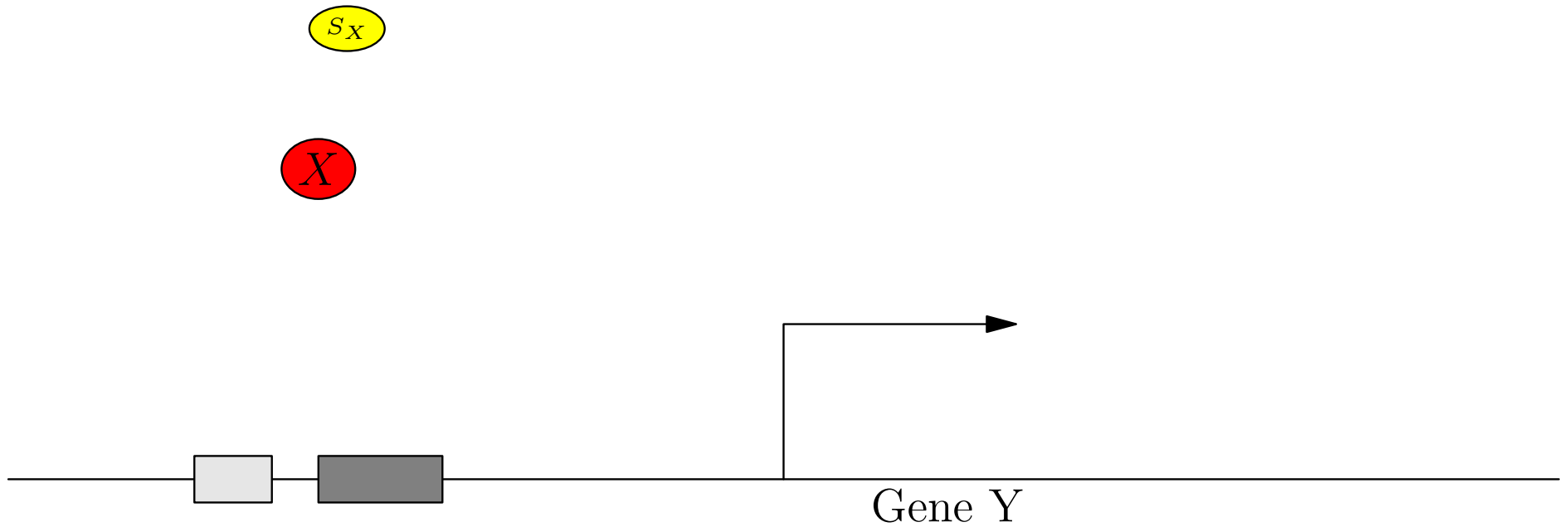
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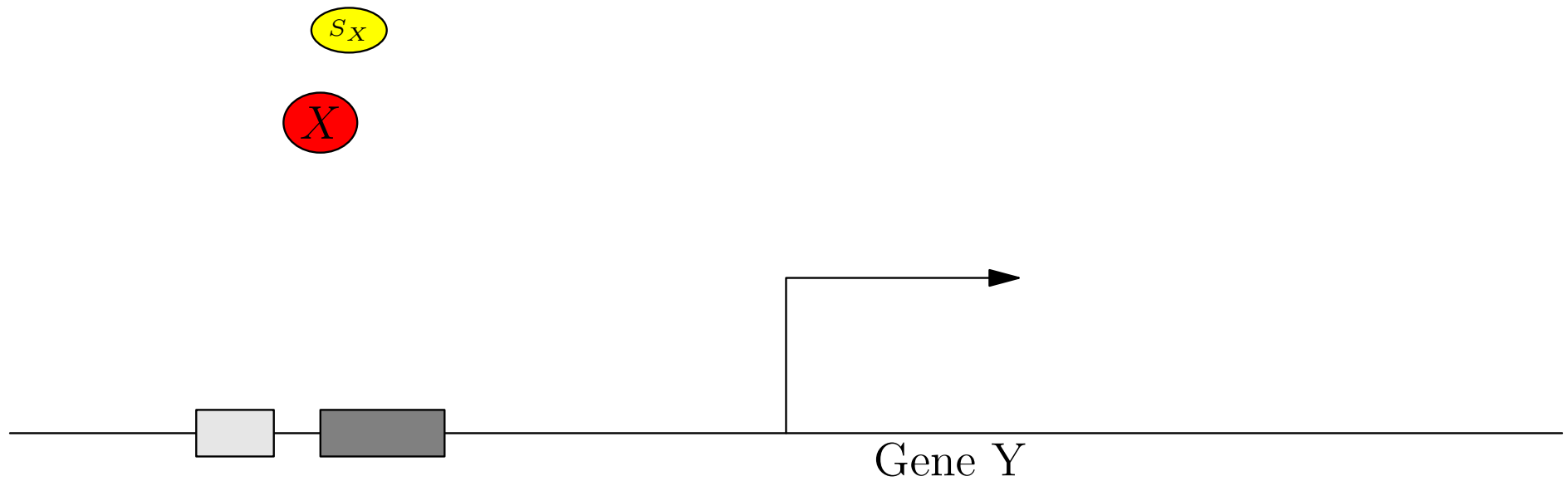
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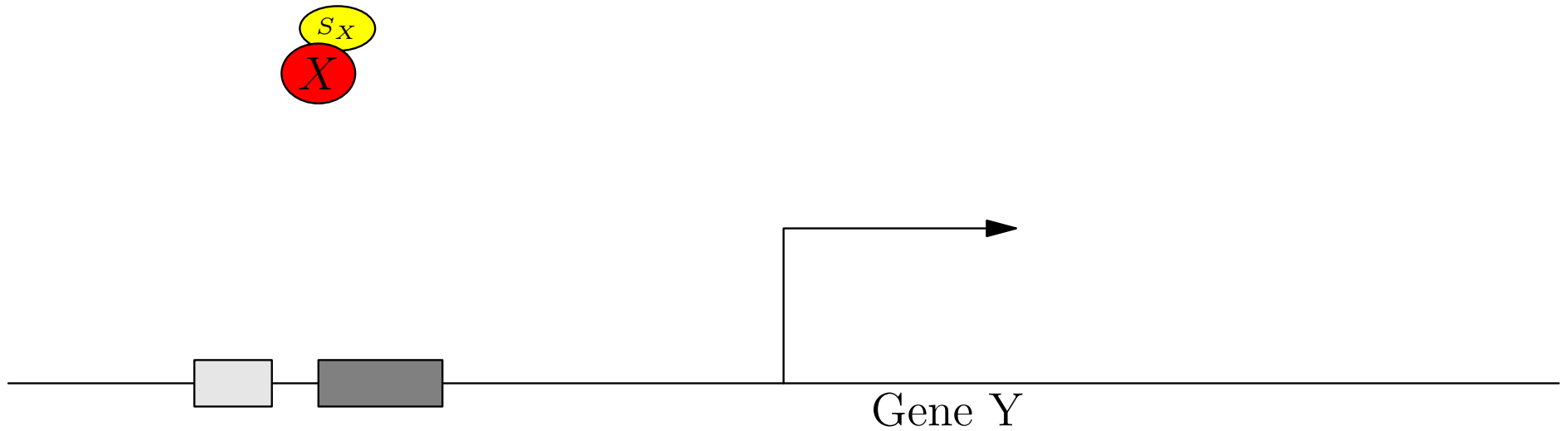
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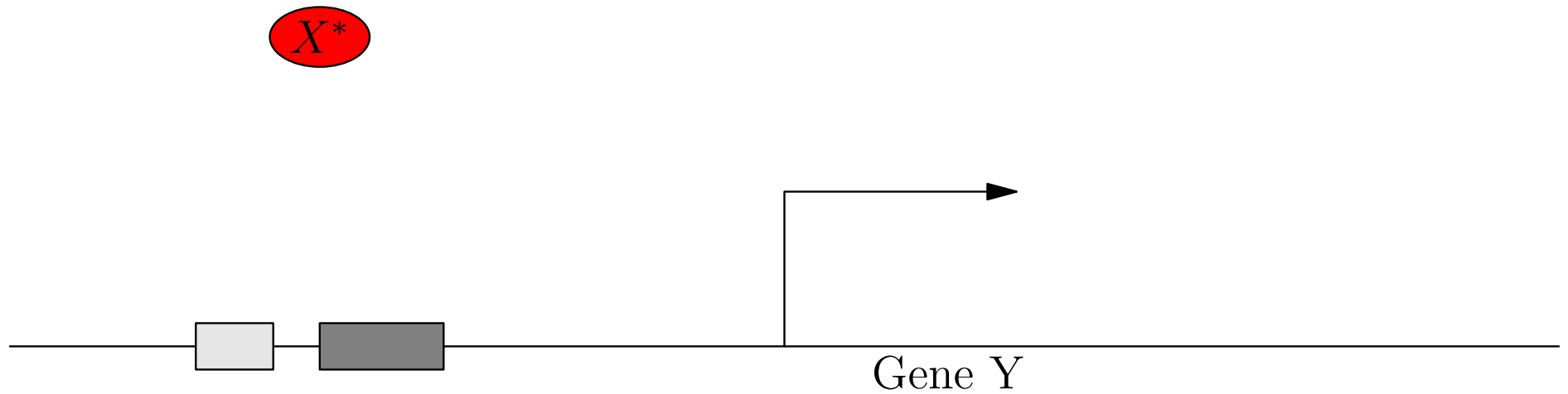
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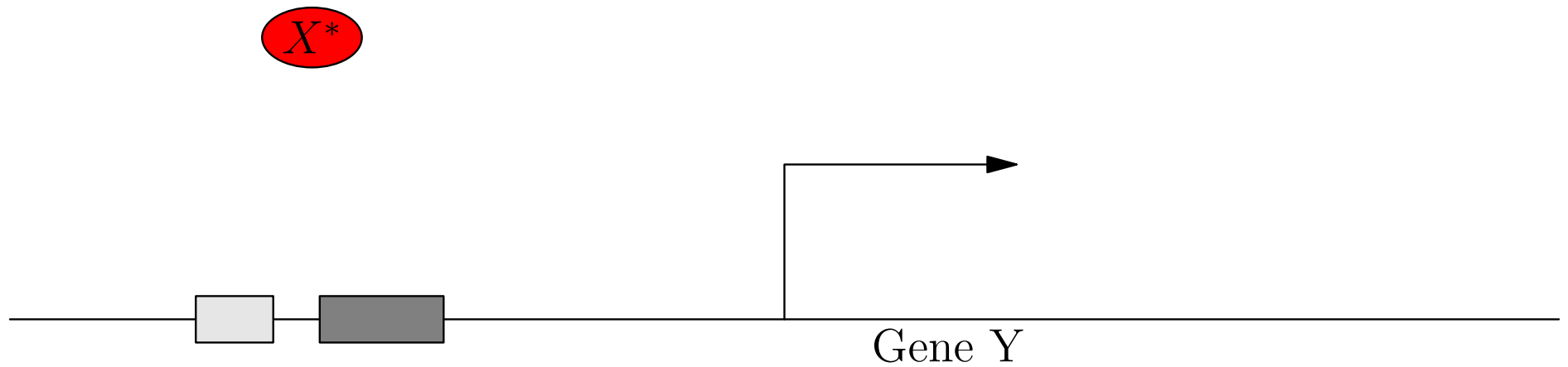
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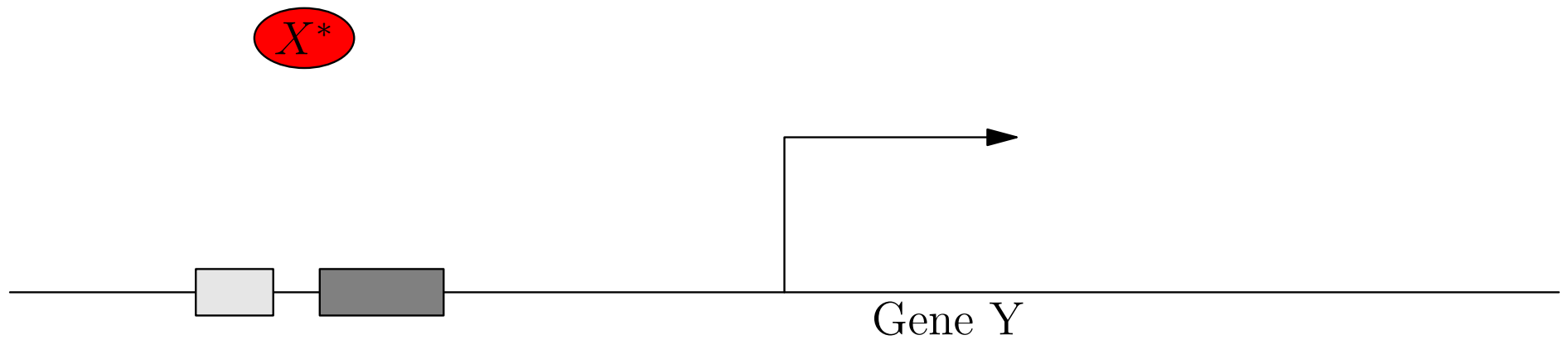
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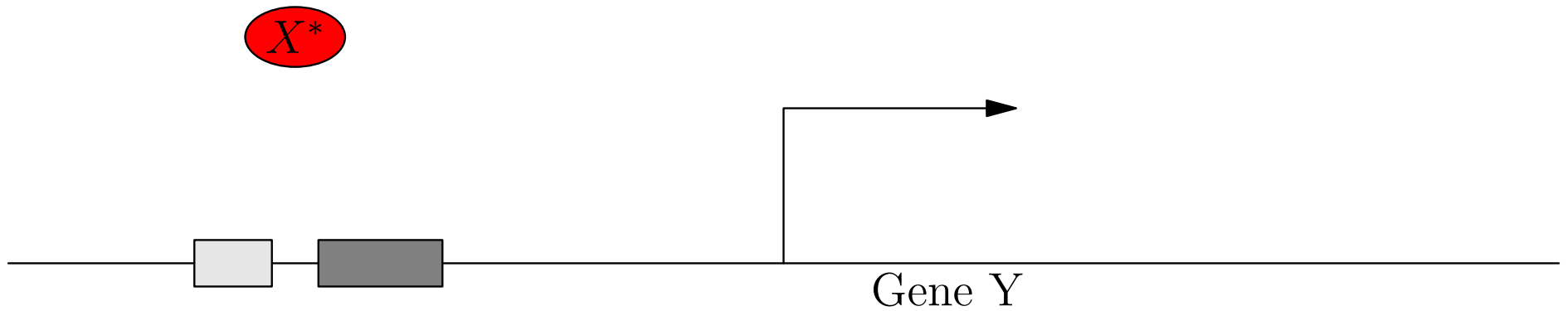
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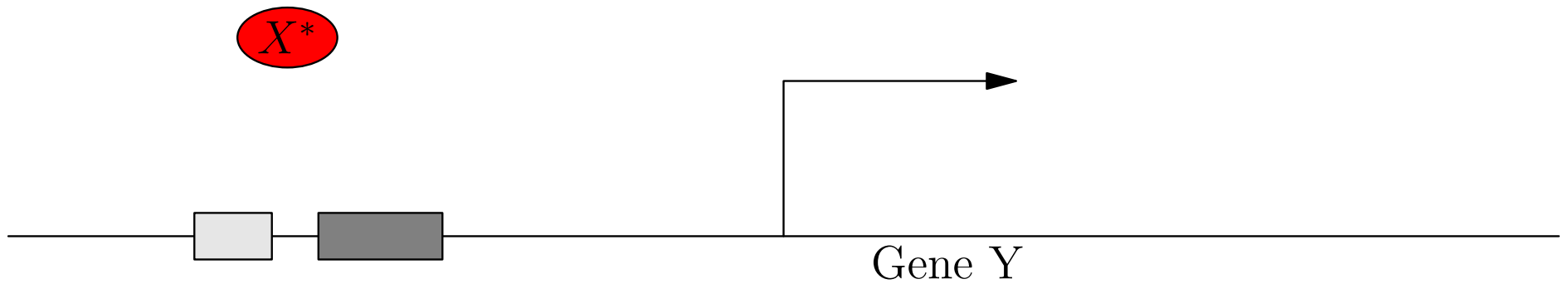
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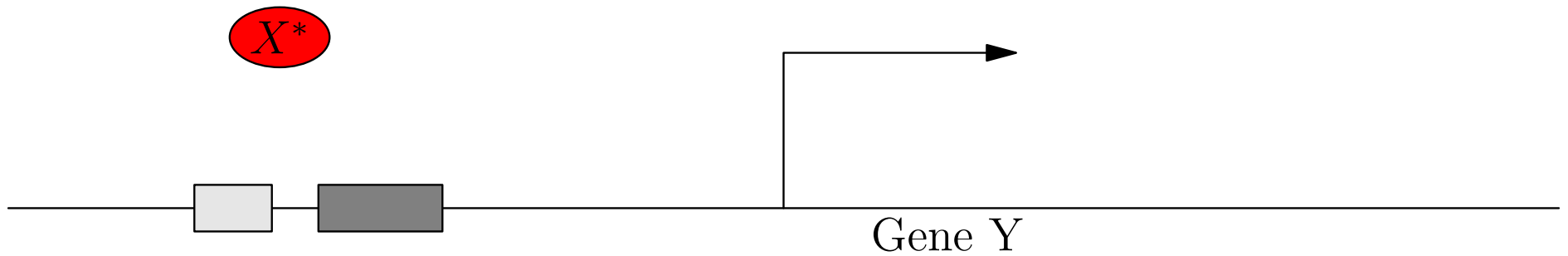
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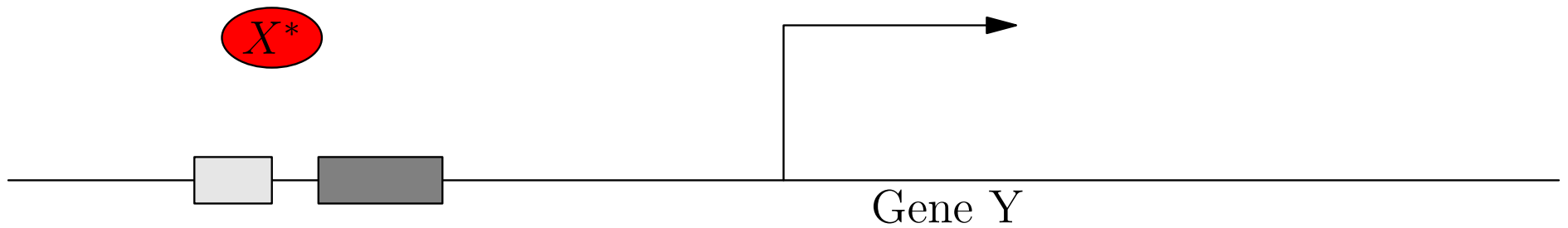
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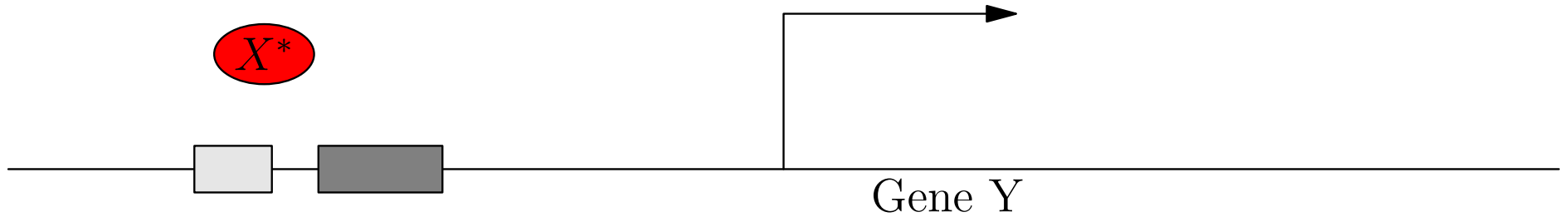
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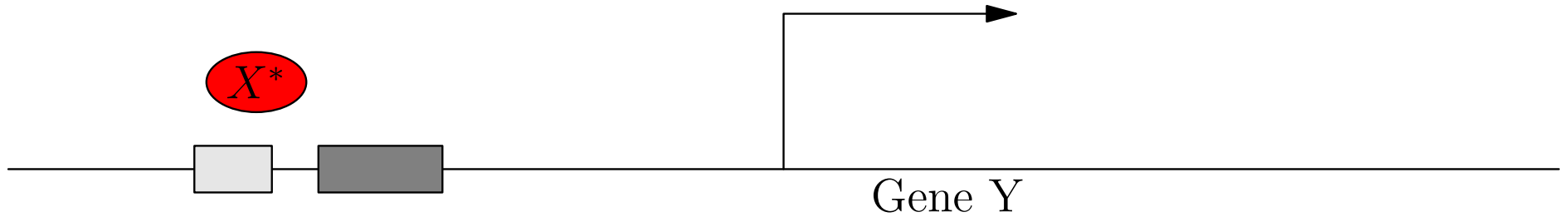
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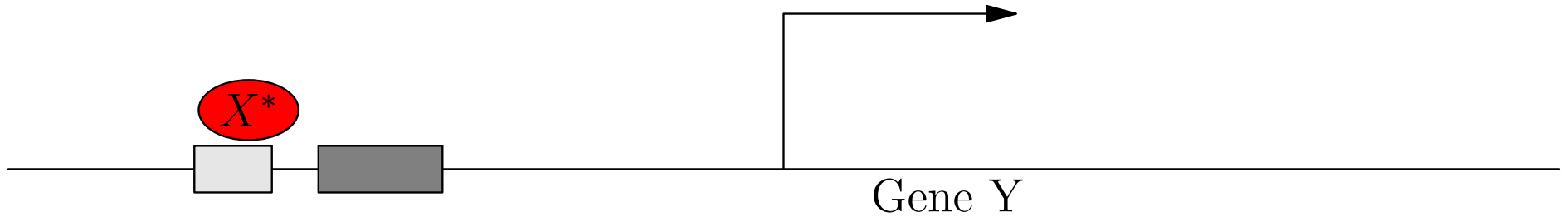
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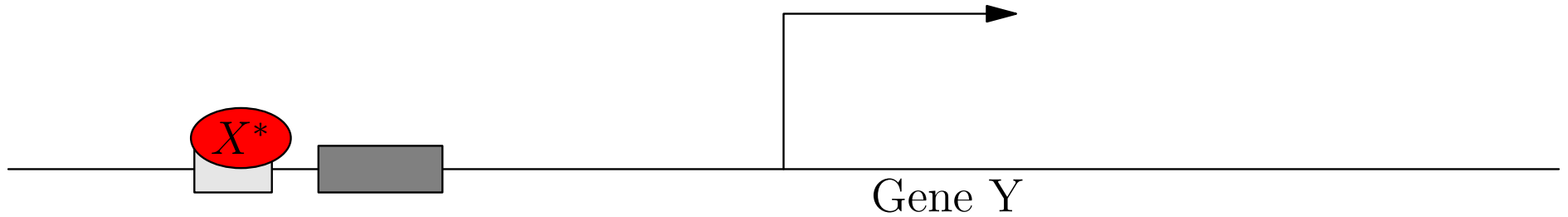
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Gene Transcription Regulation

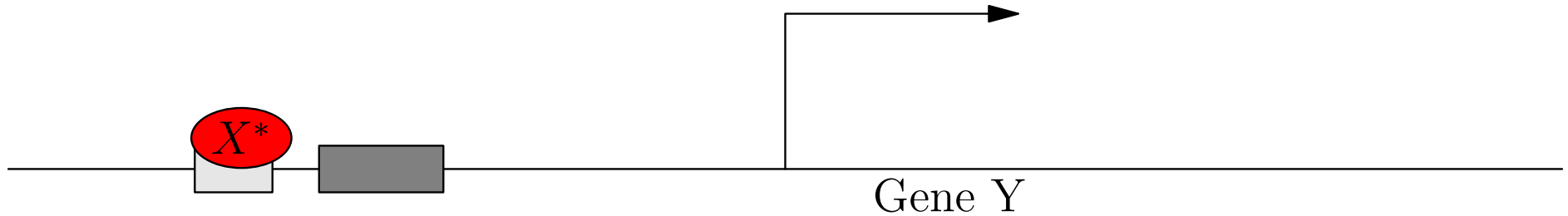
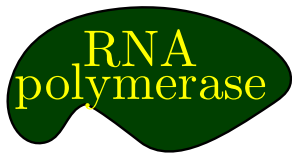


Gene Transcription Regulation



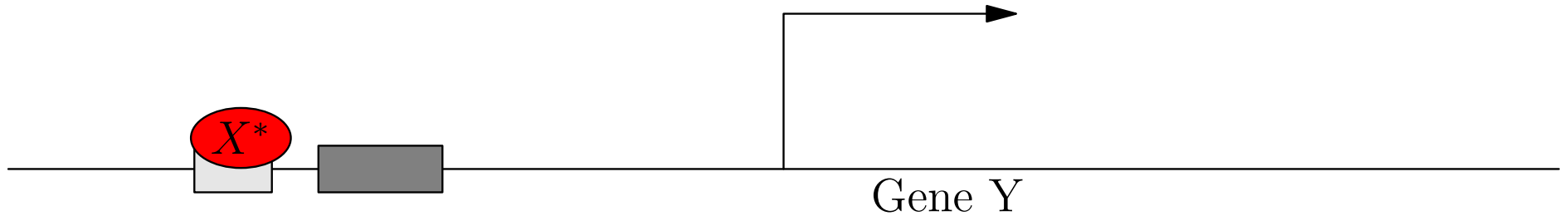
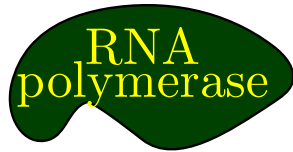
Gene Transcription Regulation

Excititory Transcription Factor



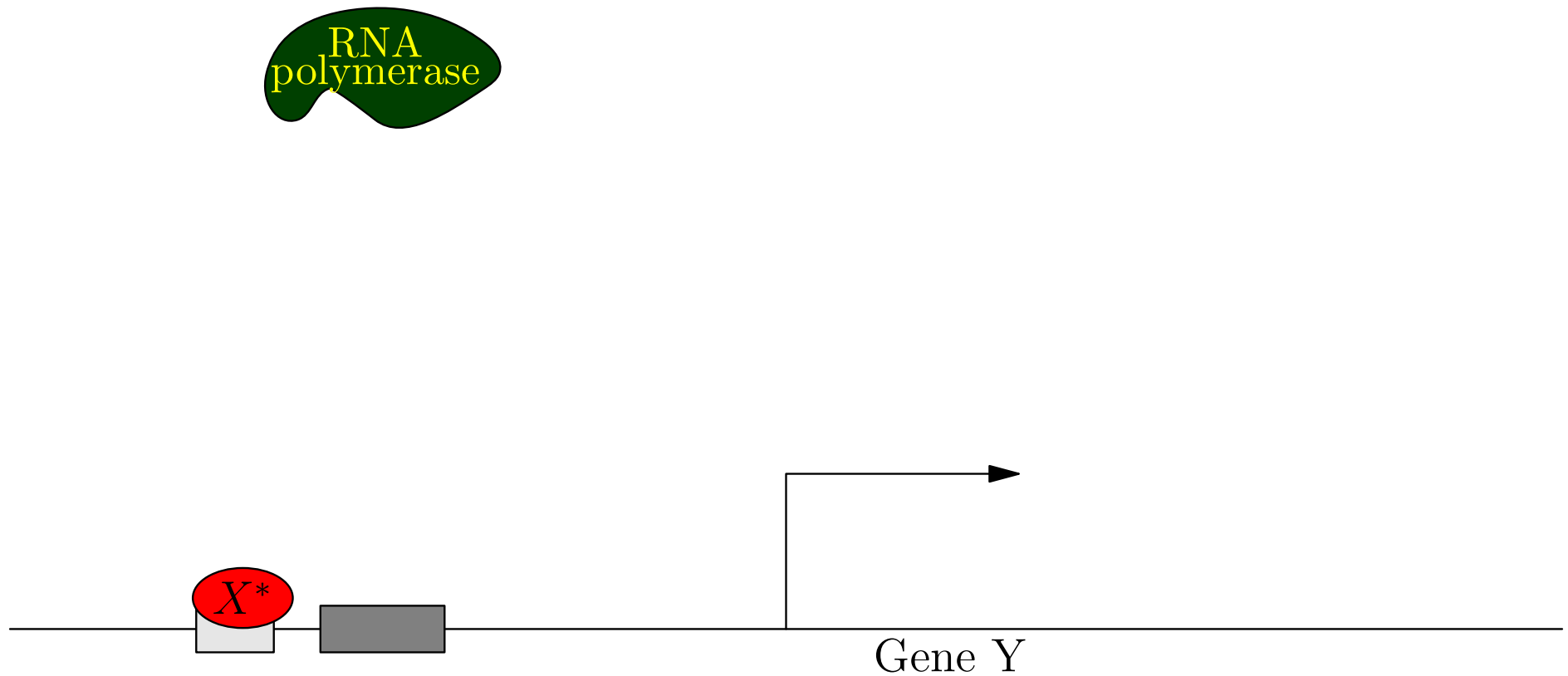
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Excititory Transcription Factor



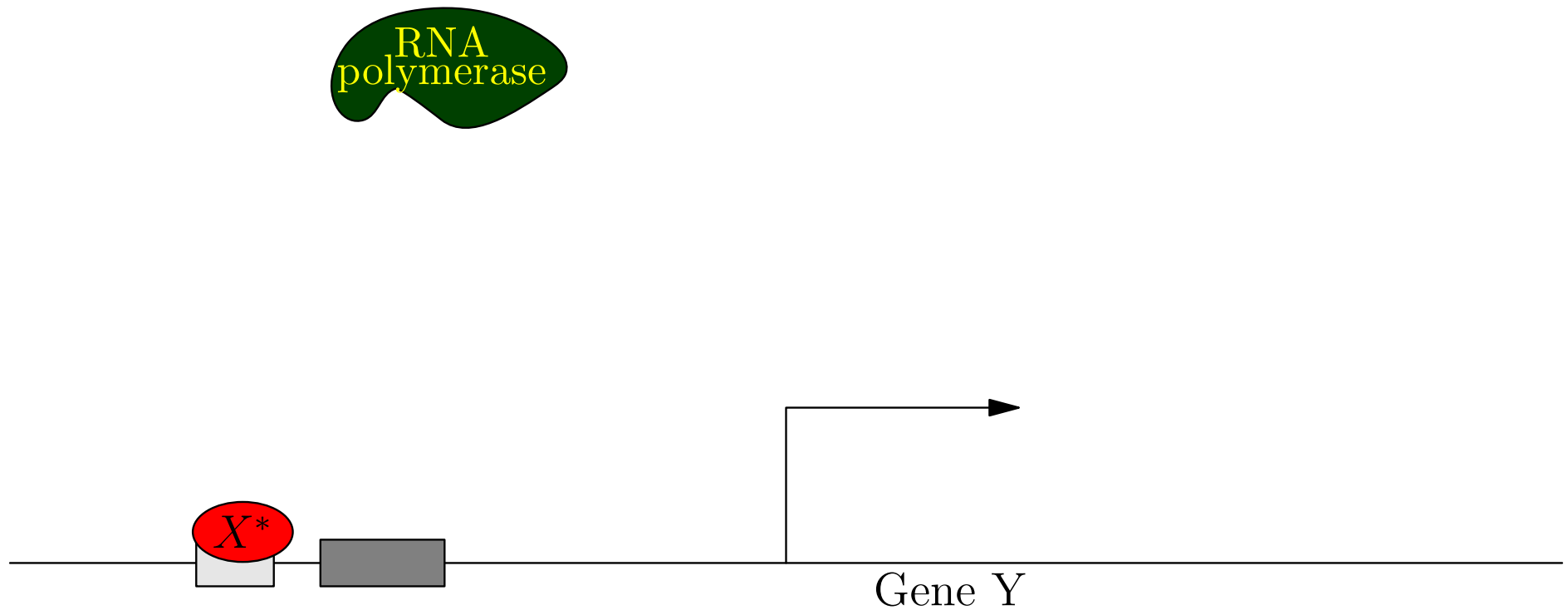
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Excititory Transcription Factor



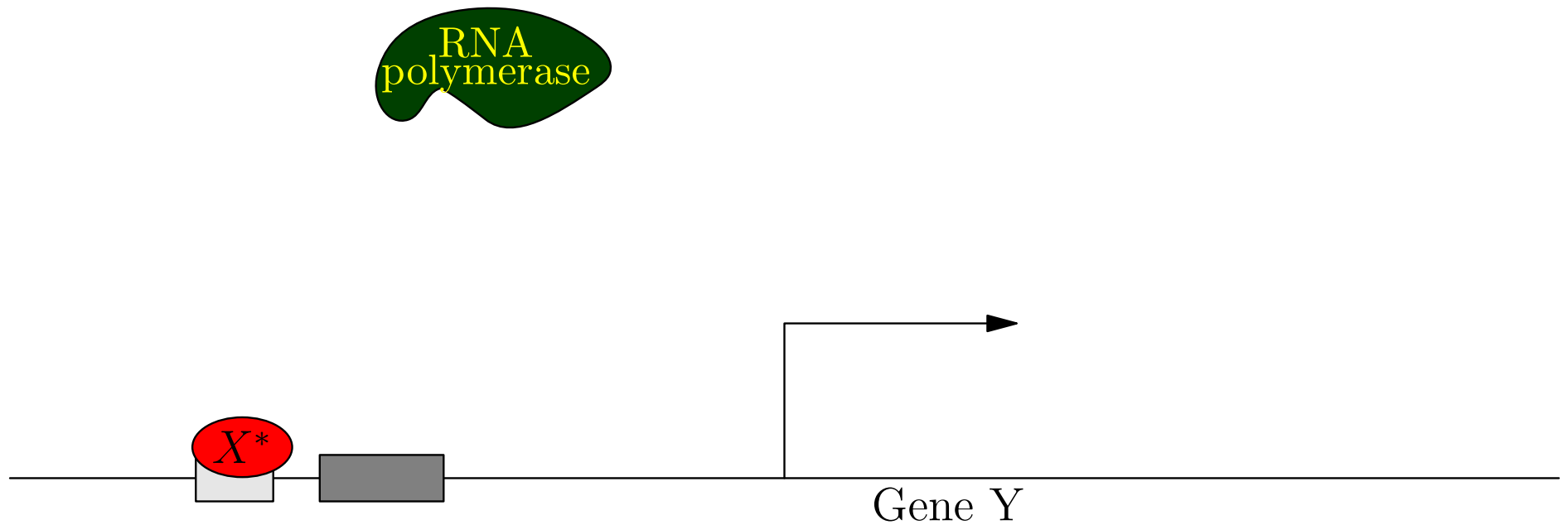
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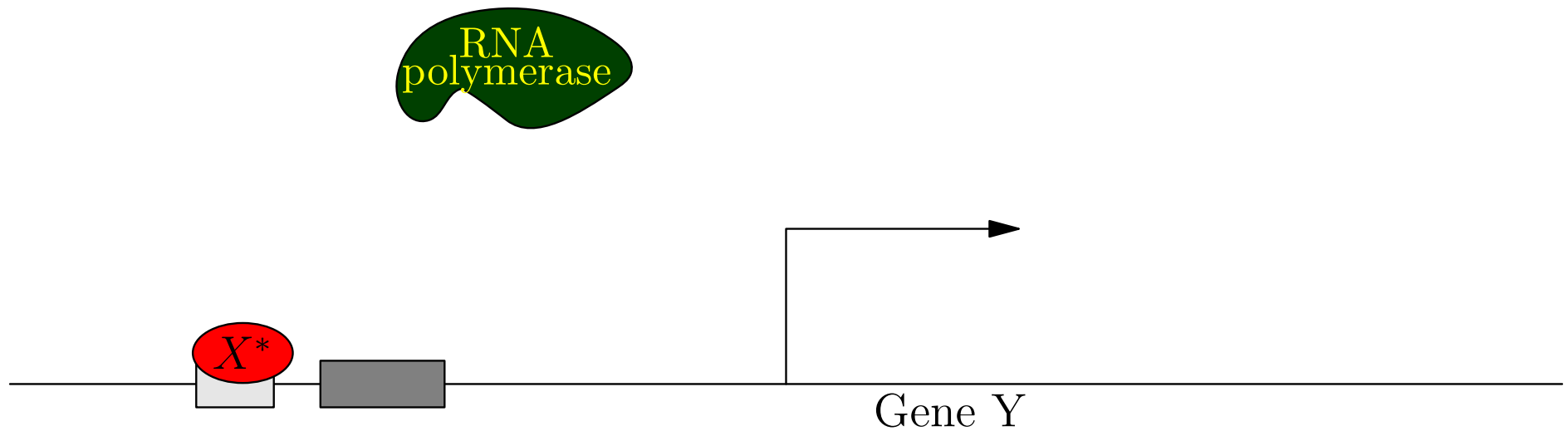
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Excititory Transcription Factor



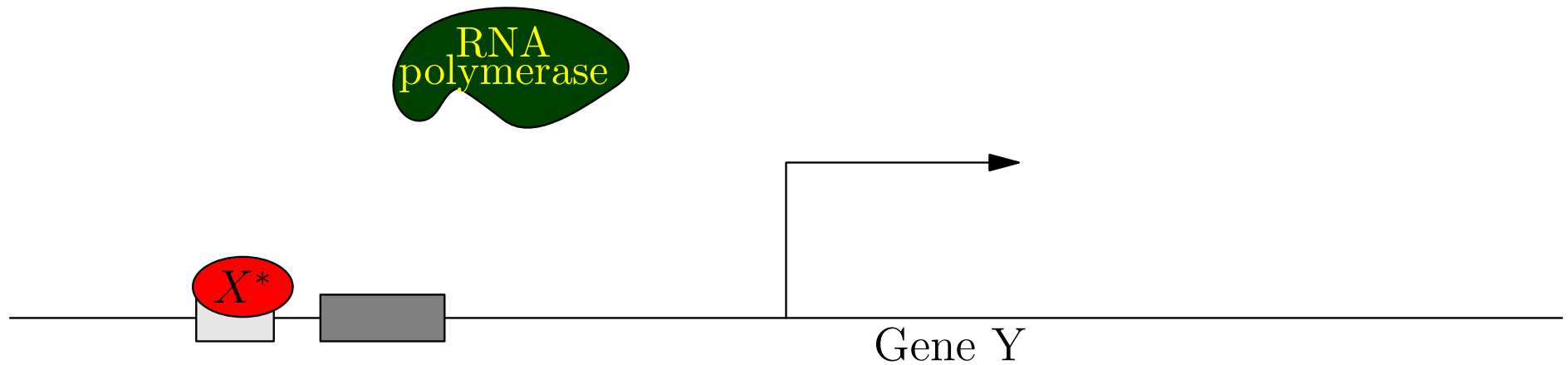
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Excititory Transcription Factor



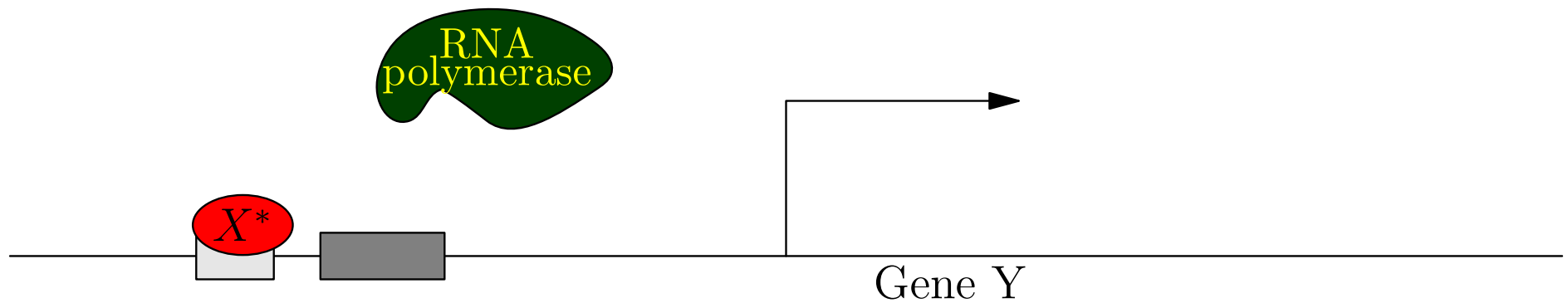
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Excititory Transcription Factor



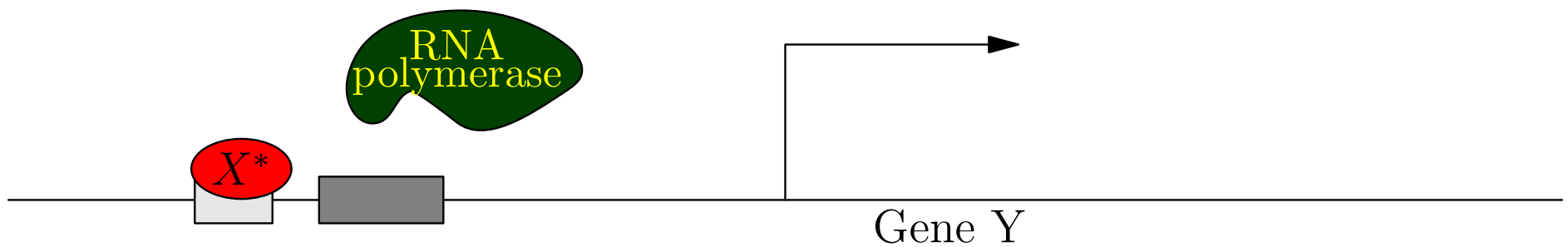
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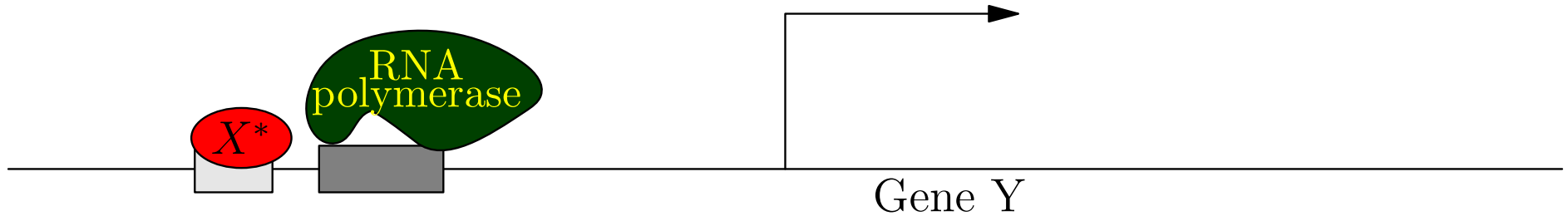
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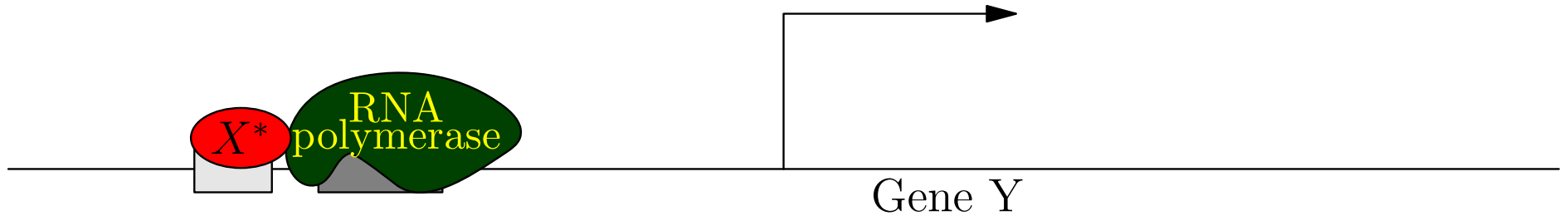
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Excititory Transcription Factor



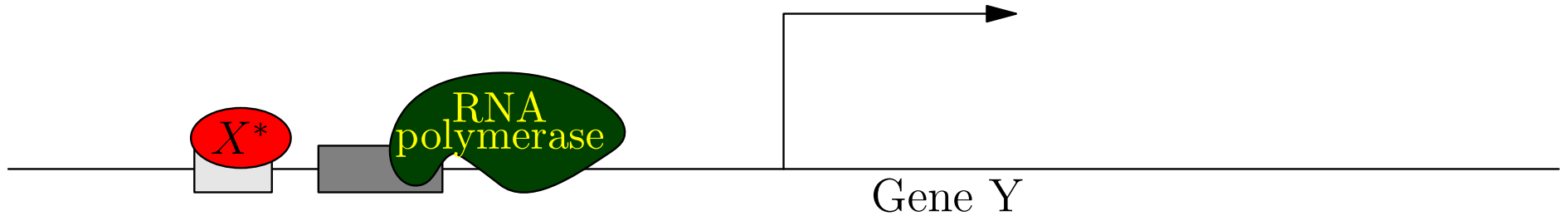
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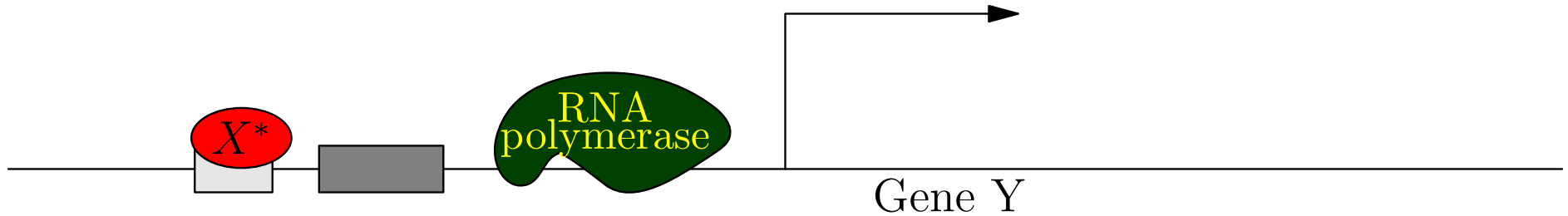
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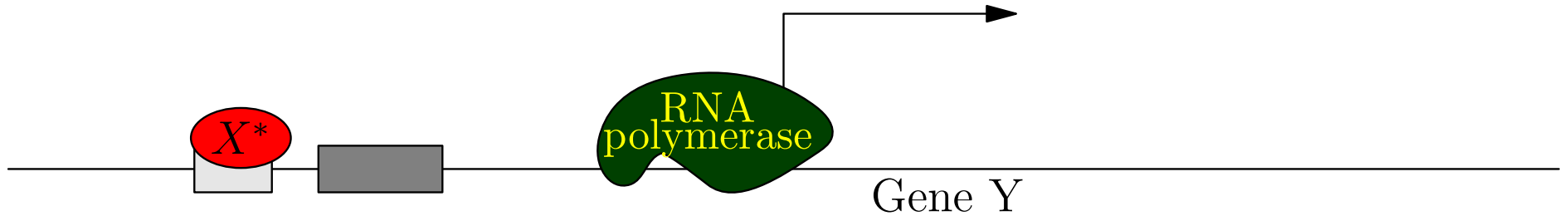
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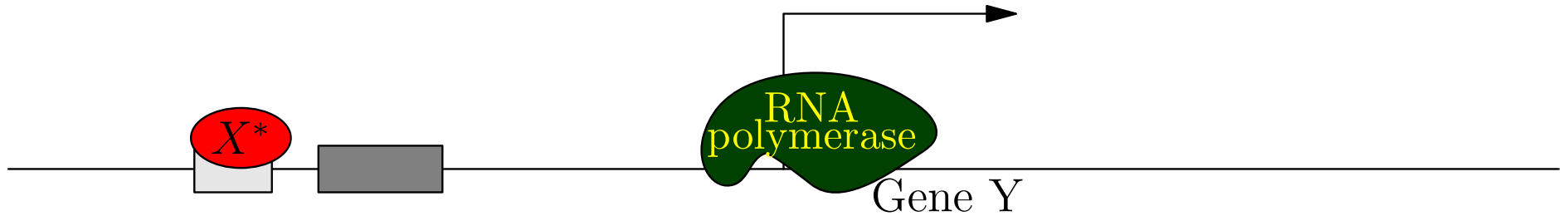
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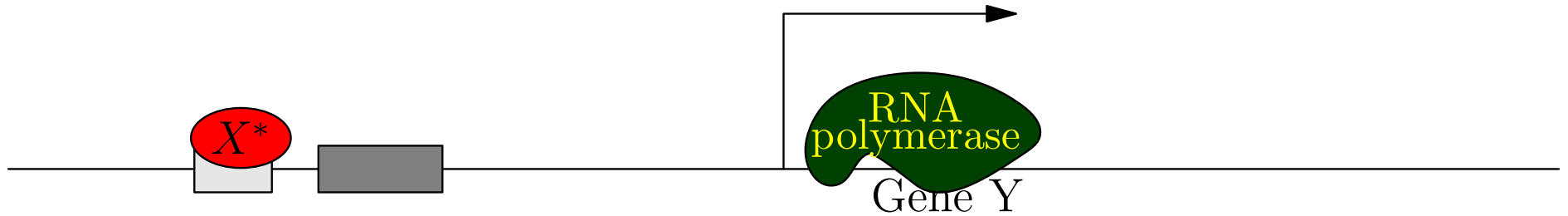
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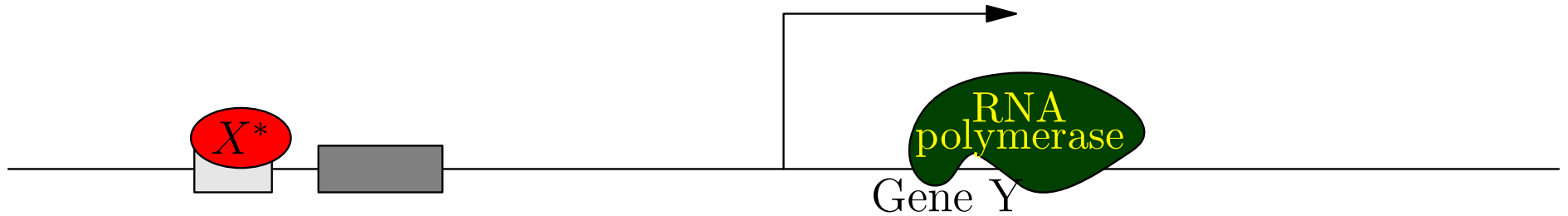
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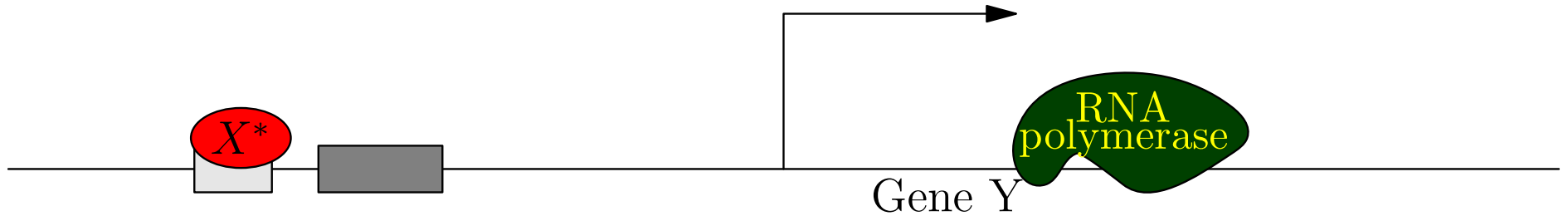
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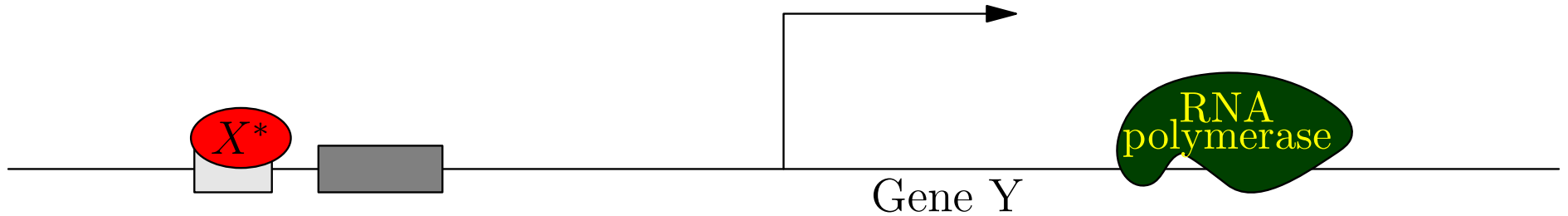
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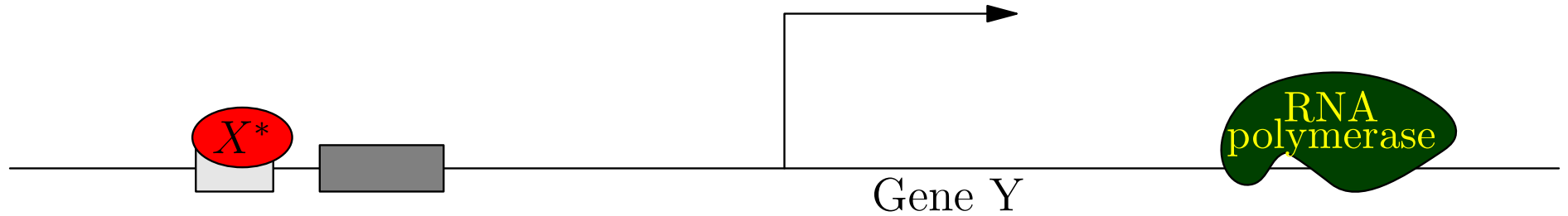
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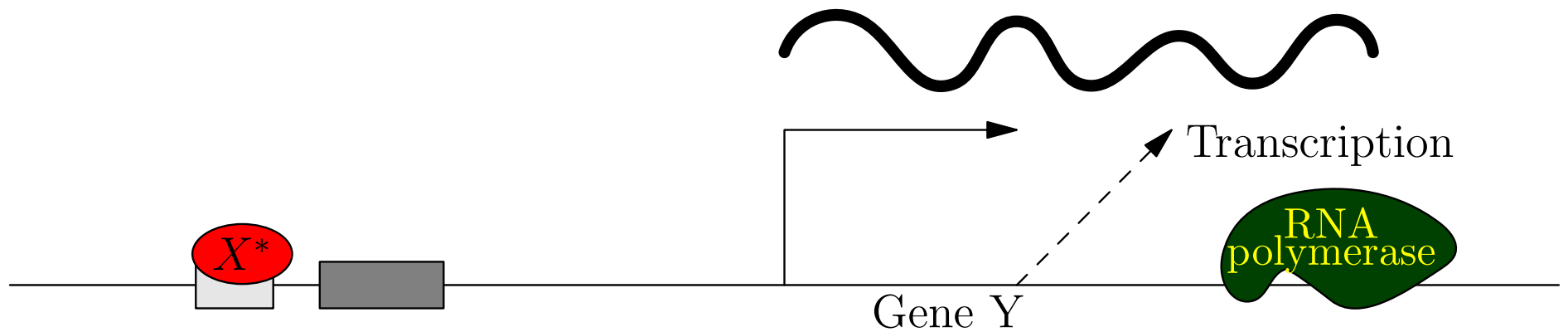
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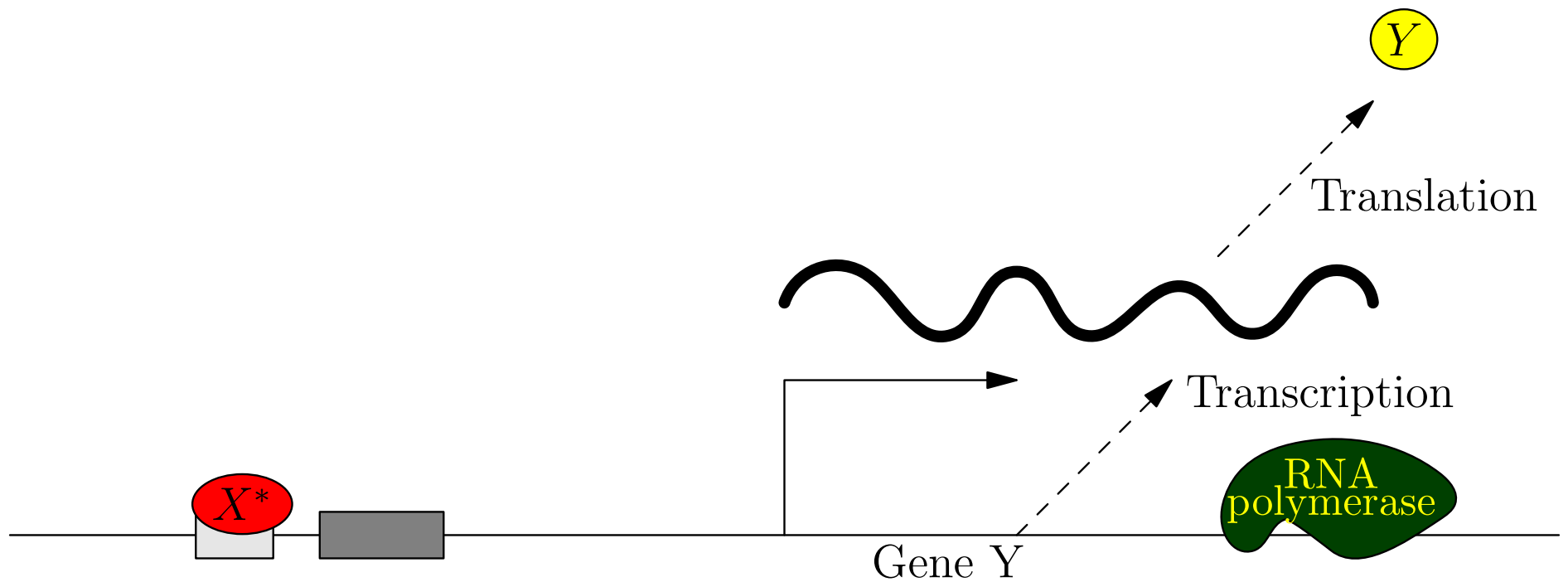
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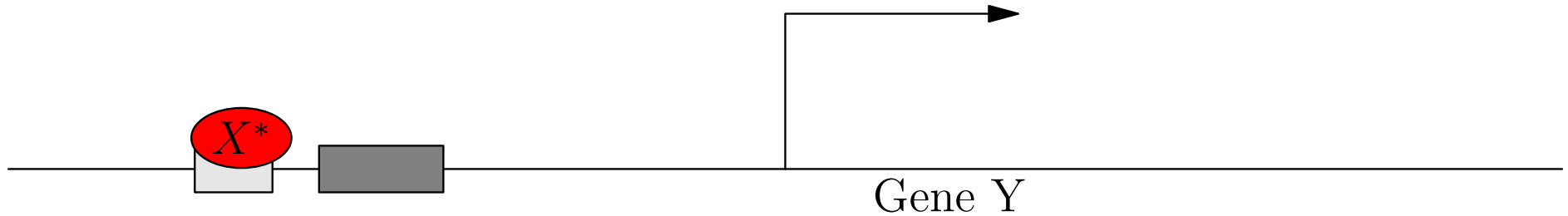
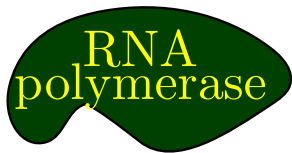
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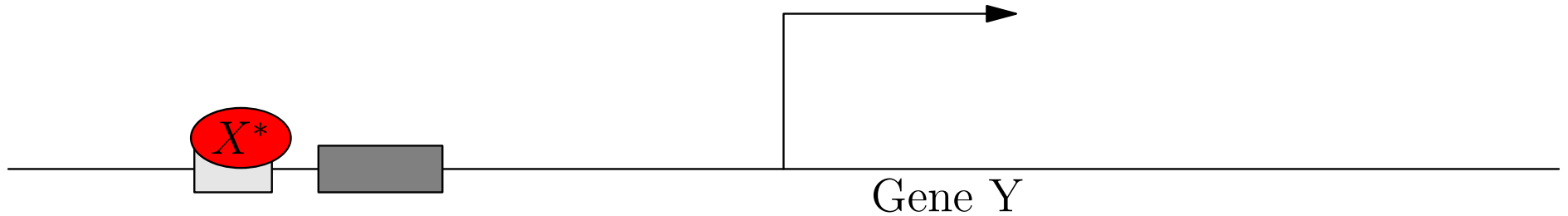
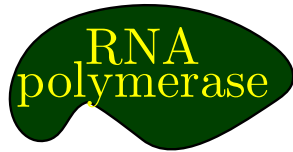
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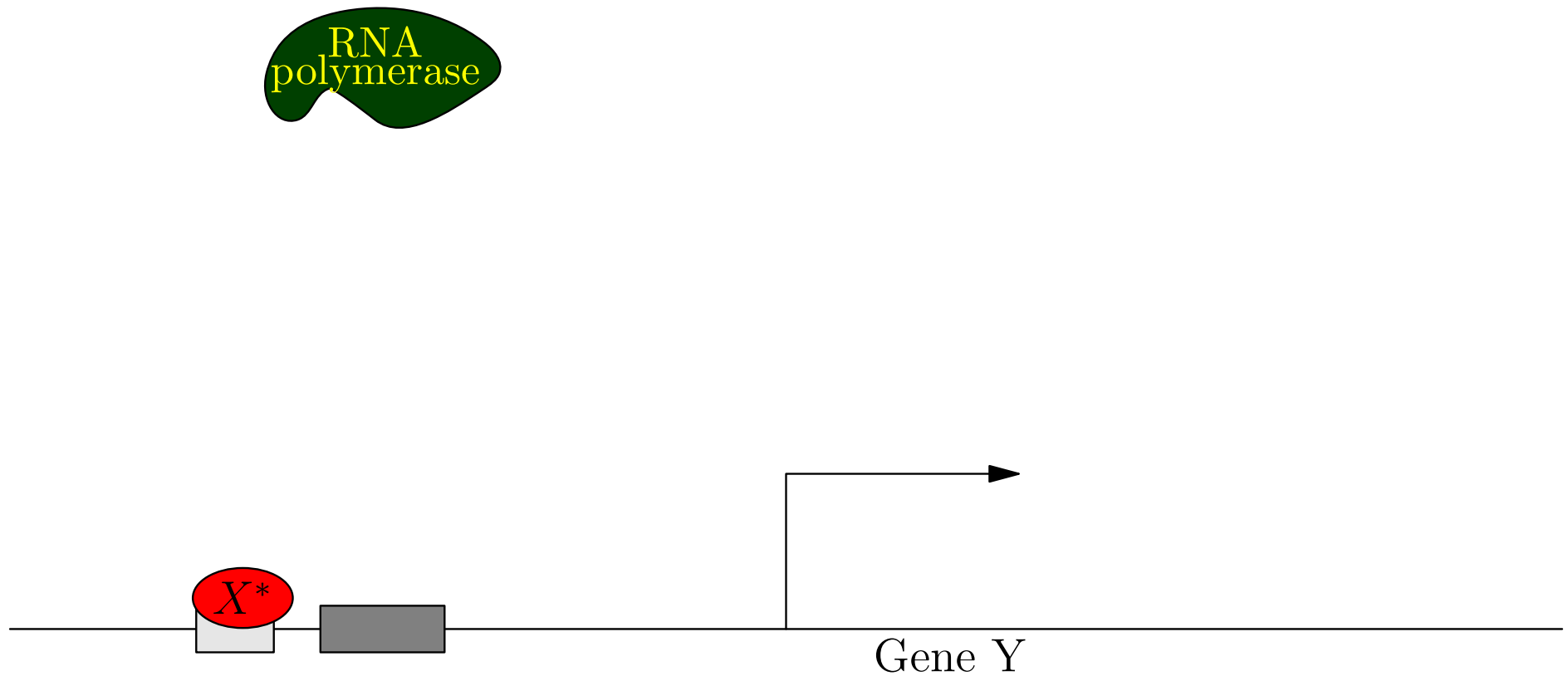
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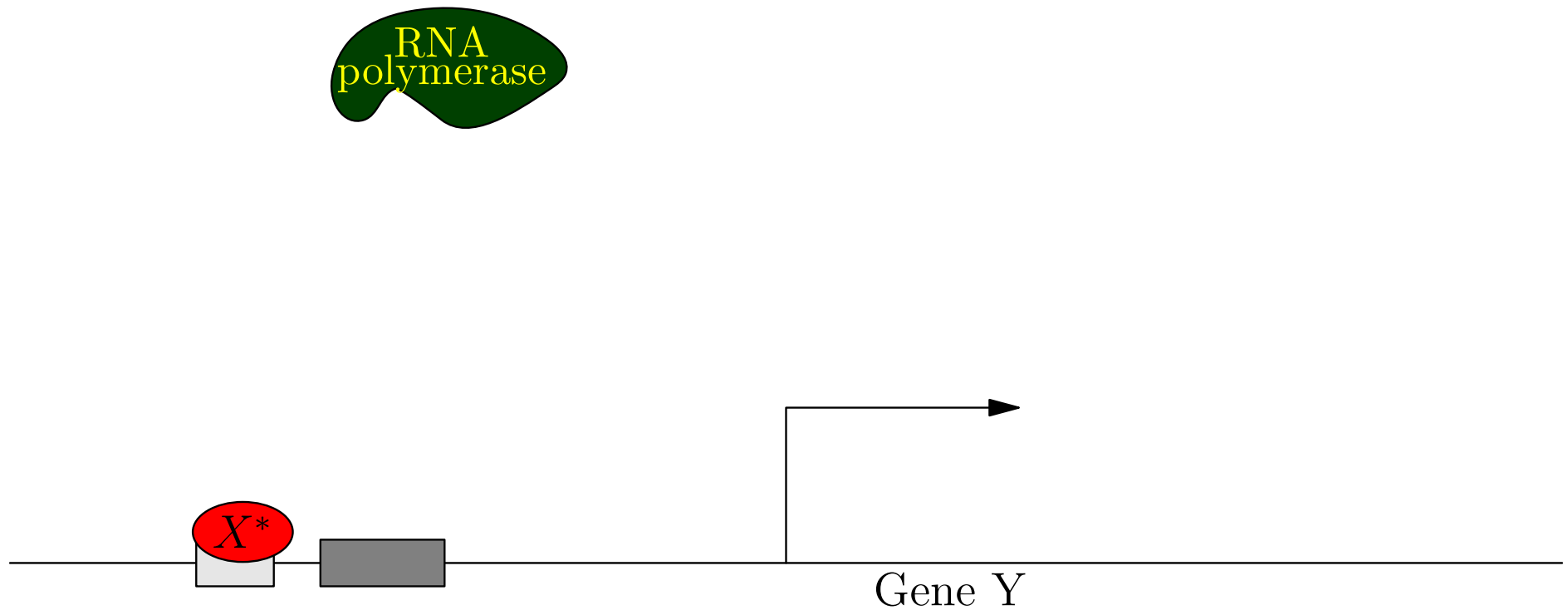
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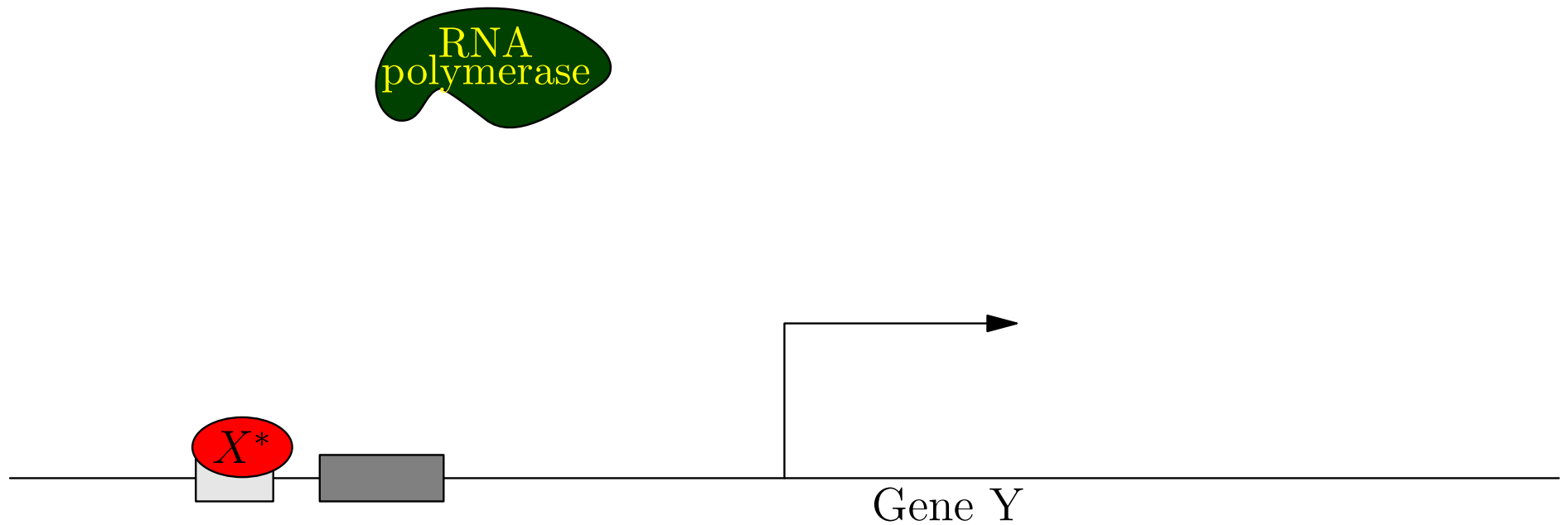
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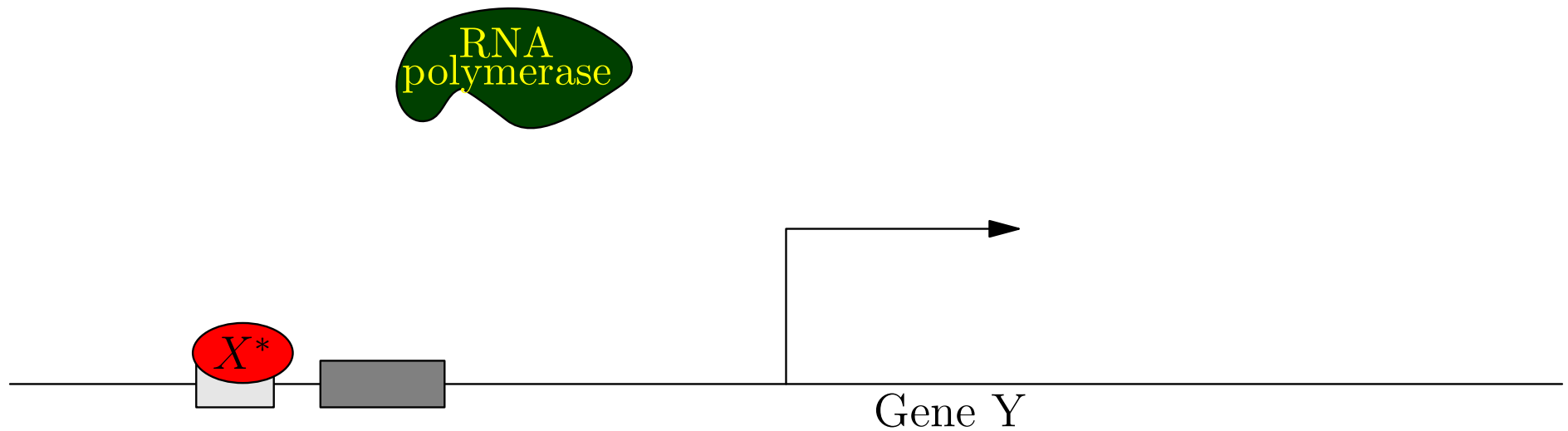
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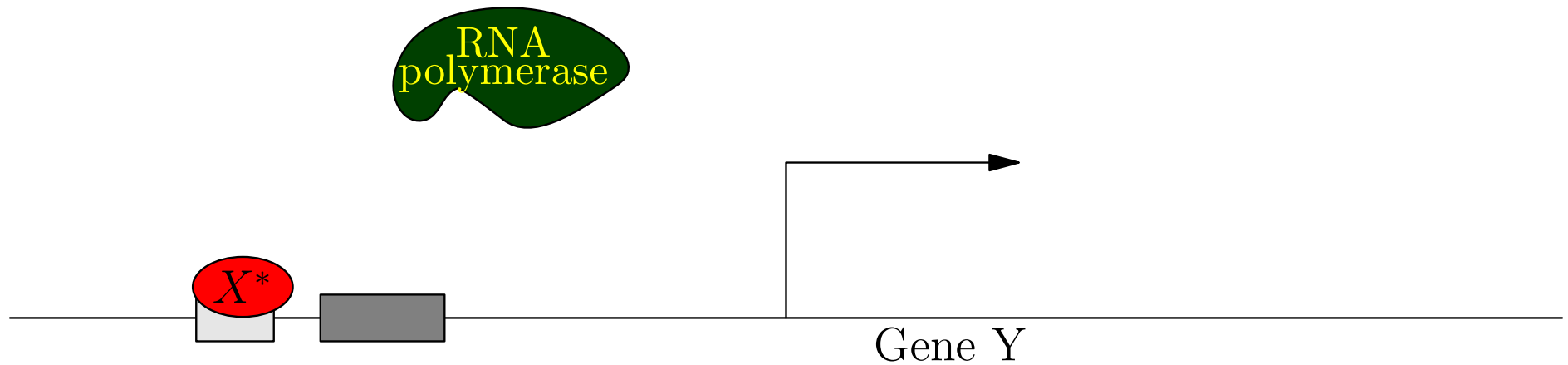
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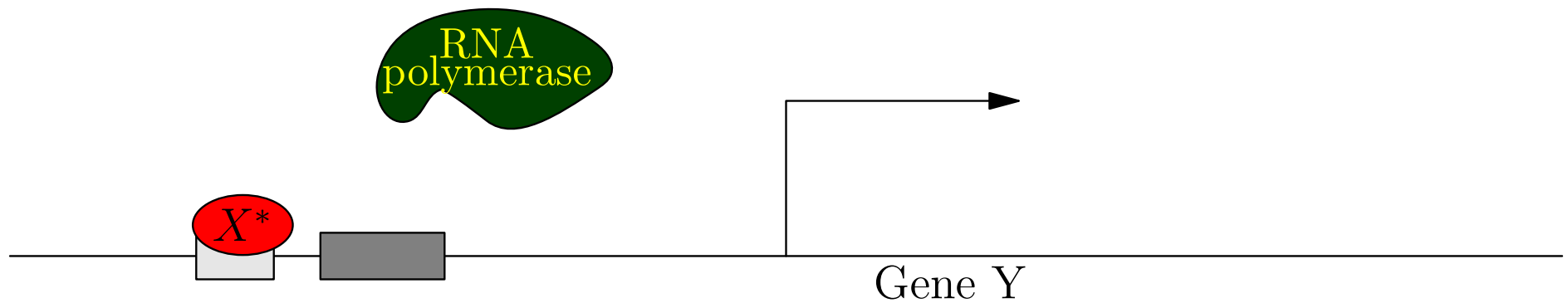
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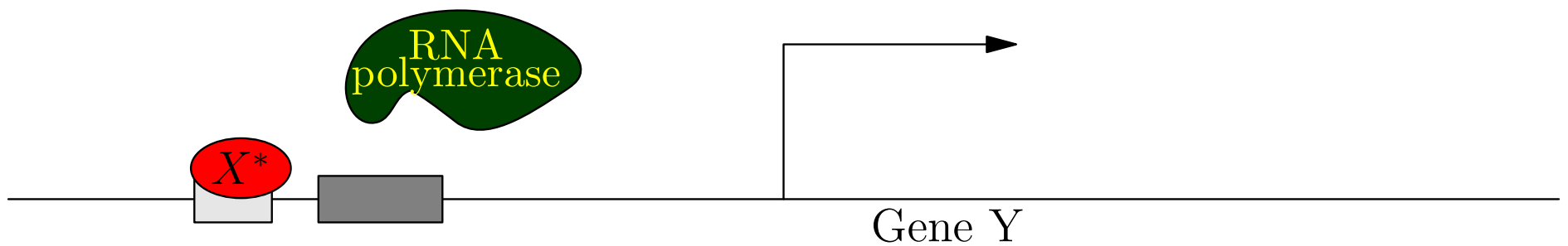
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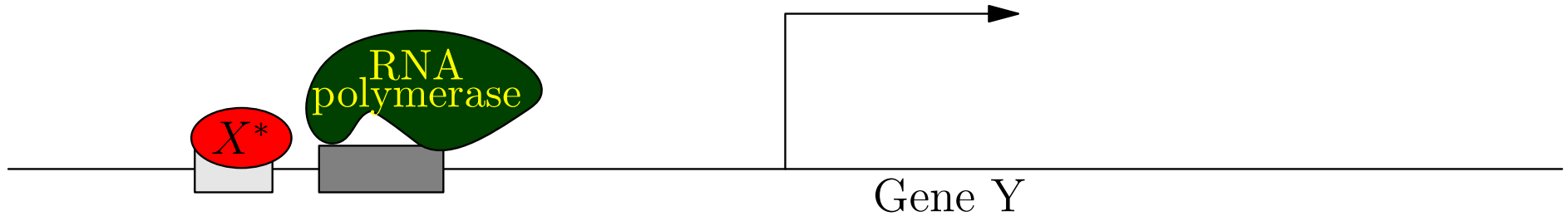
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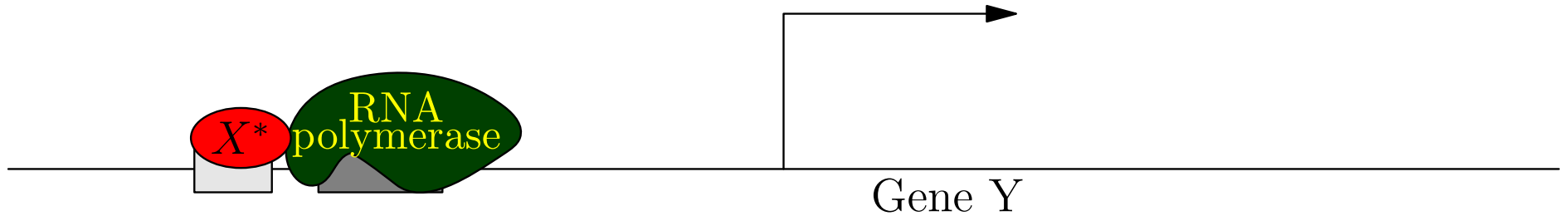
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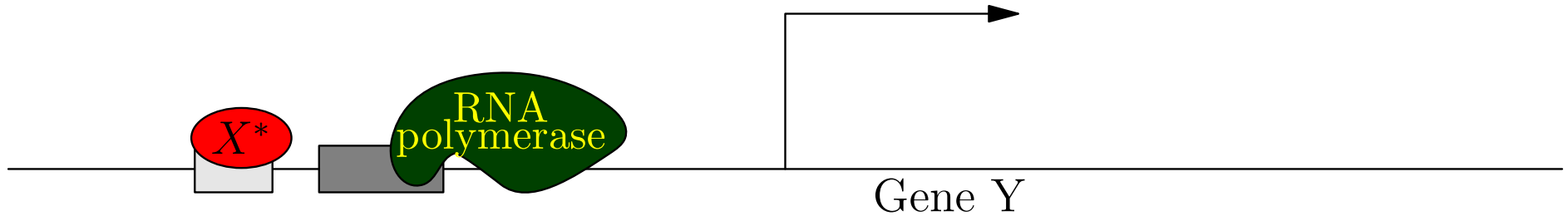
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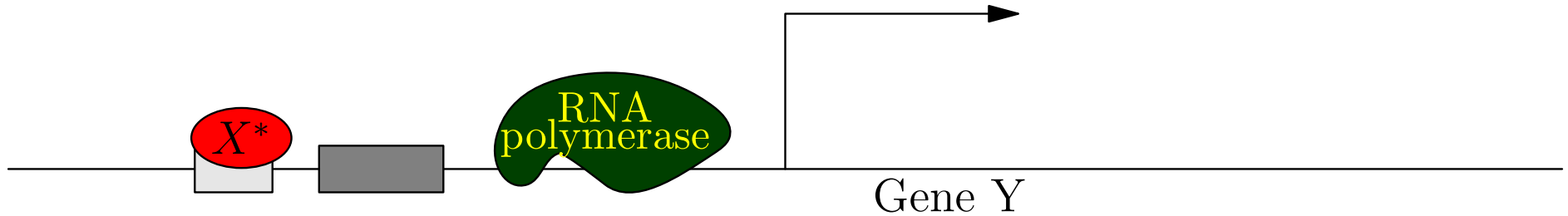
Gene Transcription Regulation

Excititory Transcription Factor



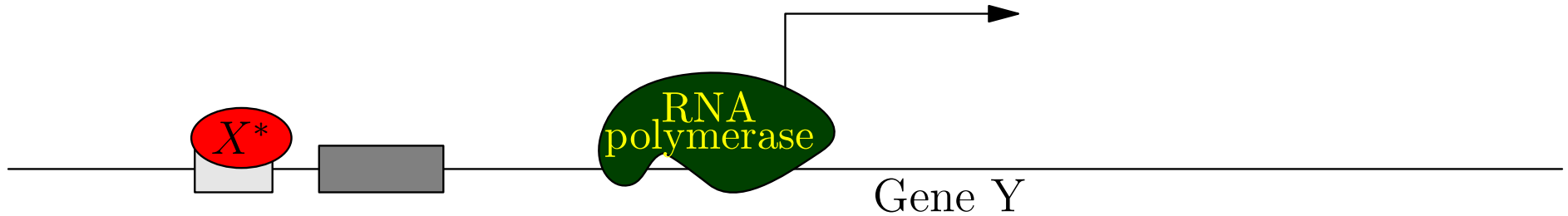
Gene Transcription Regulation

Excititory Transcription Factor



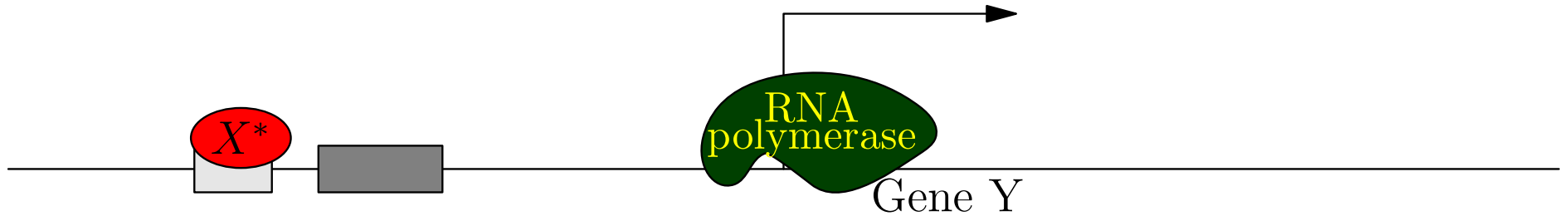
Gene Transcription Regulation

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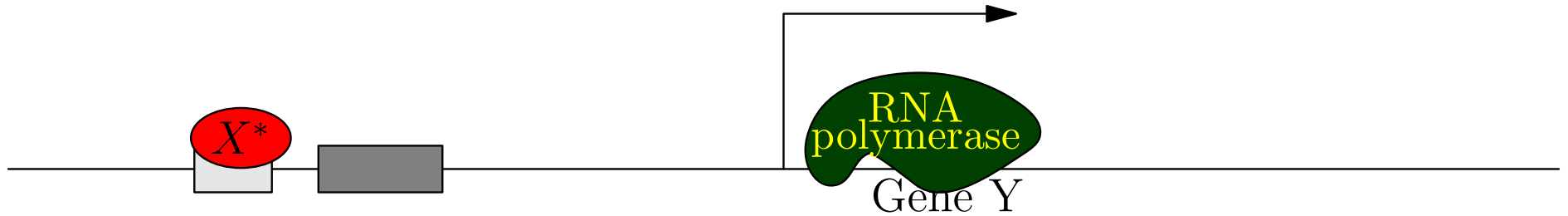
Gene Transcription Regulation

Excititory Transcription Factor



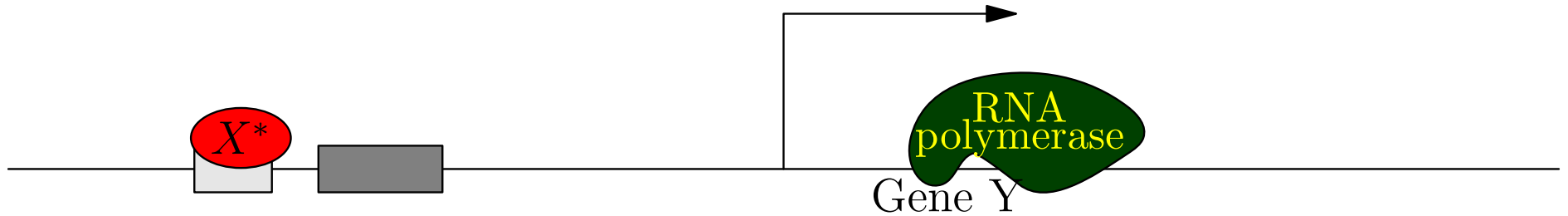
Gene Transcription Regulation

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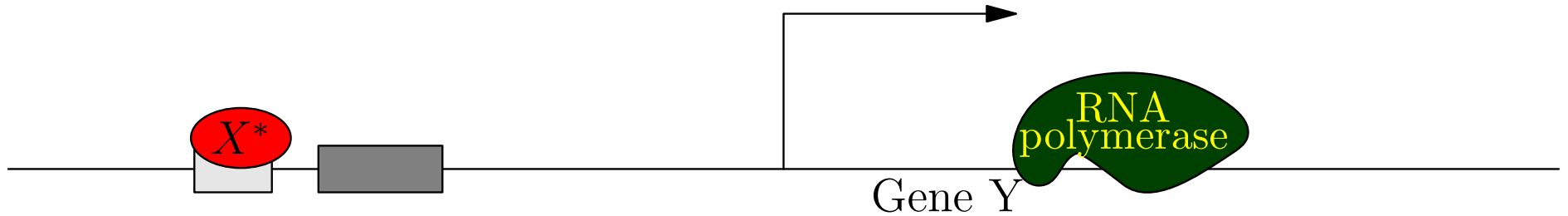
Gene Transcription Regulation

Excititory Transcription Factor



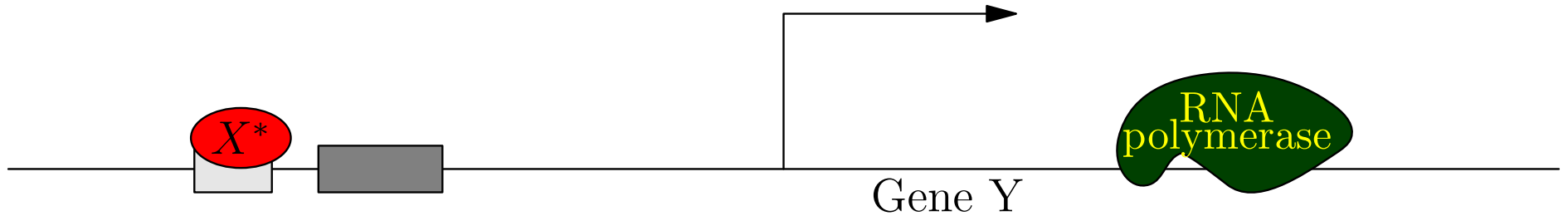
Gene Transcription Regulation

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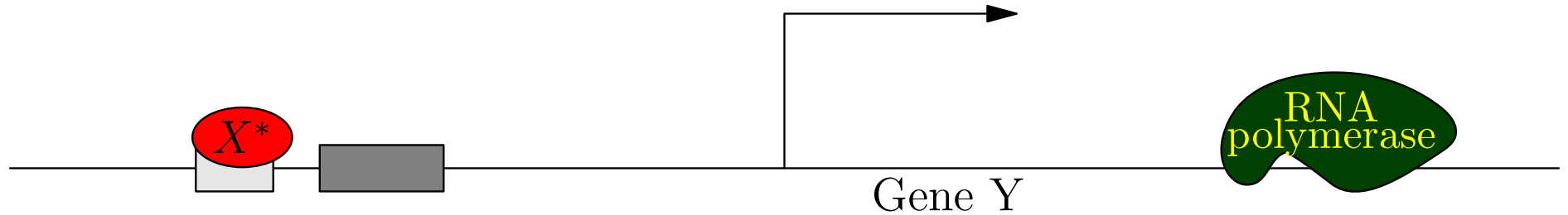
Gene Transcription Regulation

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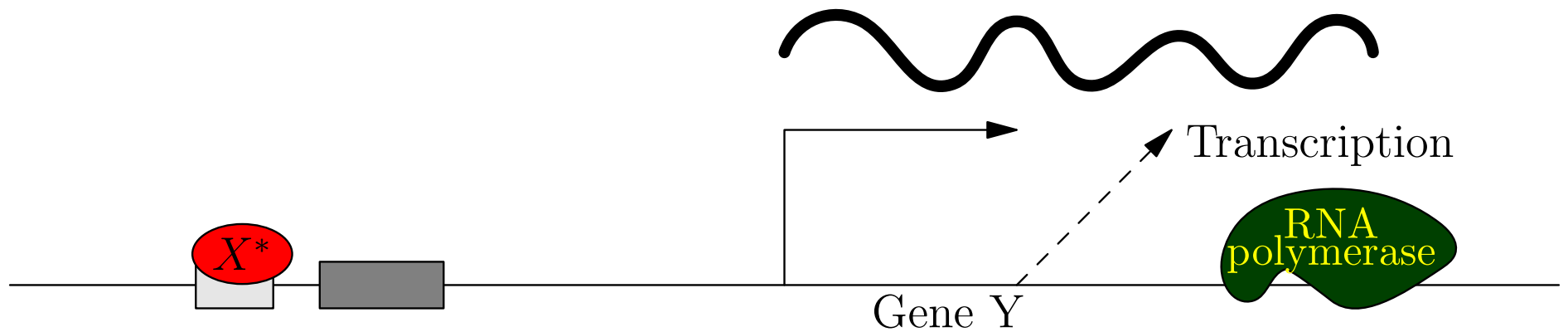
Gene Transcription Regulation

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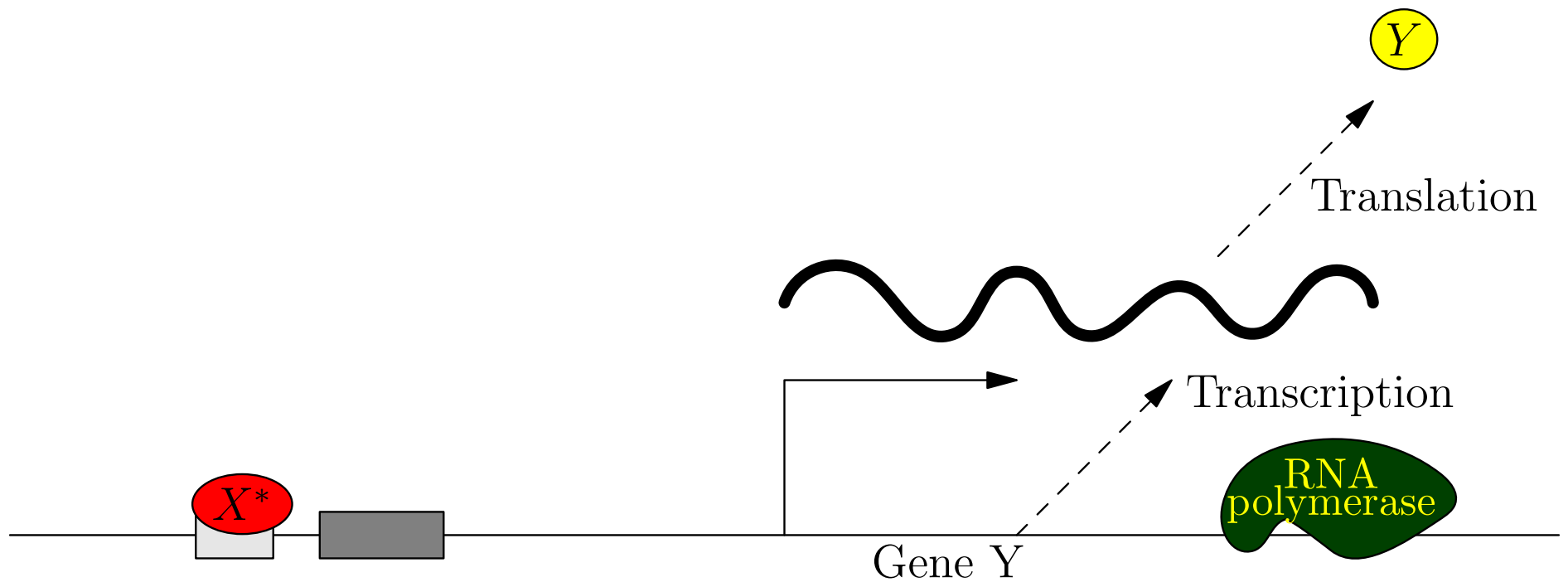
Gene Transcription Regulation

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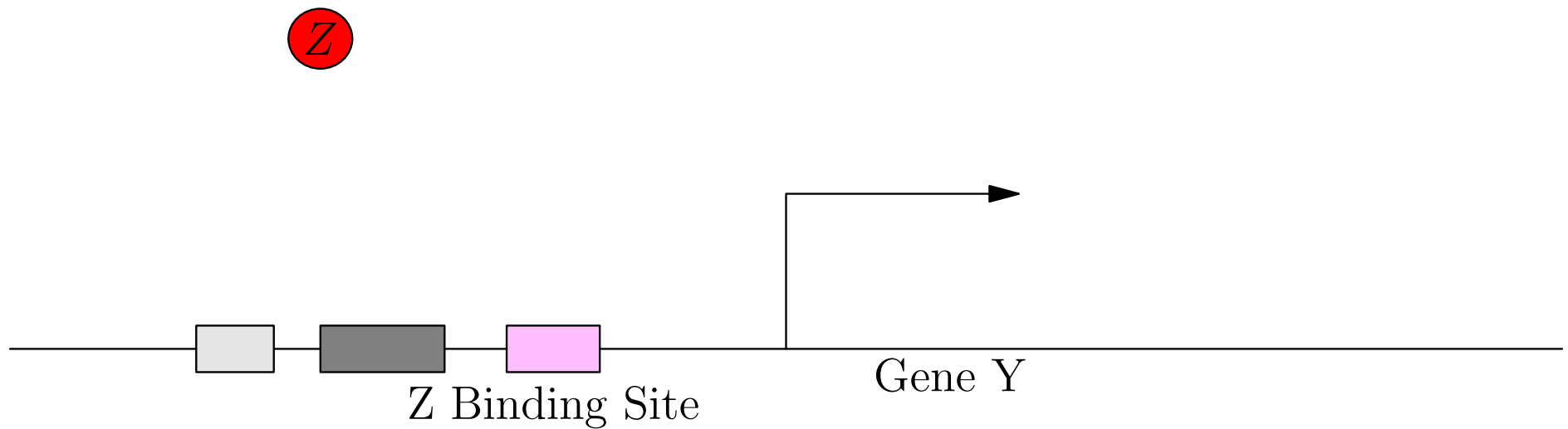


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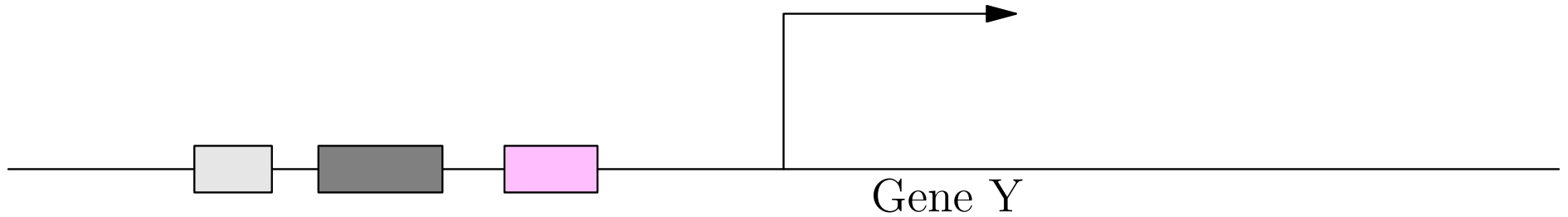
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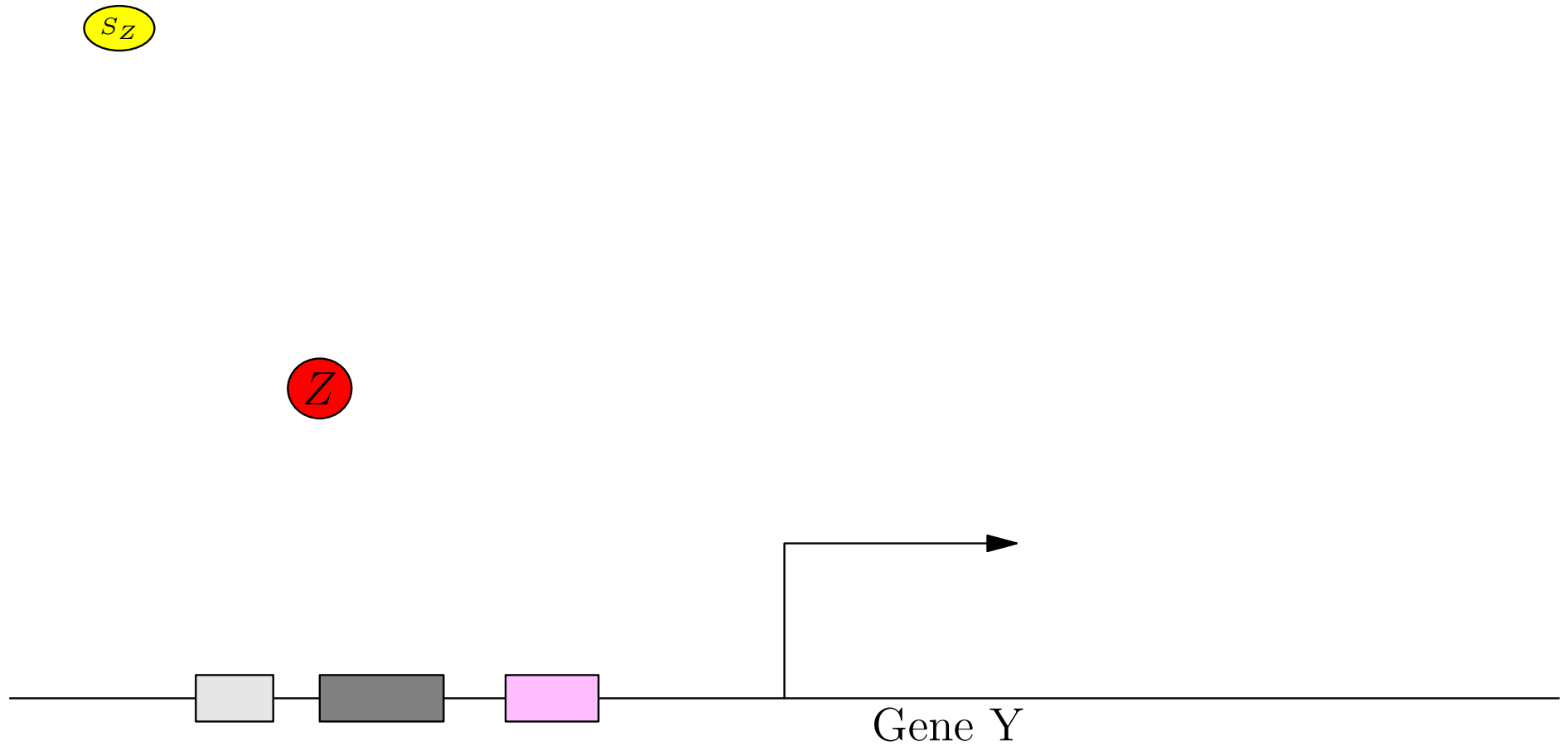
Gene Transcription Regulation

S_Z

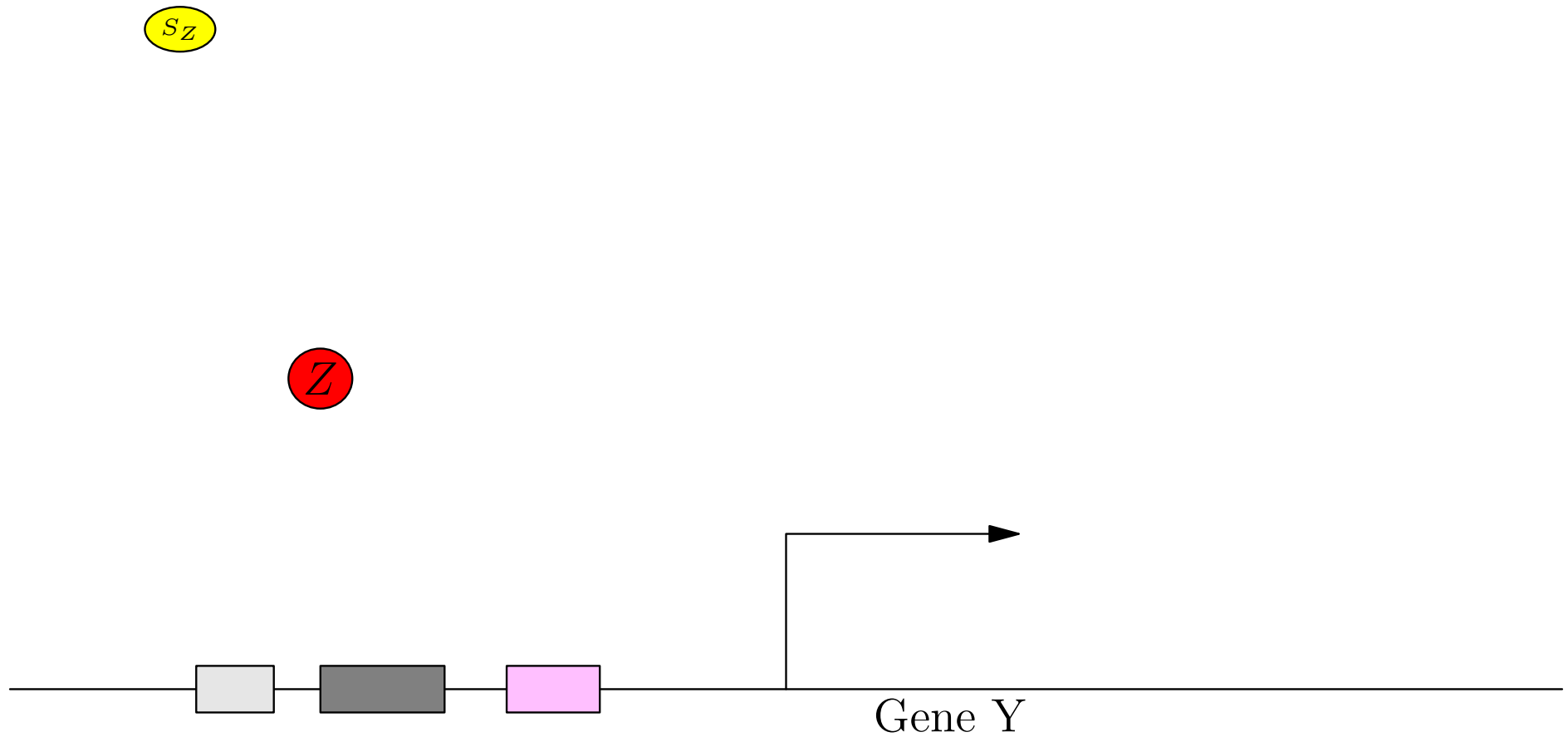
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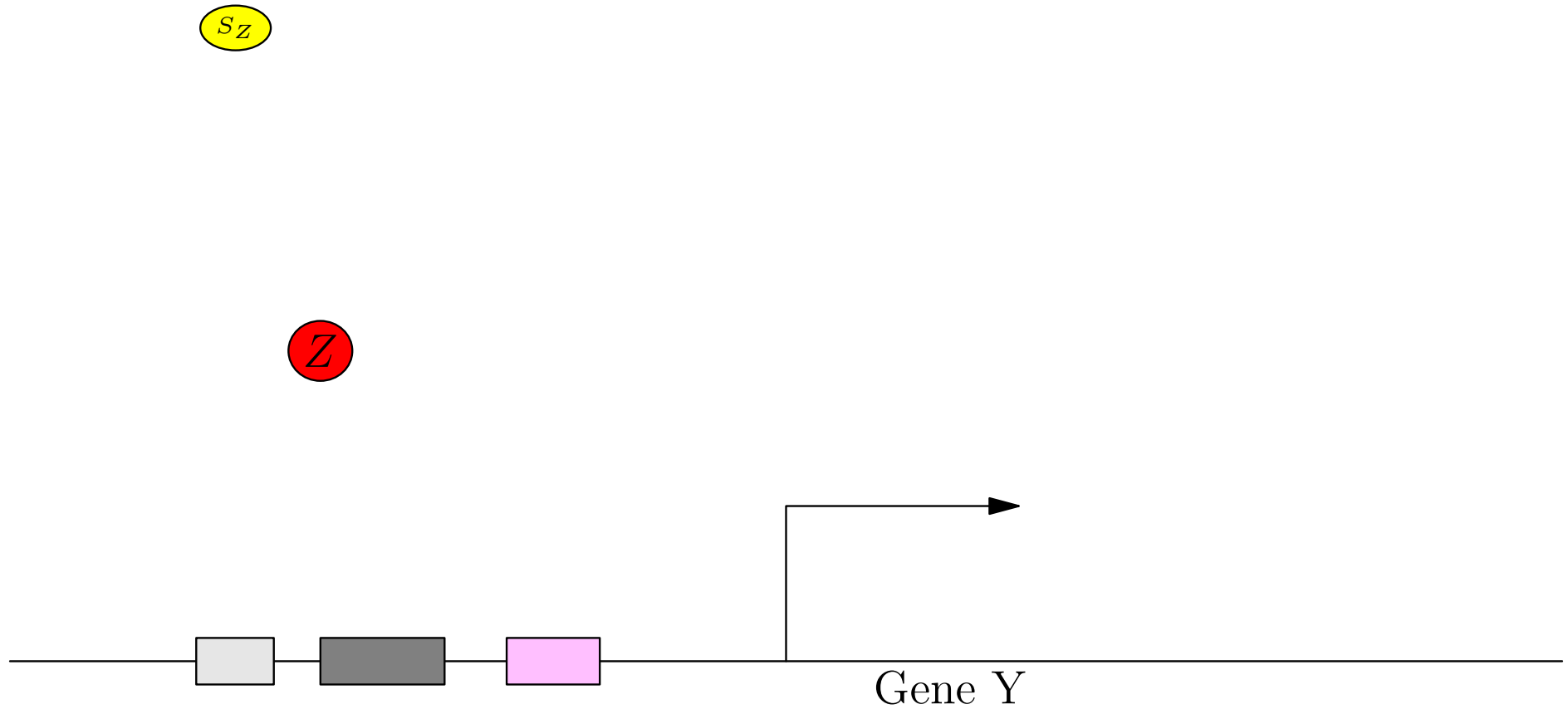
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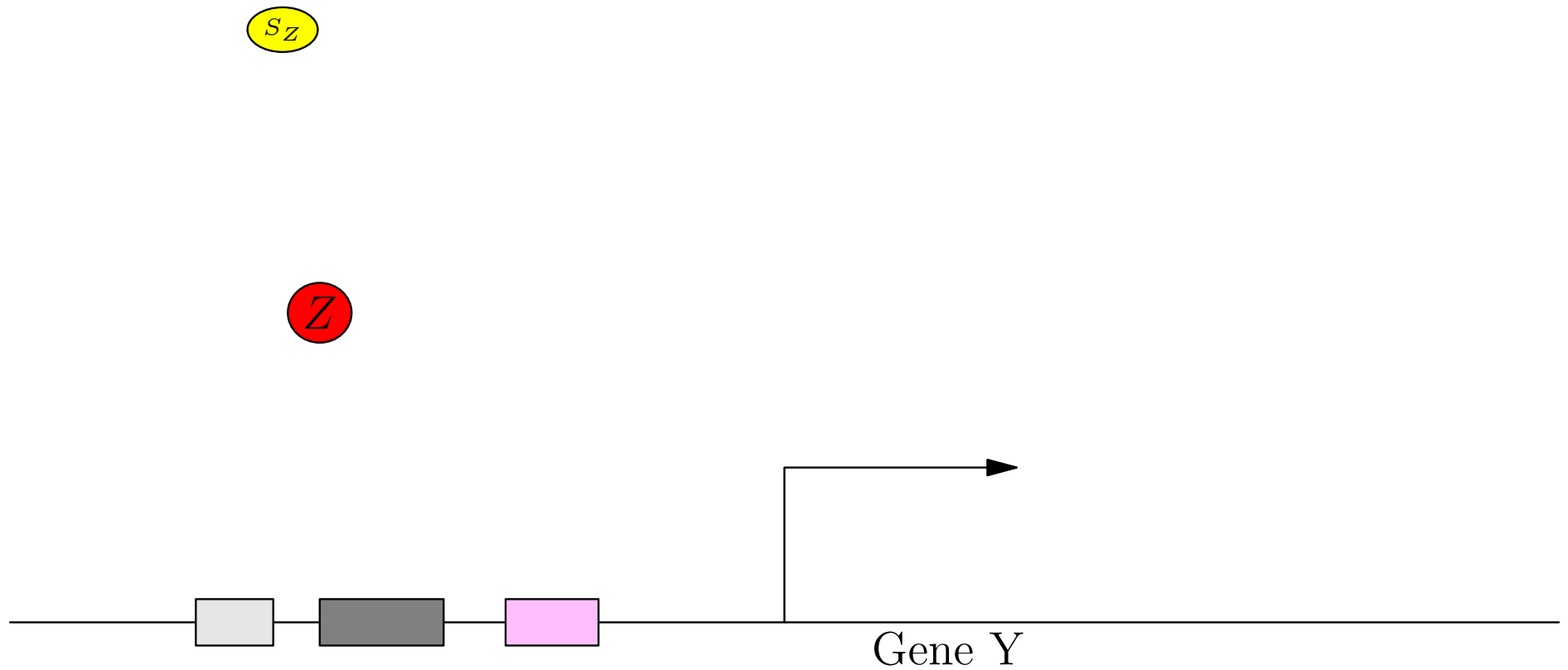
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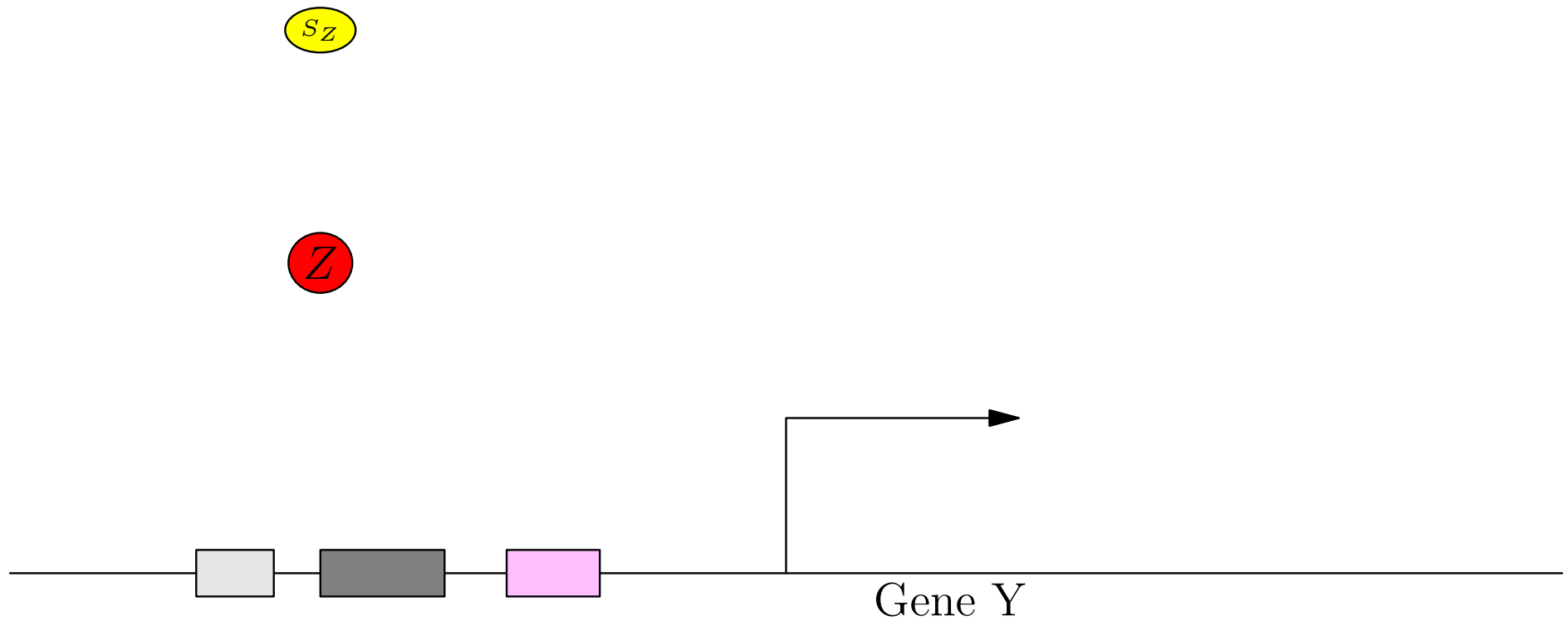
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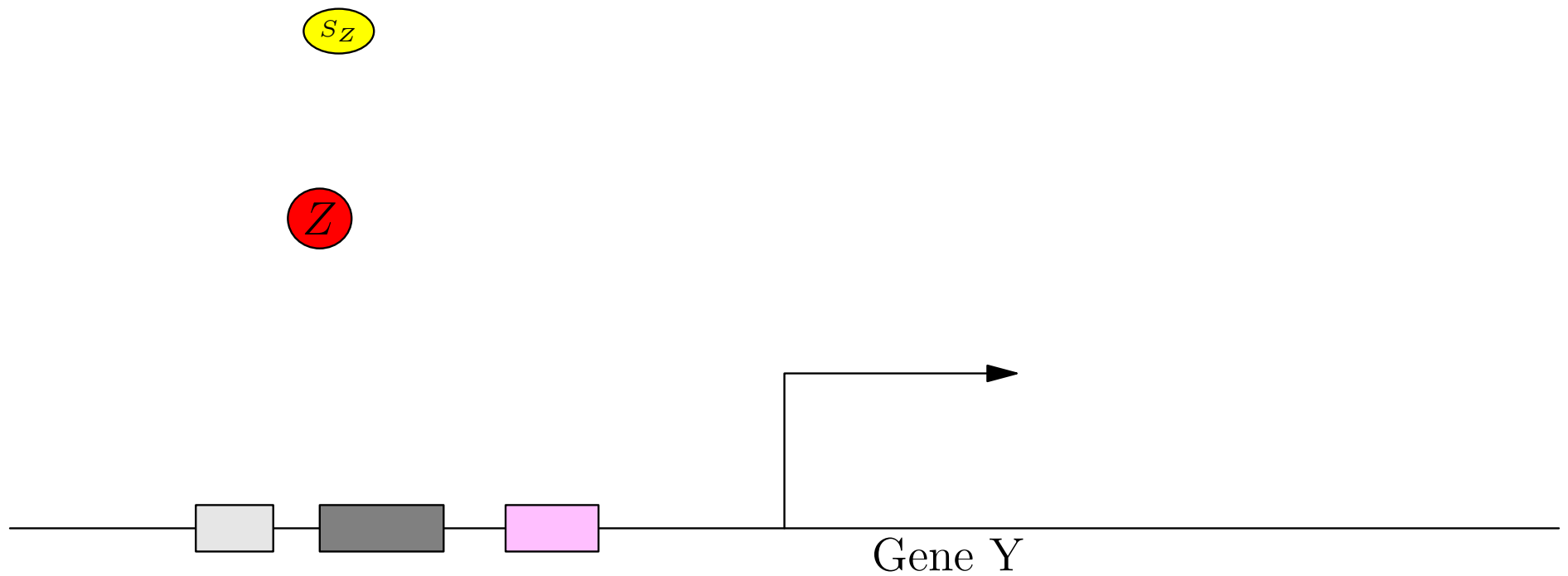
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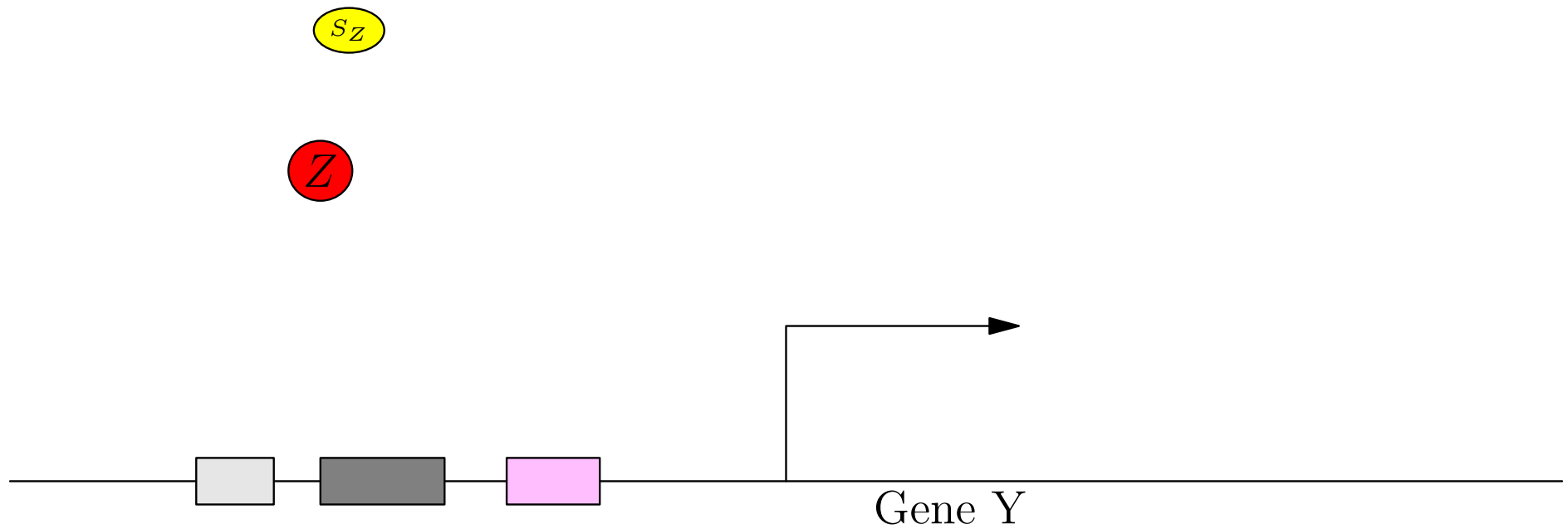
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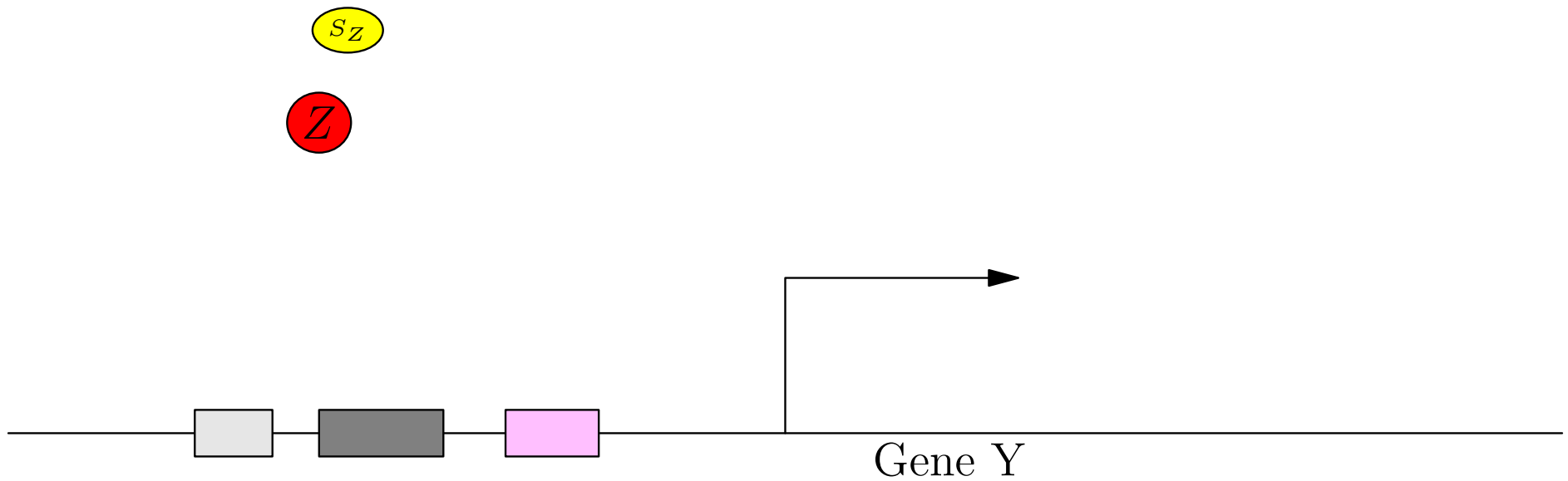
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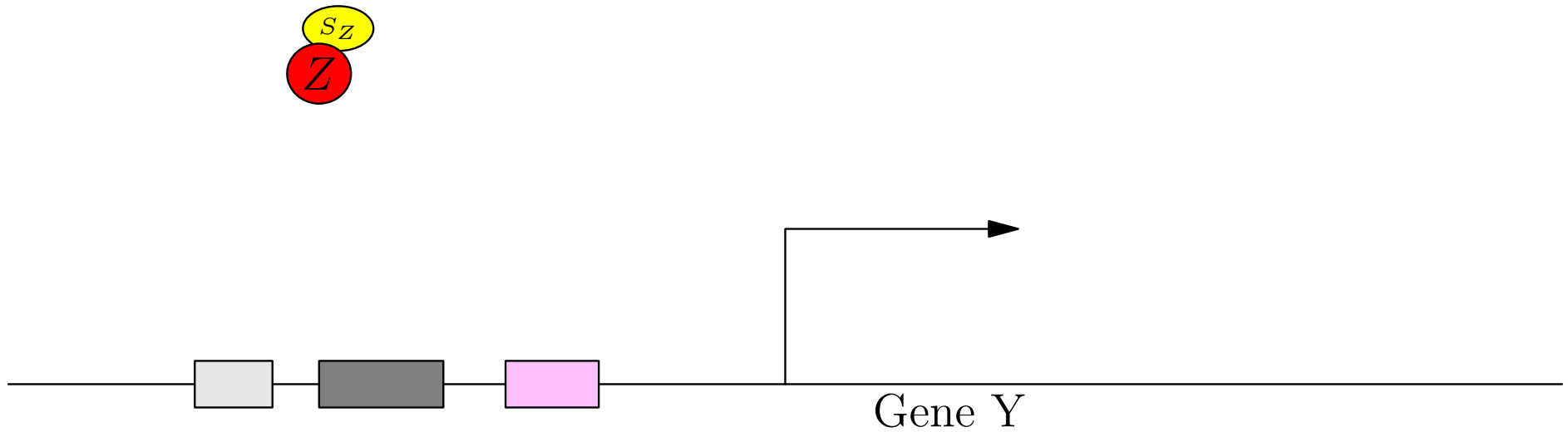
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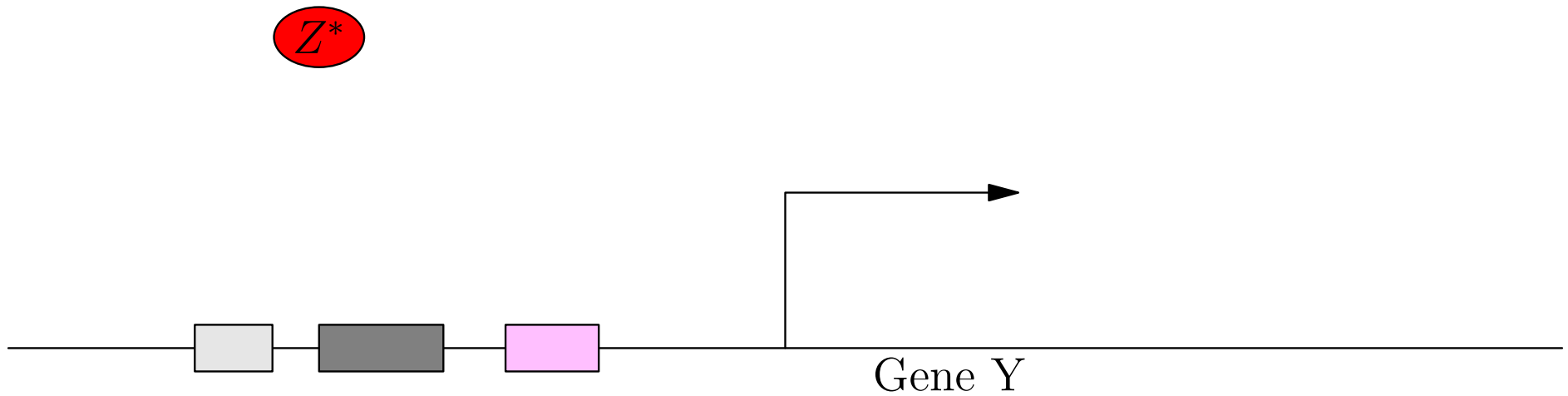
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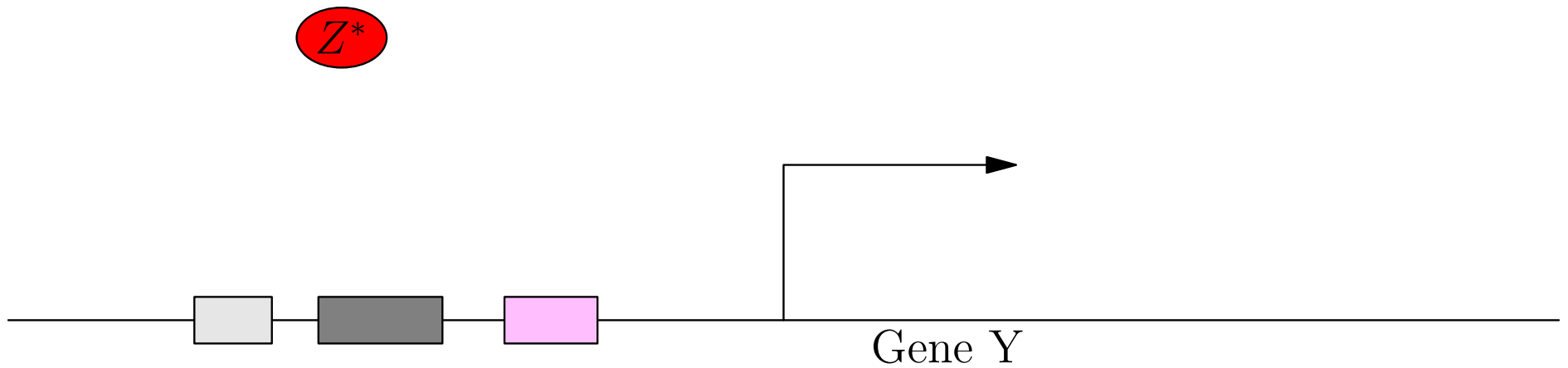
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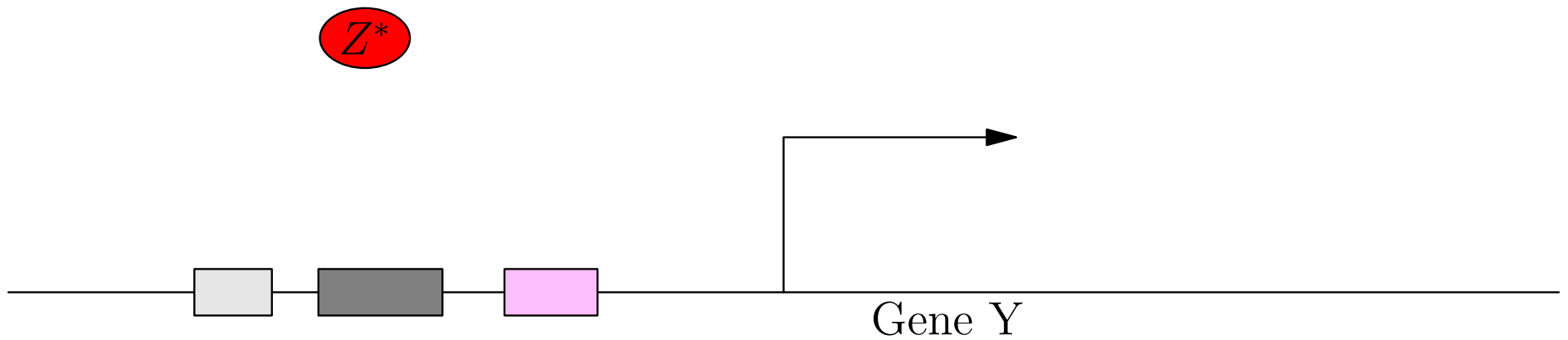
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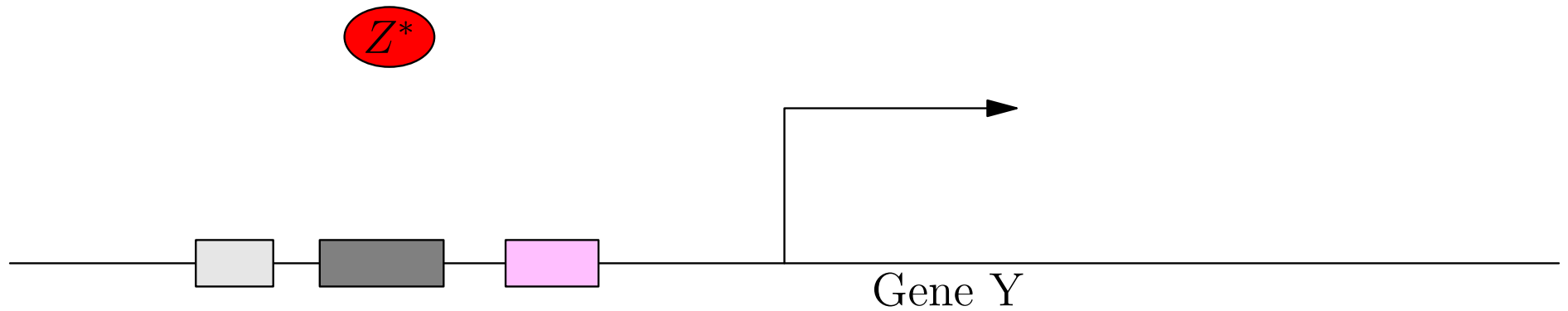
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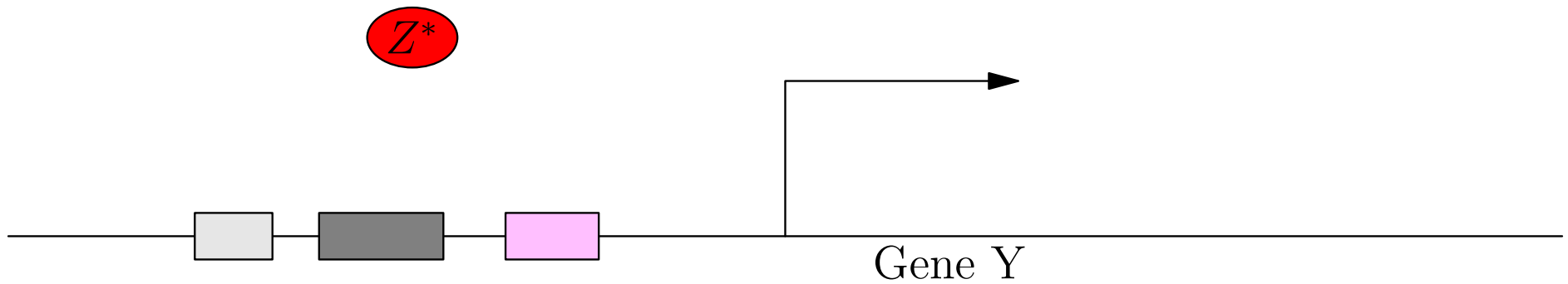
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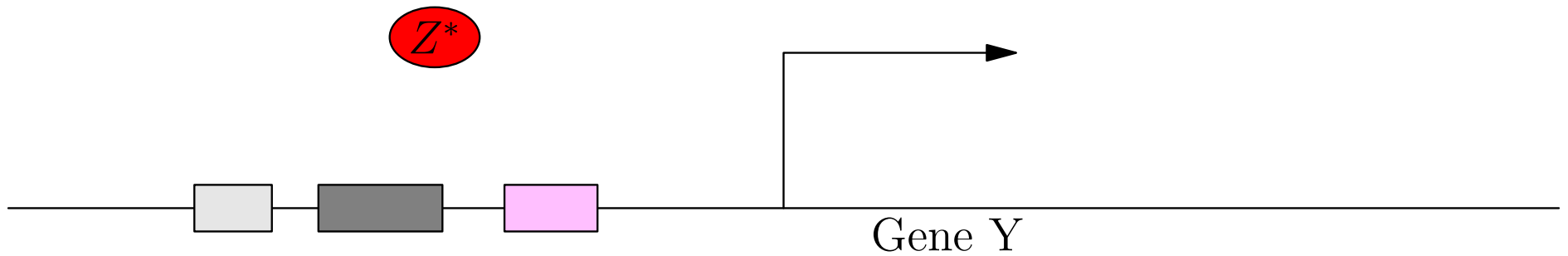
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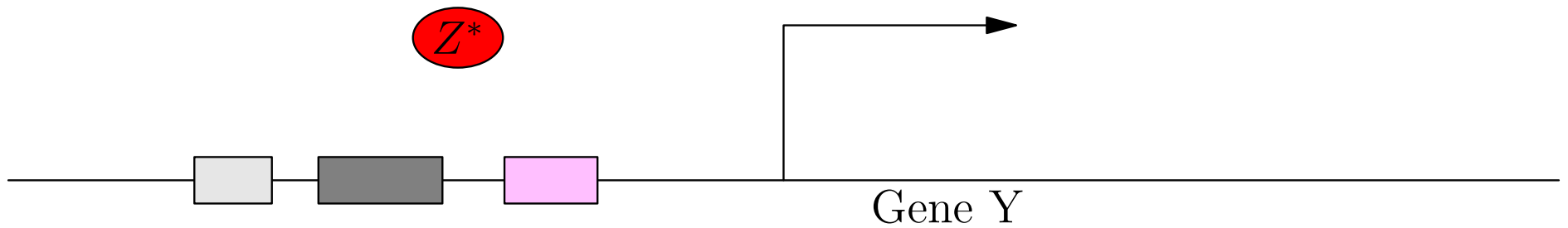
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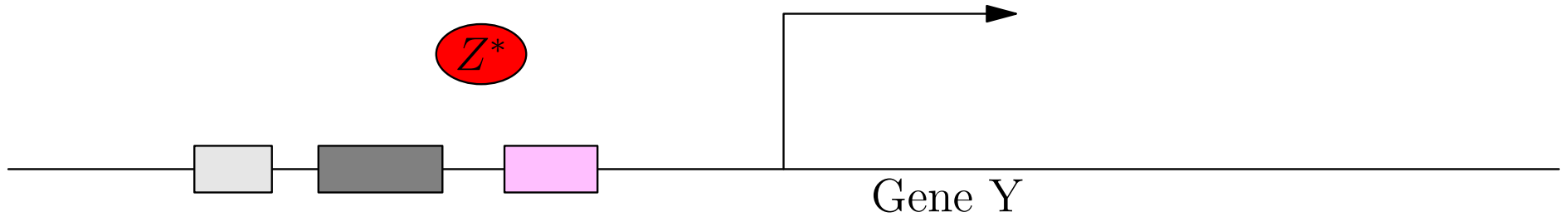
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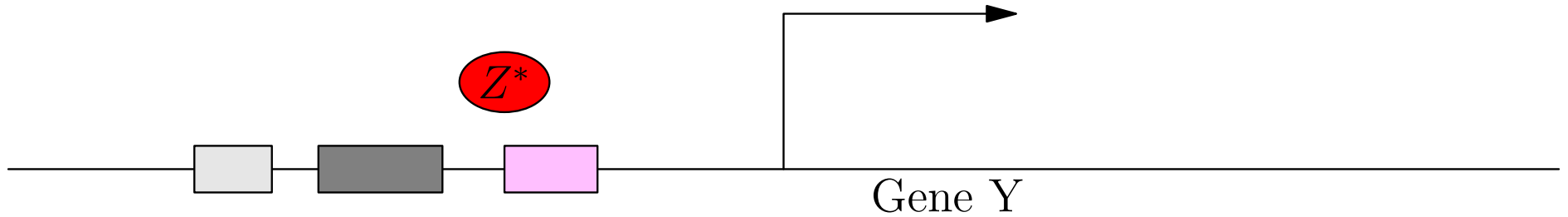
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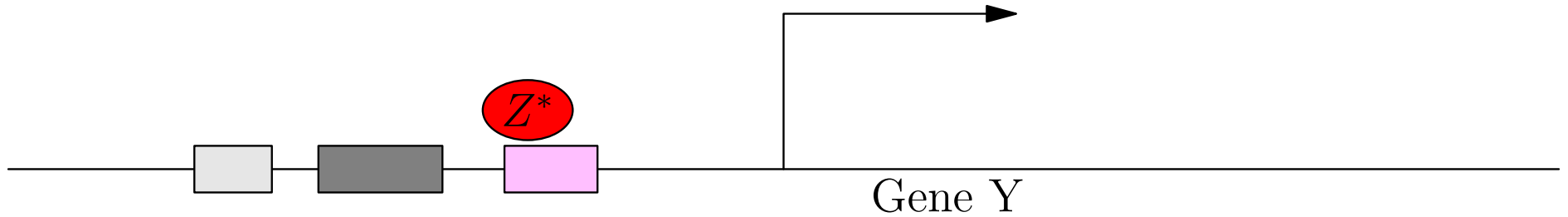
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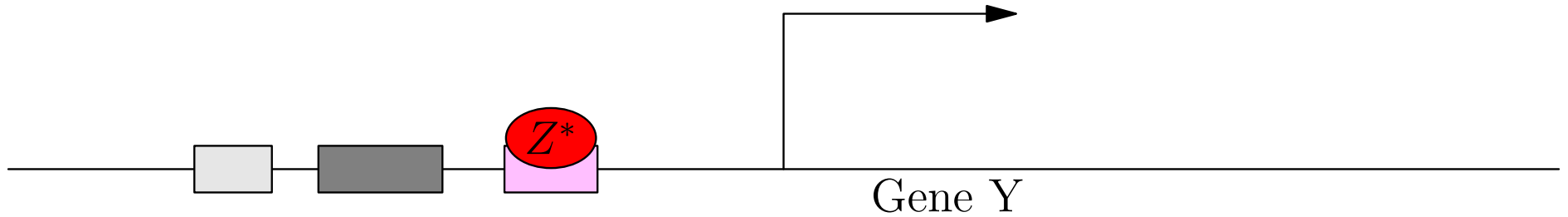
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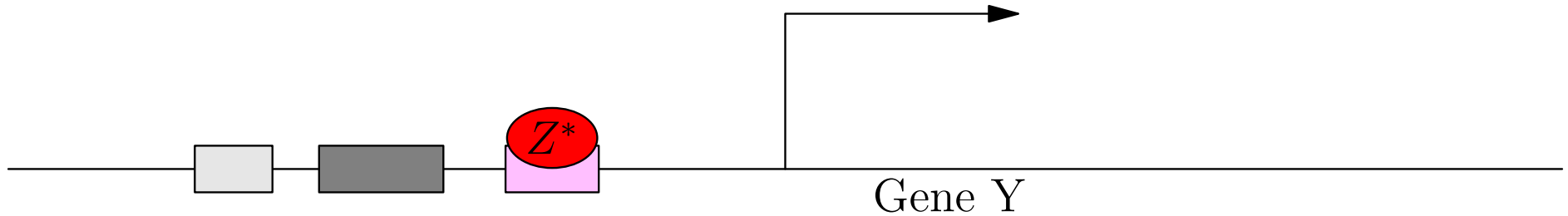
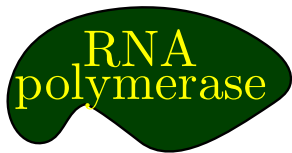


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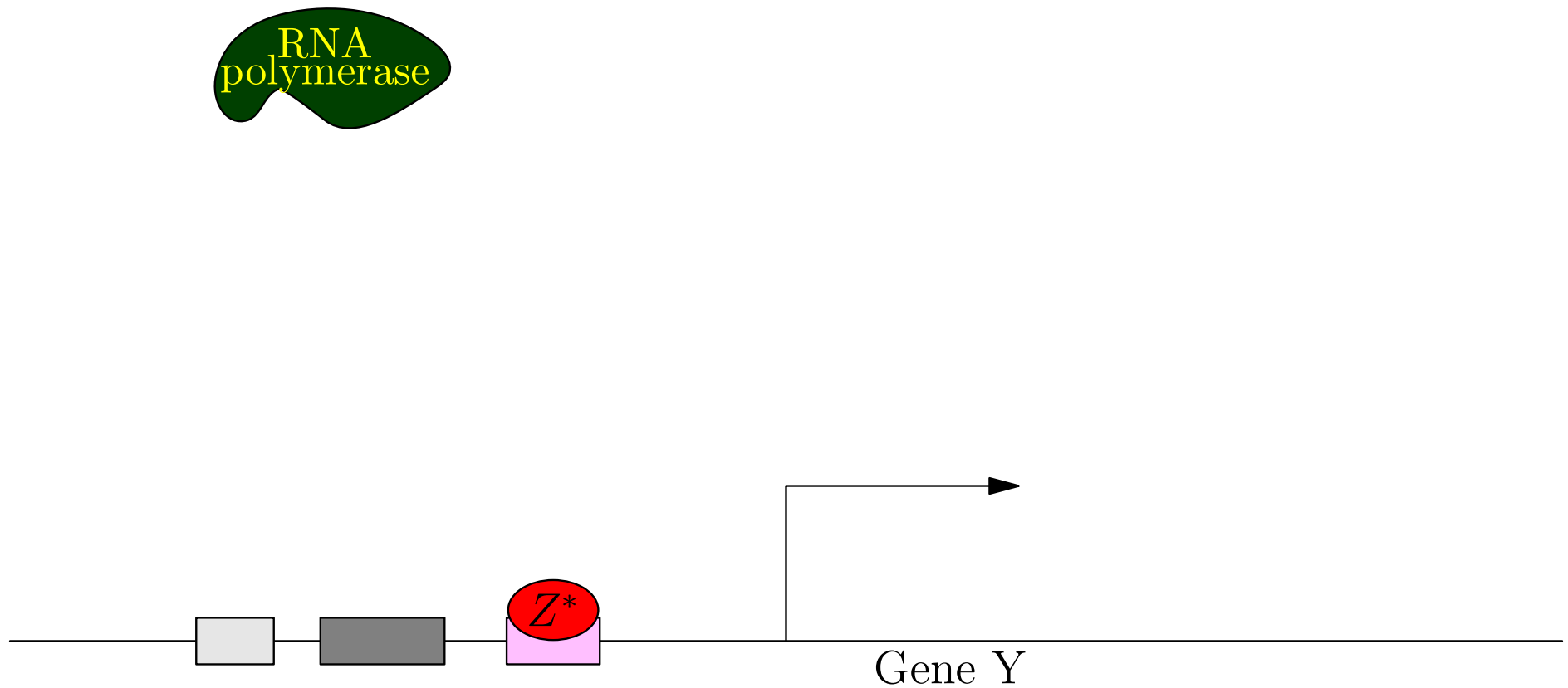
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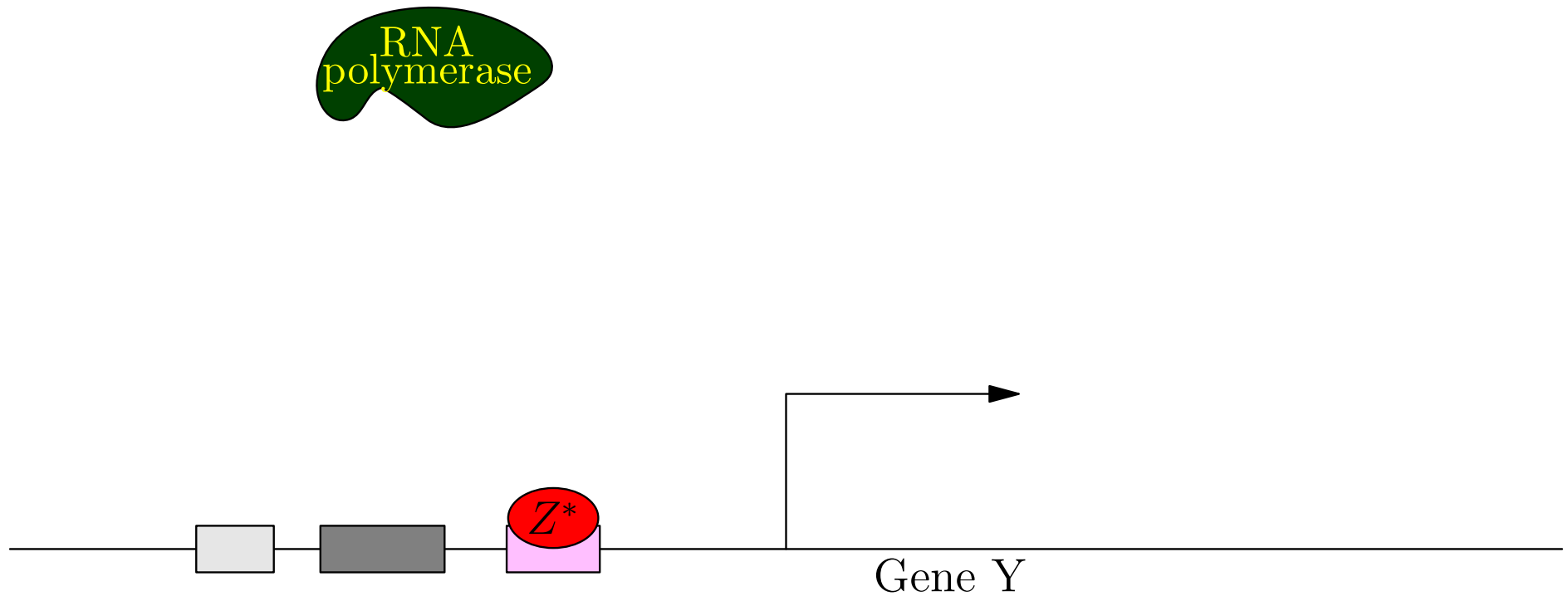
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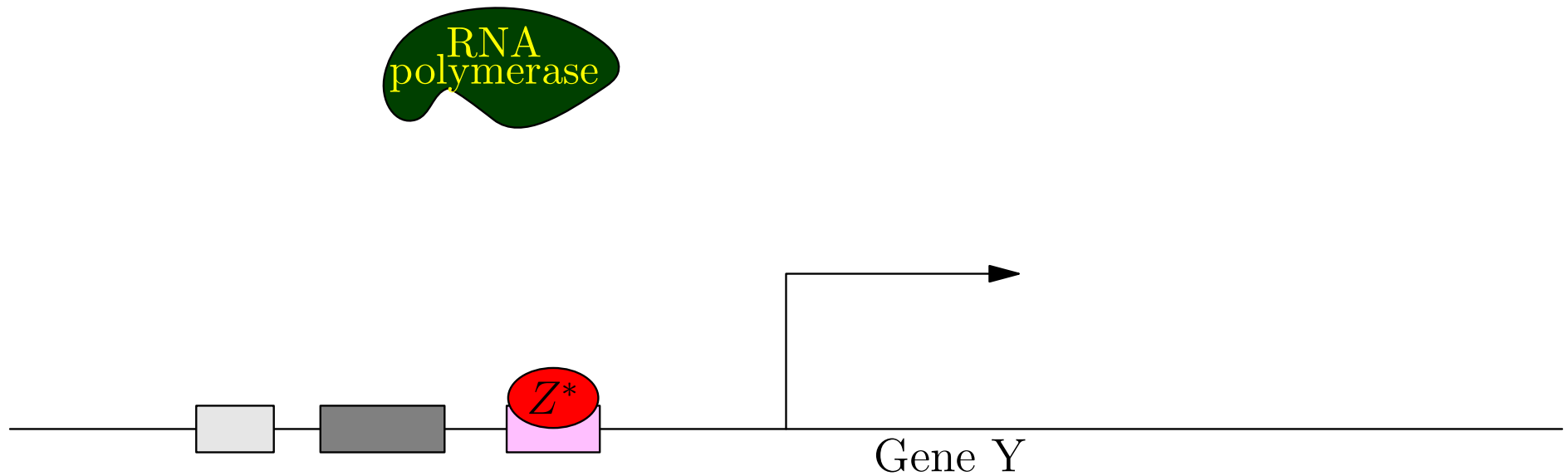
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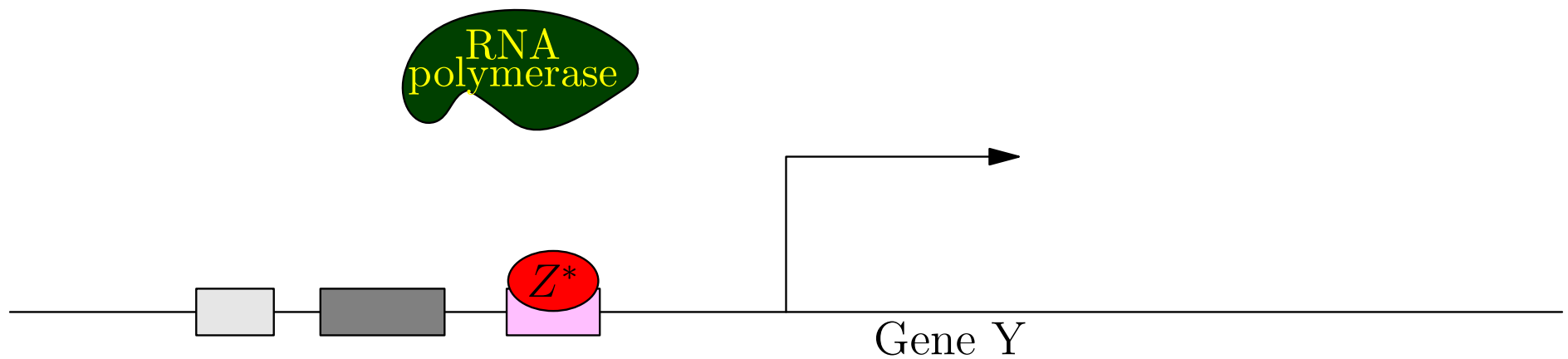
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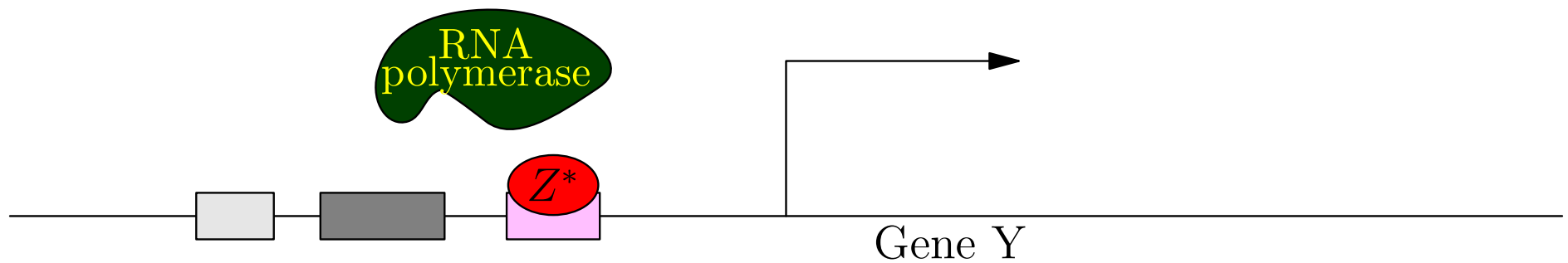
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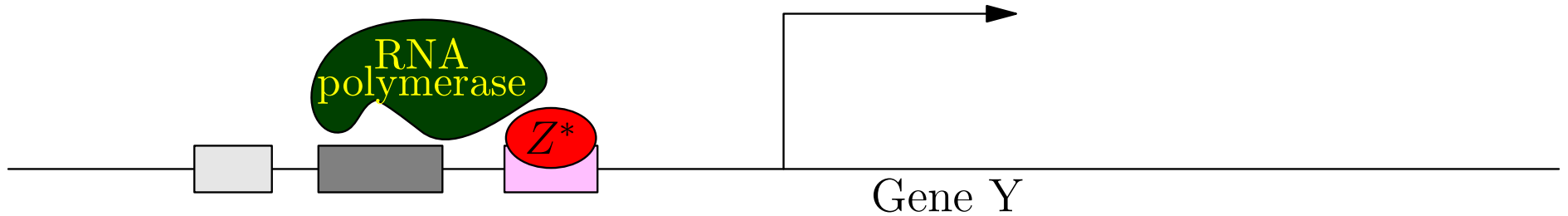
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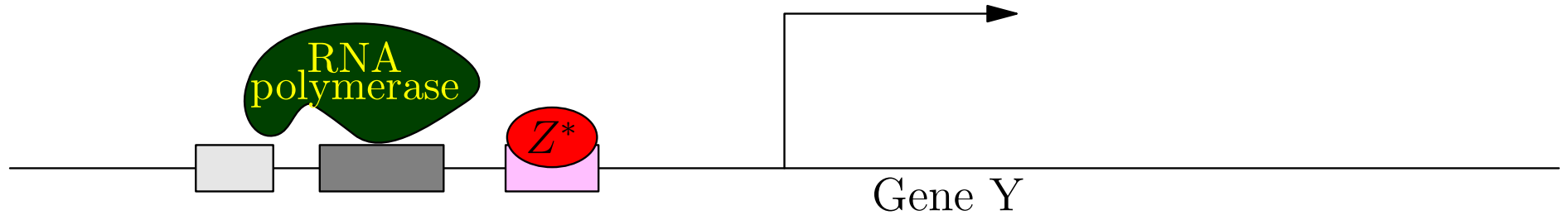
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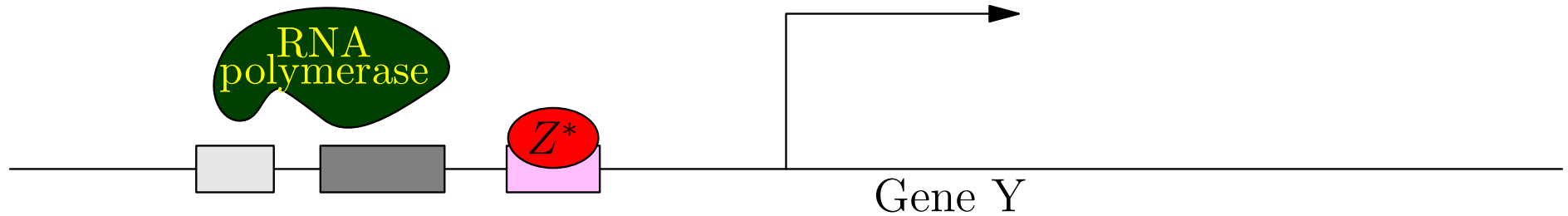
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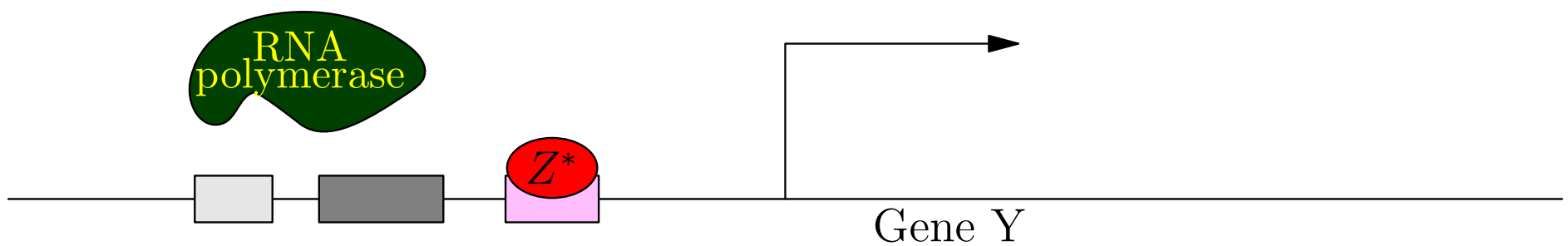
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Time Scales

- Reactions happen on different time scales
 - ★ Binding of small (signalling) molecules: $\sim 1 \text{ msec}$
 - ★ Binding of active transcription factor: $\sim 1 \text{ sec}$
 - ★ Transcription + translation of gene: $\sim 5 \text{ min}$
 - ★ 50% change in concentration of protein: $\sim 1 \text{ hour}$
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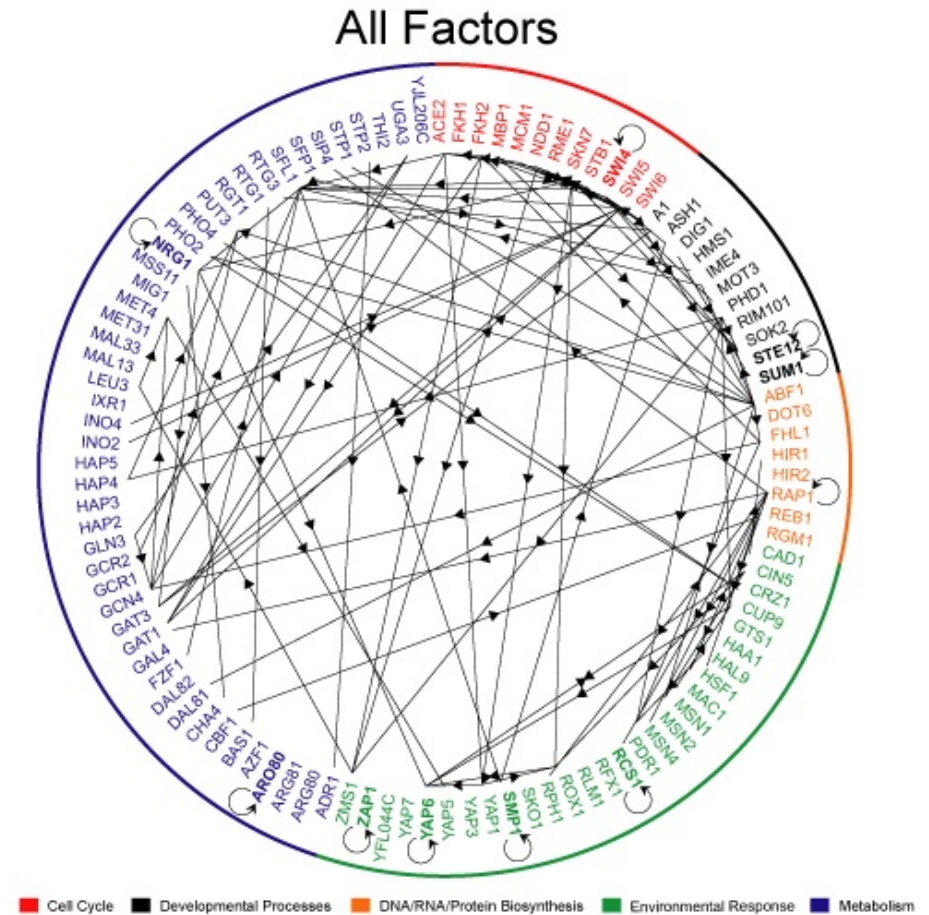
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Outline

1. Gene Regulation
2. Chemical Equations
3. Motifs



Chemical Equations

- We can write down the mechanism that underlie gene regulation in terms of chemical equations



- These are massively simplified (but we can ignore reactions that happen quickly)
- The rate of reaction depends on the number/concentration of the molecules on the left-hand side and the reaction rate k_i

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- Chemical equations happen at random, so the rate equations describe the average changes



- Let $[X]$ denote the (mean) concentration of X then

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- $\Omega = VA$, $[X] = X/\Omega$ and $[Y] = Y/\Omega$

Fluctuations

- In chemistry we are used to there being so many molecules that the average rate equations provide a very good approximation of what really happens
- This is often true in biology, but not always
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- Proteins can have multiple sub-units that each bind with a signalling molecule
- A complex might only be stable when the protein has bound some number of signalling molecules

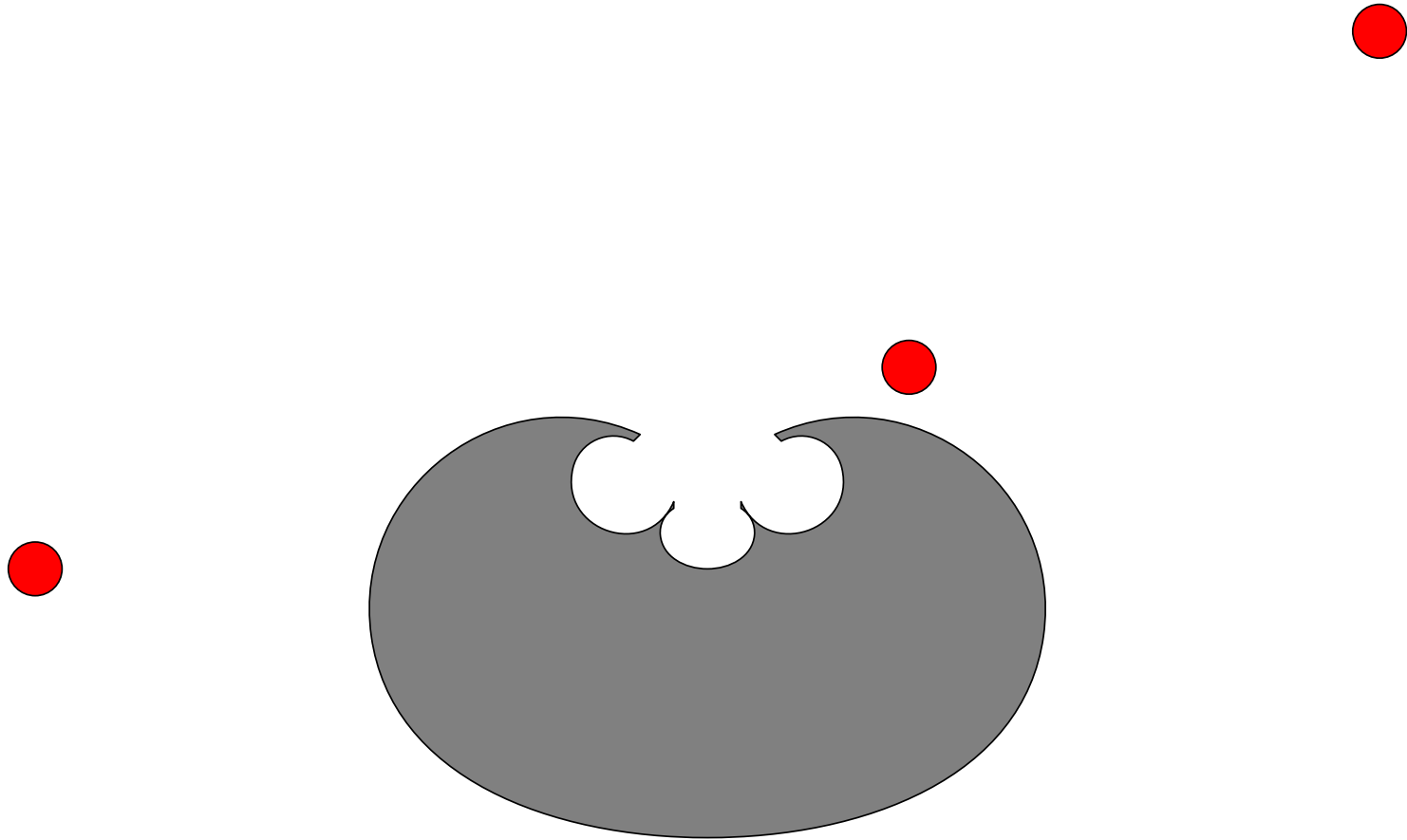
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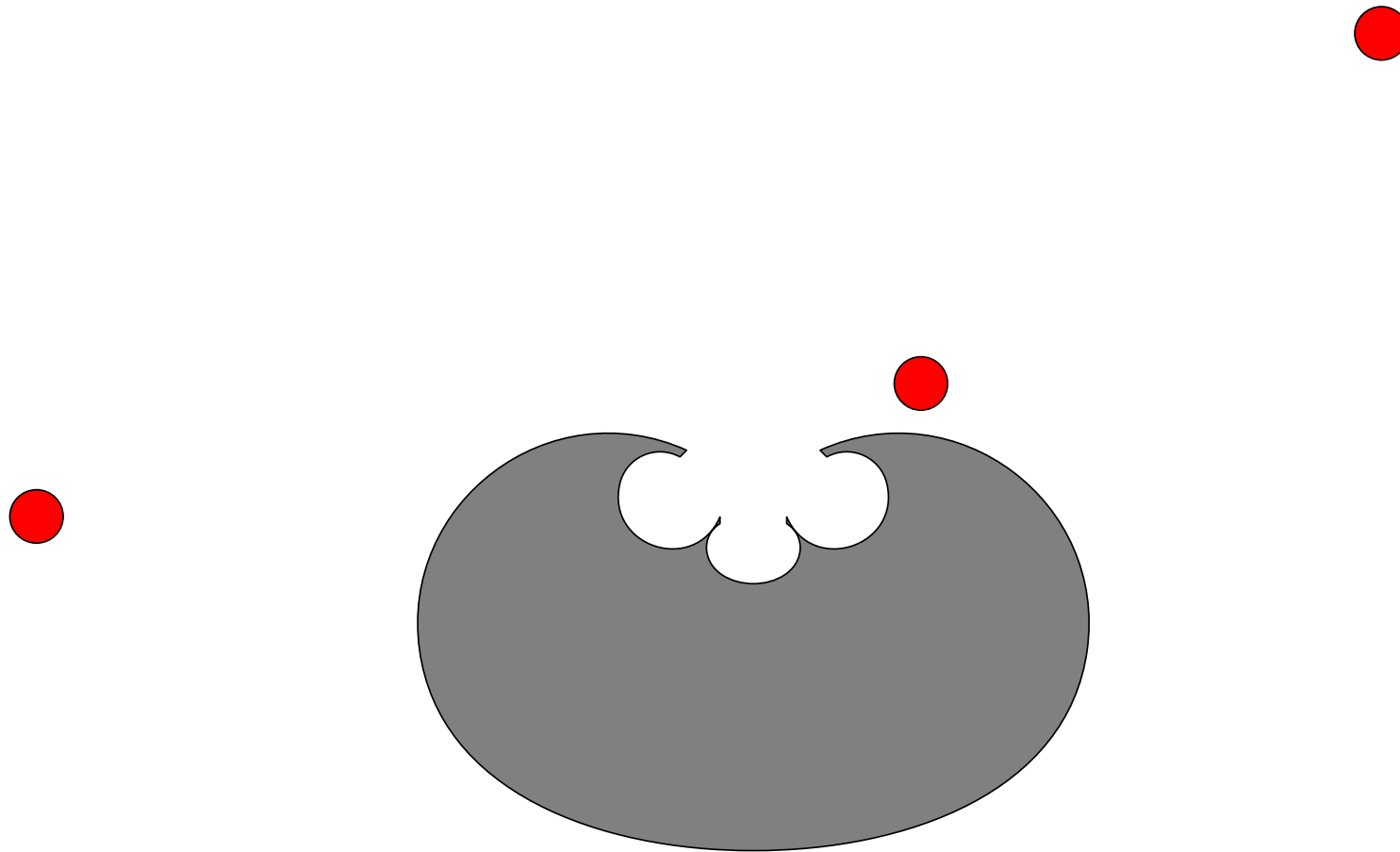
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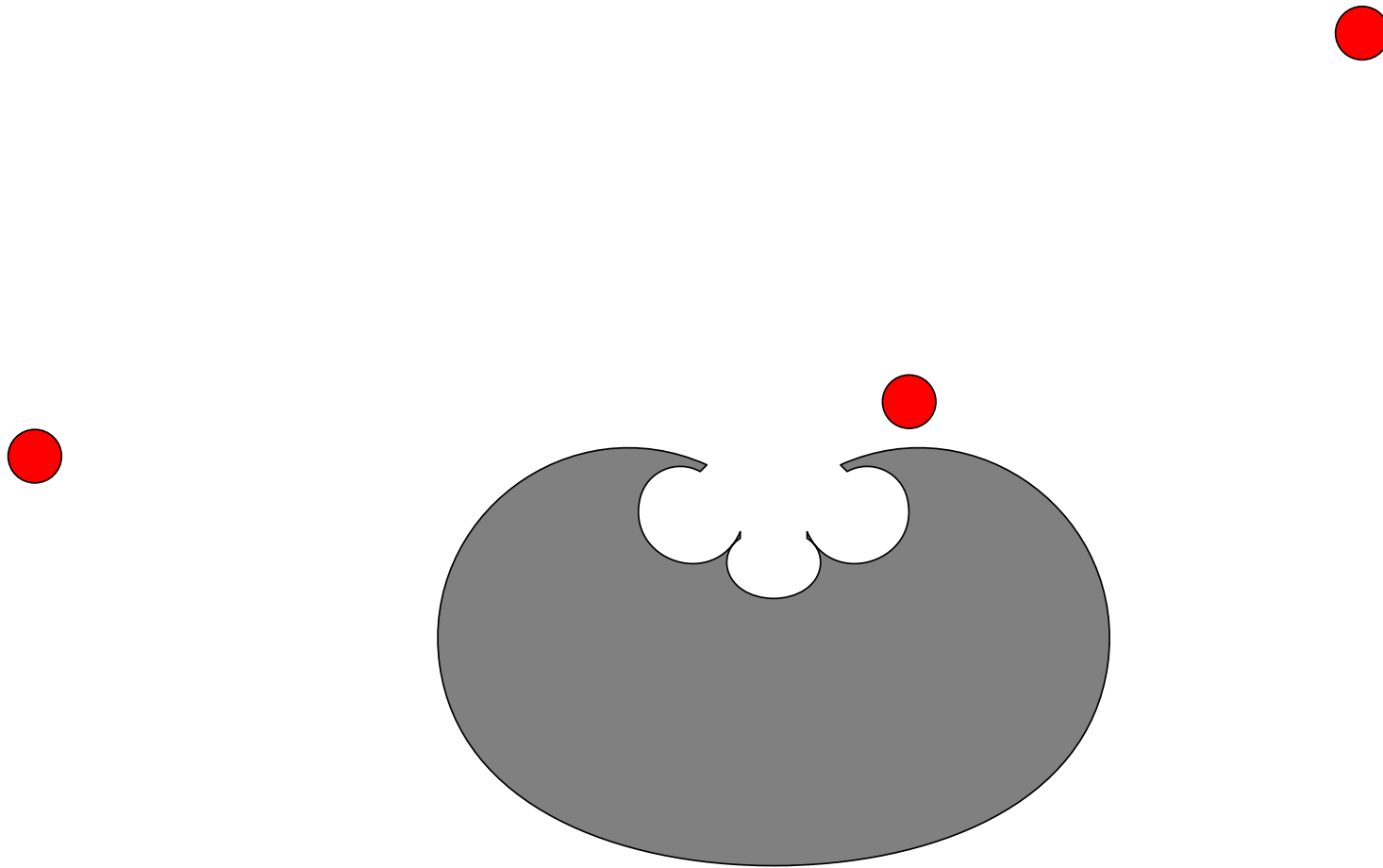
Multi-signalling



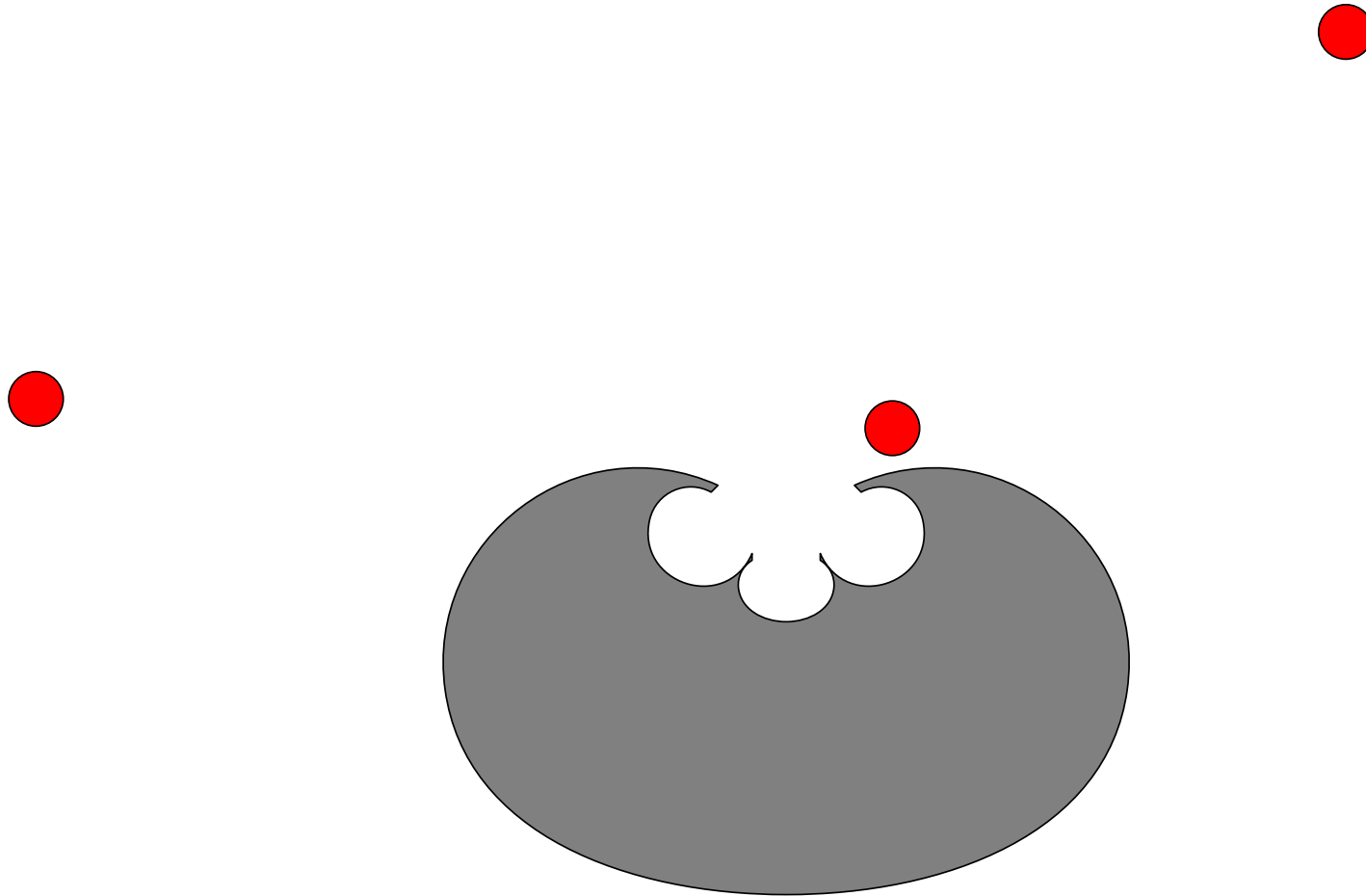
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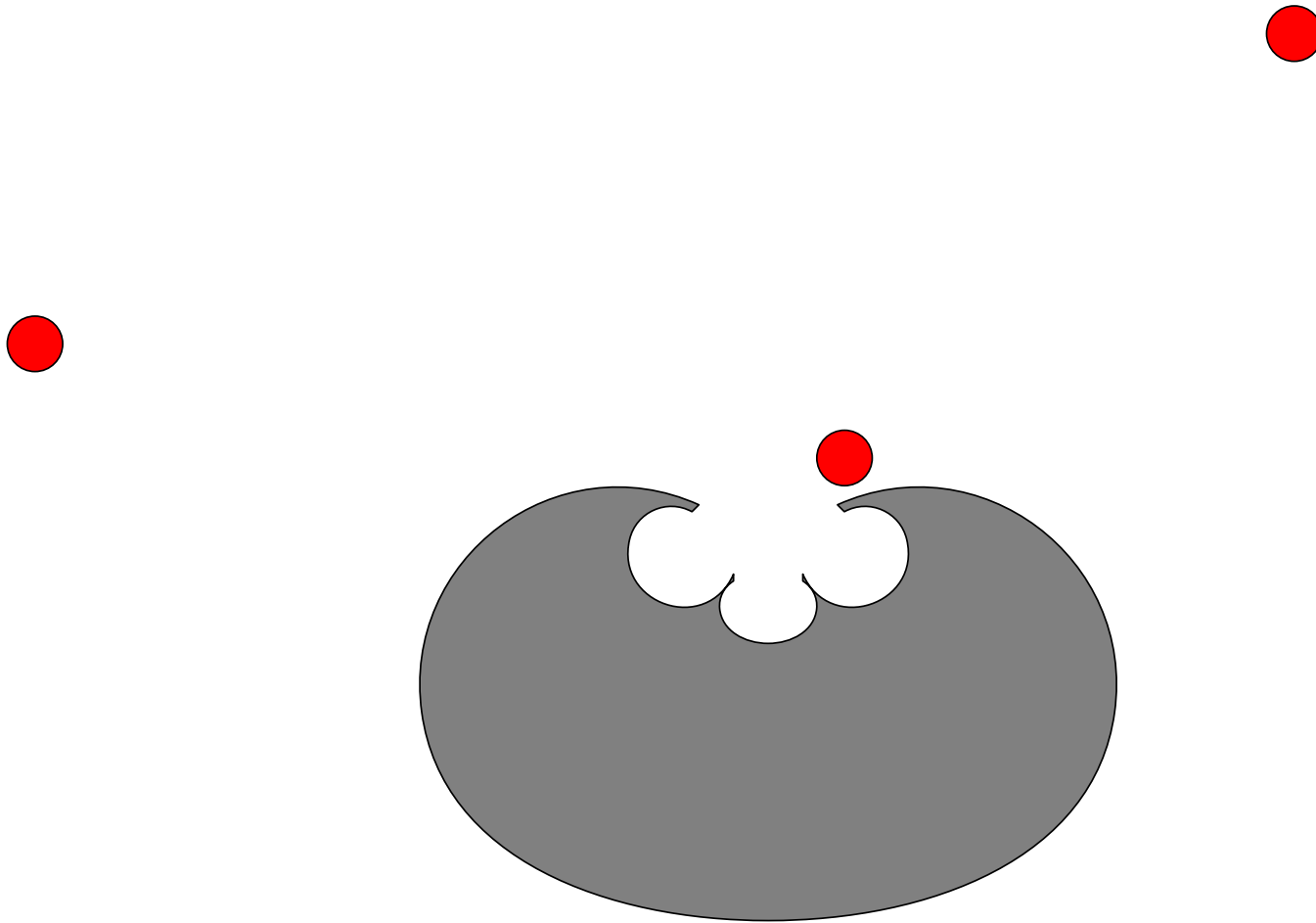
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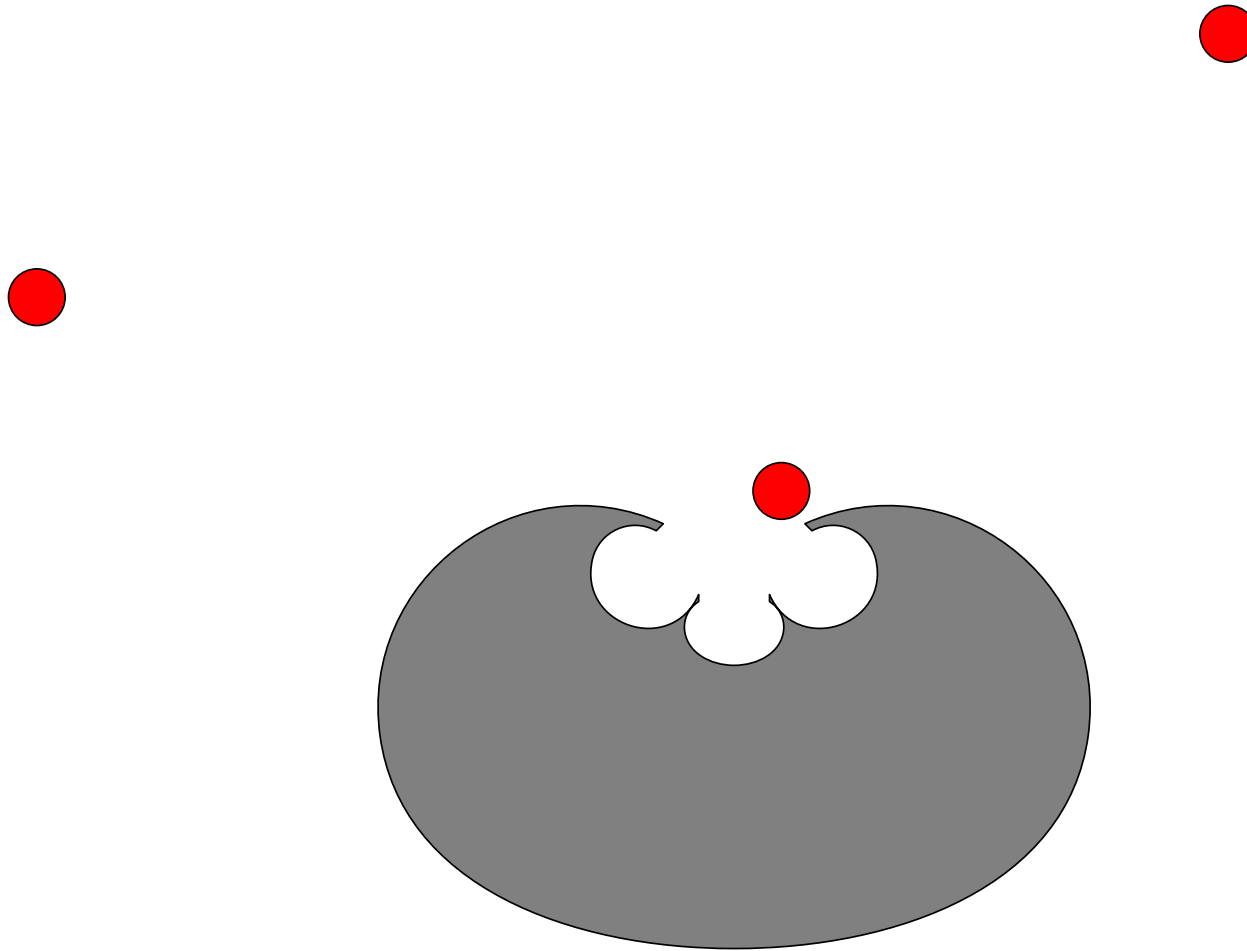
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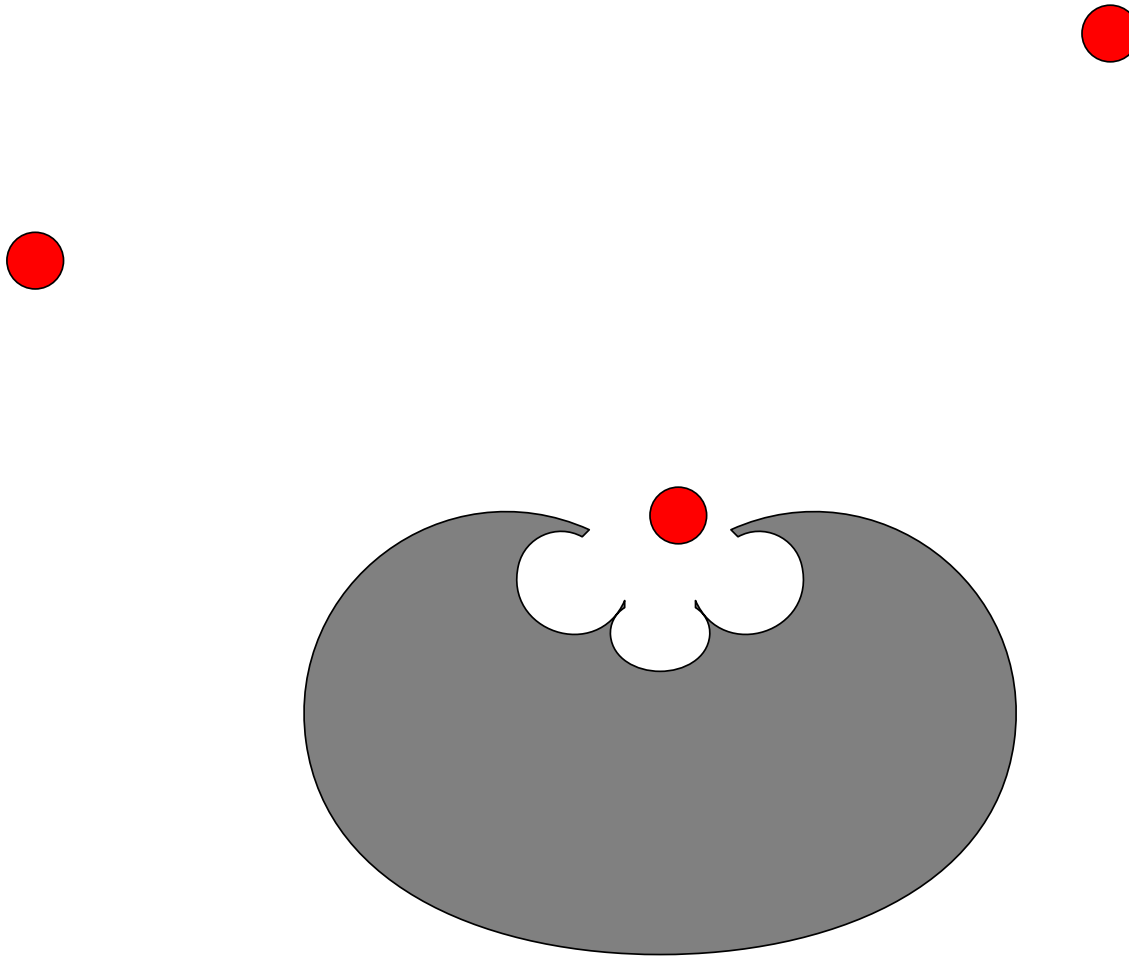
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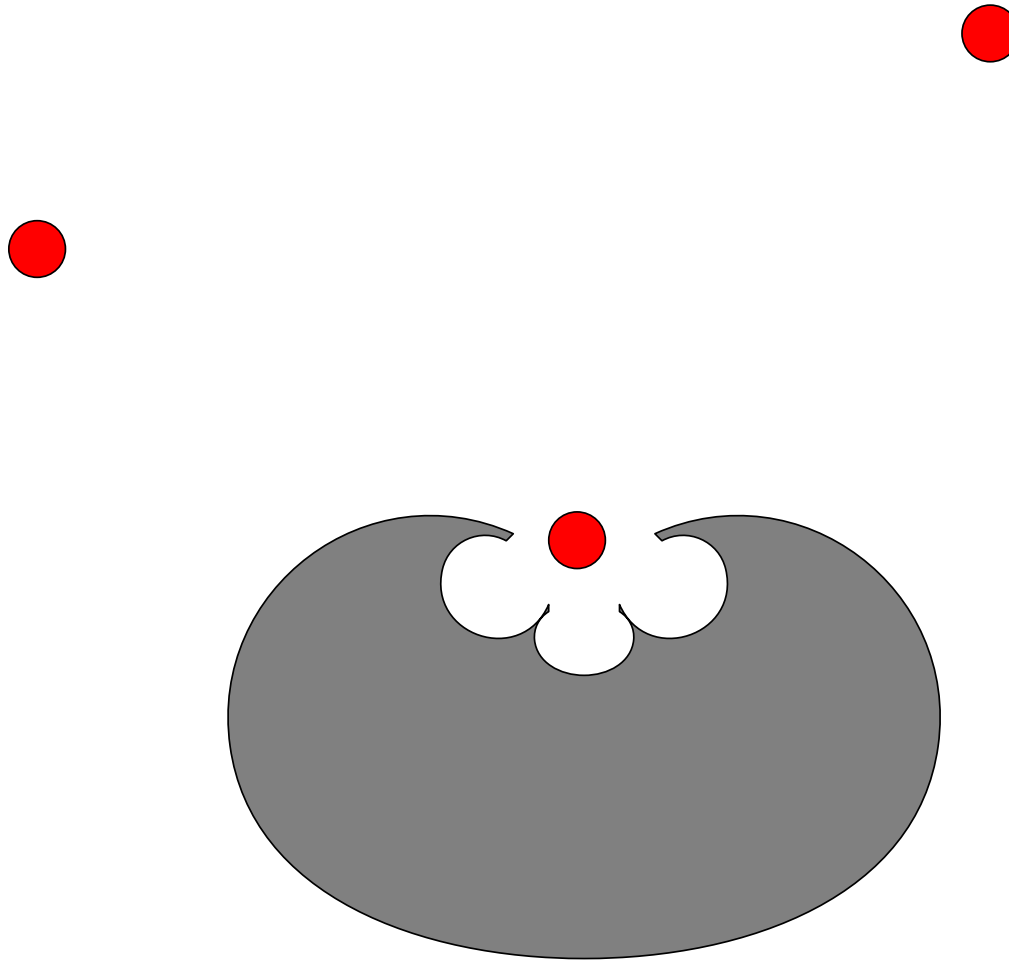
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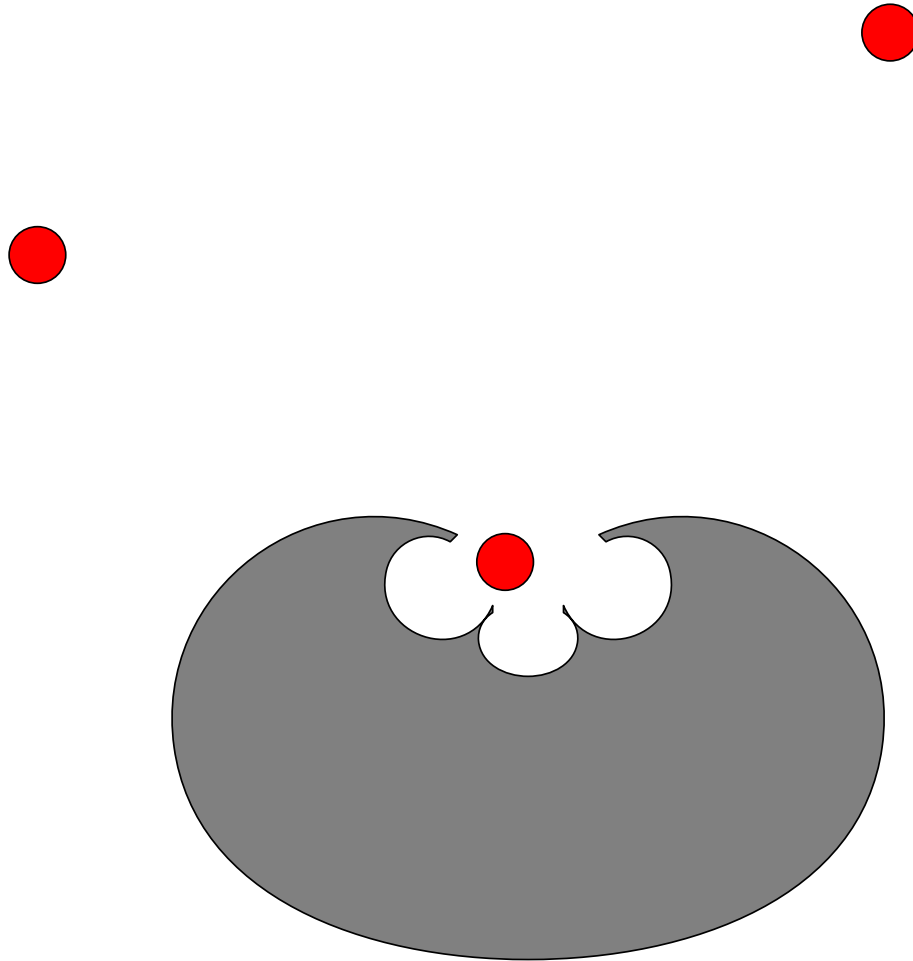
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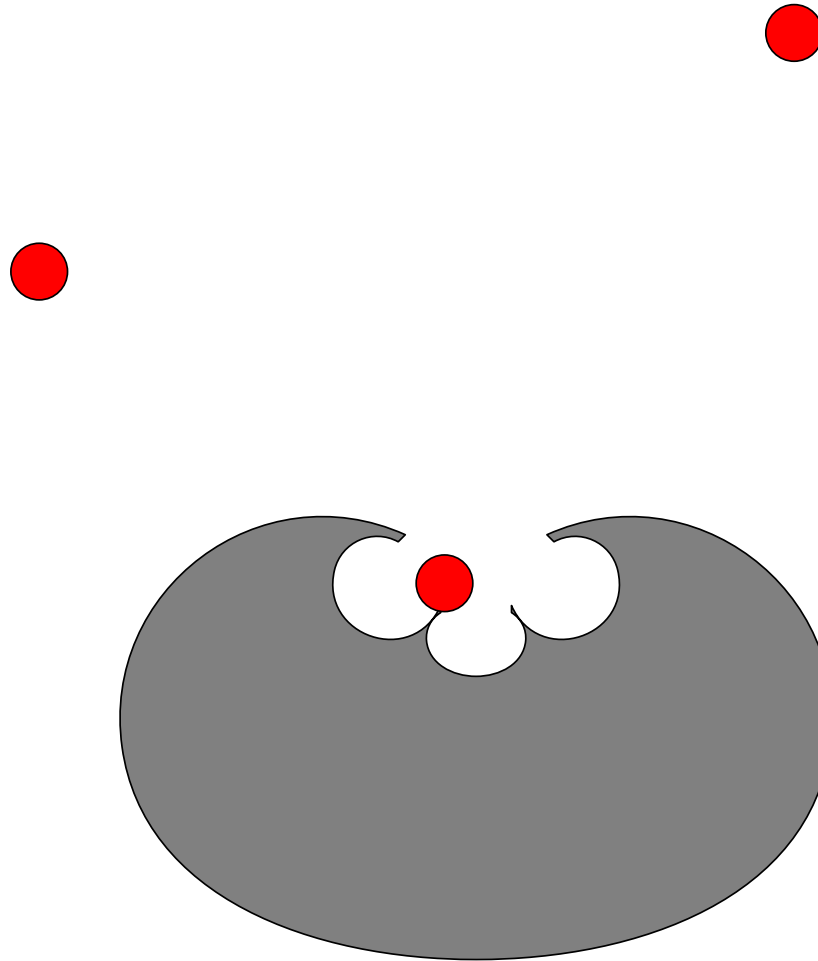
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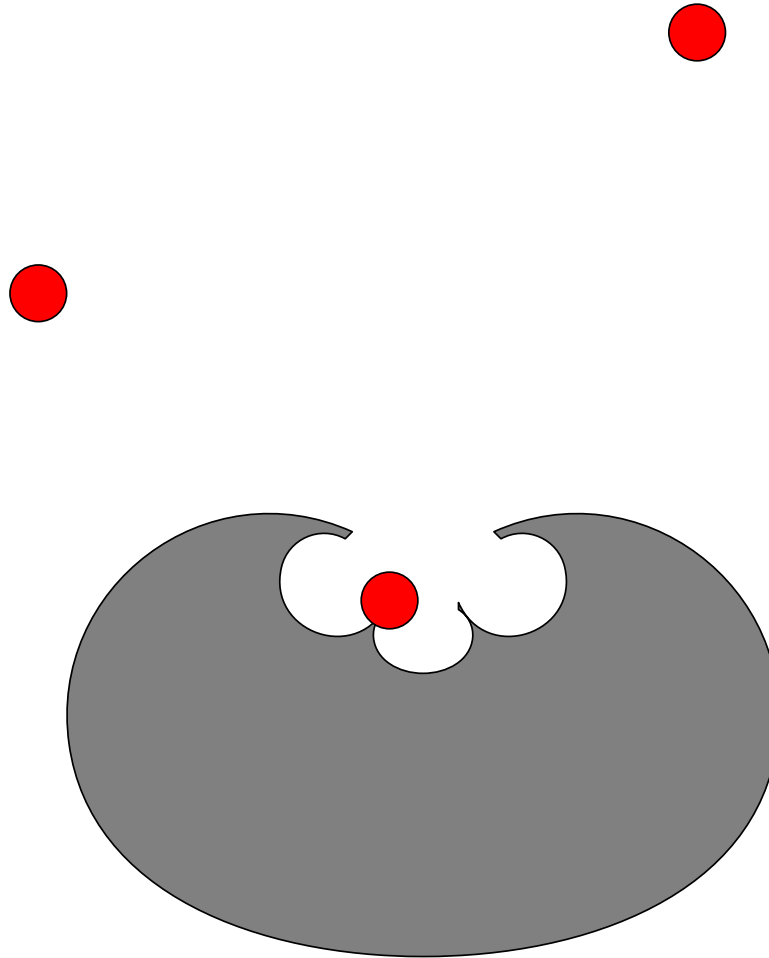
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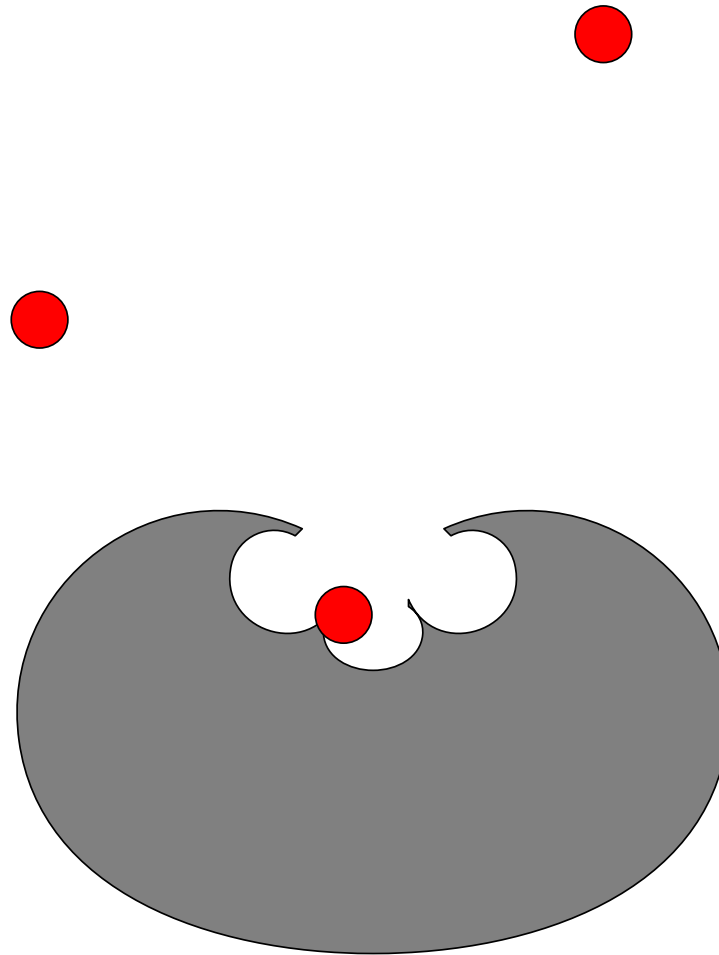
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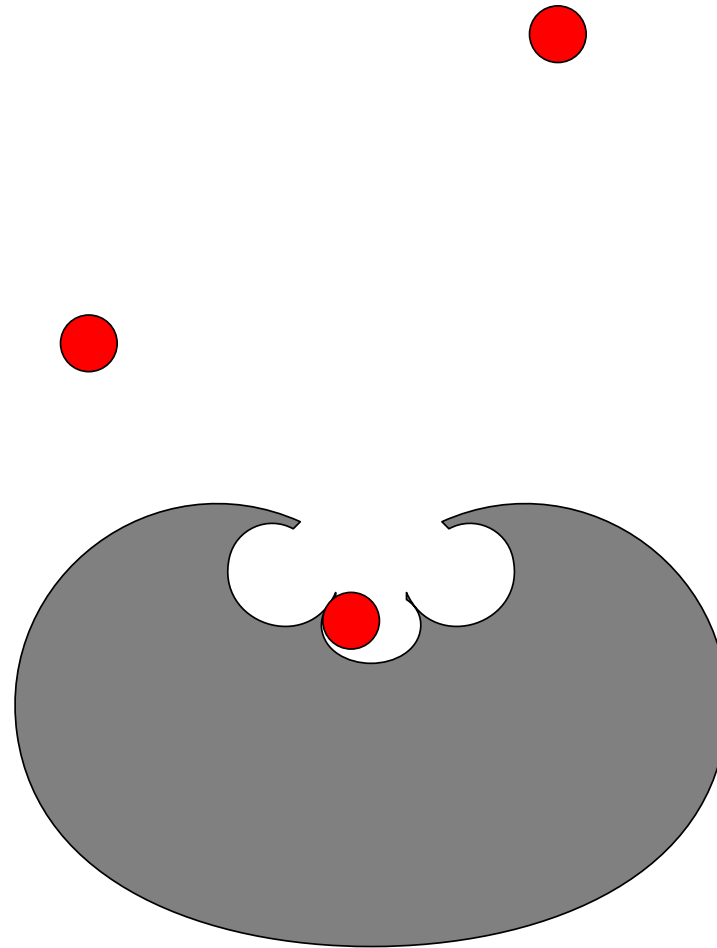
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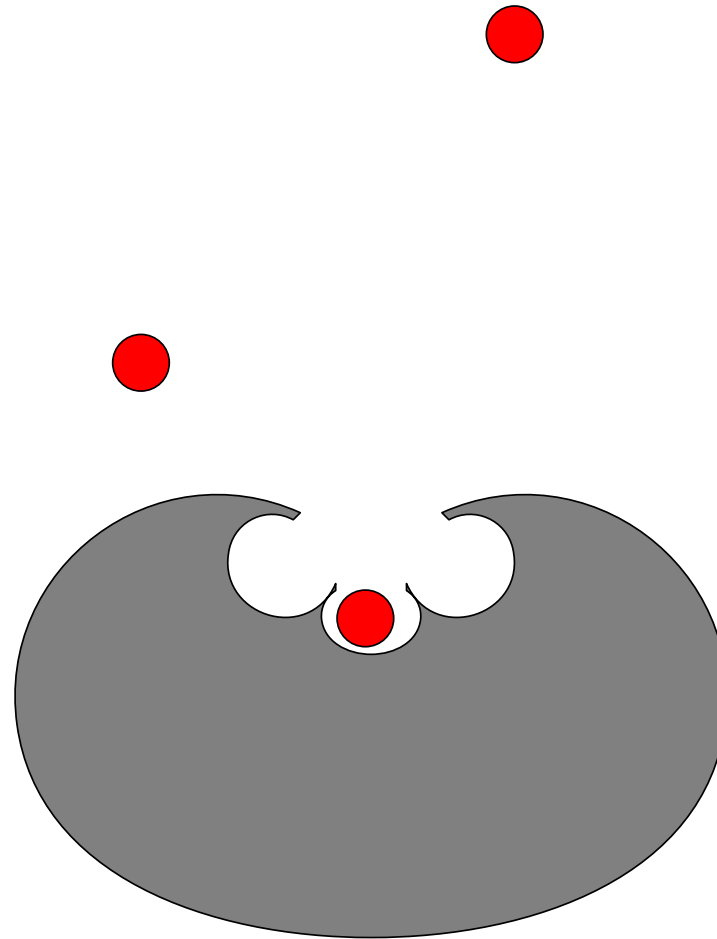
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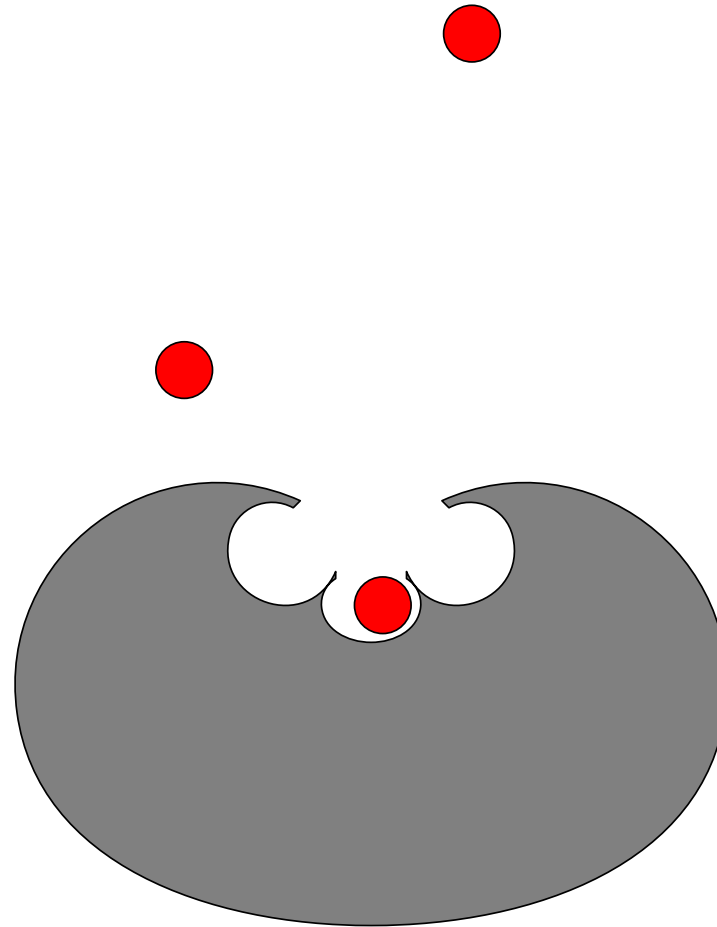
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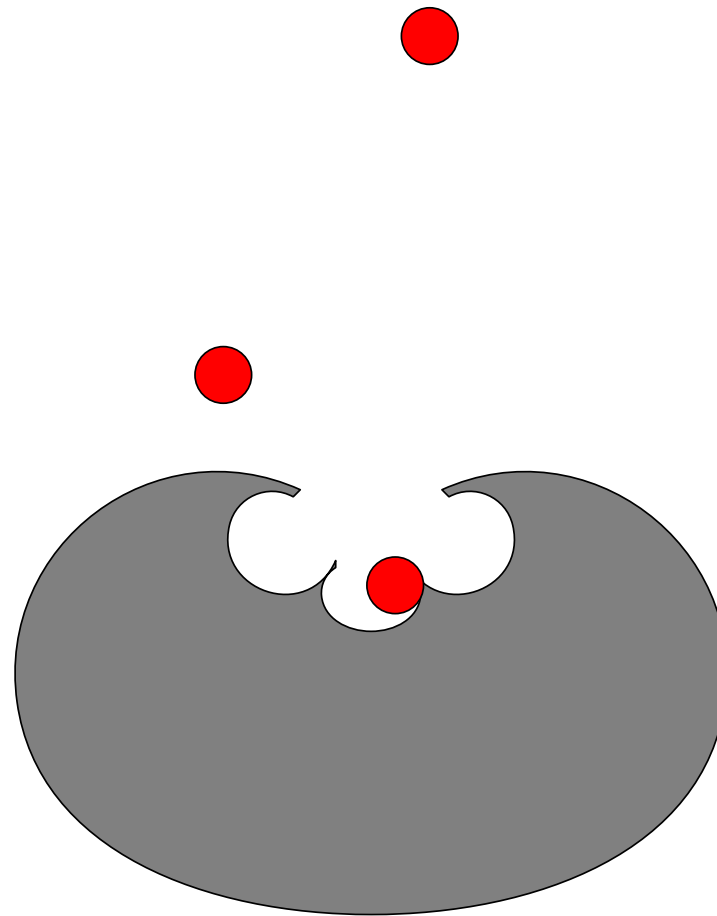
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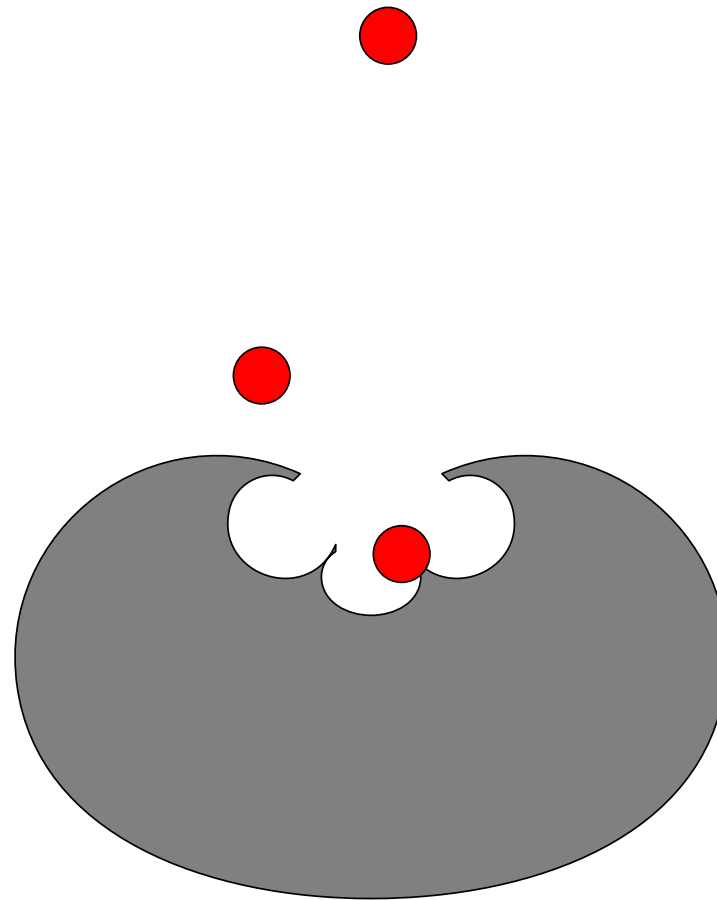
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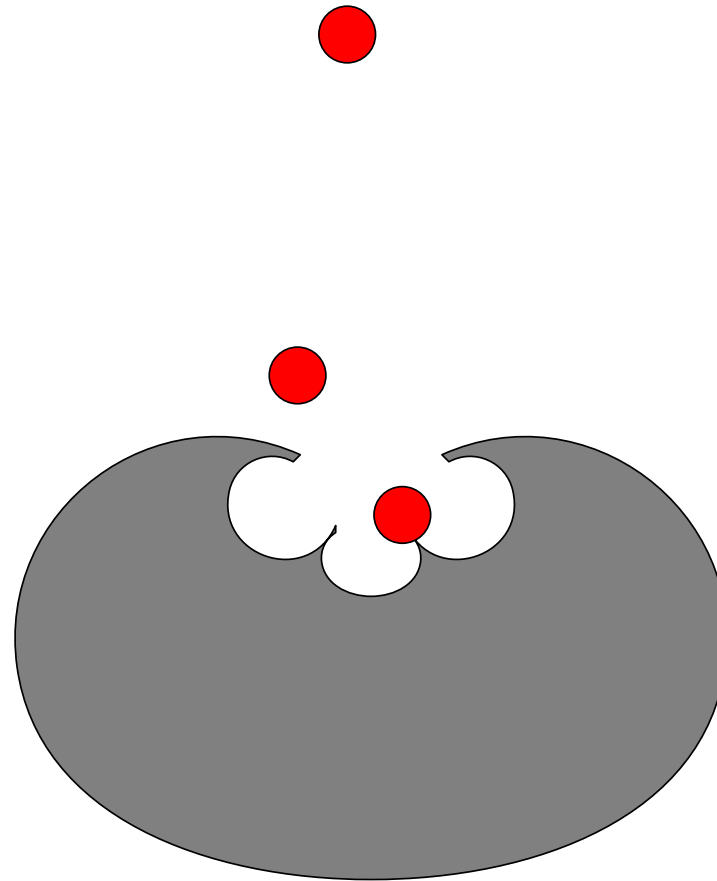
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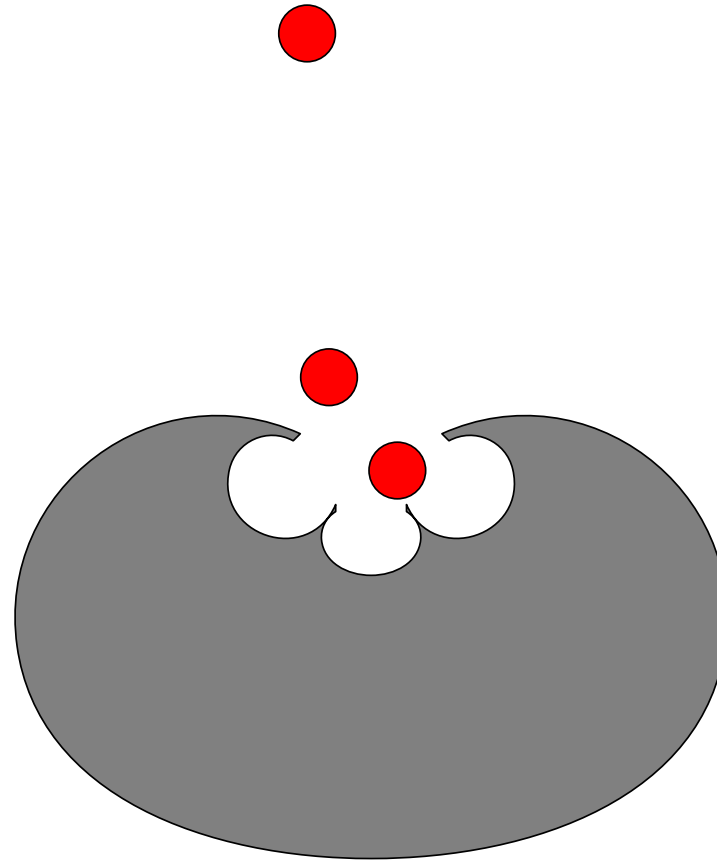
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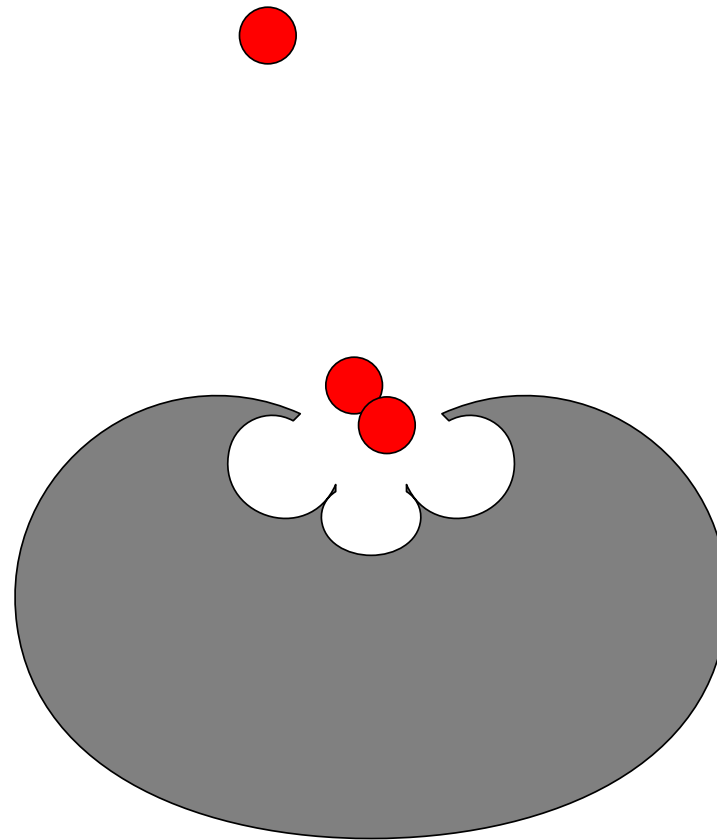
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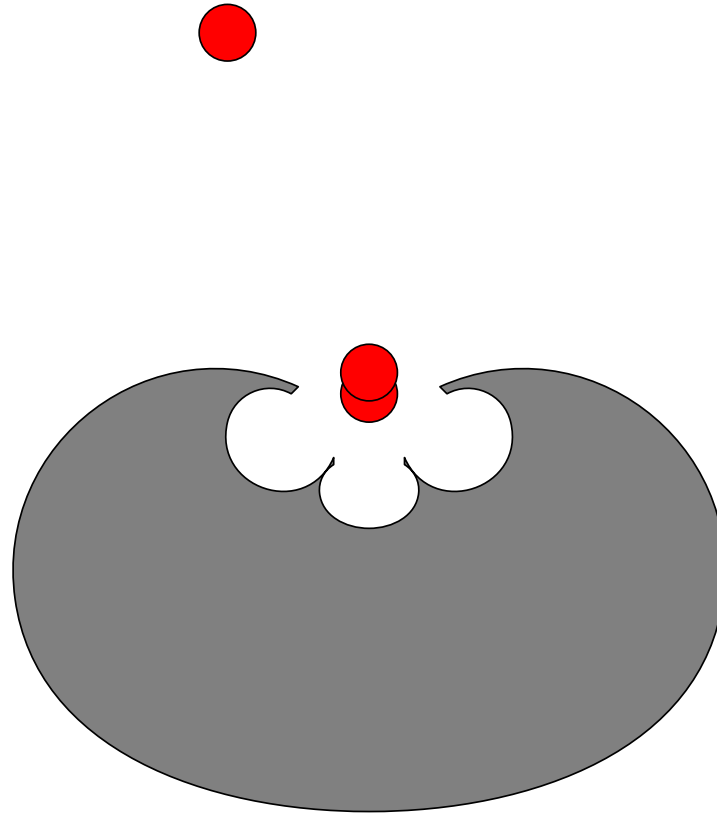
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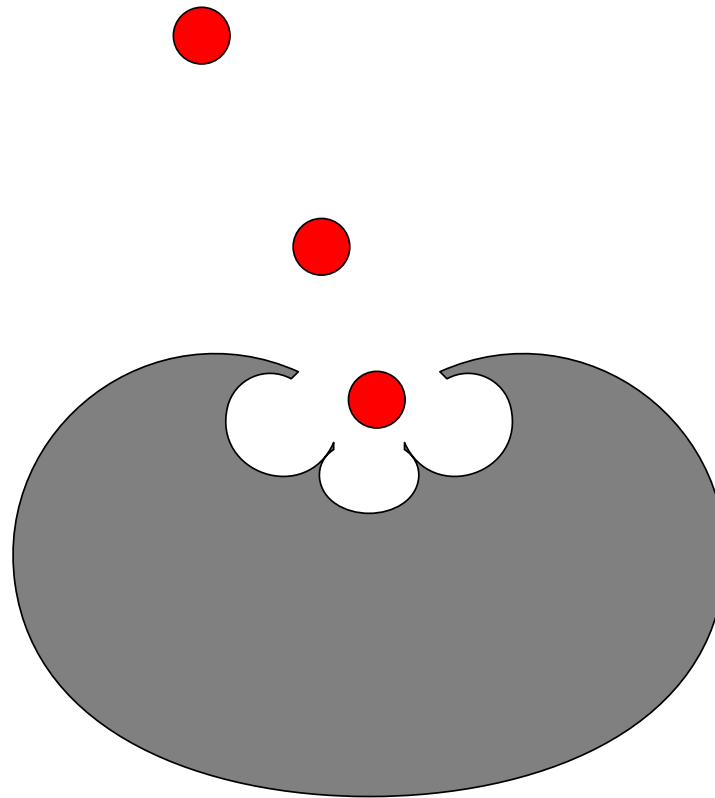
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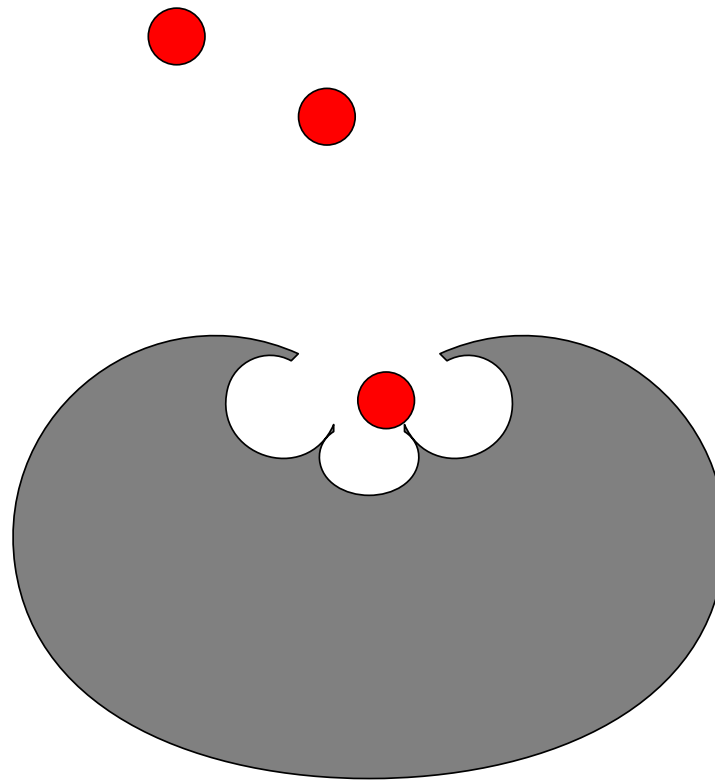
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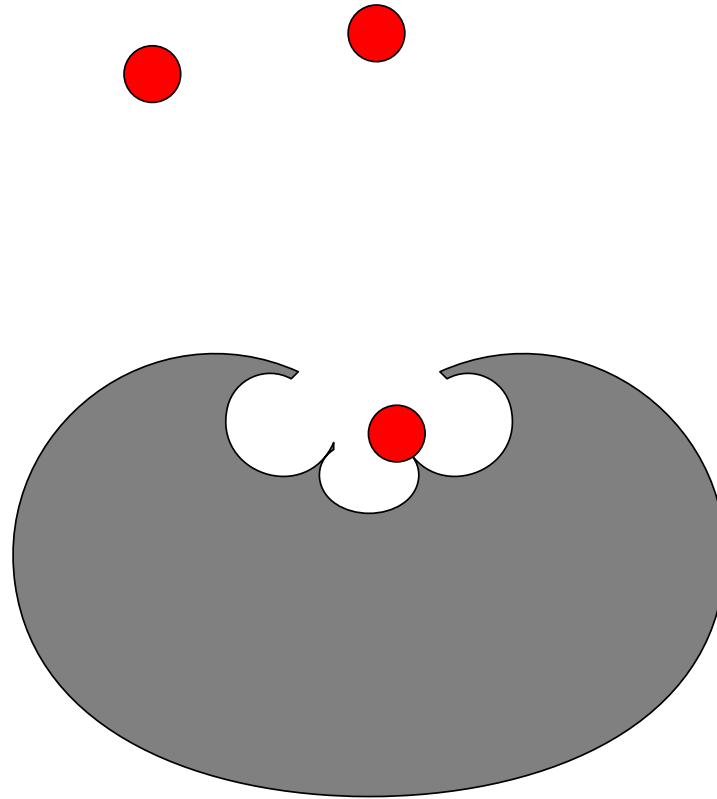
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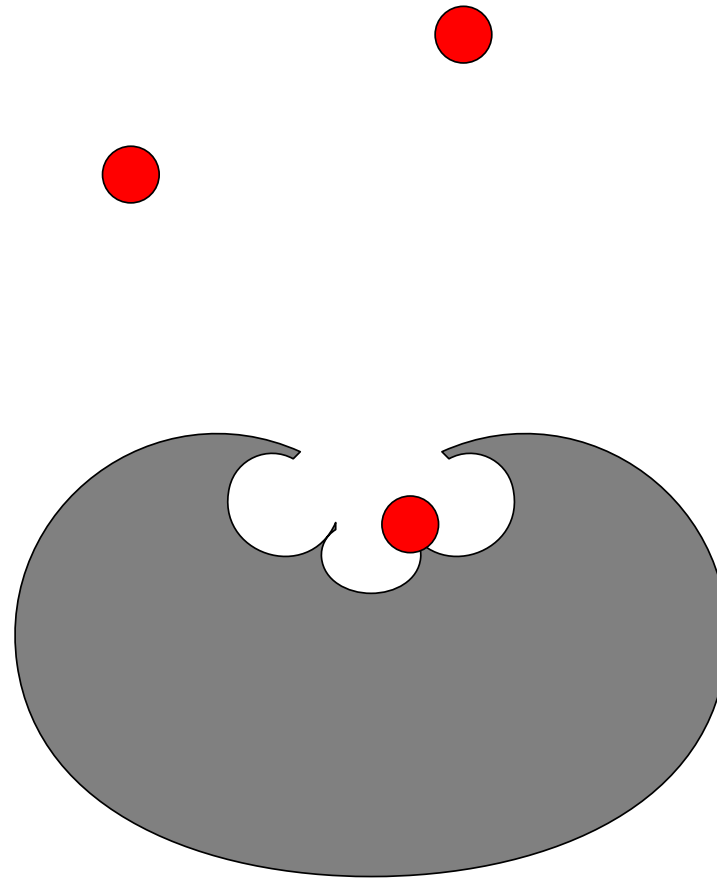
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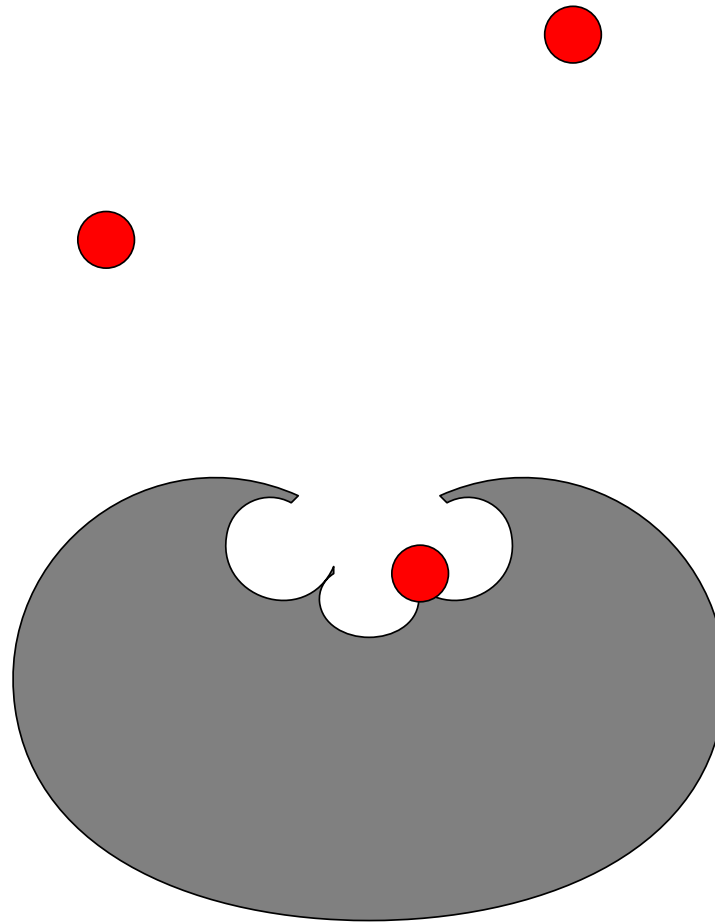
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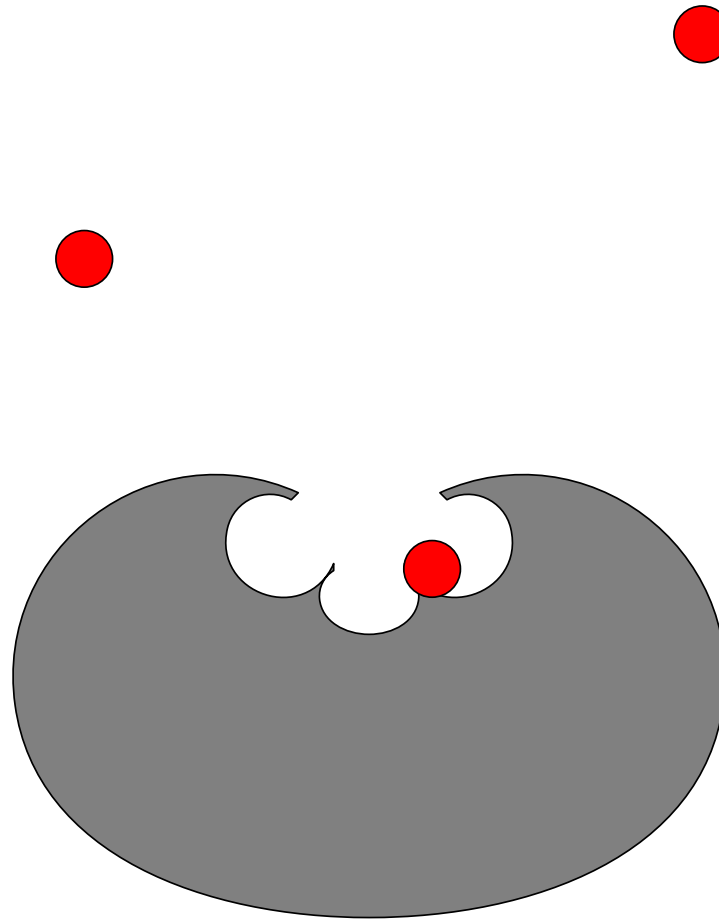
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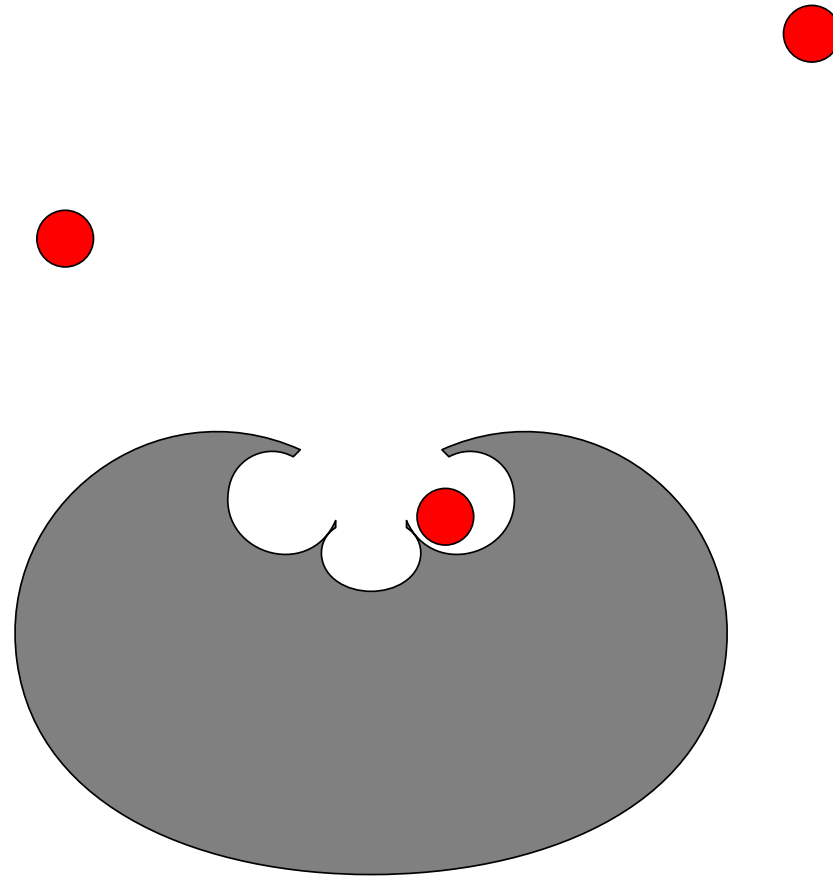
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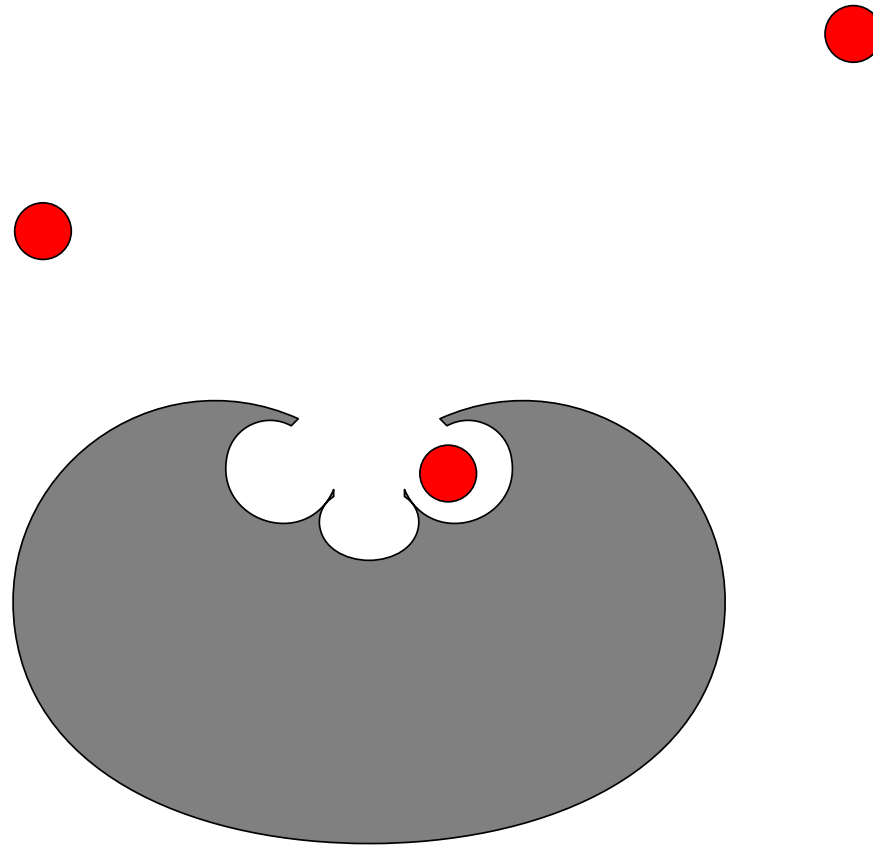
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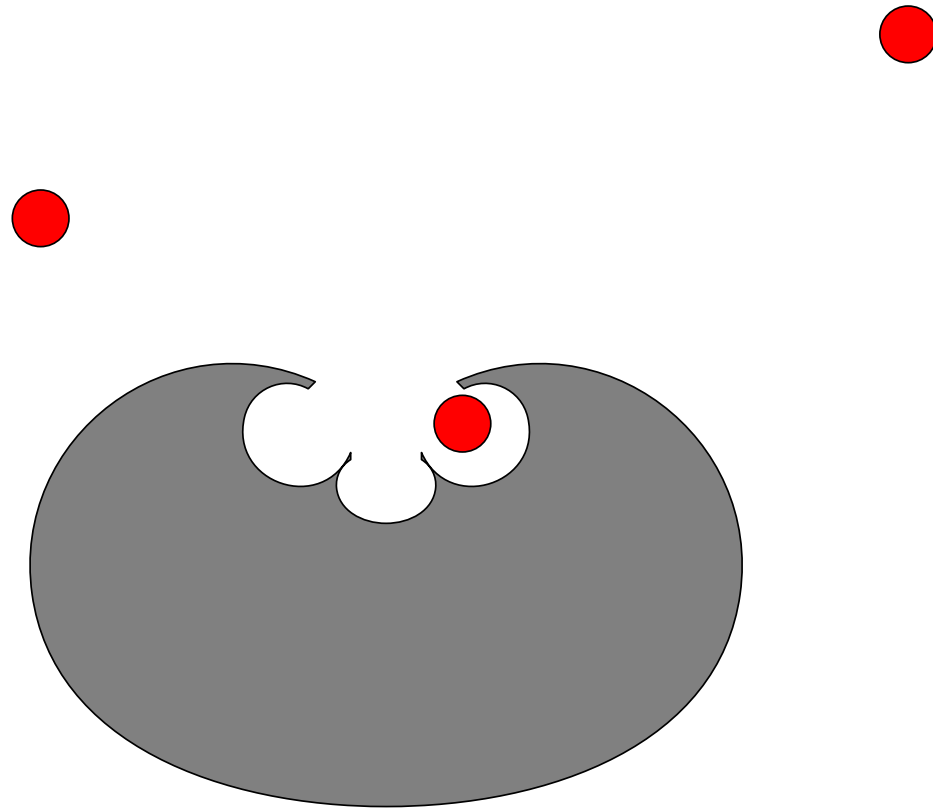
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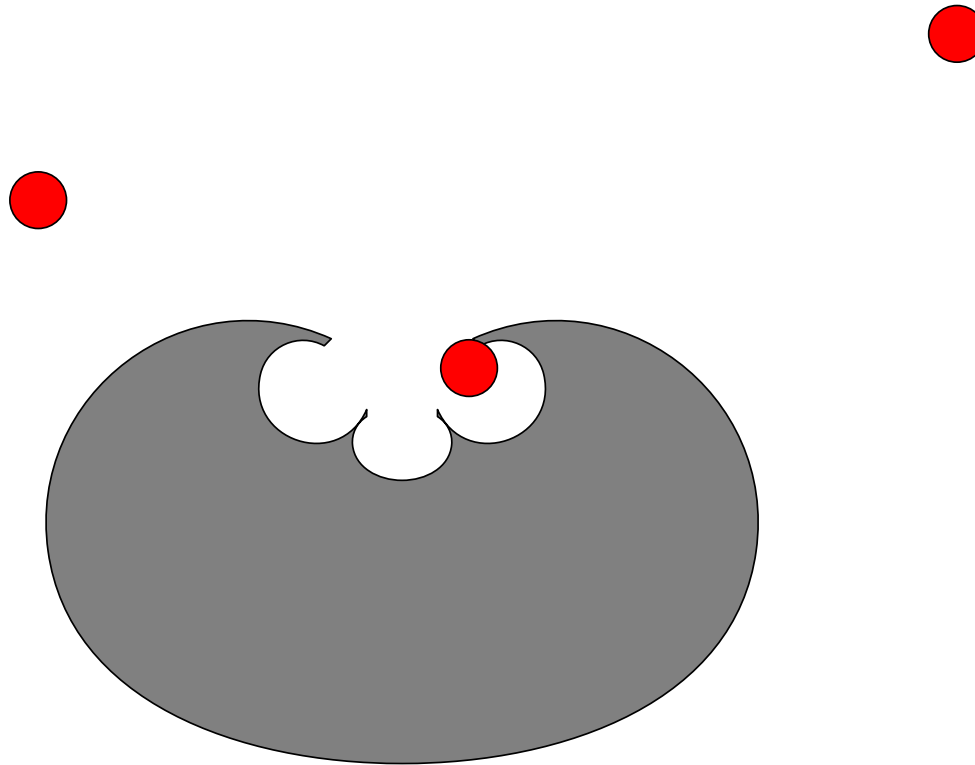
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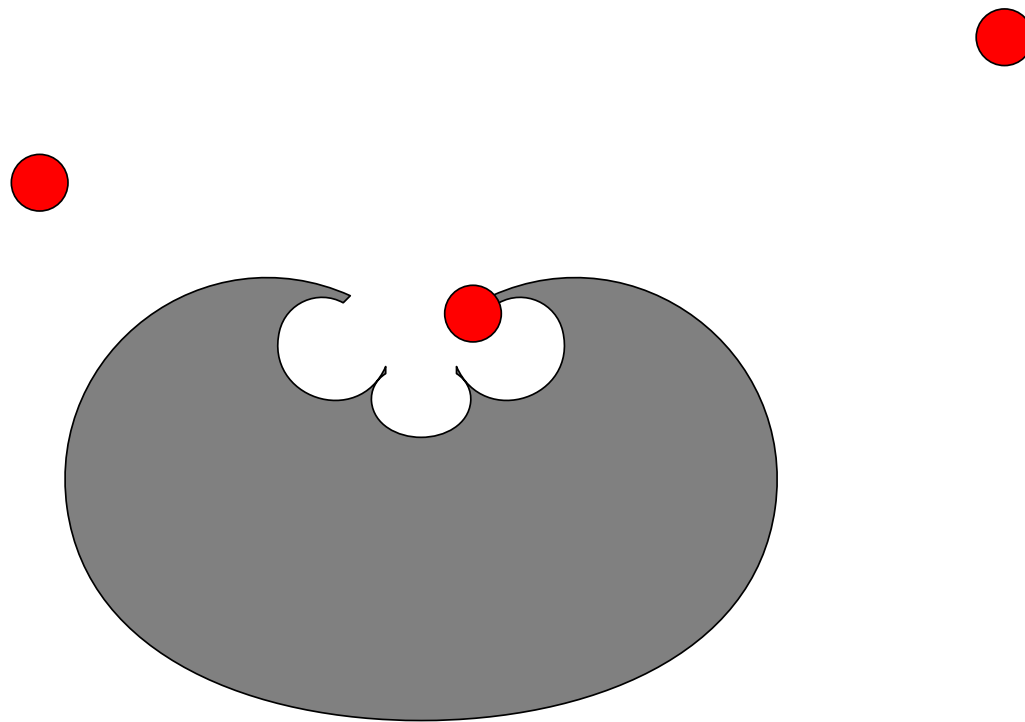
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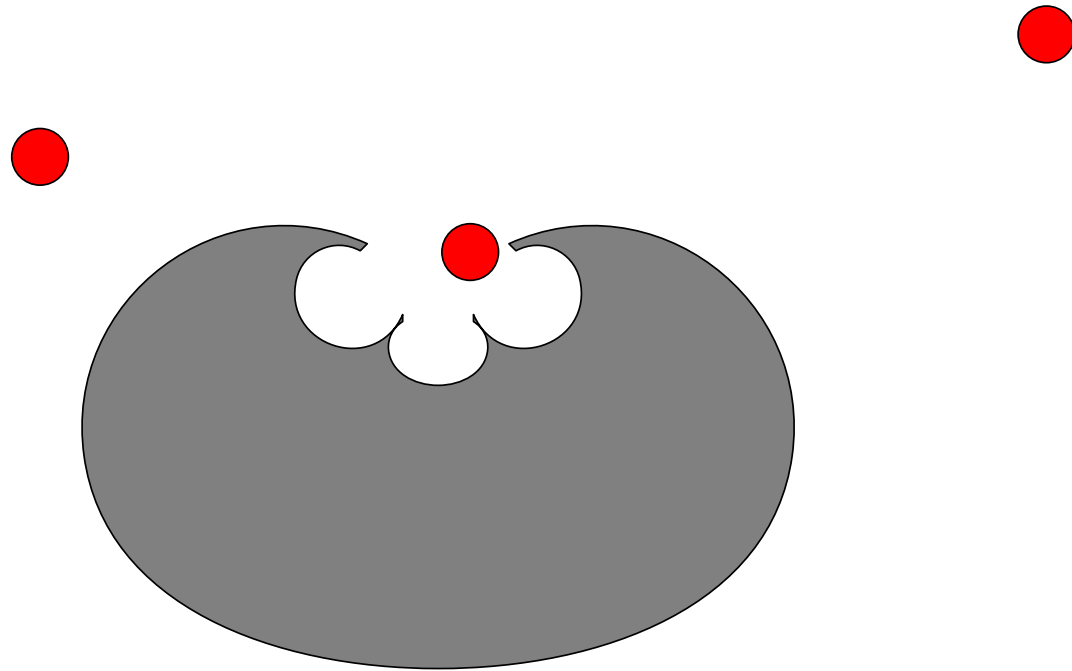
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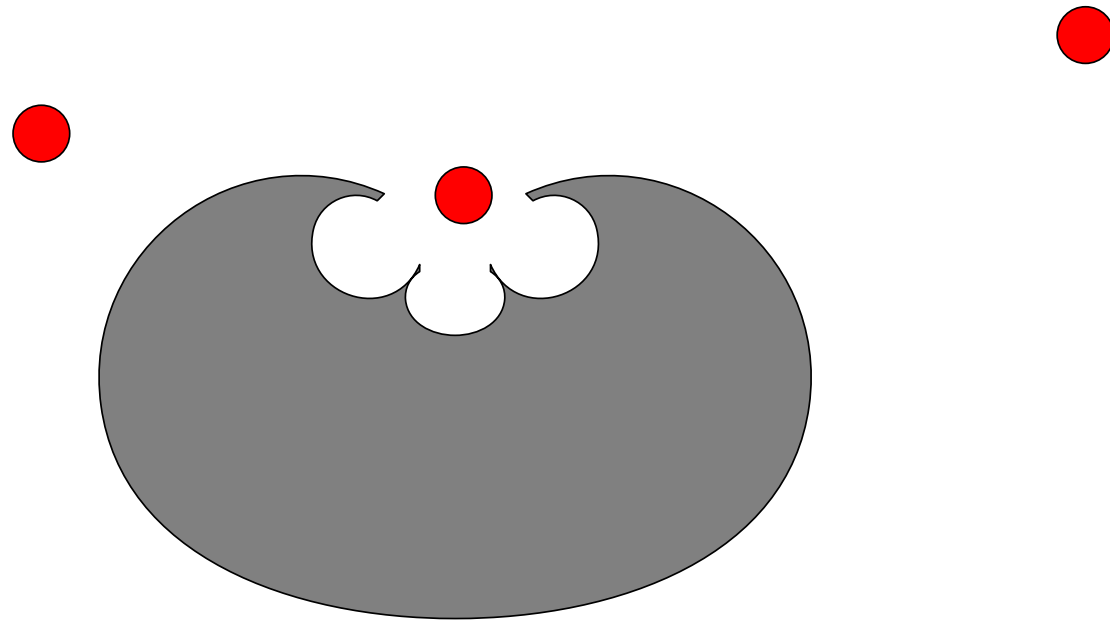
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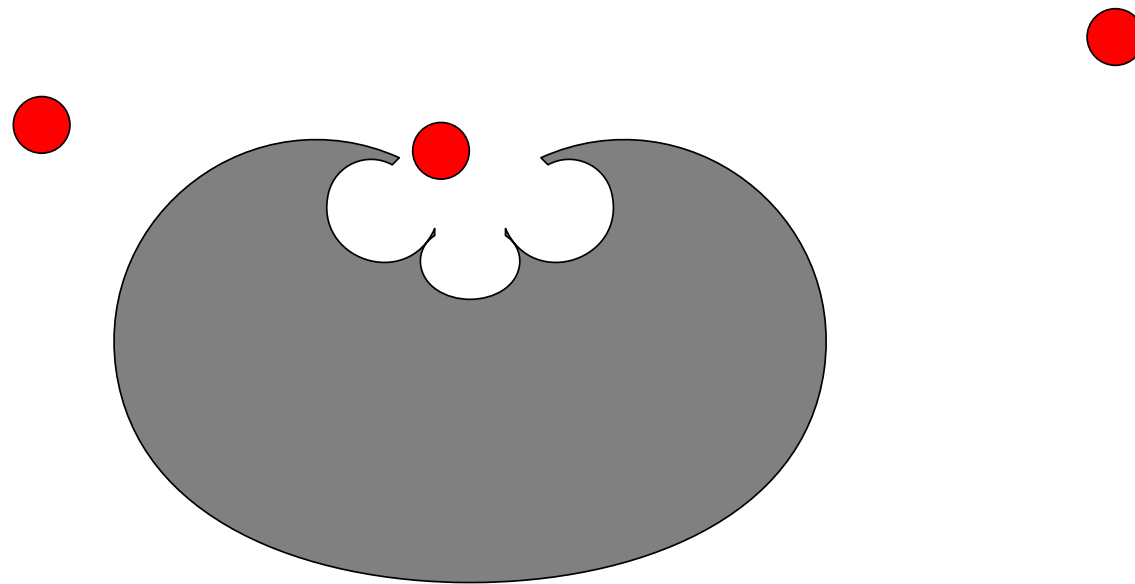
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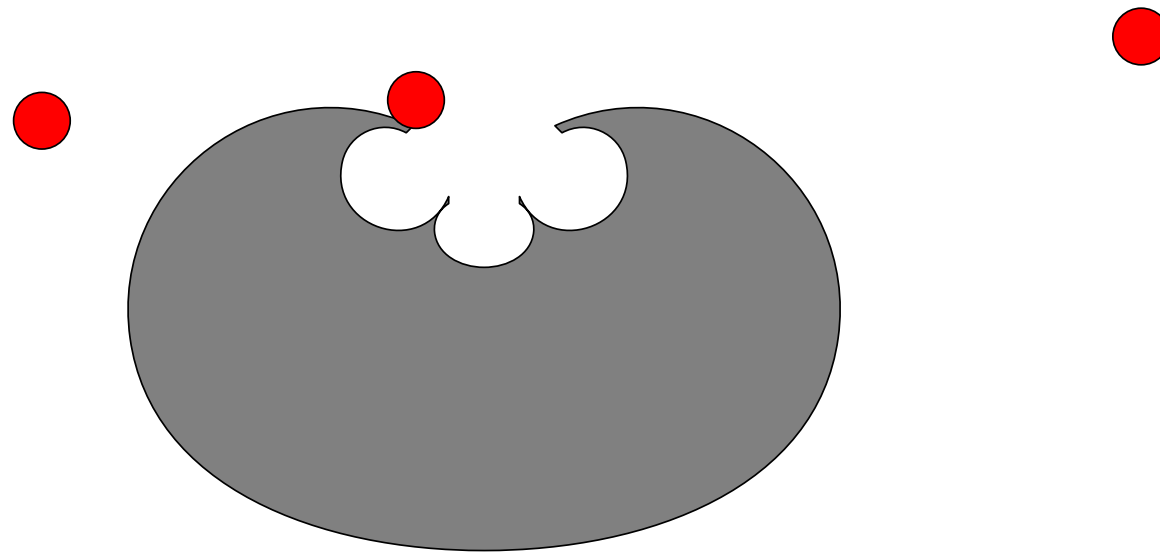
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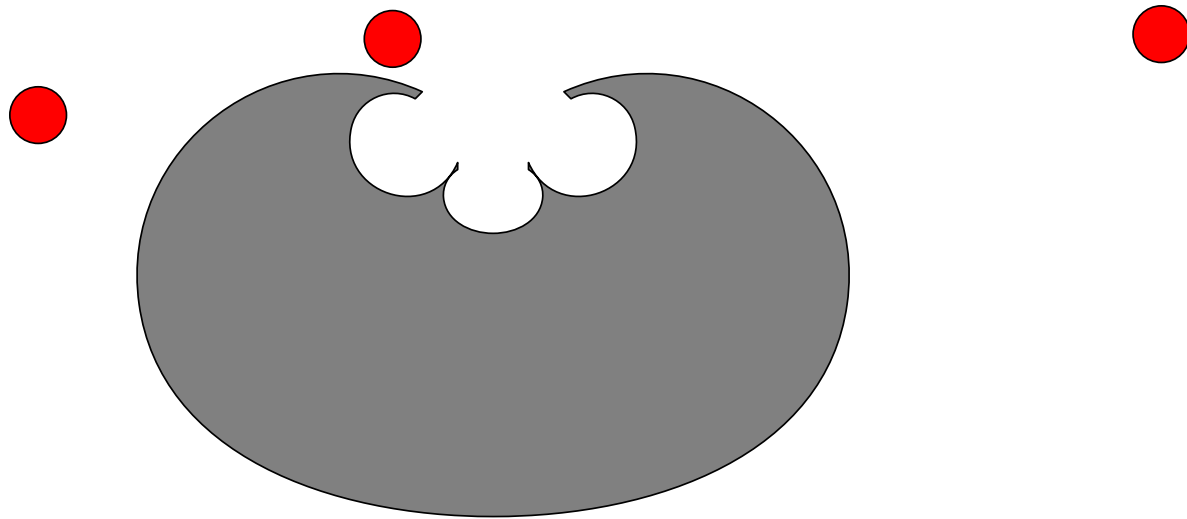
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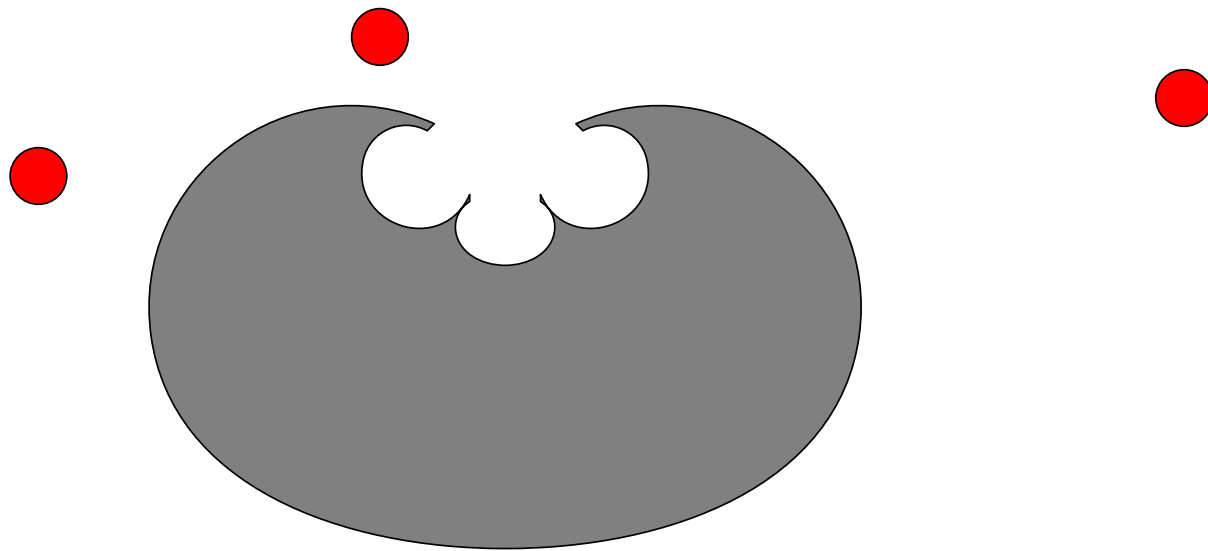
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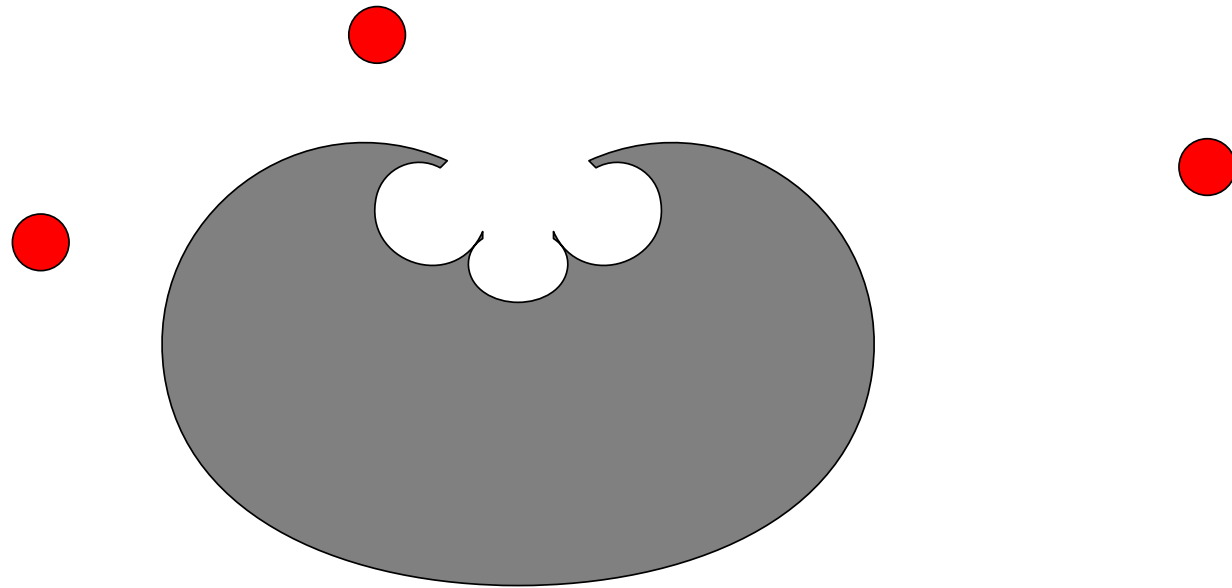
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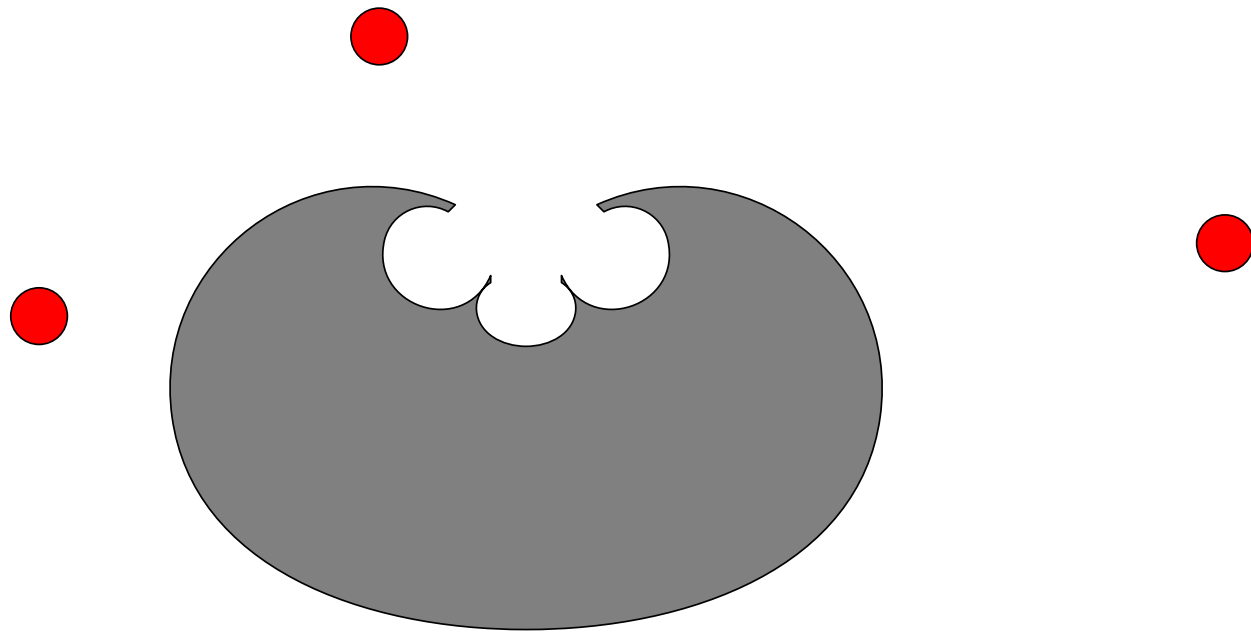
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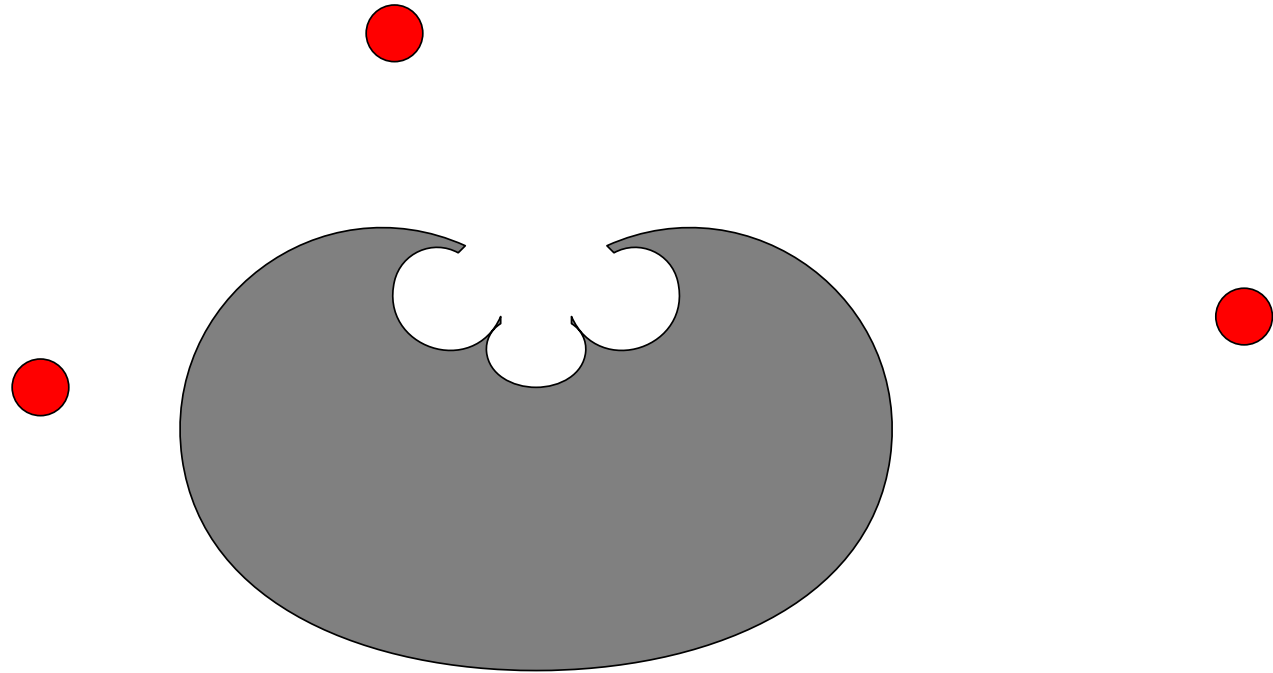
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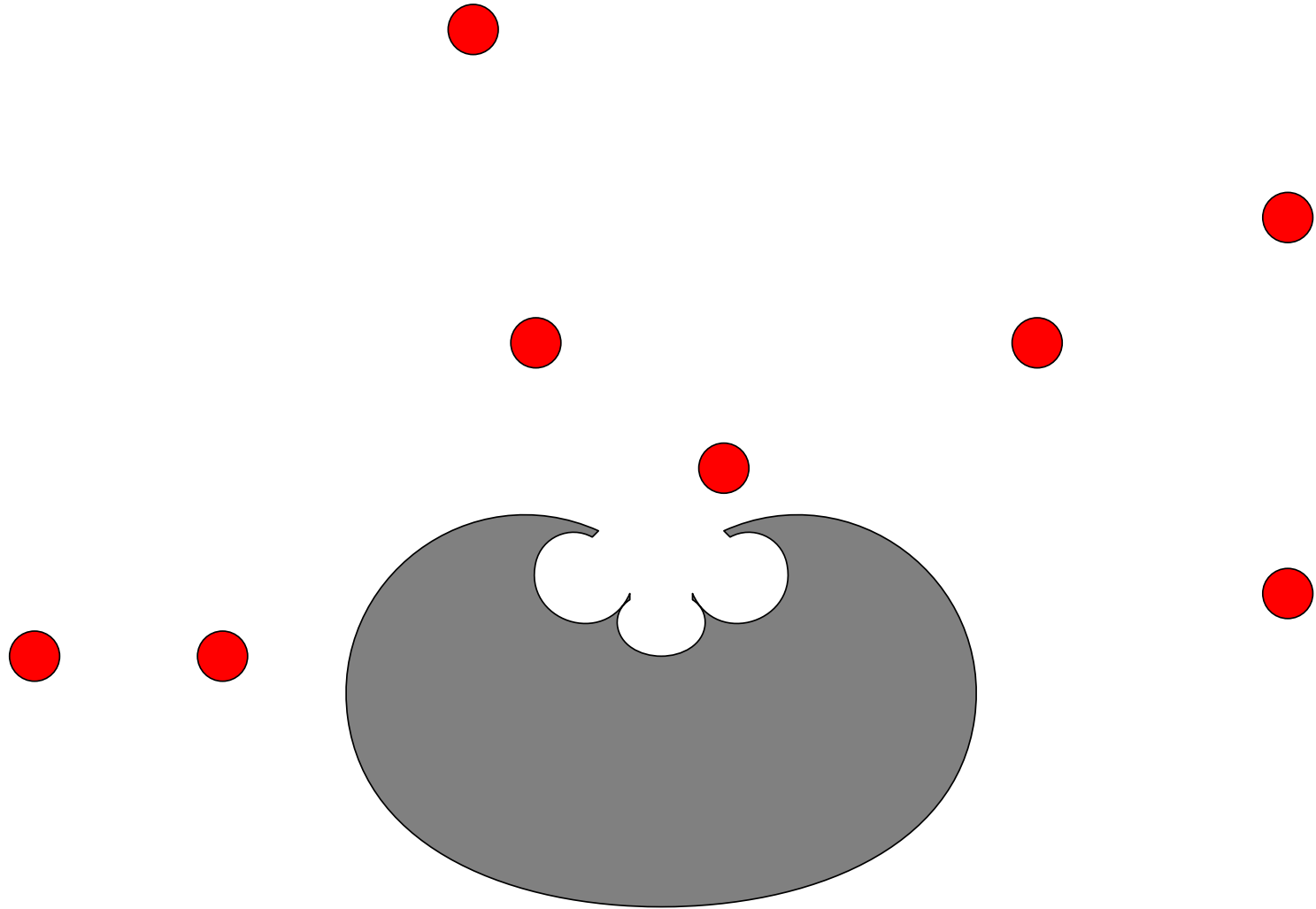
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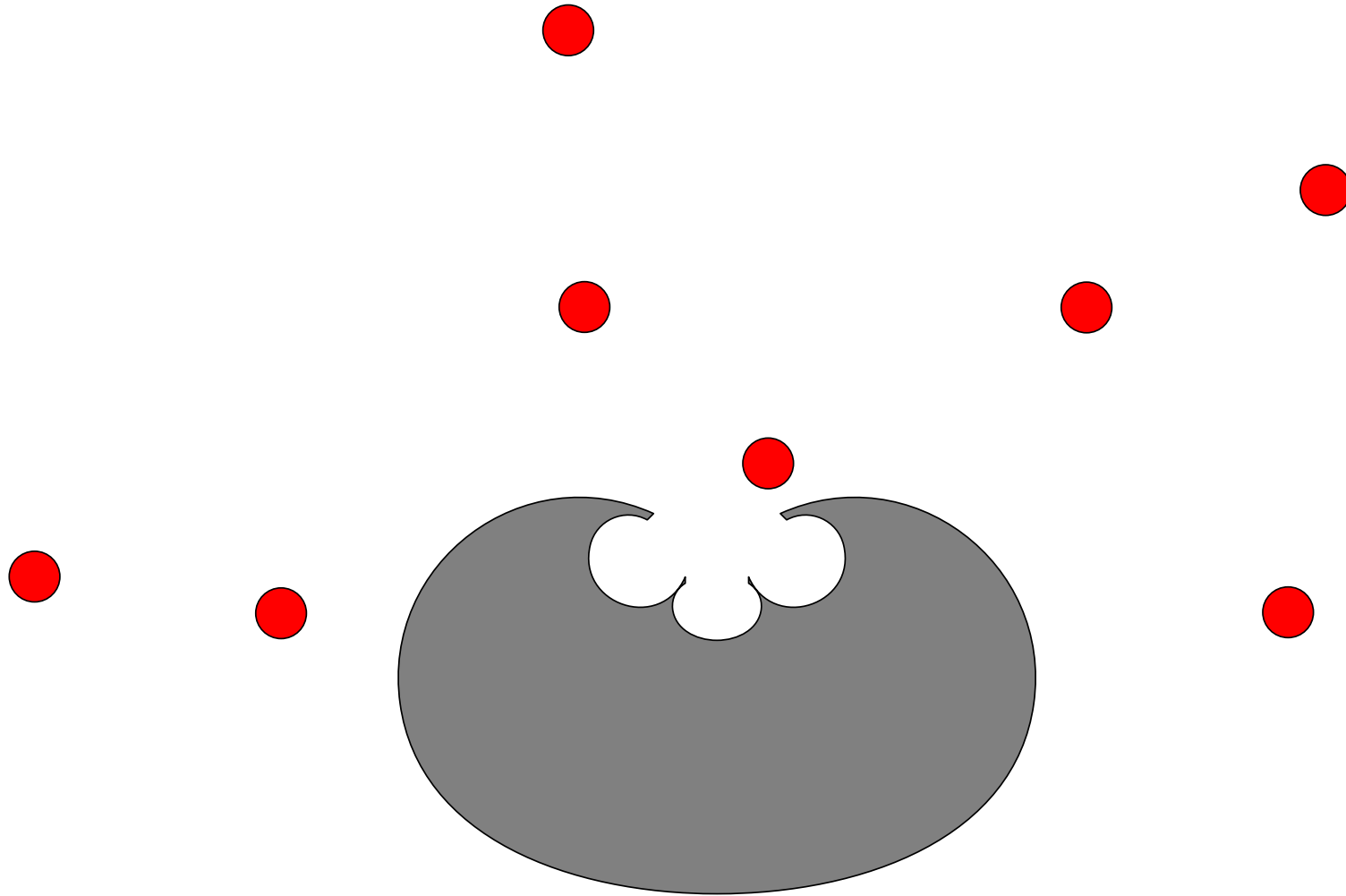
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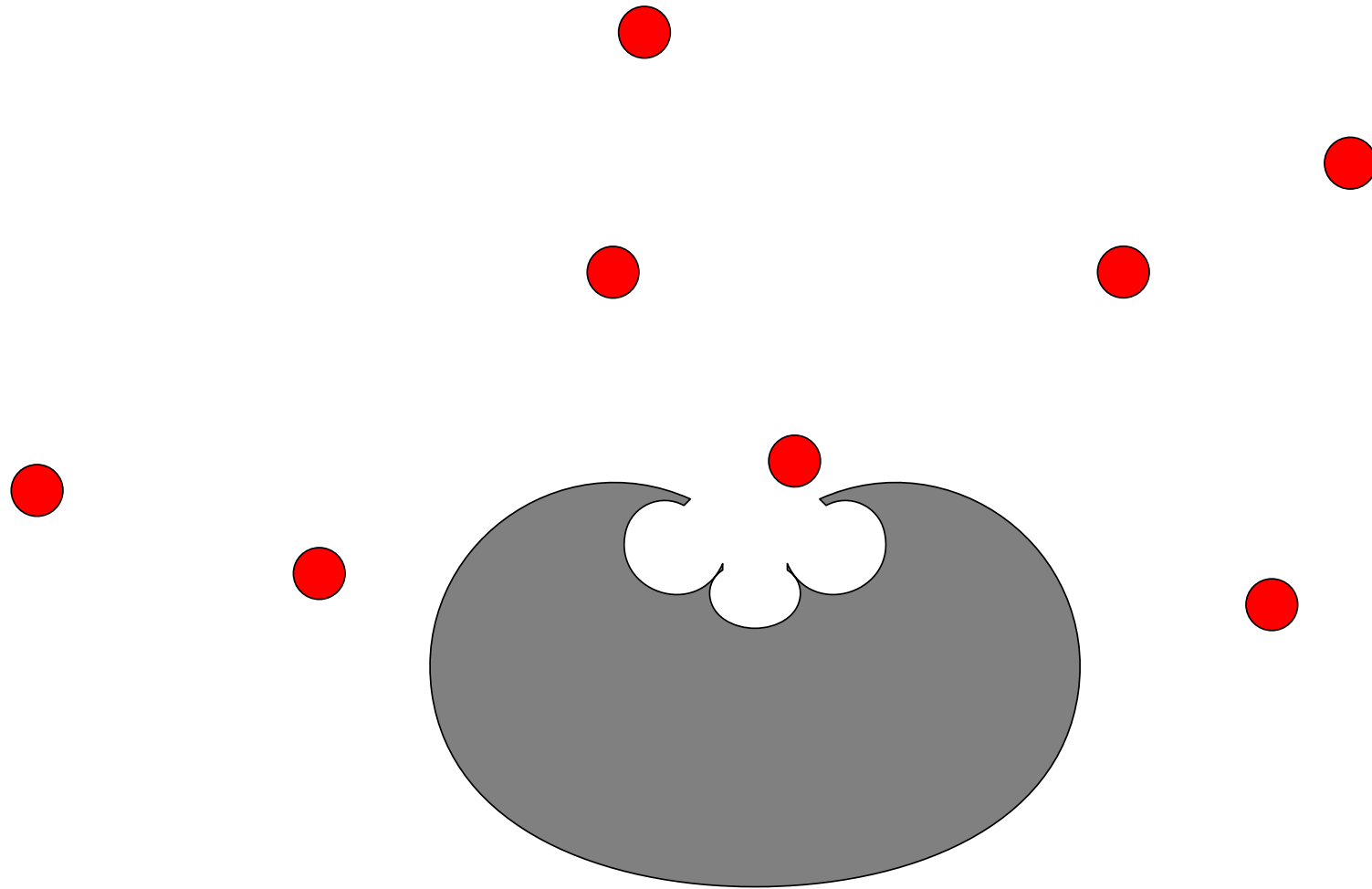
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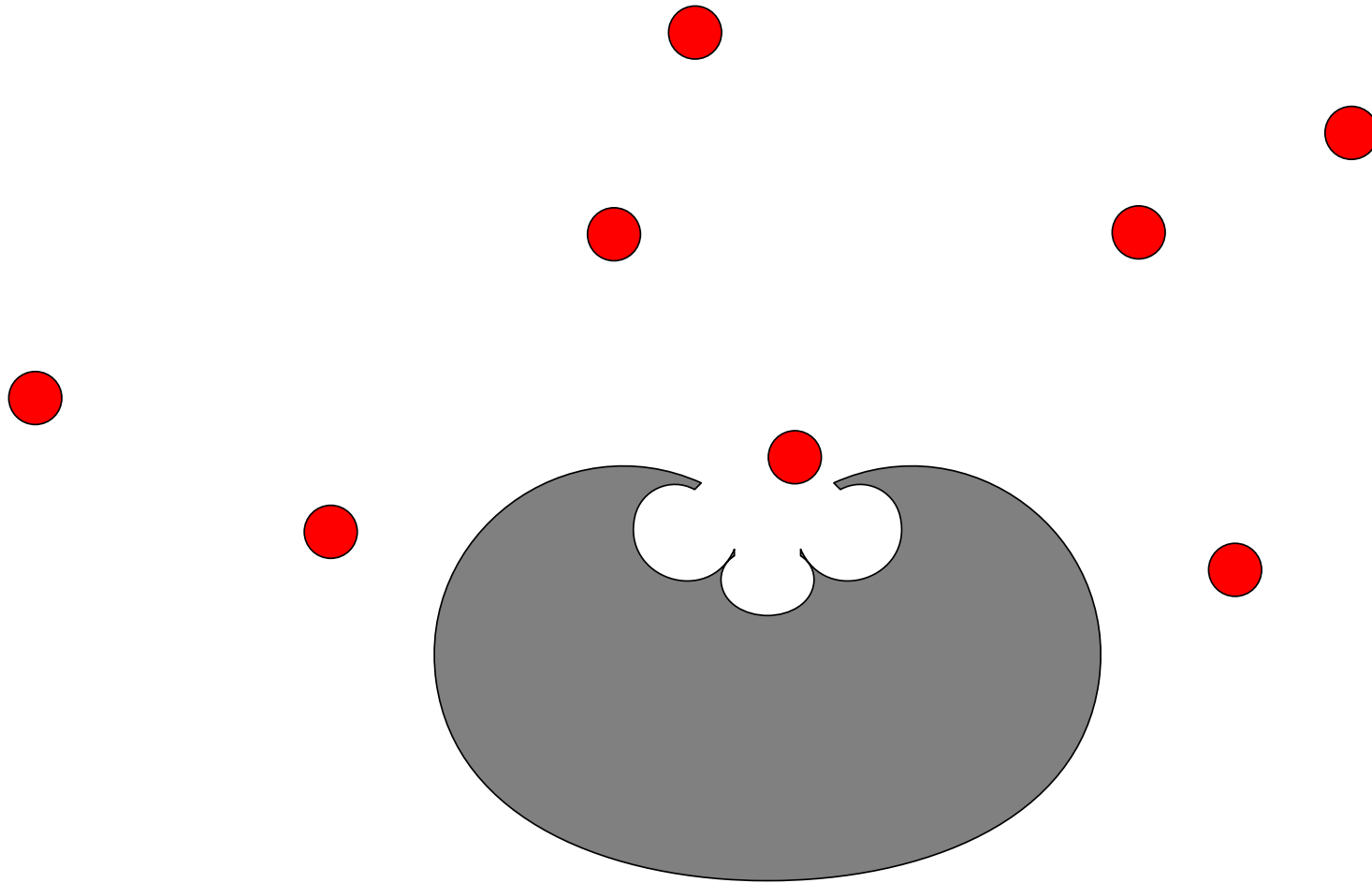
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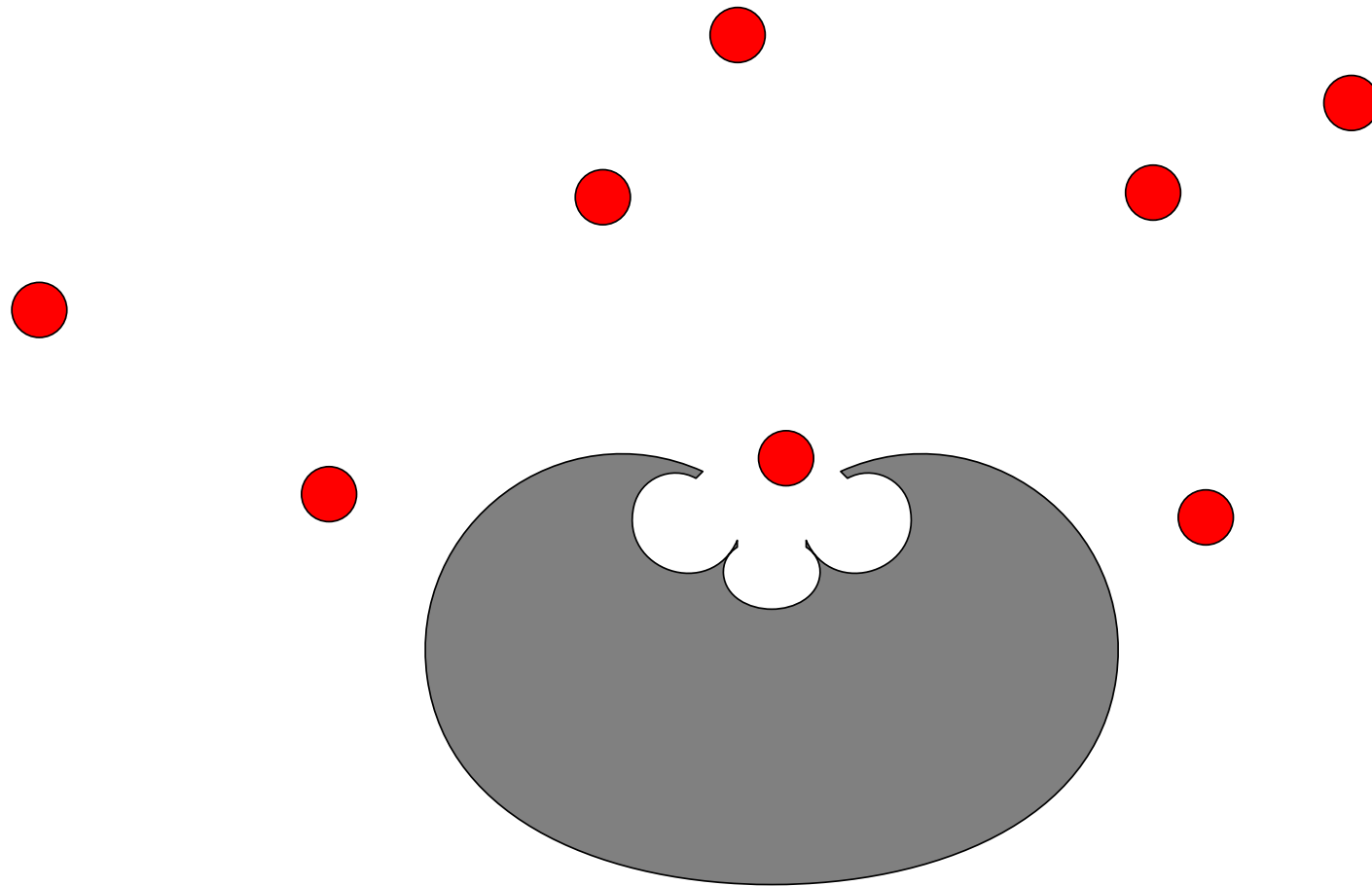
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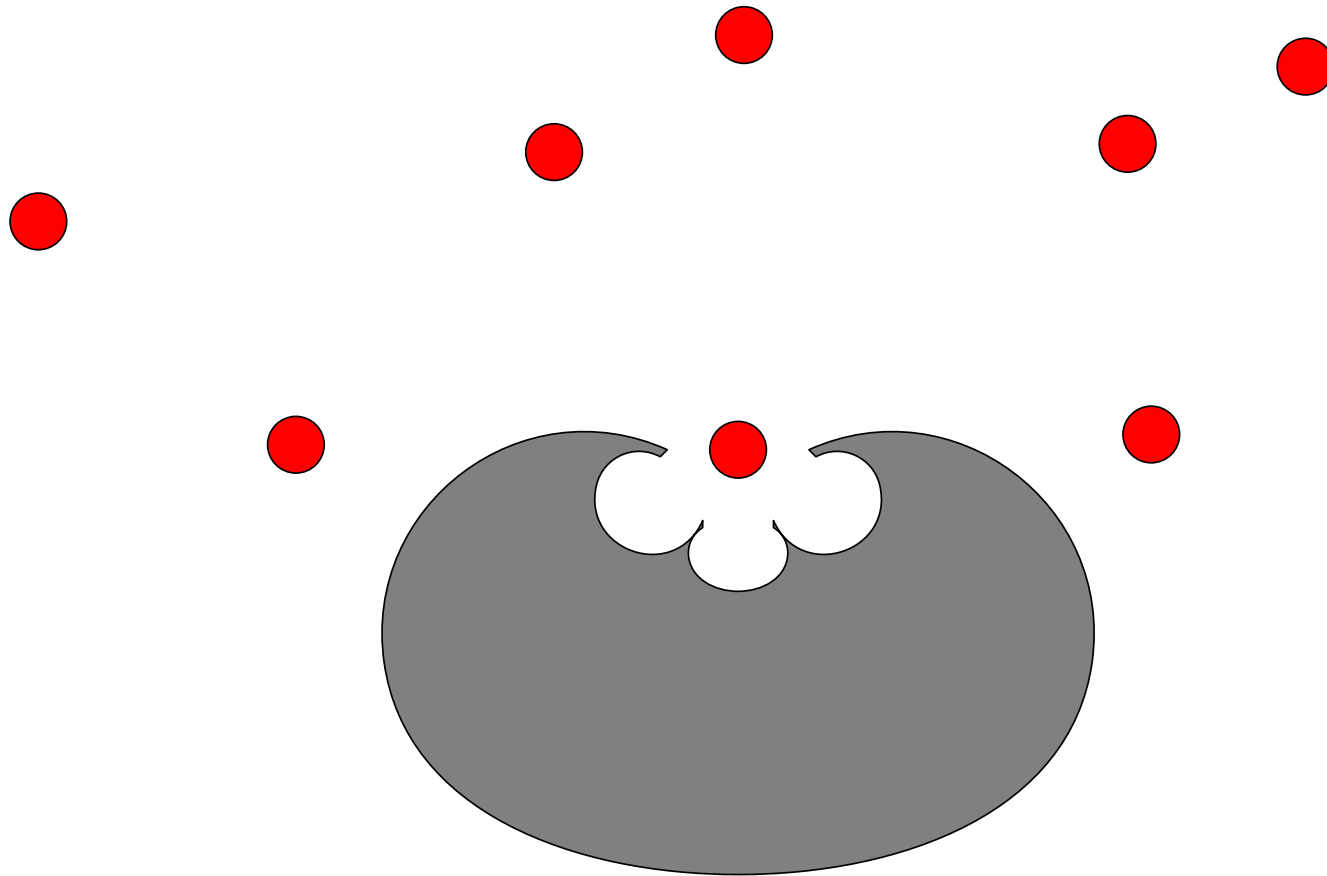
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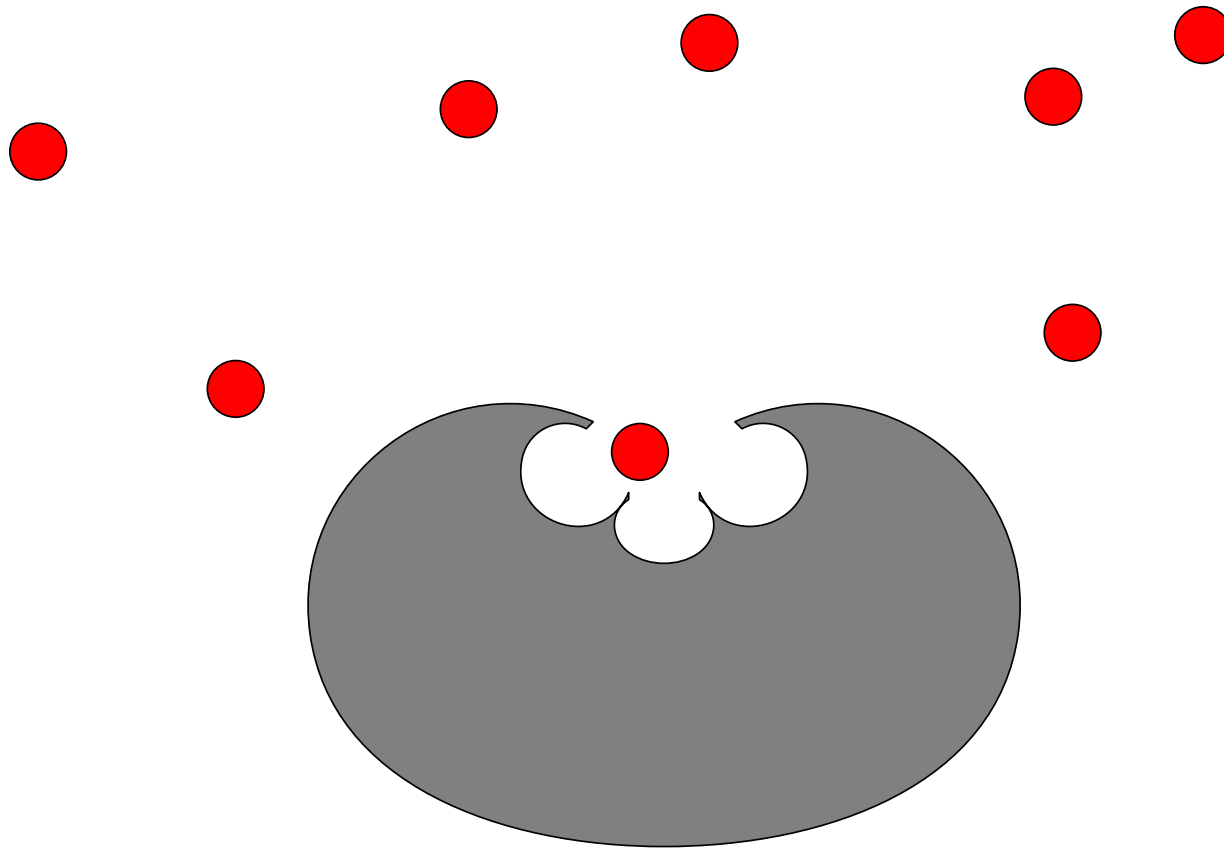
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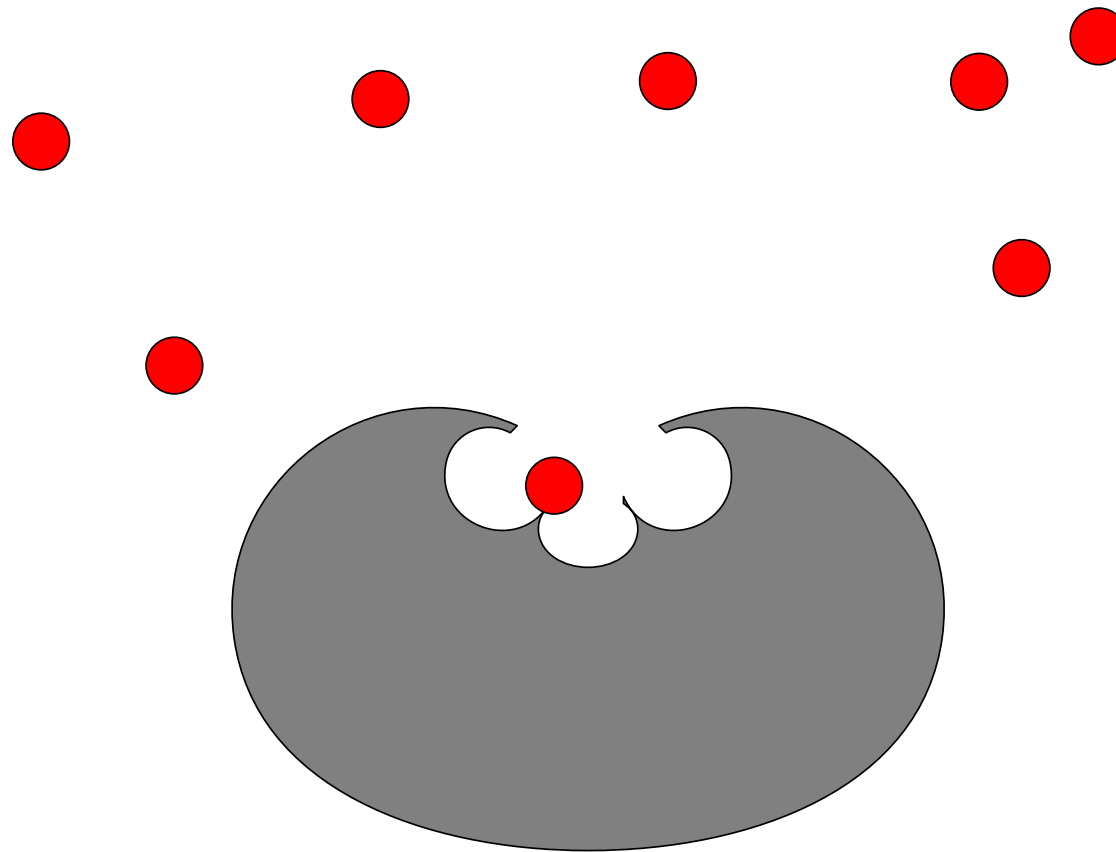
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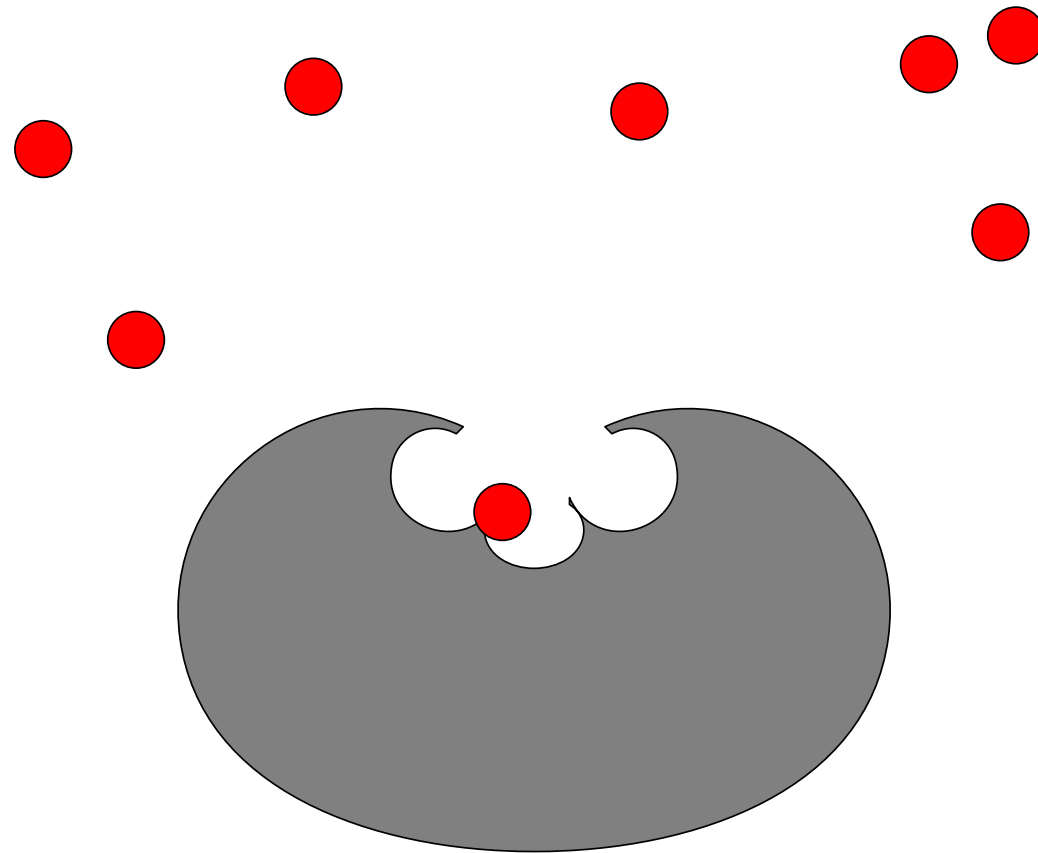
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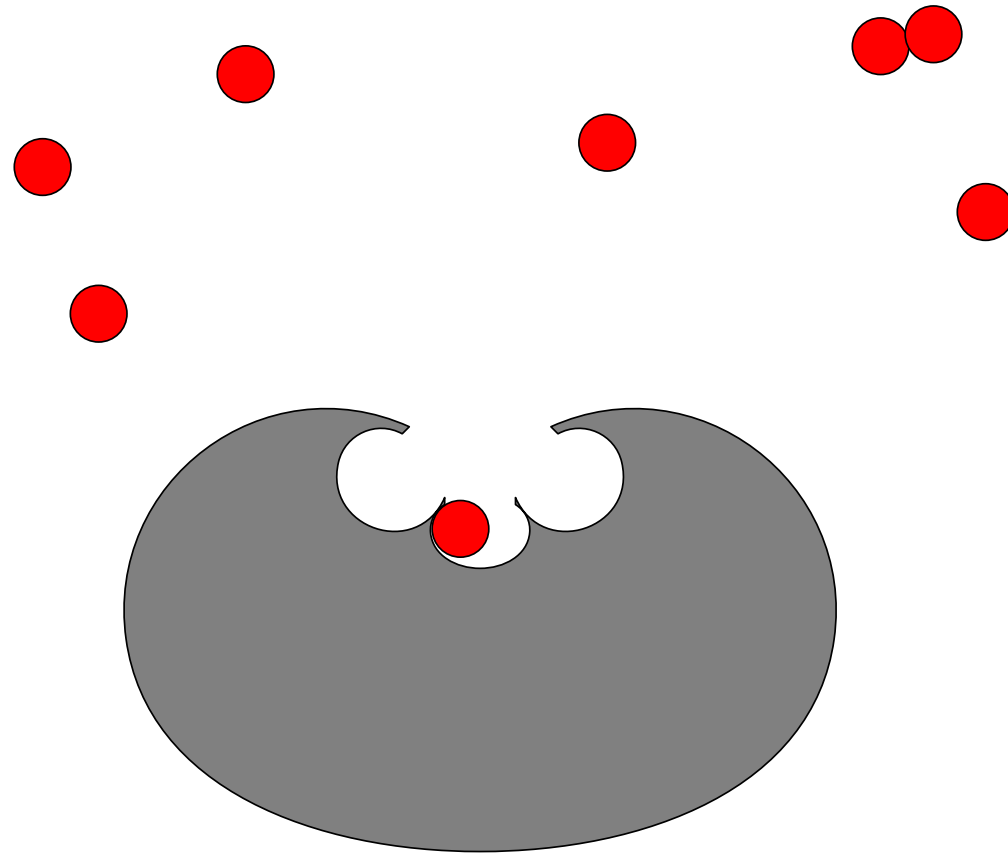
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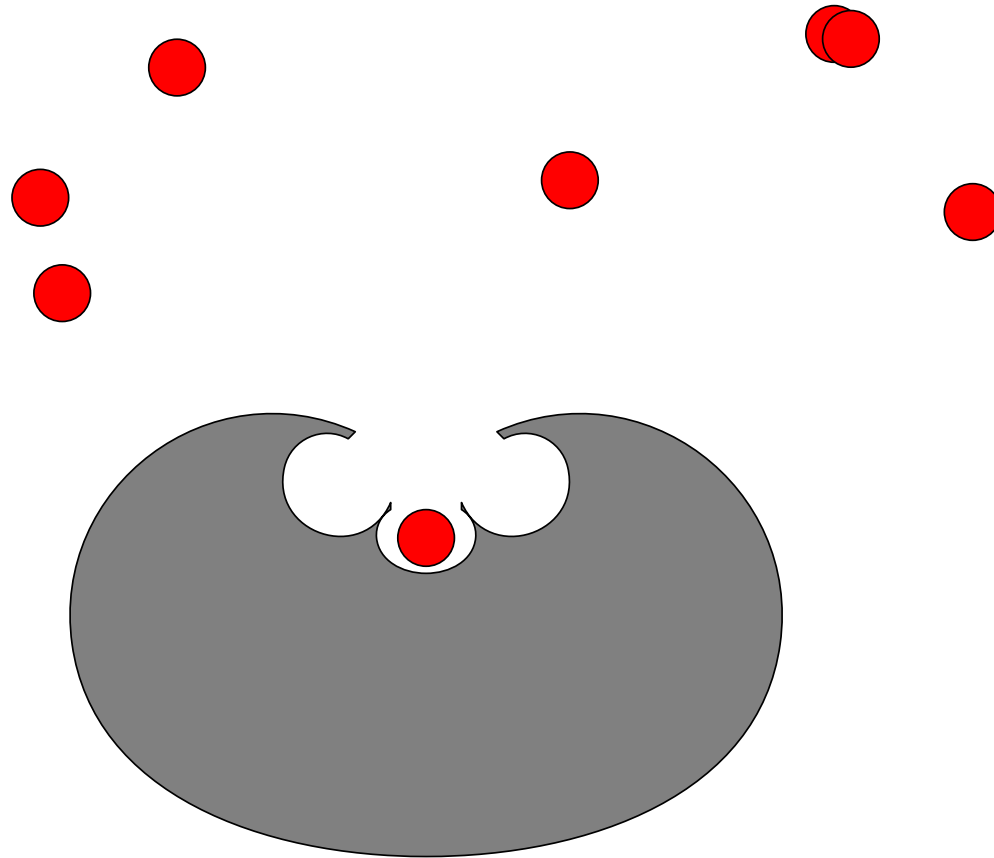
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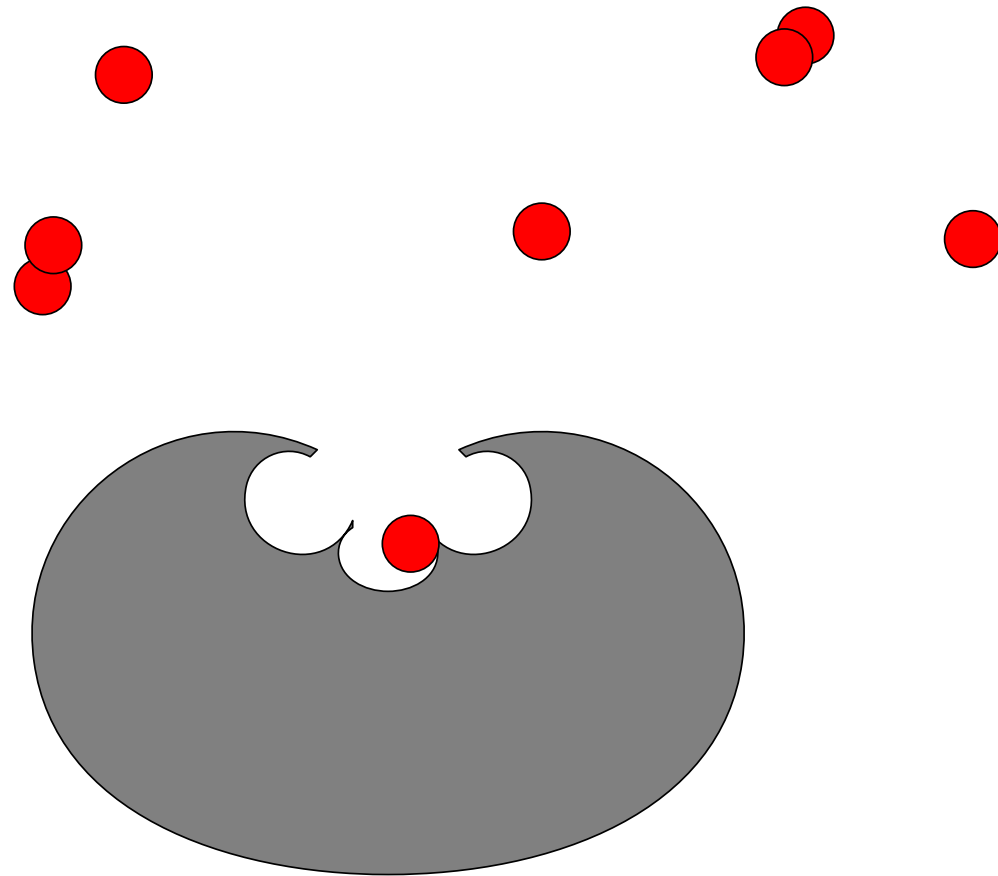
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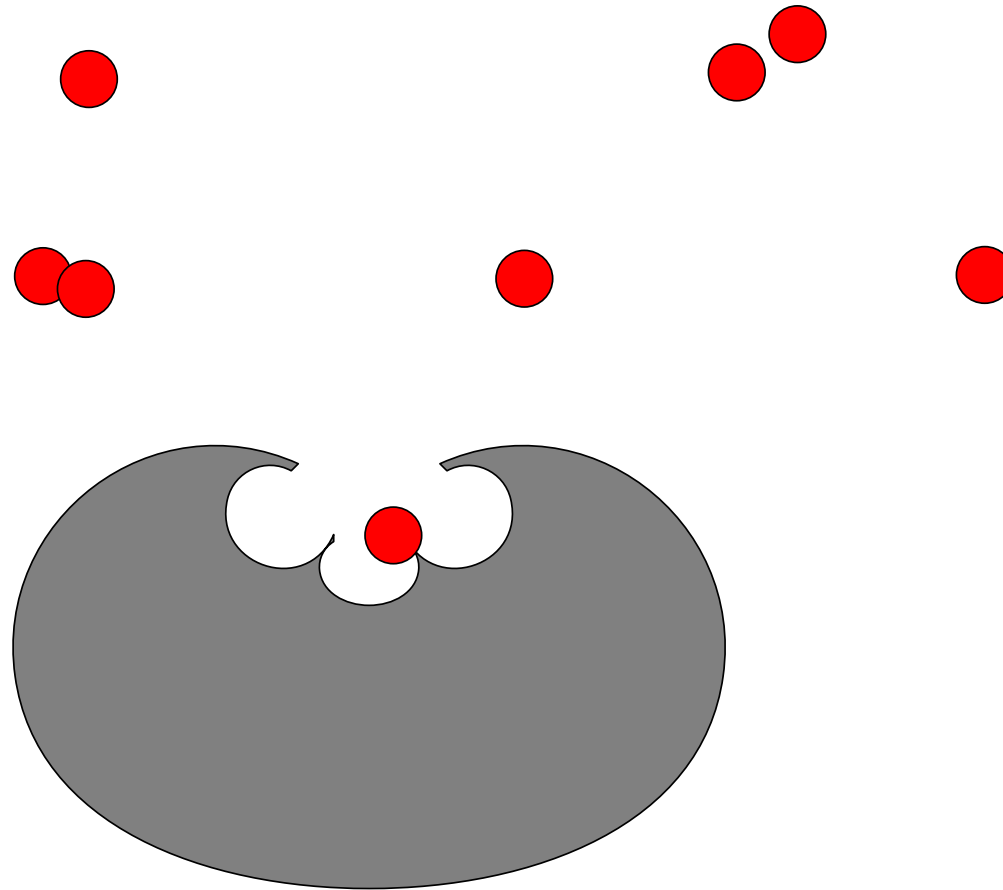
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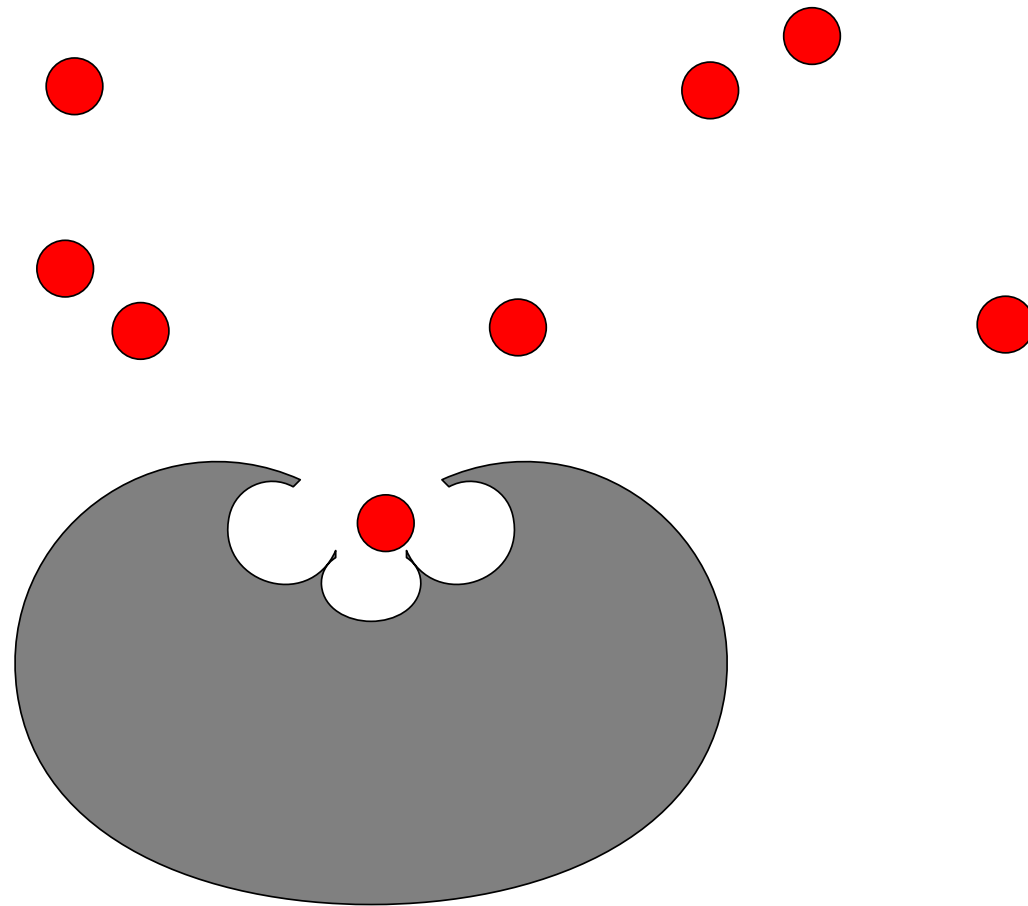
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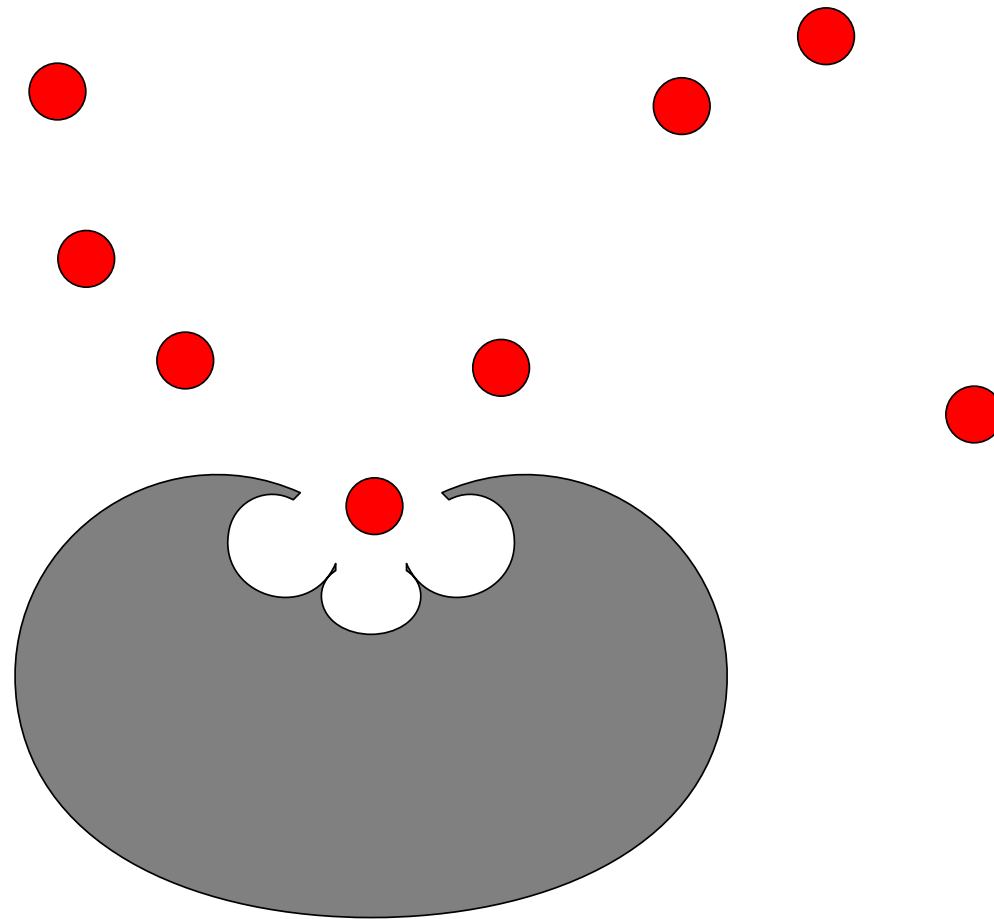
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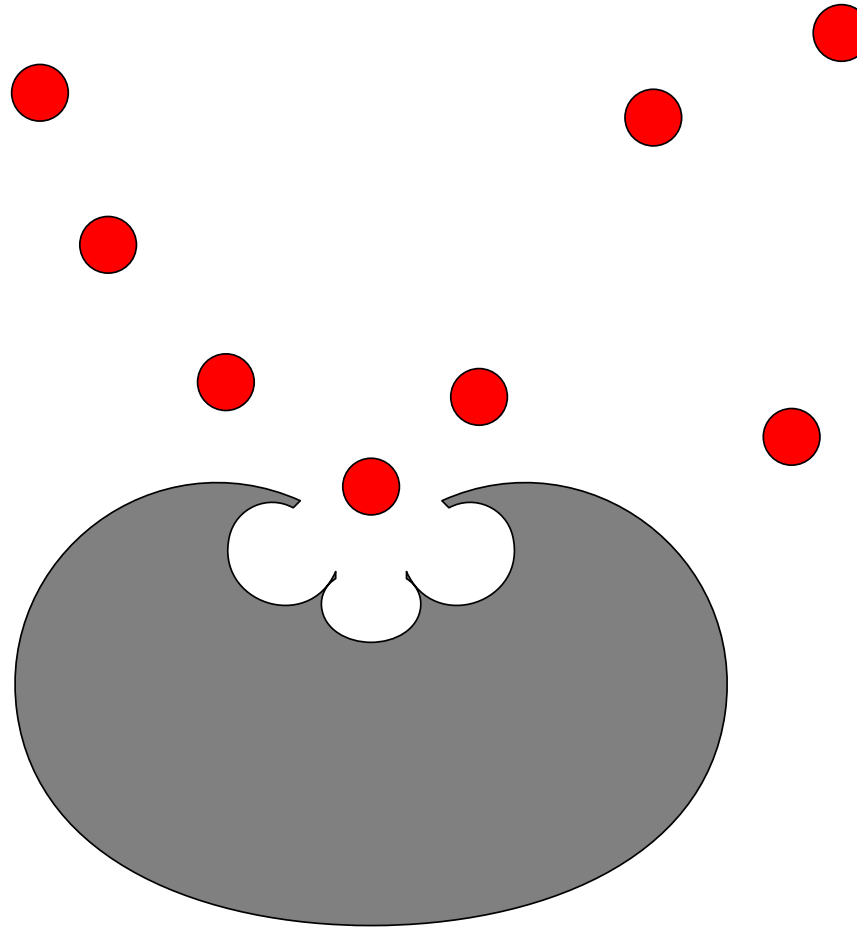
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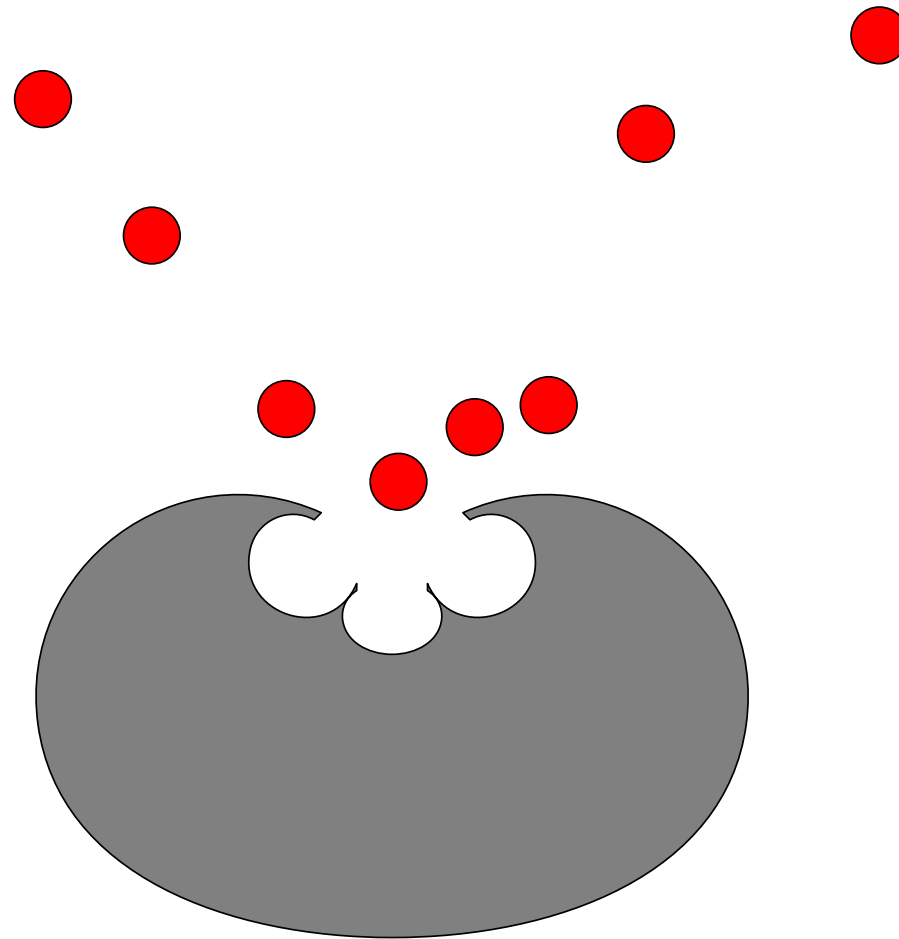
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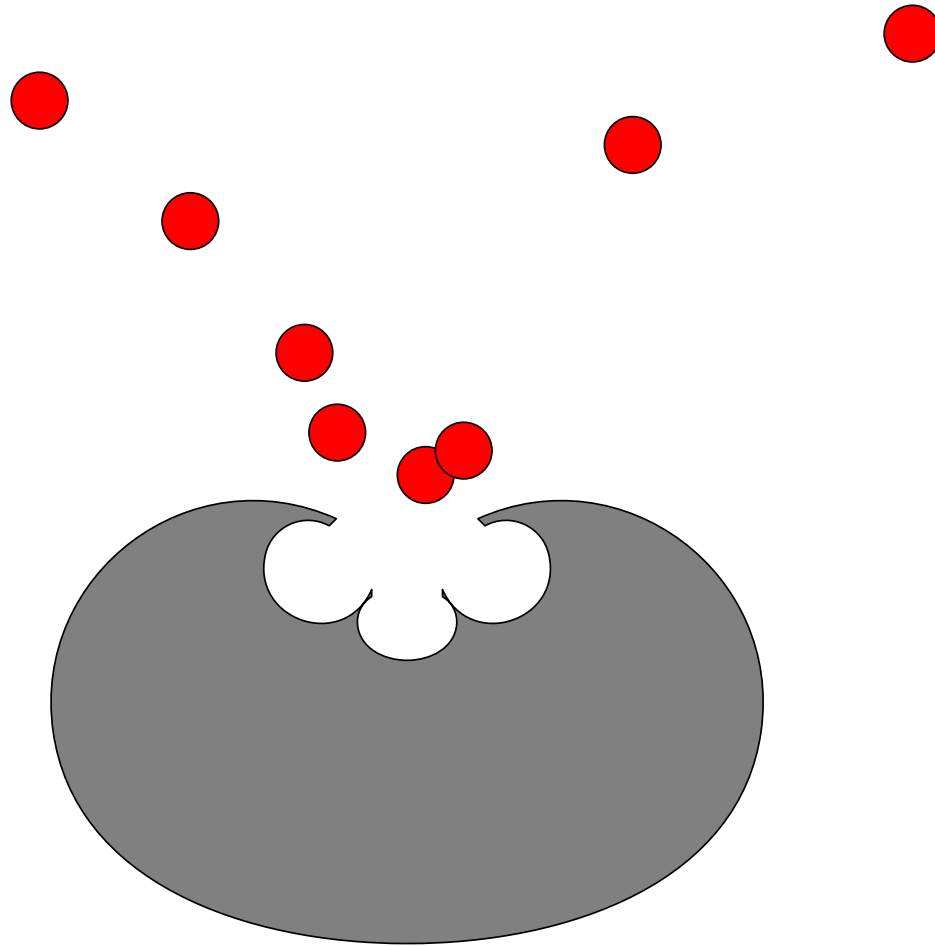
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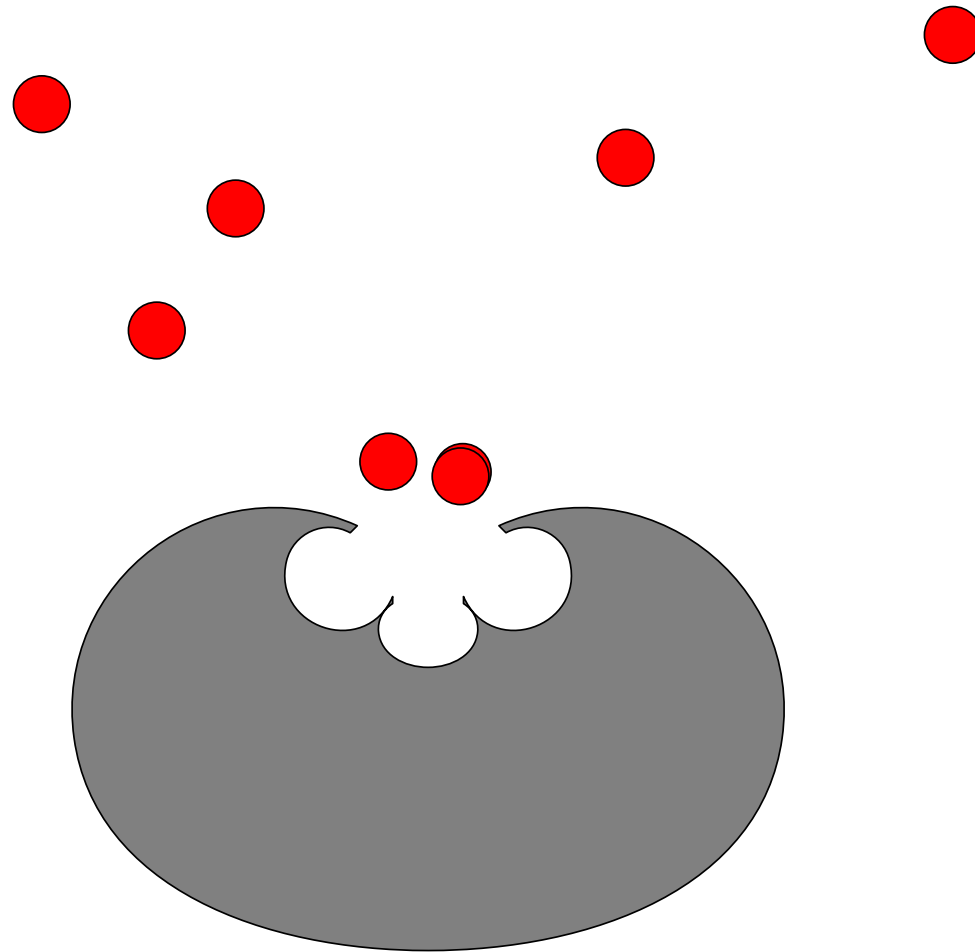
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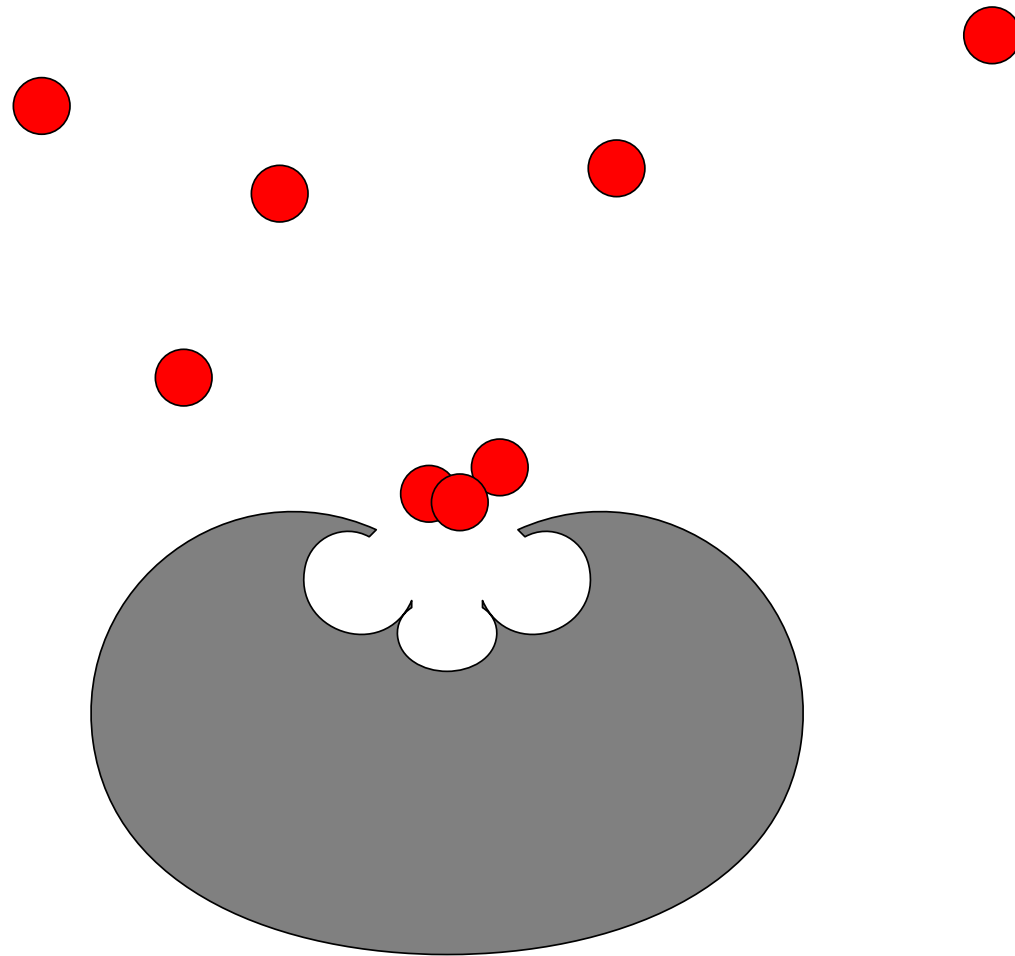
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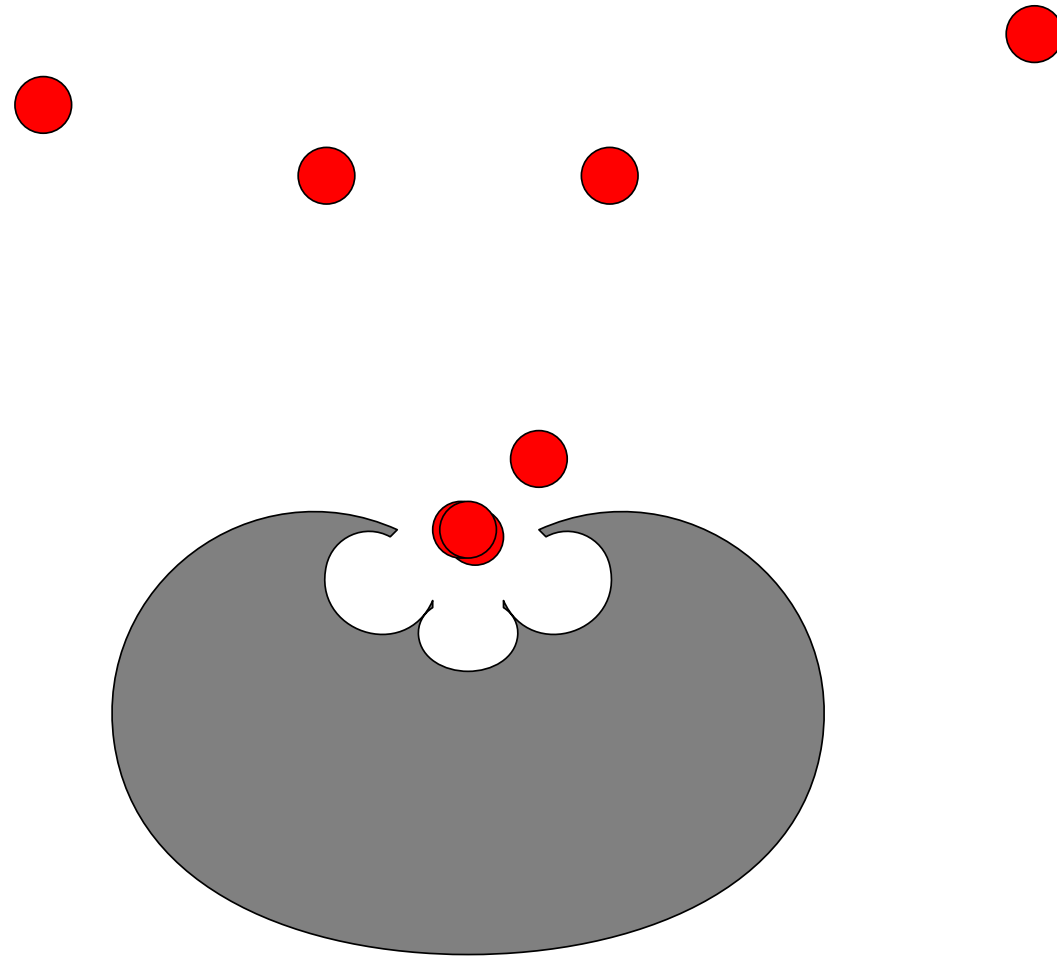
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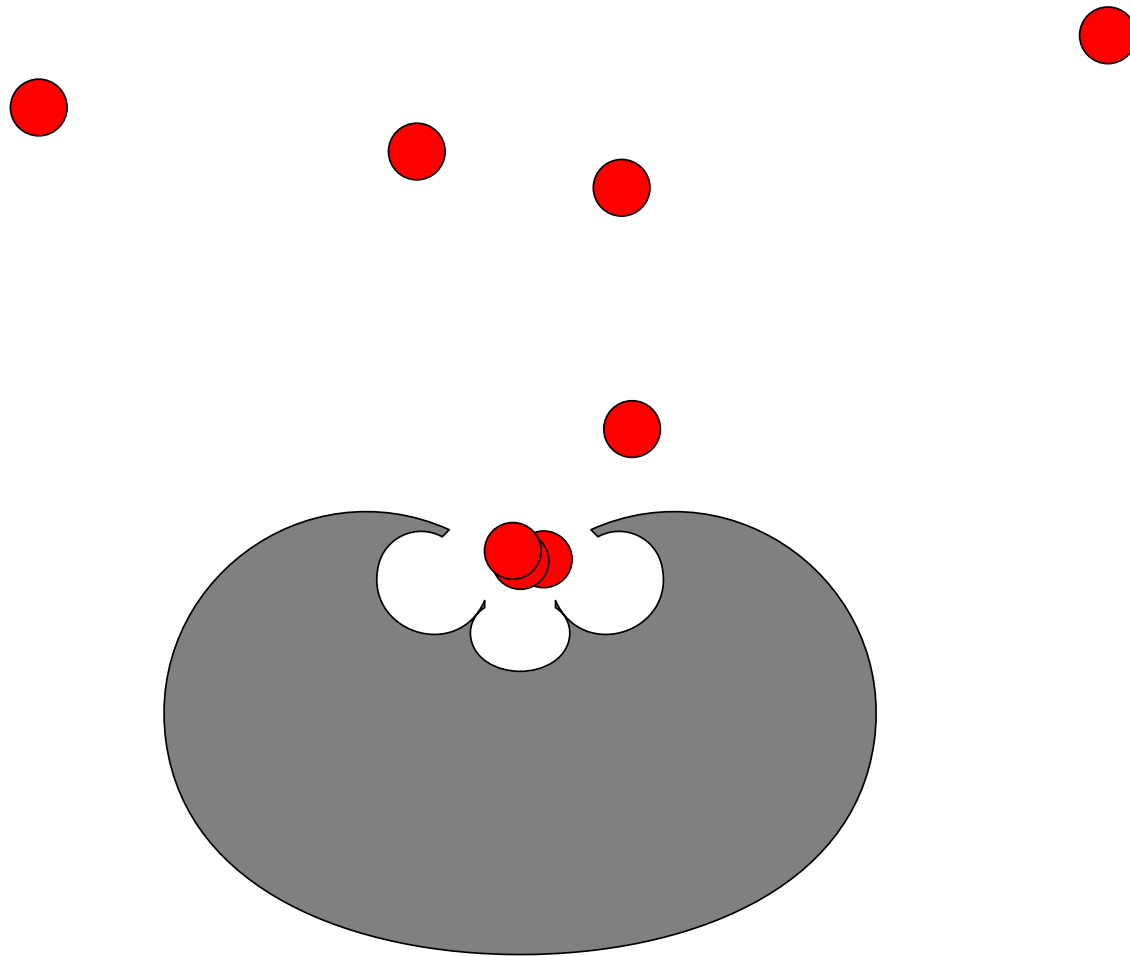
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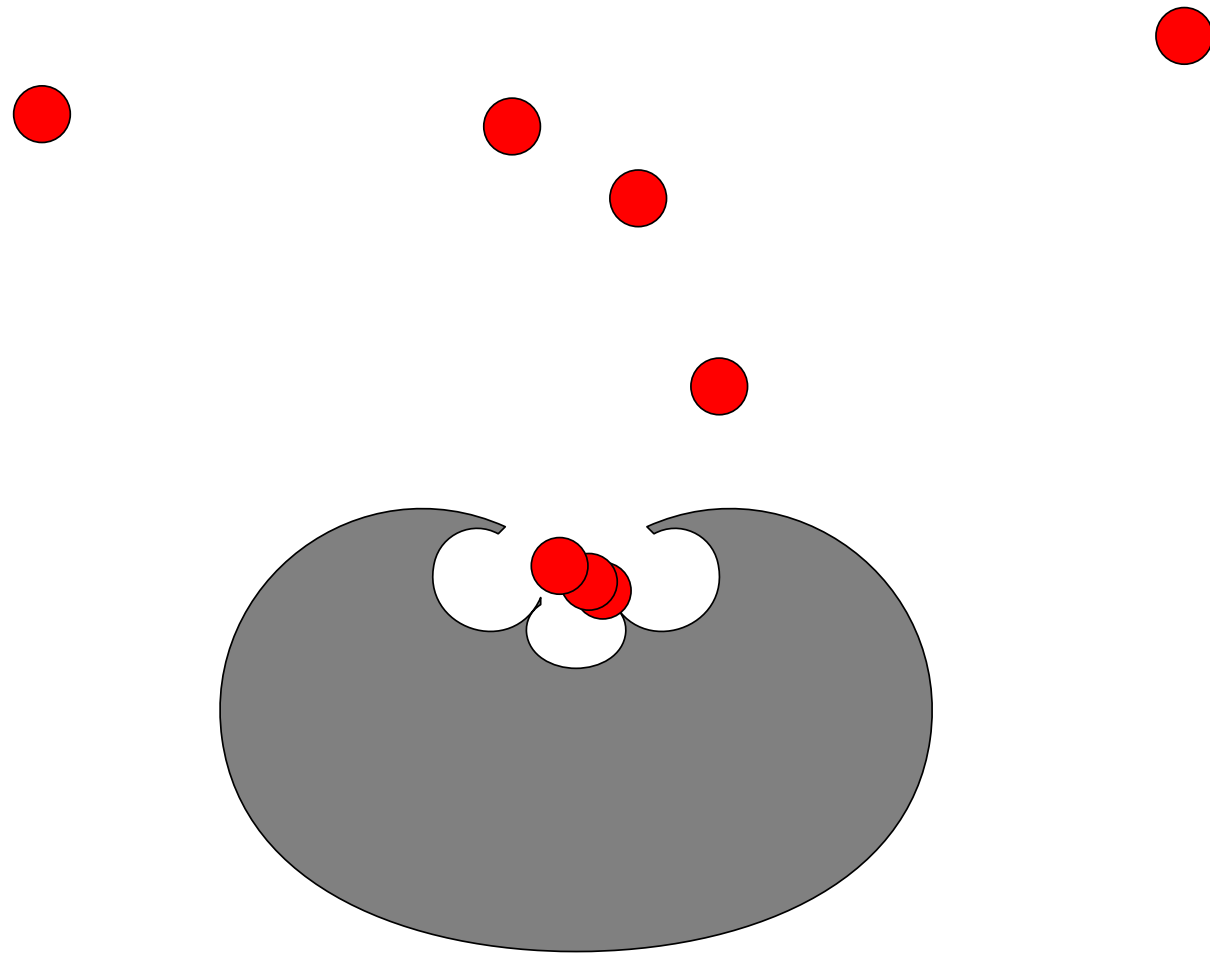
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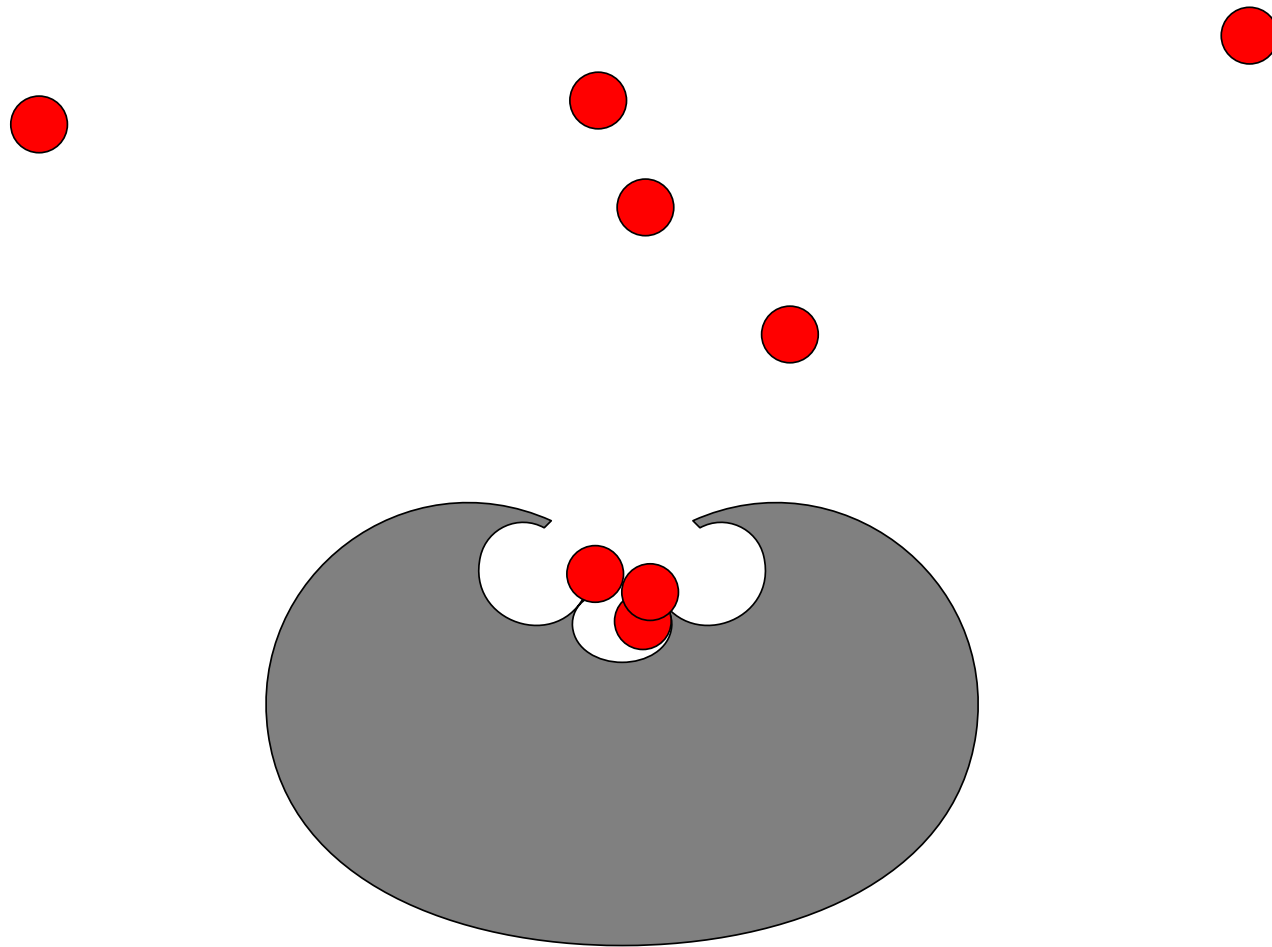
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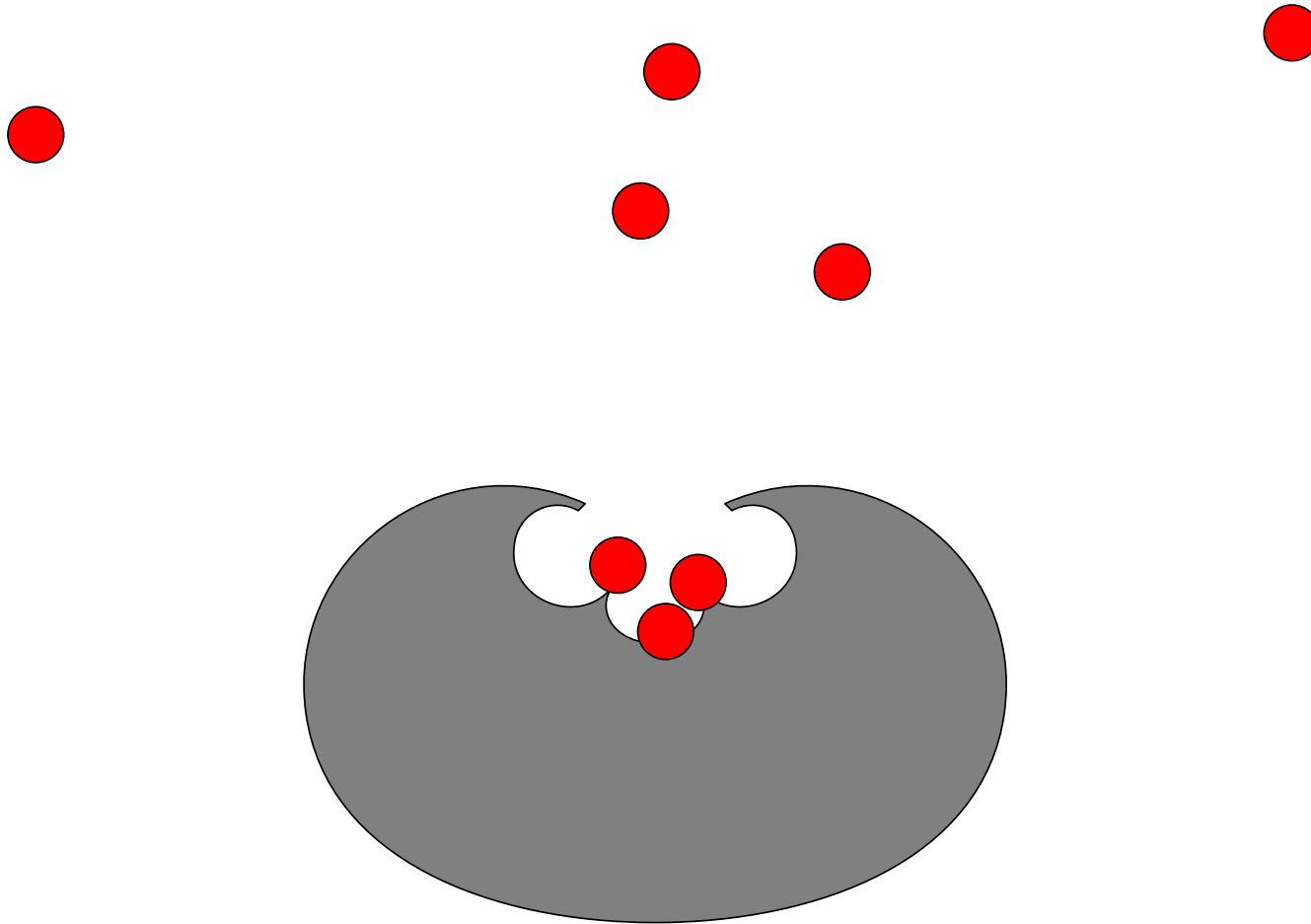
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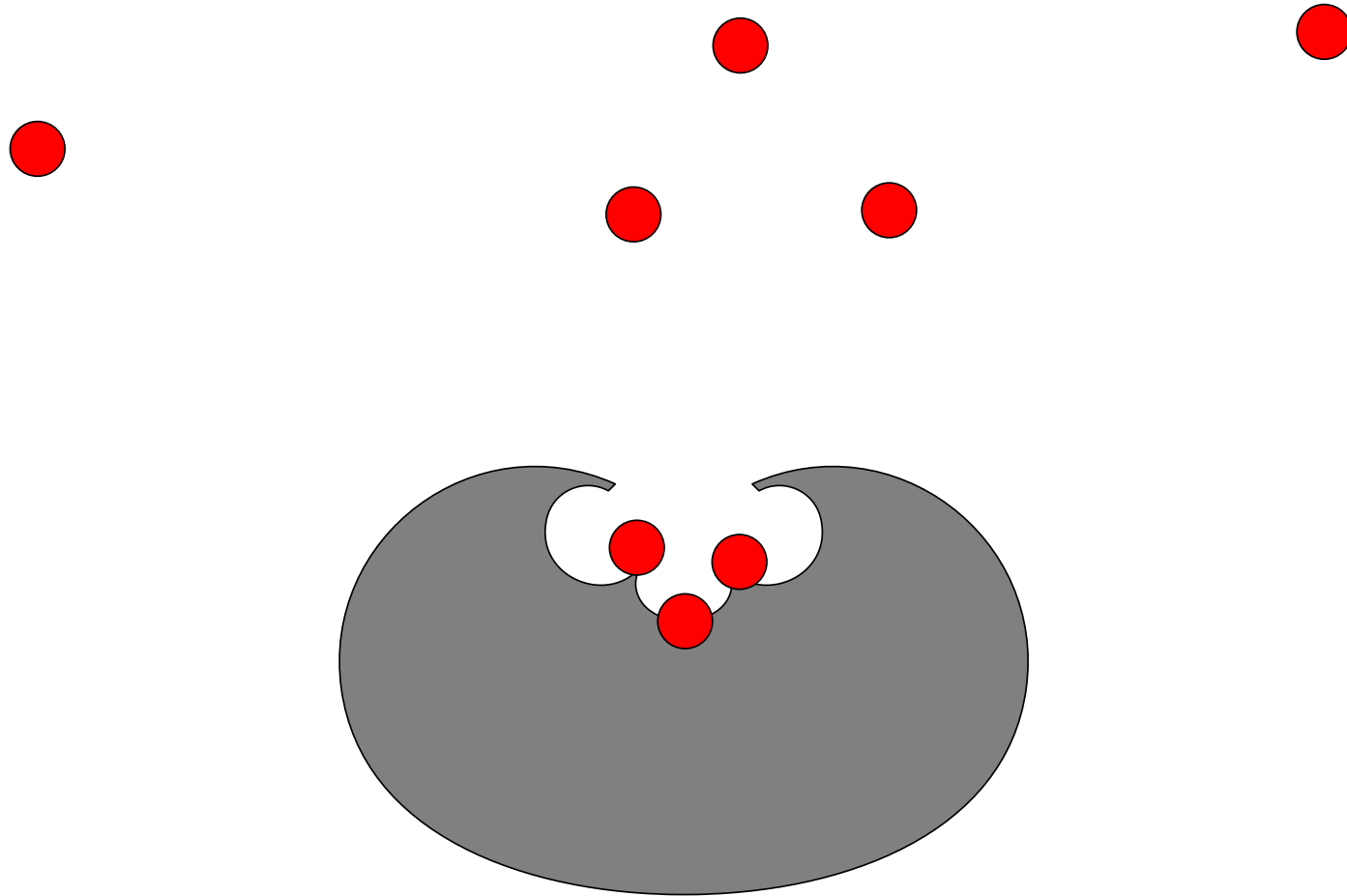
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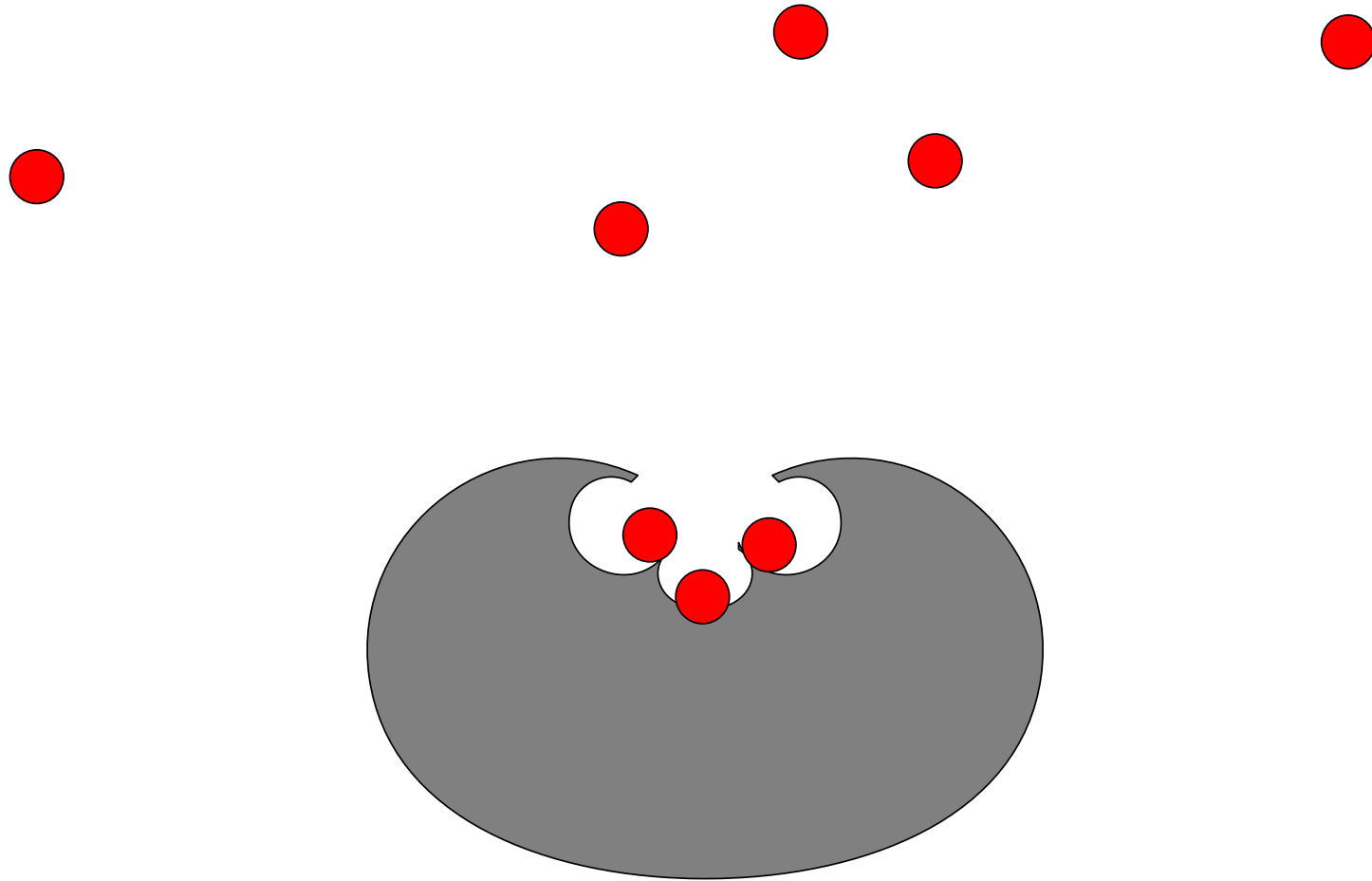
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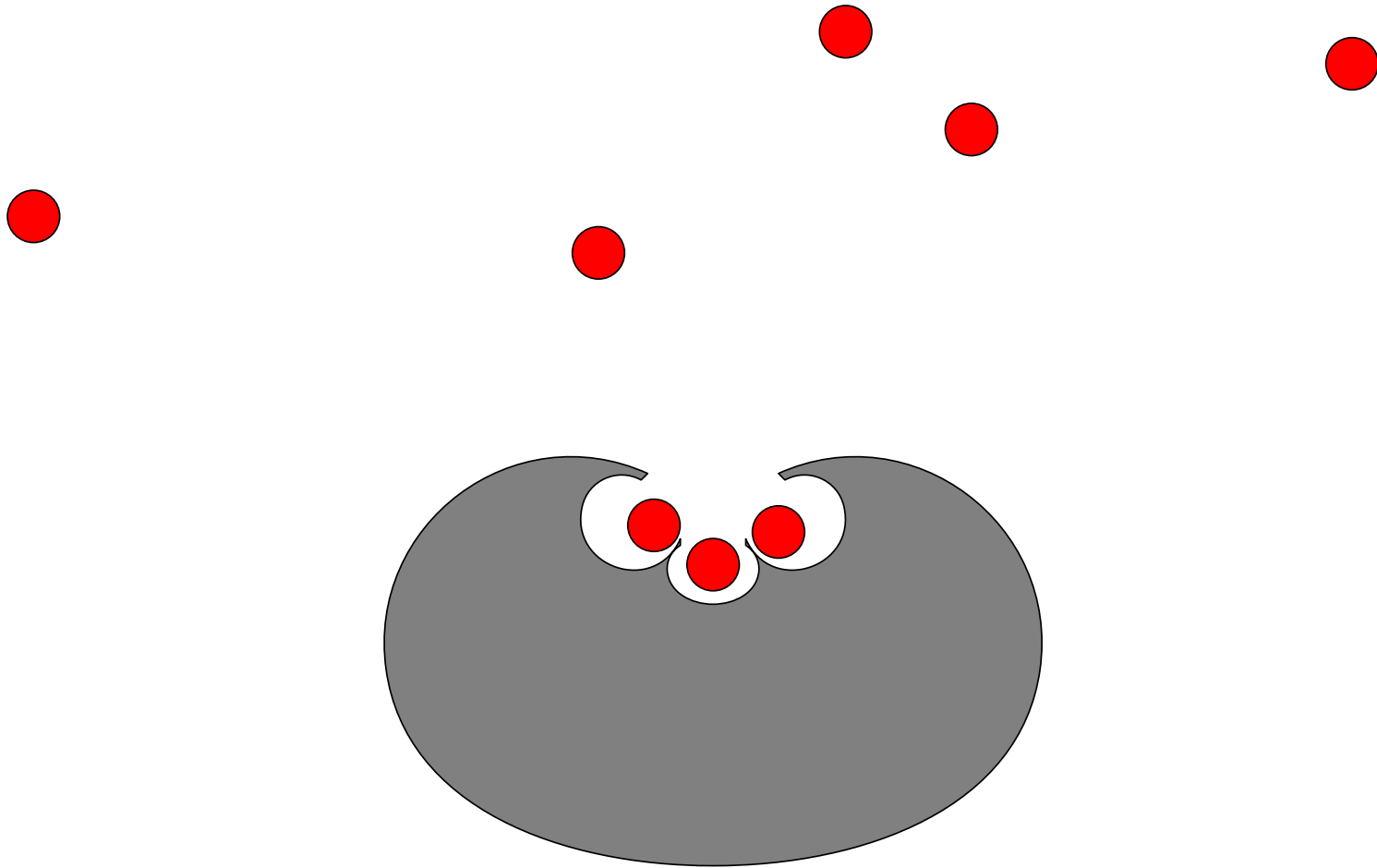
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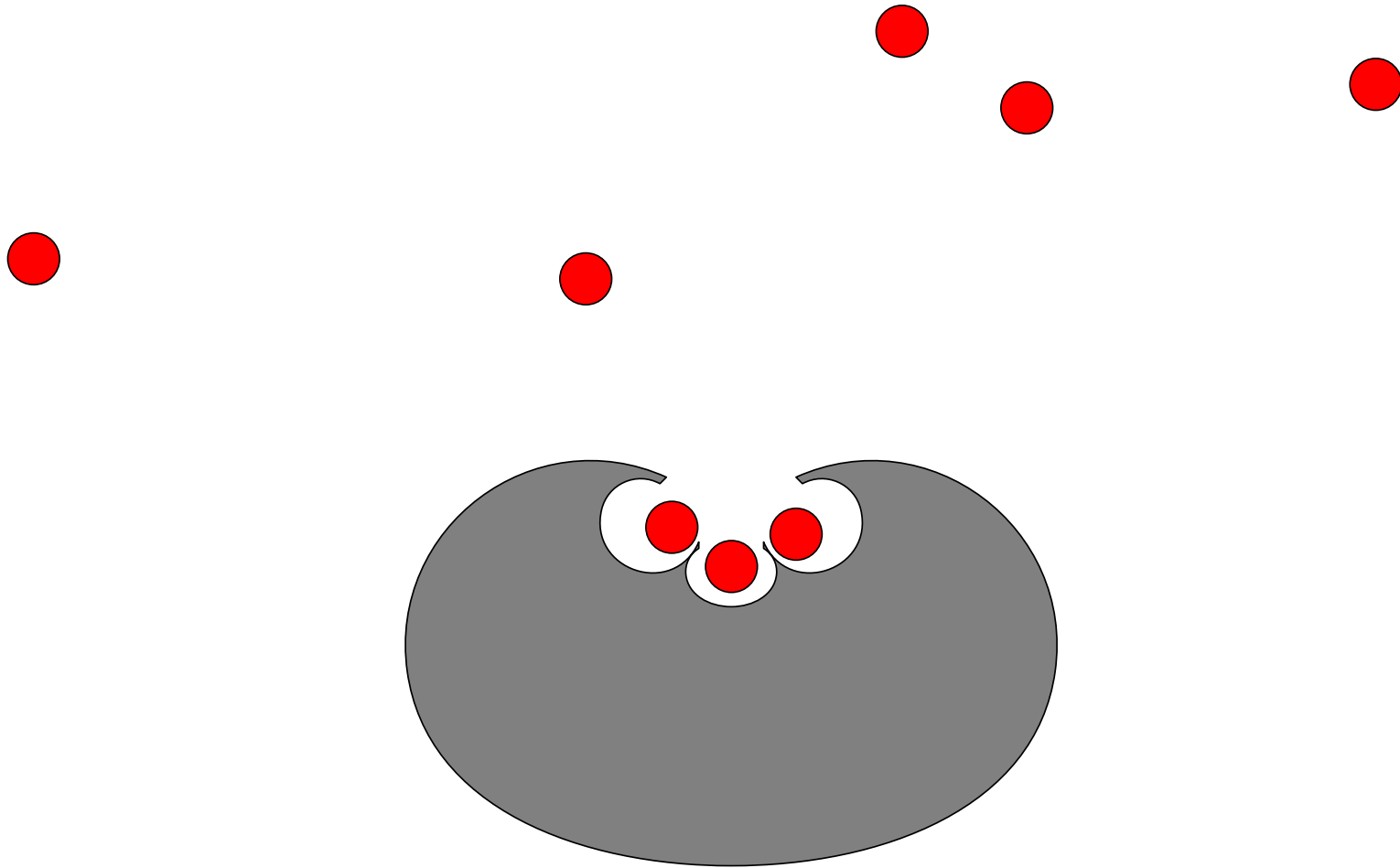
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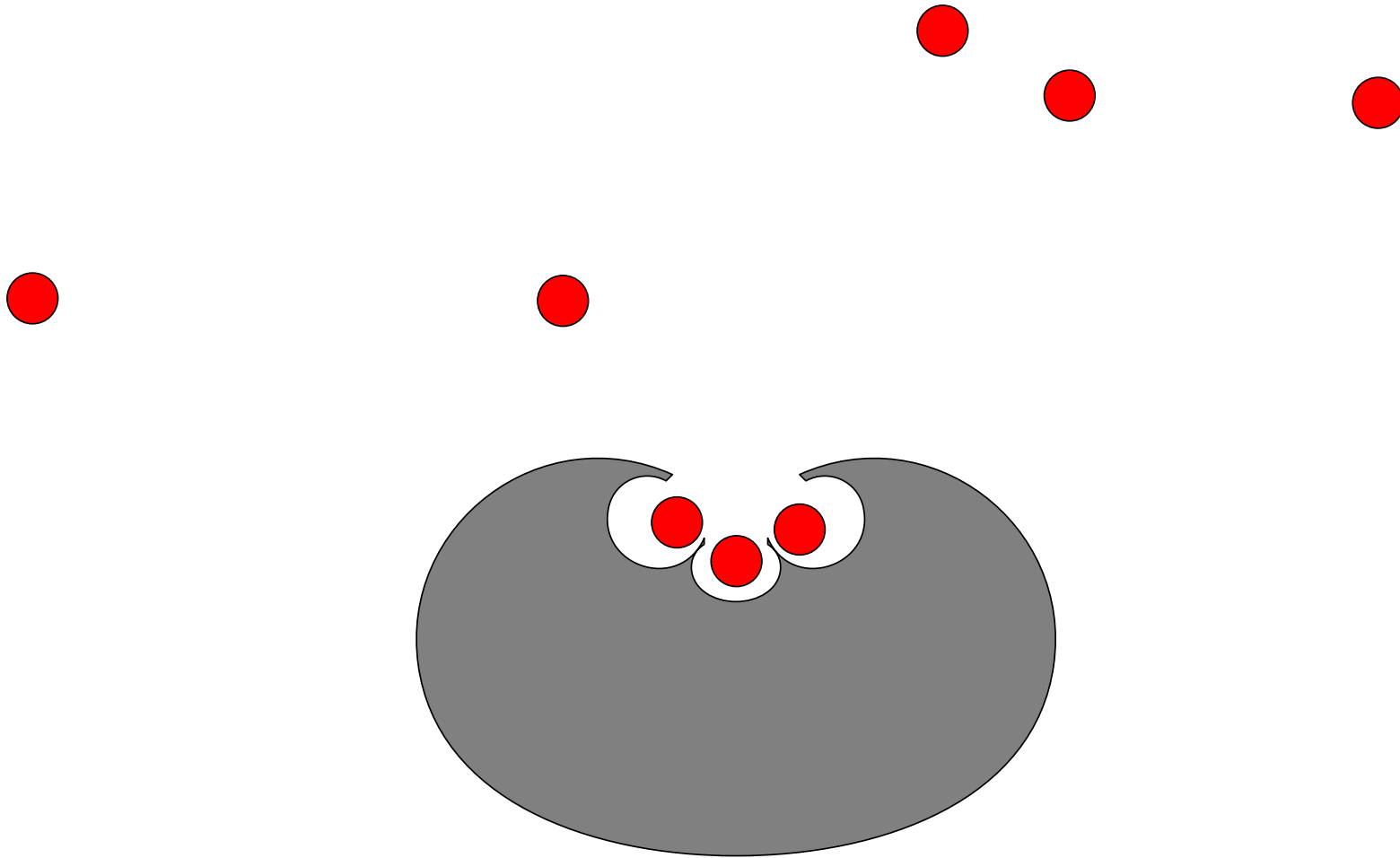
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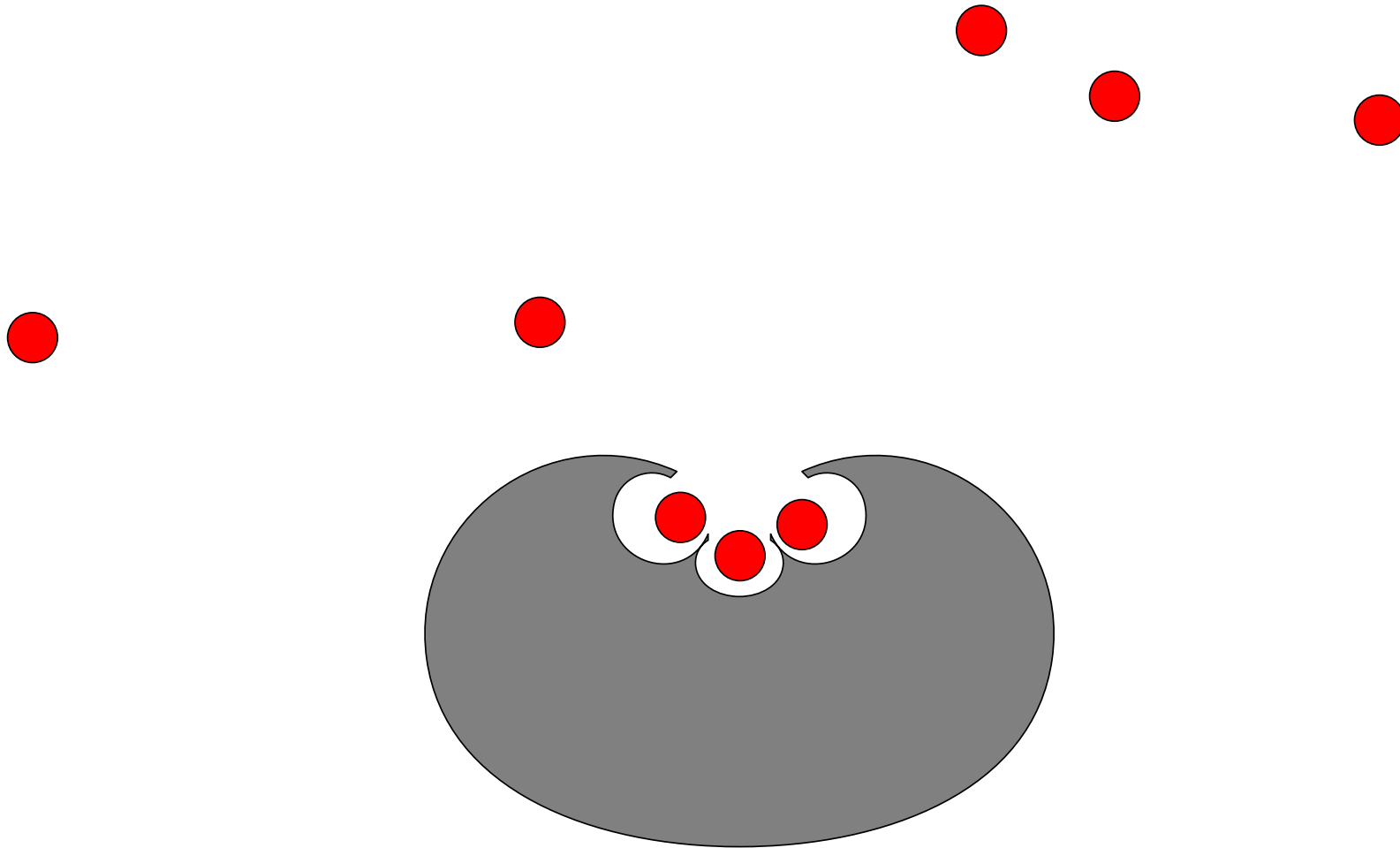
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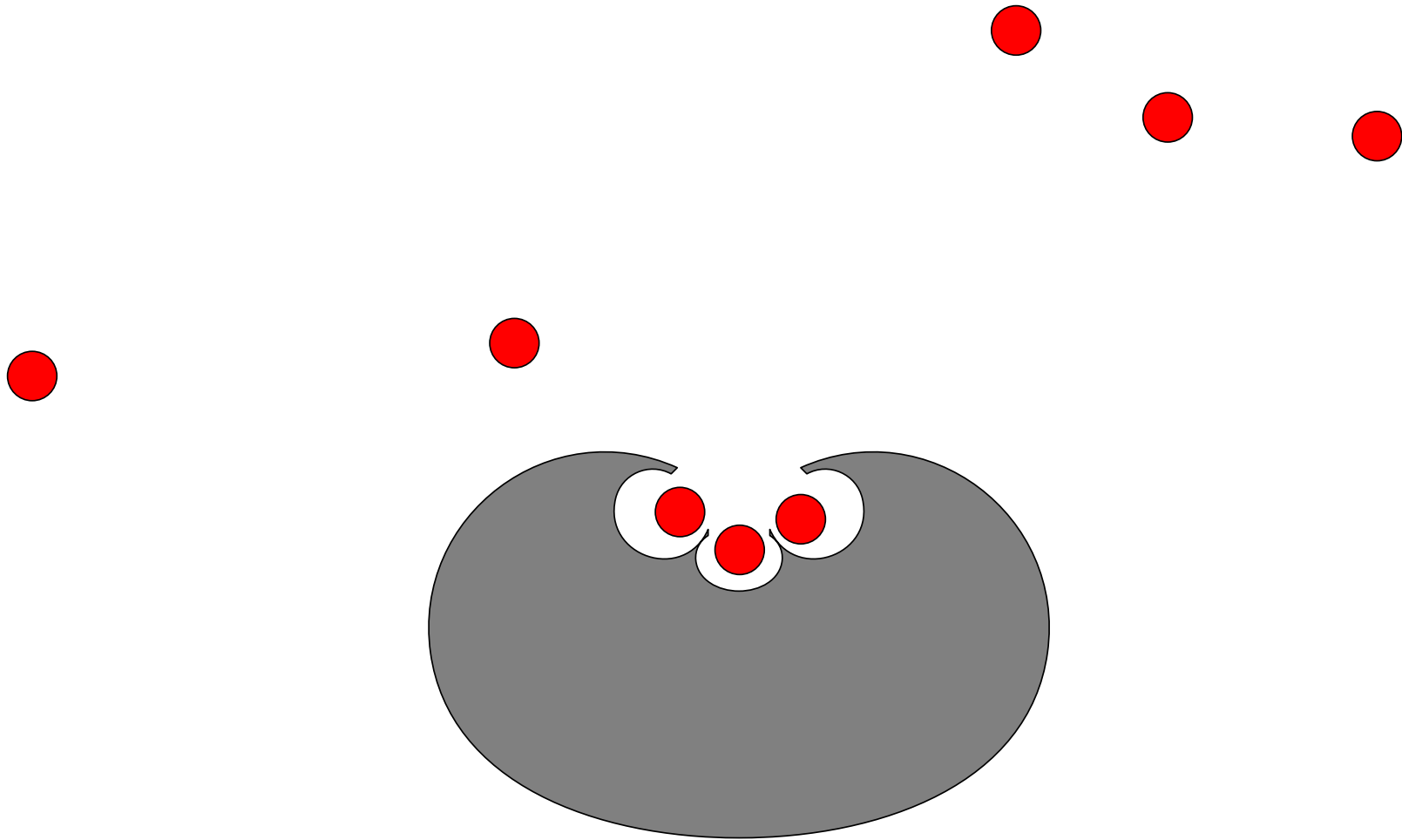
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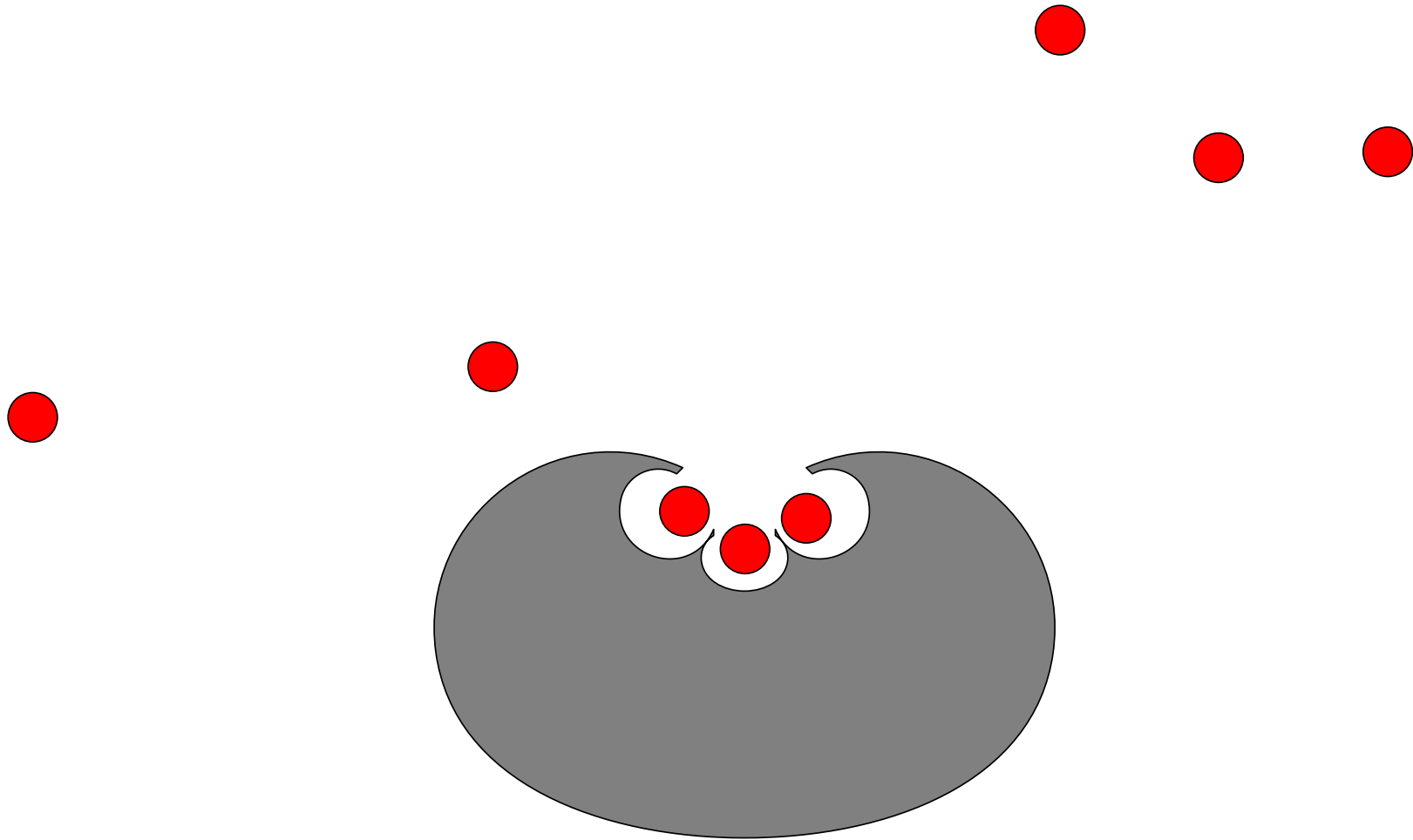
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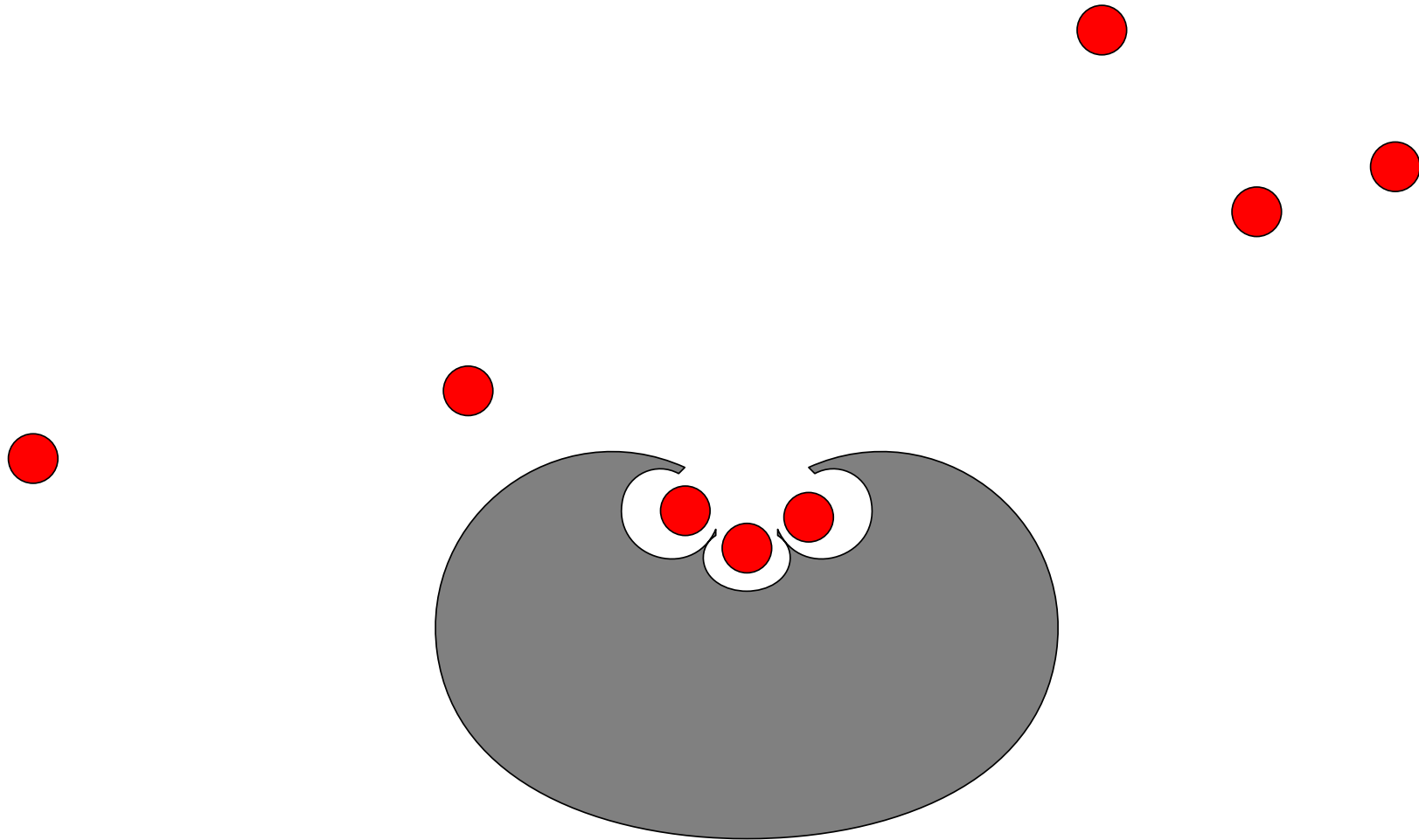
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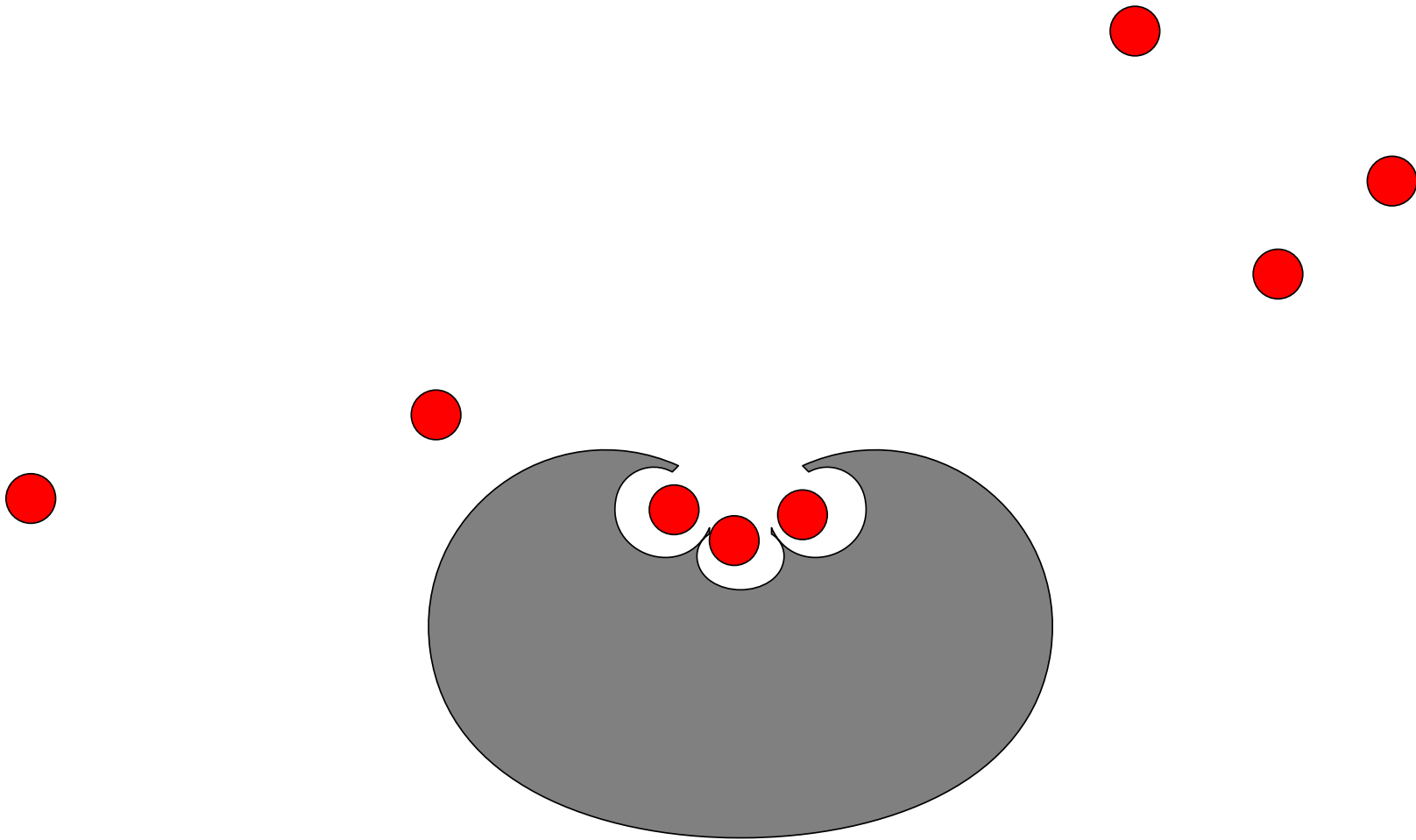
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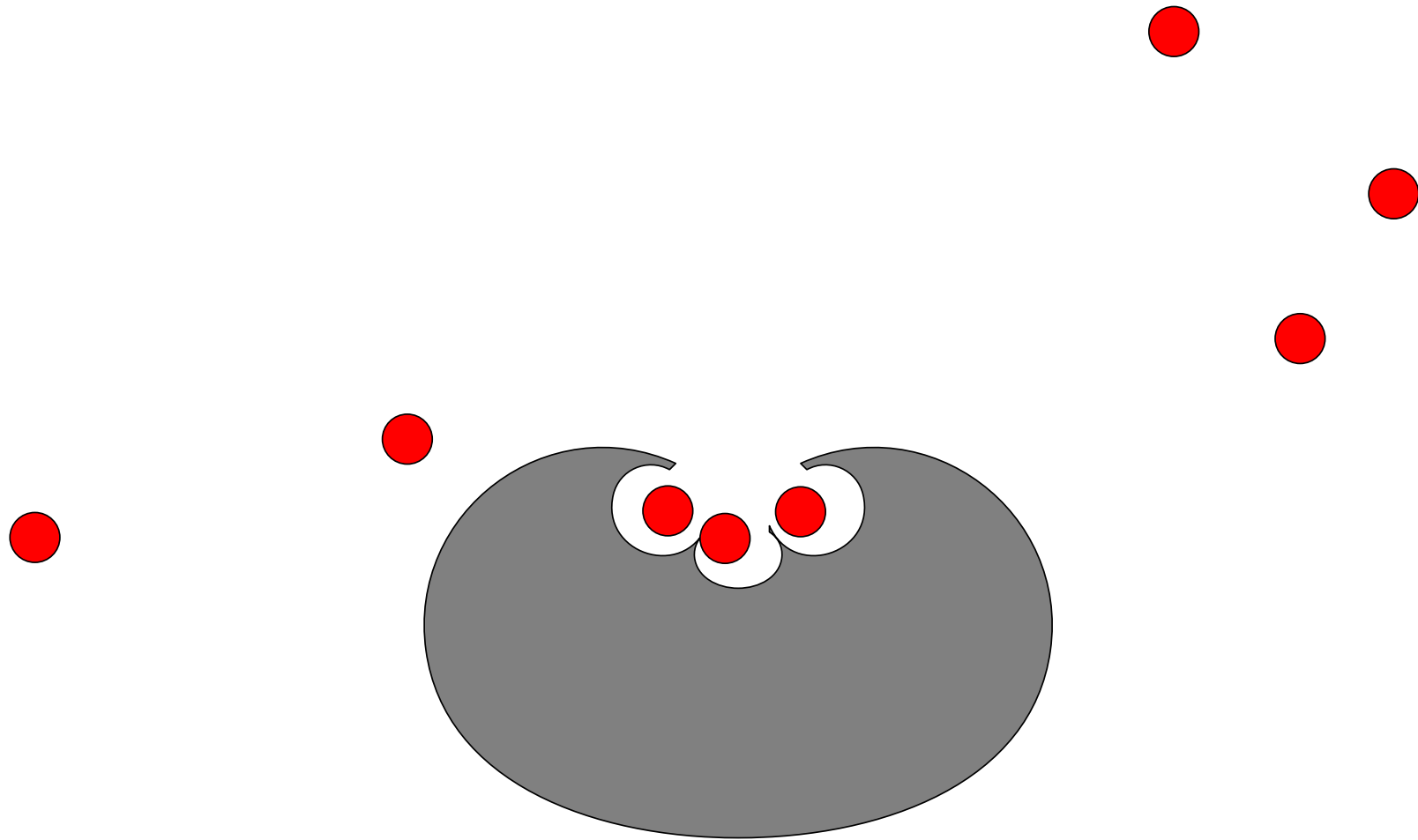
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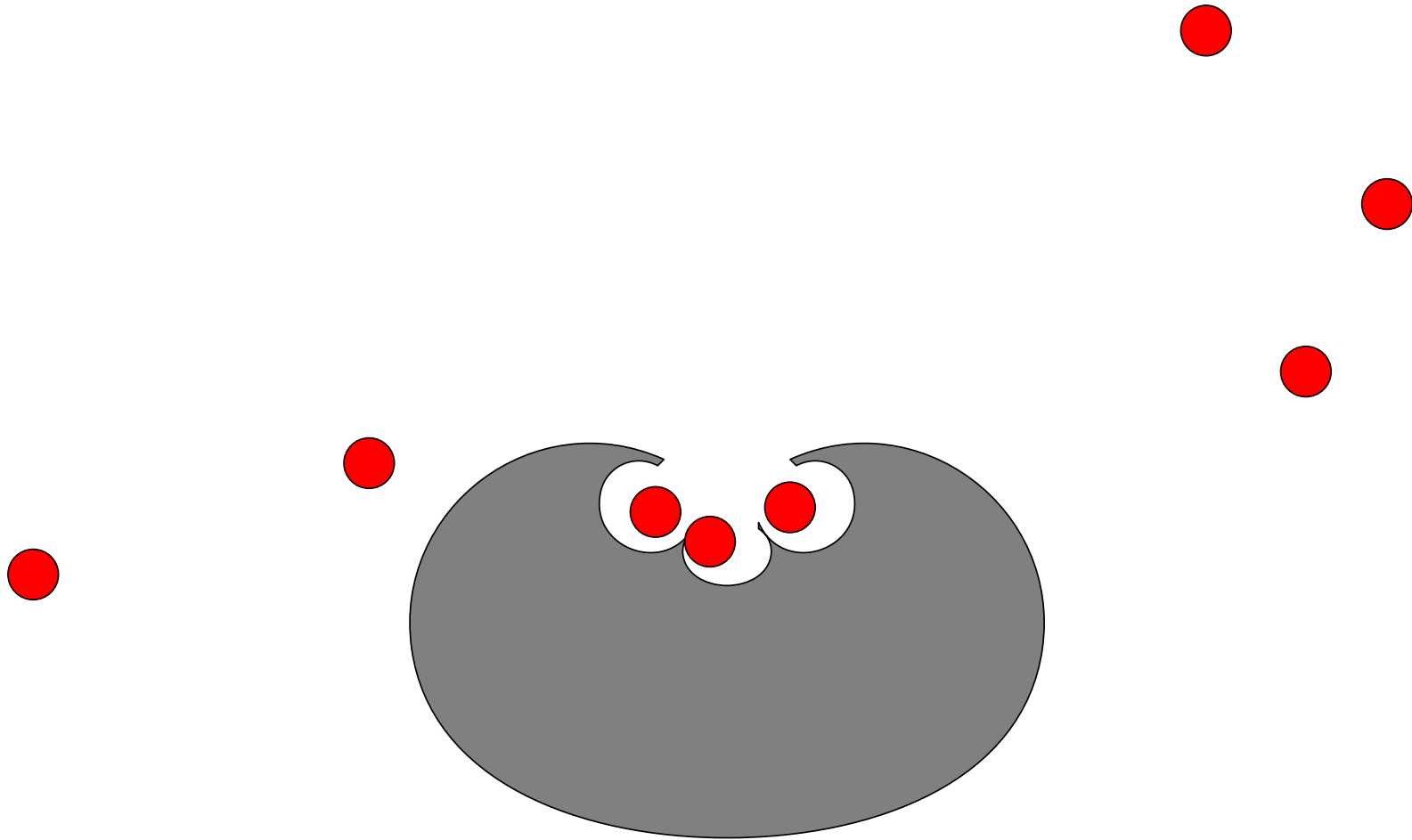
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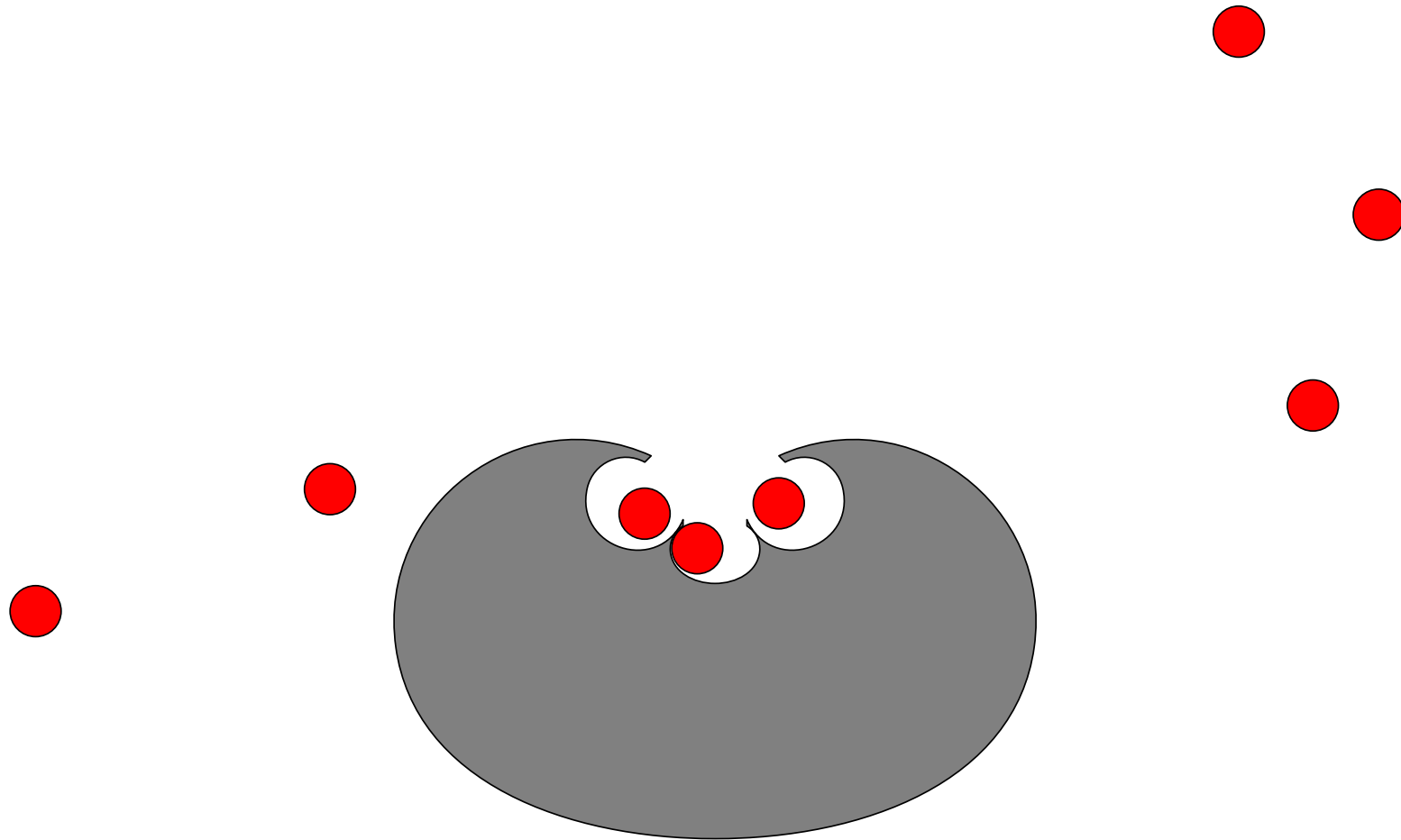
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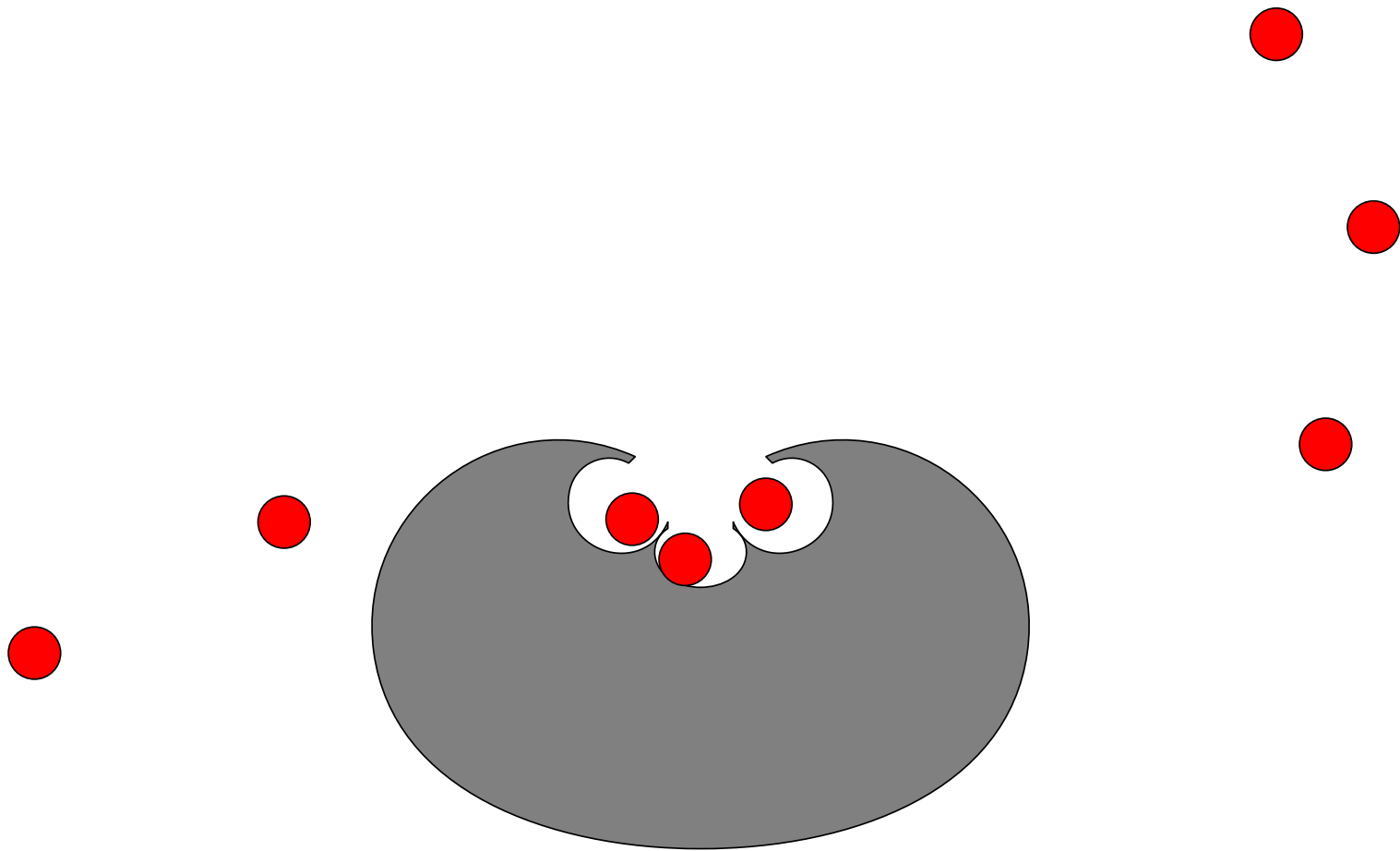
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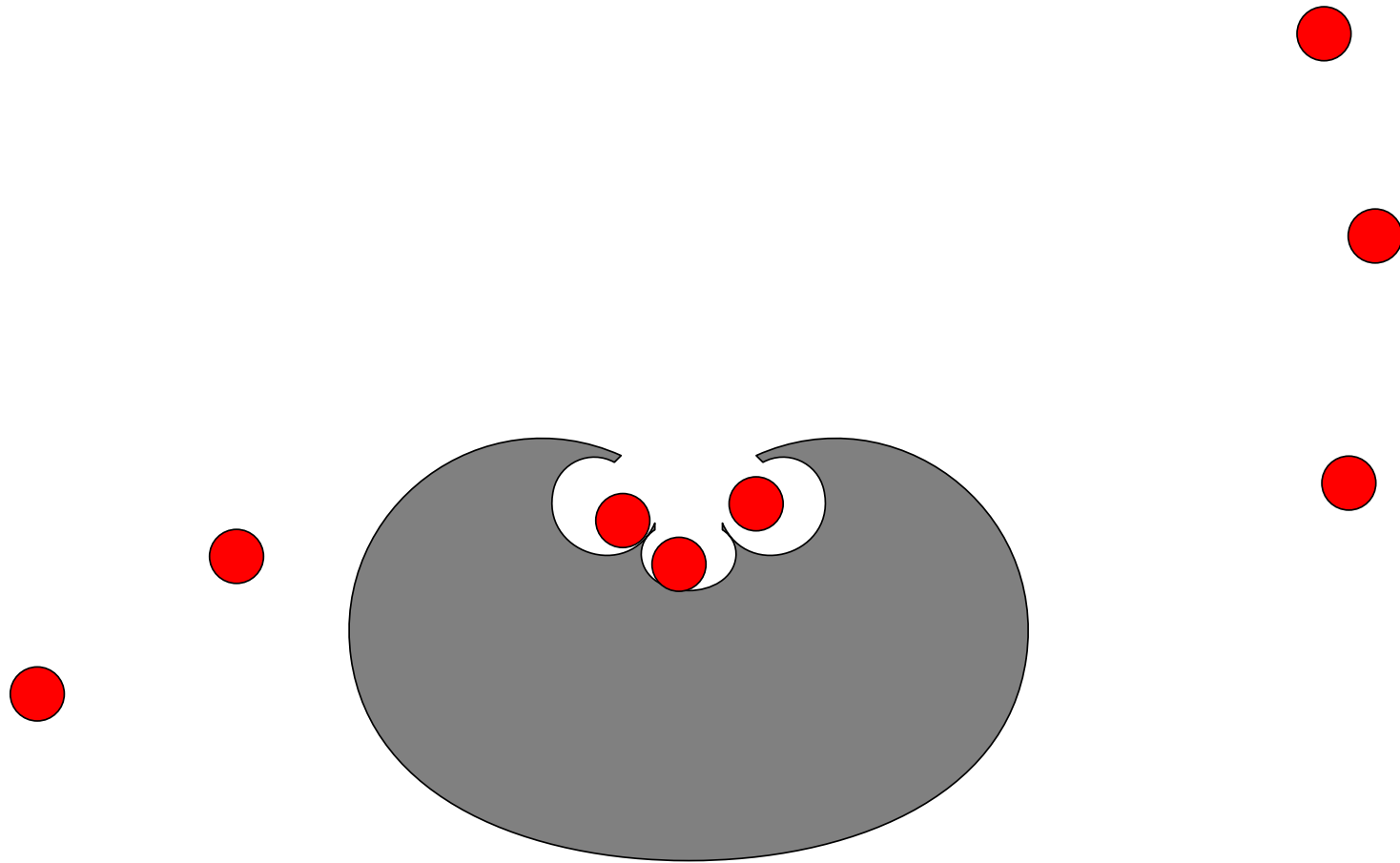
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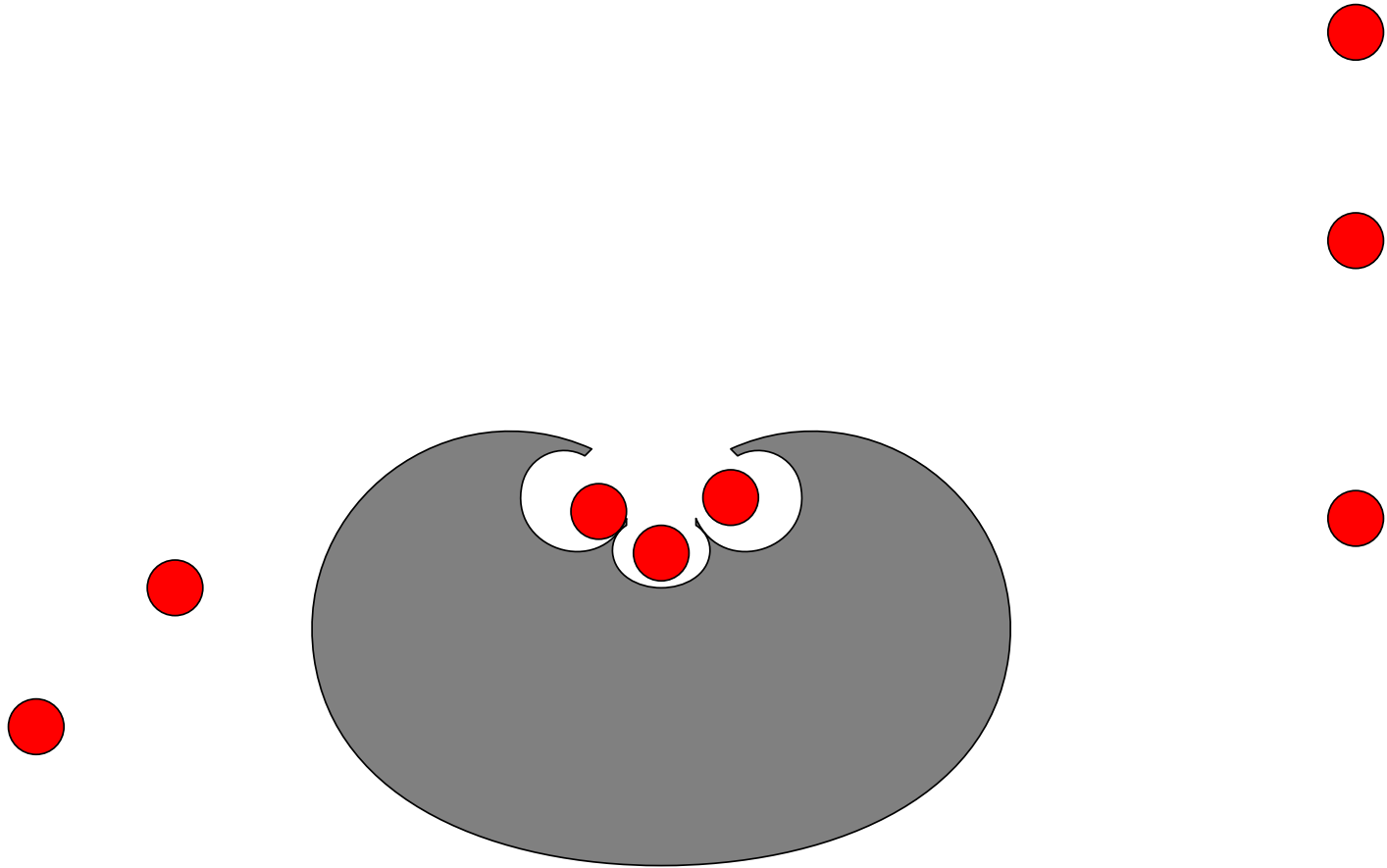
Multi-signalling



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Hill Equation

- We assume that we form a complex with n signalling molecules S_X



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- Which we can model using

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Dividing through by k_{on}

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$$\frac{k_{off}}{k_{on}} [C] = ([X_T] - [C]) [S_X]^n$$

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$$\frac{k_{\text{off}}}{k_{\text{on}}} [C] = ([X_T] - [C]) [S_X]^n$$

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$$[C] = \frac{[X_T] [S_X]^n}{K_X^n + [S_X]^n}$$

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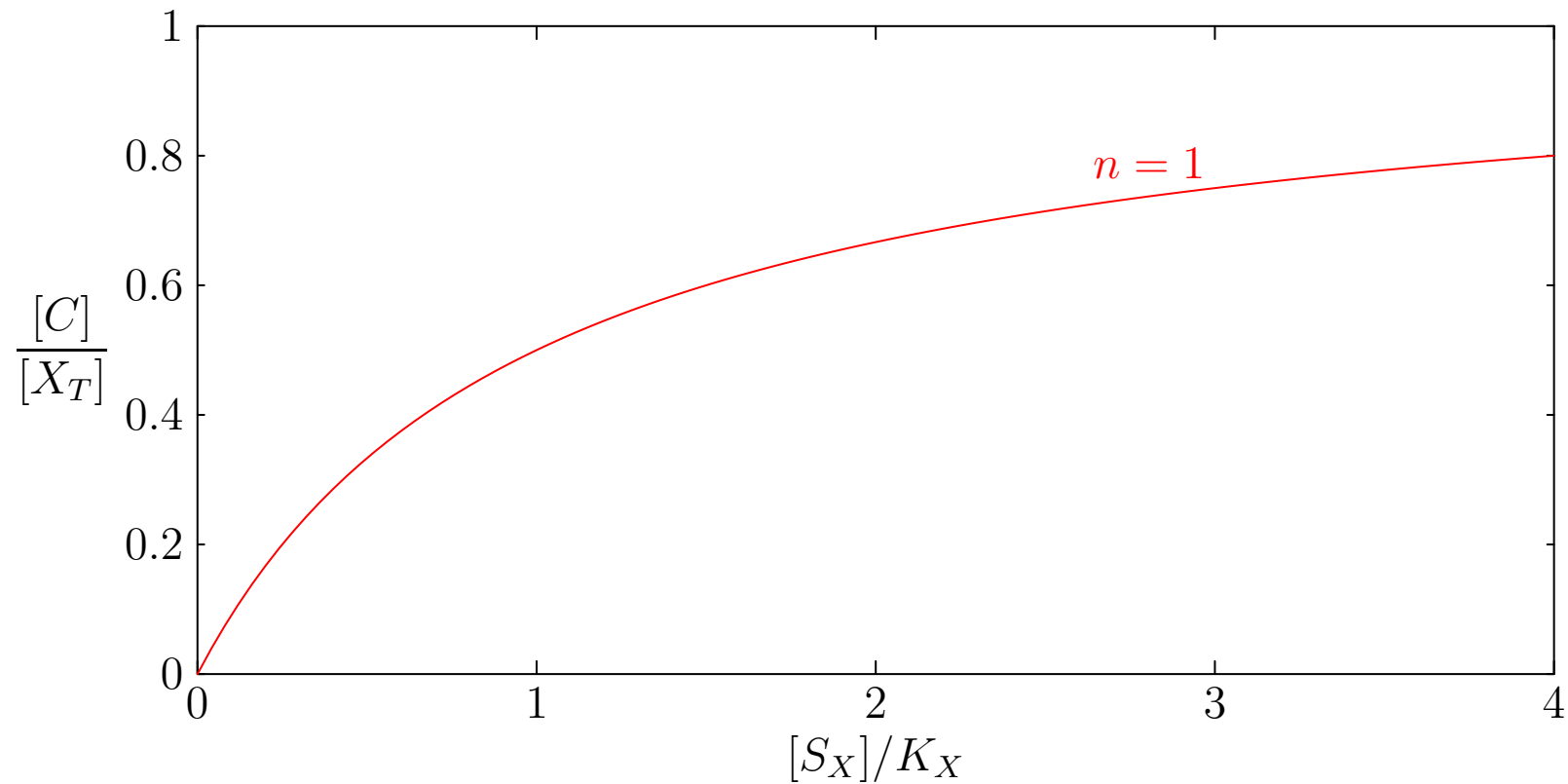


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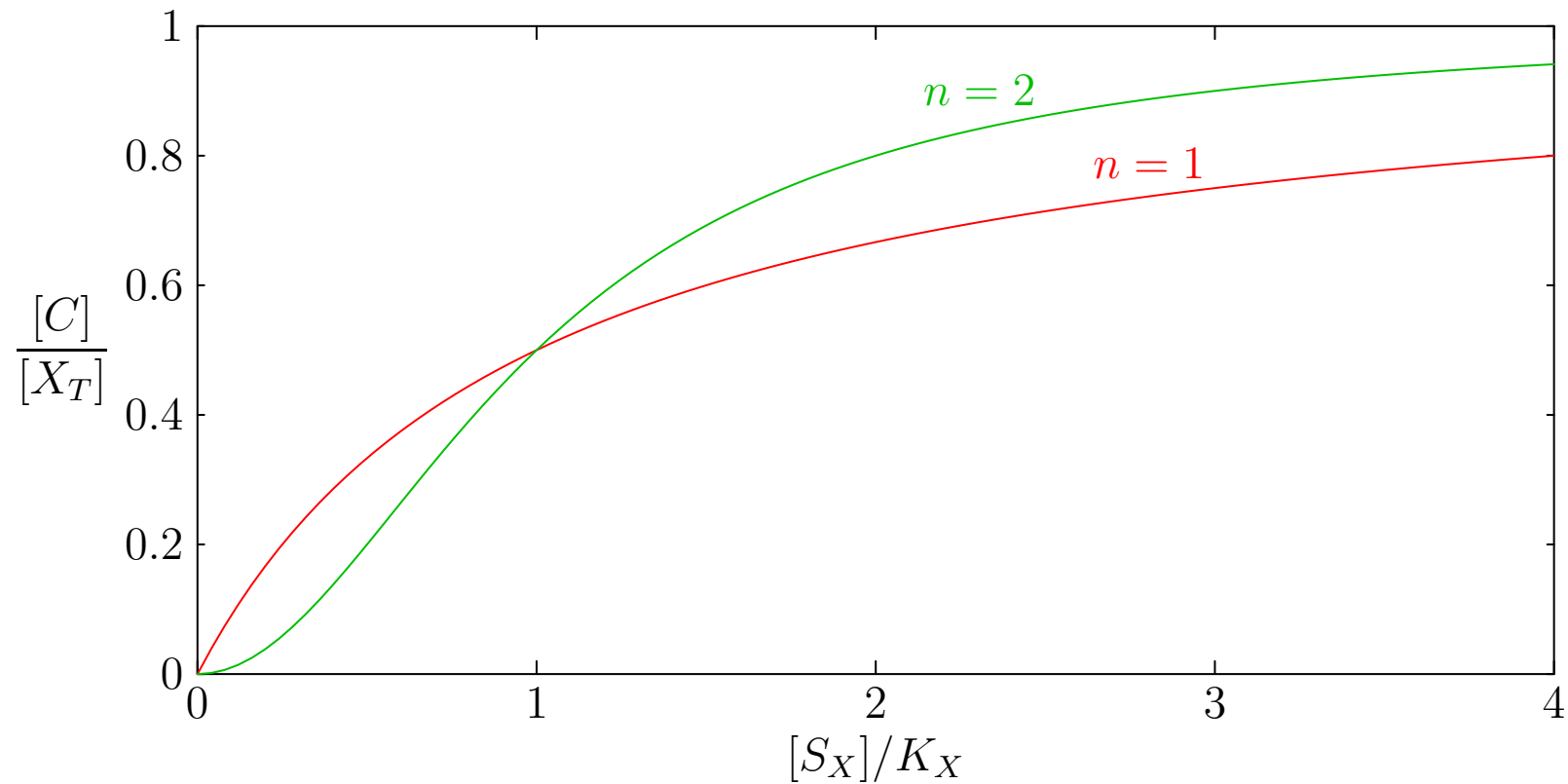
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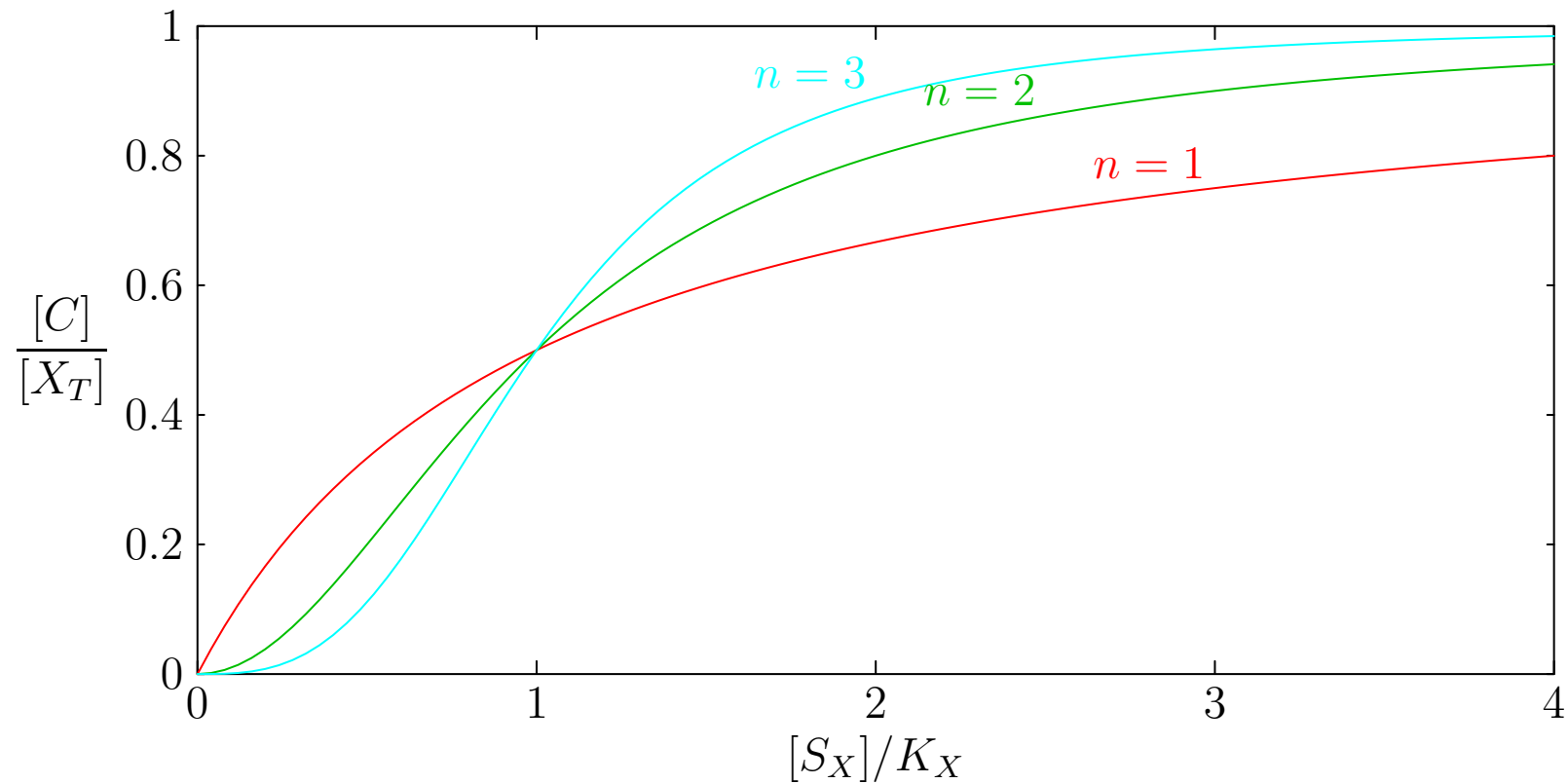
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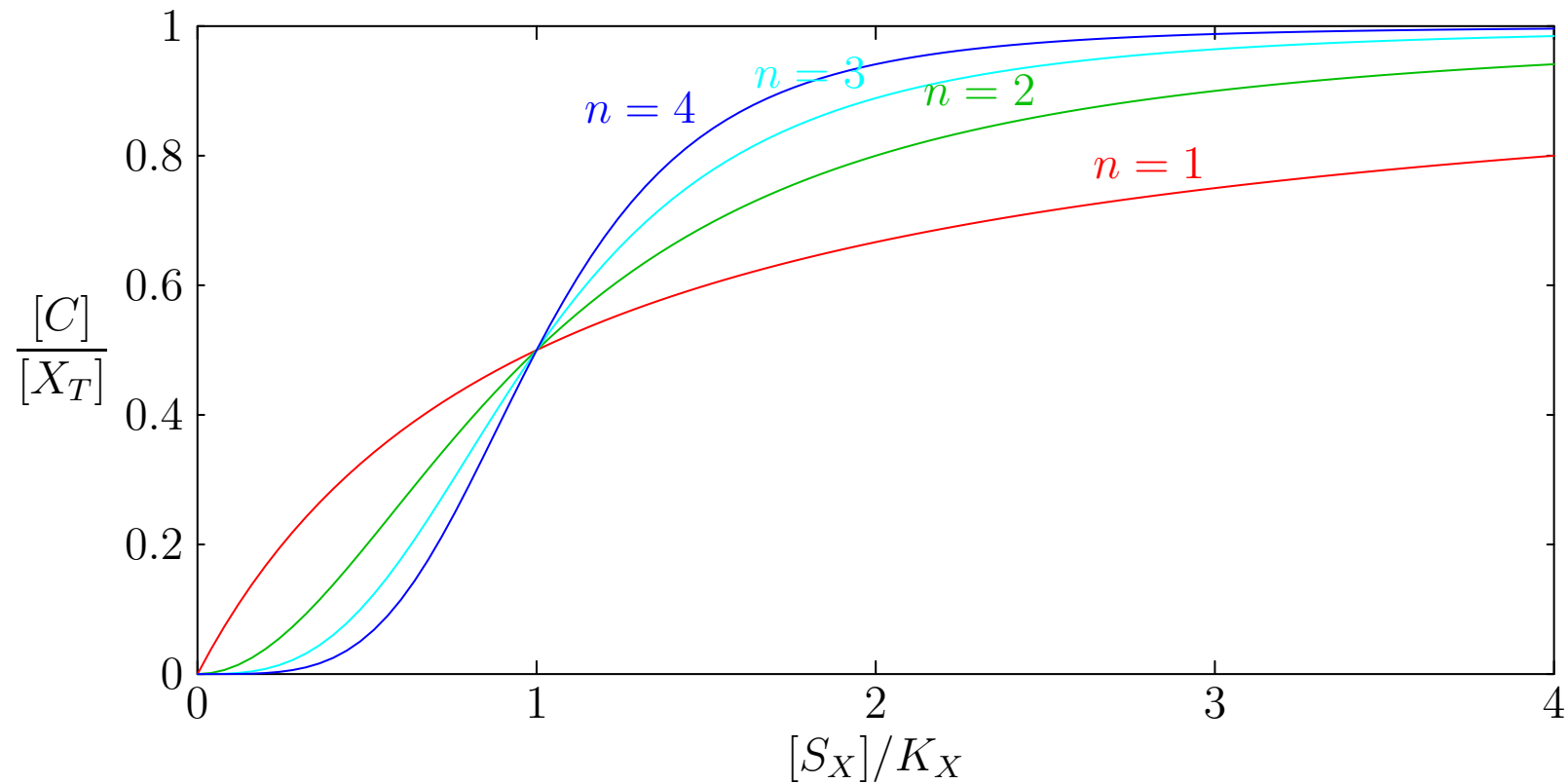
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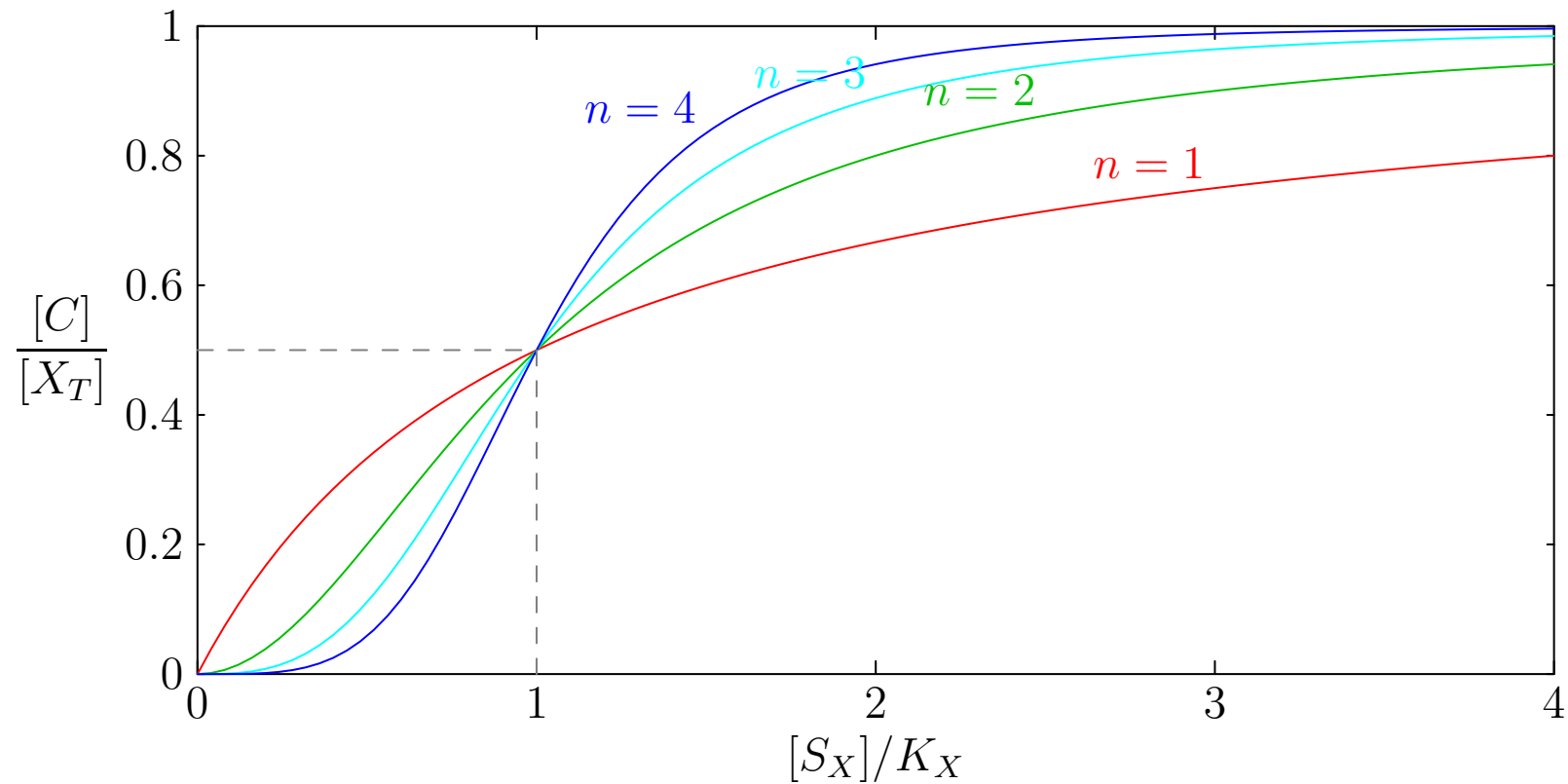
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Activator and Repressor

- A protein that acts to increase the expression of a protein we call an **activator**
- The activation coefficient is often given by the Hill-equation

$$f([X]) = \frac{\beta [X]^n}{K^n + [X]^n}$$

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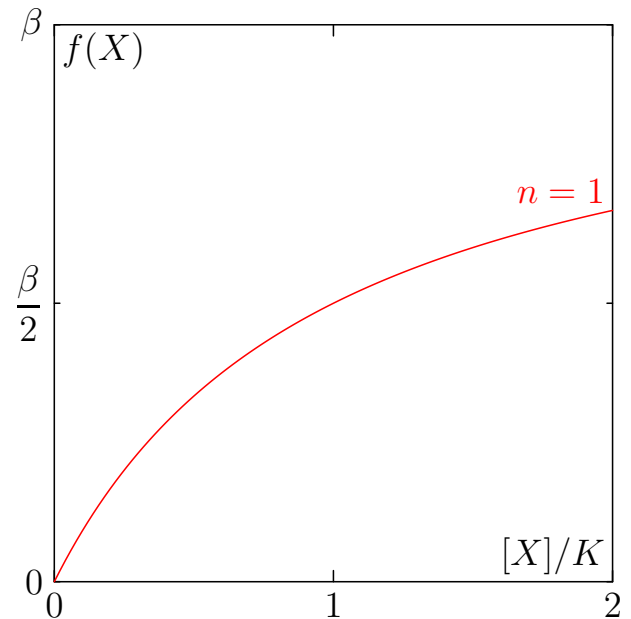
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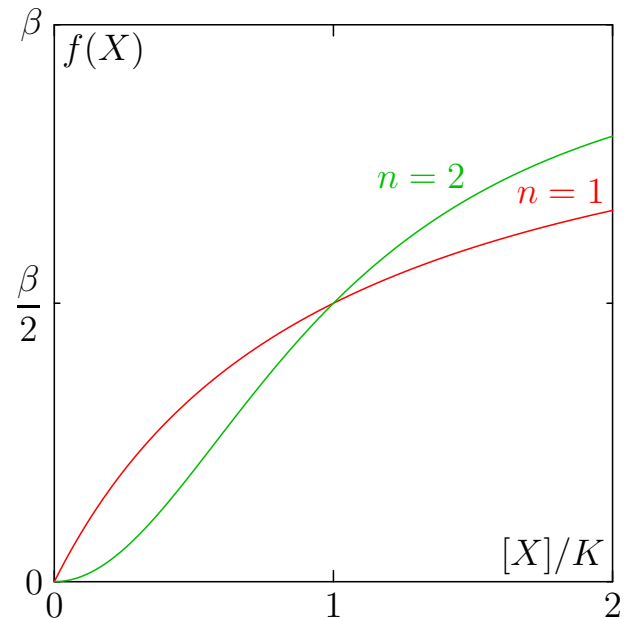
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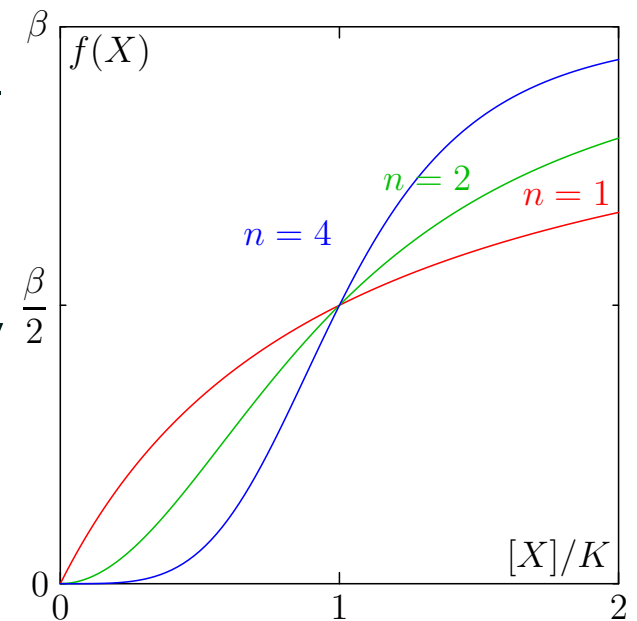
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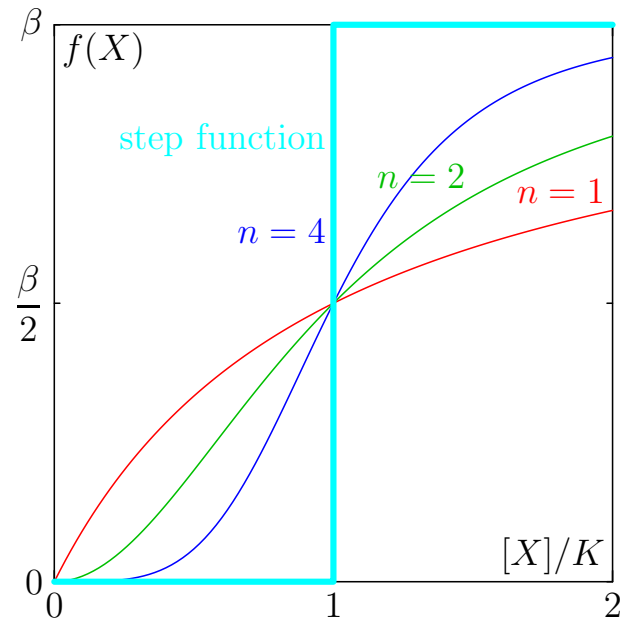
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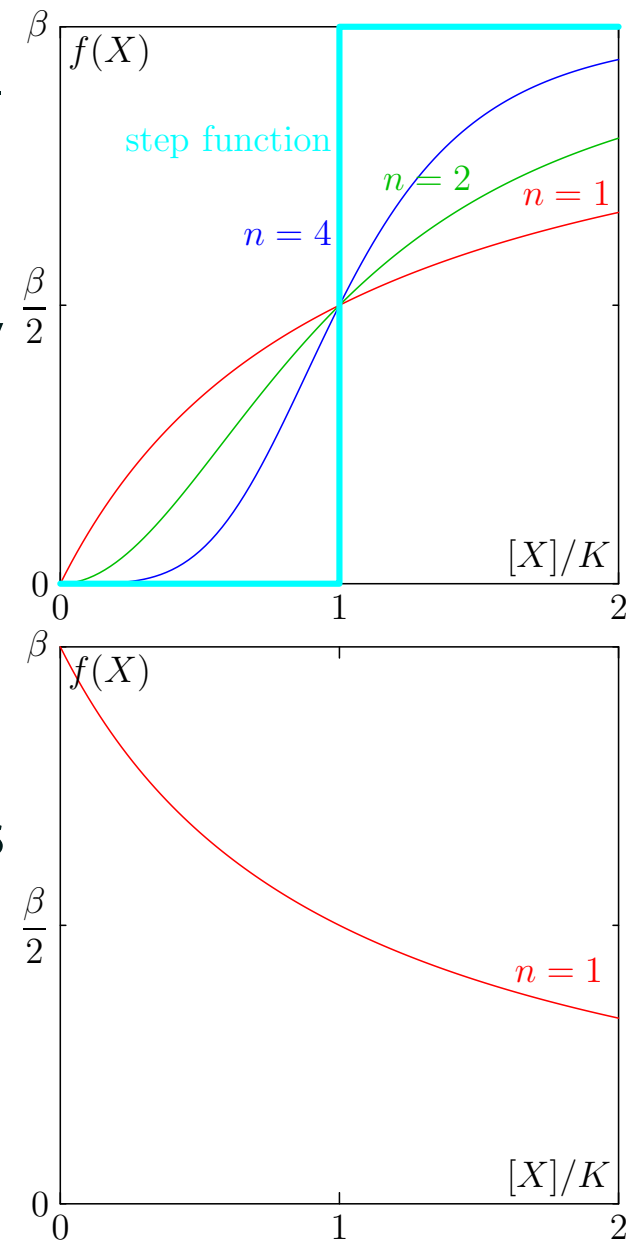
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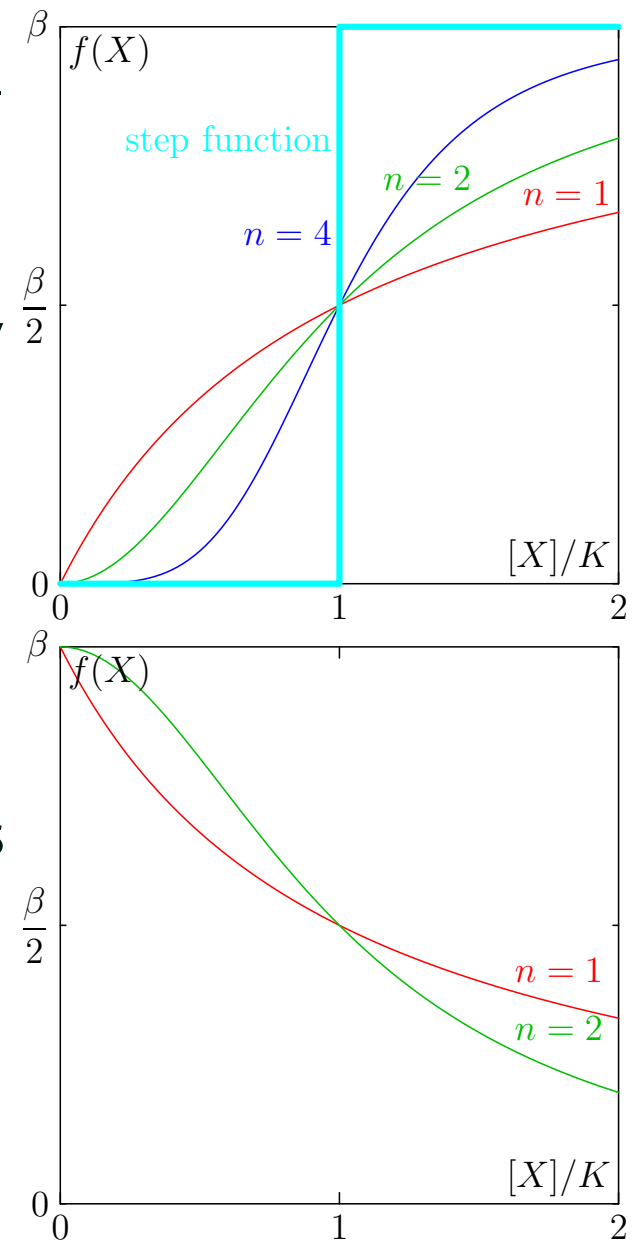
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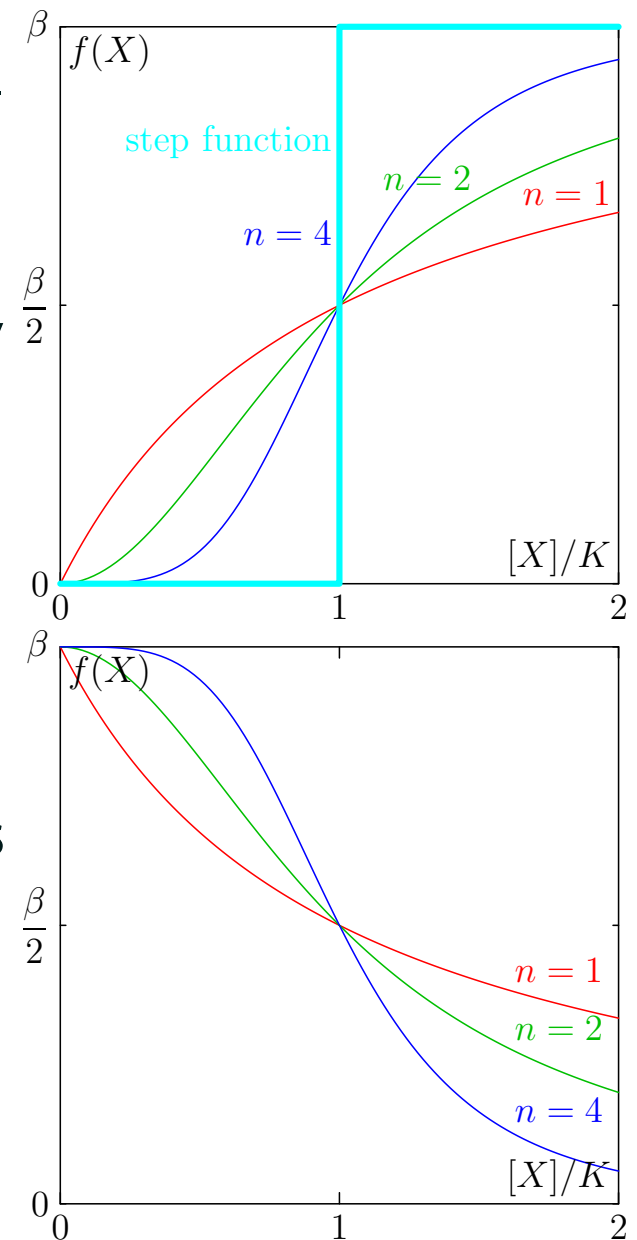
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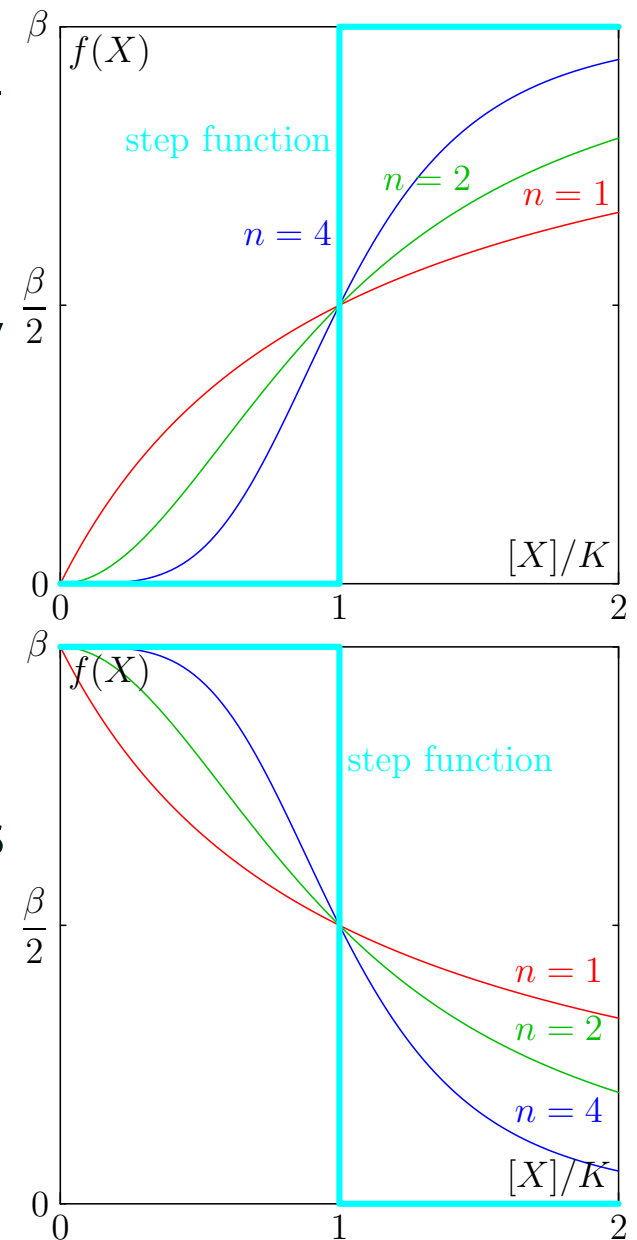
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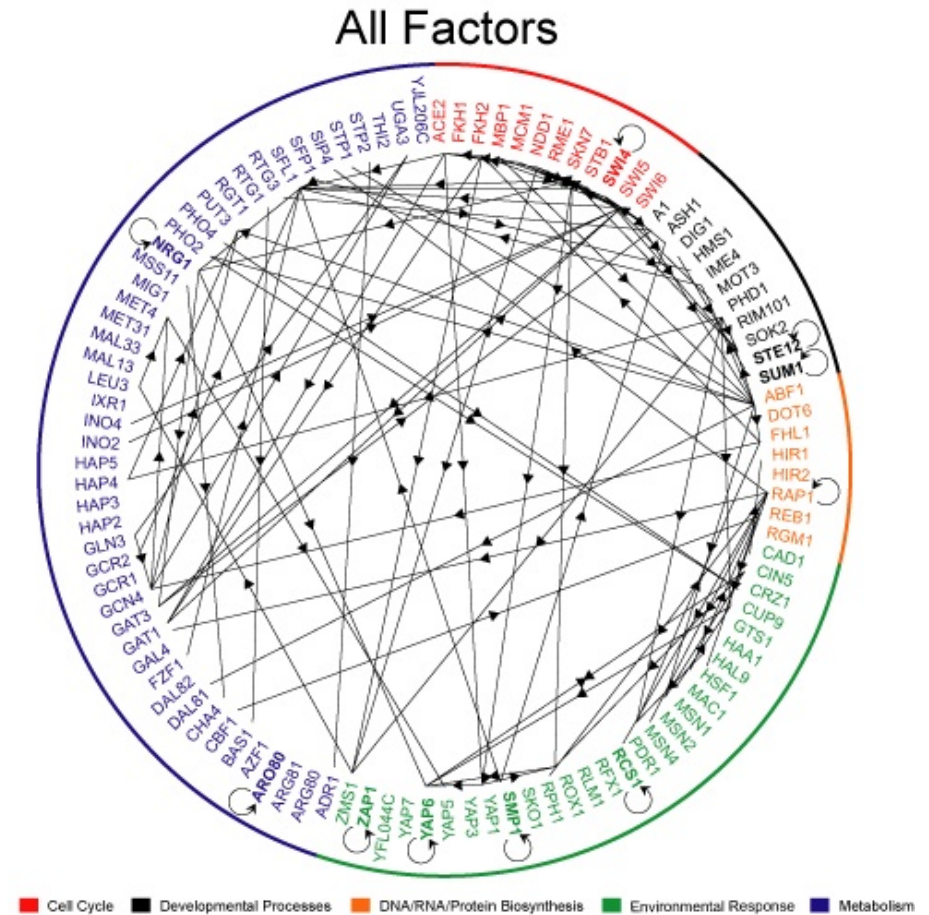
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Outline

1. Gene Regulation
2. Chemical Equations
3. **Motifs**



Understanding Gene Regulation Networks

- We can infer interactions between proteins as we described in a previous lecture
- Gene regulation networks tend to be rather complicated
- To make sense of them we wish to look for patterns, “**motifs**”, that occur very often
- We look at interactions that occur more often than would be expected to happen in a random network

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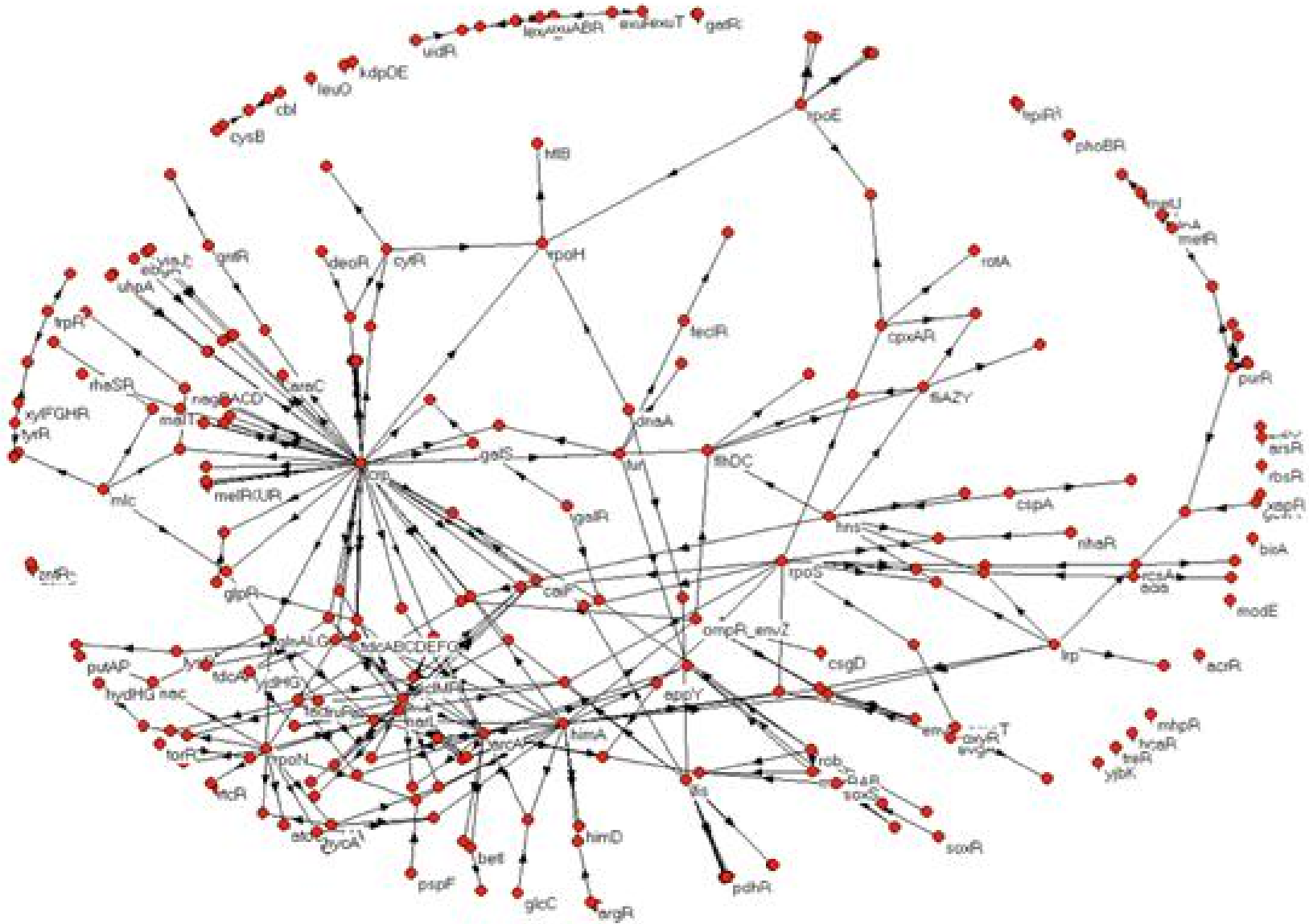
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E-Coli GRN



Auto-regulation

- In E-coli we have $N = 424$ nodes and $E = 519$ edges
- The total number of directed edges (including self-edges) is N^2
- The number of self edges is N so the probability of a randomly chosen edge being a self-edge is $p = 1/N$
- The probability that if we choose E edges k will be self-edges is

$$\mathbb{P}(k|E, p) = \text{Binom}(k|E, p) = \binom{E}{k} p^k (1 - p)^{E-k}$$

- This has mean $\mathbb{E}[N_{\text{self}}] = E p \approx 1.2$ and standard deviation $\sigma = \sqrt{E p (1 - p)} \approx 1.1$
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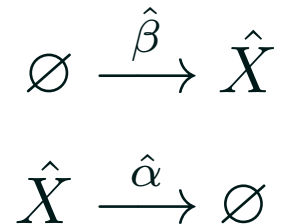
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- To understand why auto-regulation is interesting let us first consider the case with no negative feedback

- We assume



- So on average

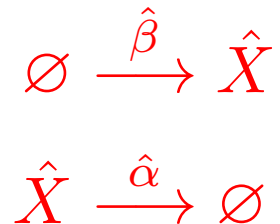
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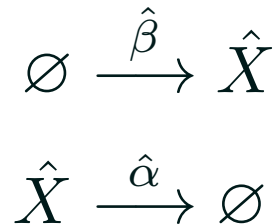
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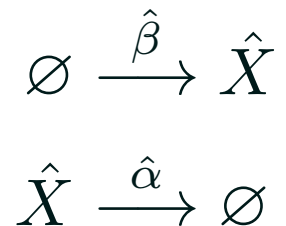
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- The full dynamics is given by the solution to the differential equation and depend on the initial condition, $[\hat{X}(0)]$
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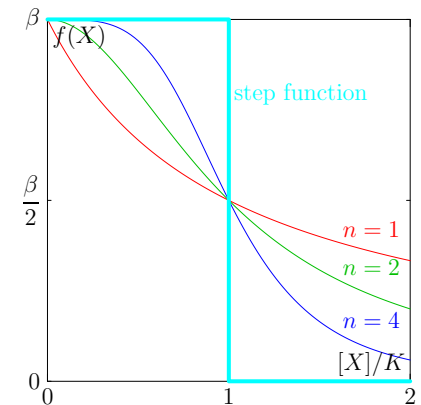
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- Where we have a Hill-type repressor

$$f([X]) = \frac{\beta}{1 + ([X]/K)^n} \approx \beta \llbracket [X] < K \rrbracket$$



- Starting from $[X(0)] = 0$ initially $[X(t)] = \beta t$
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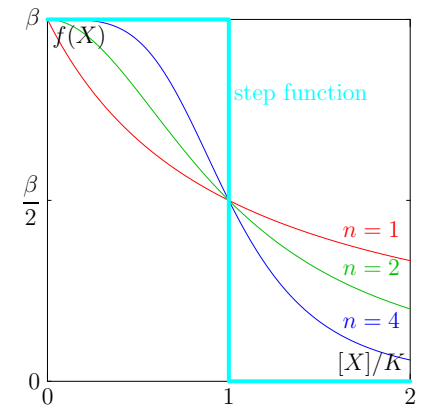
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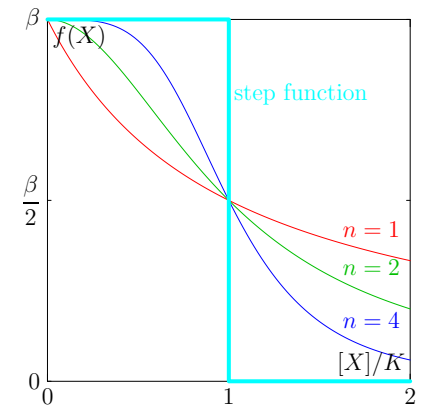
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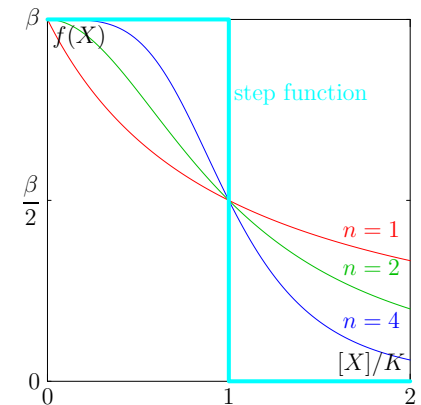
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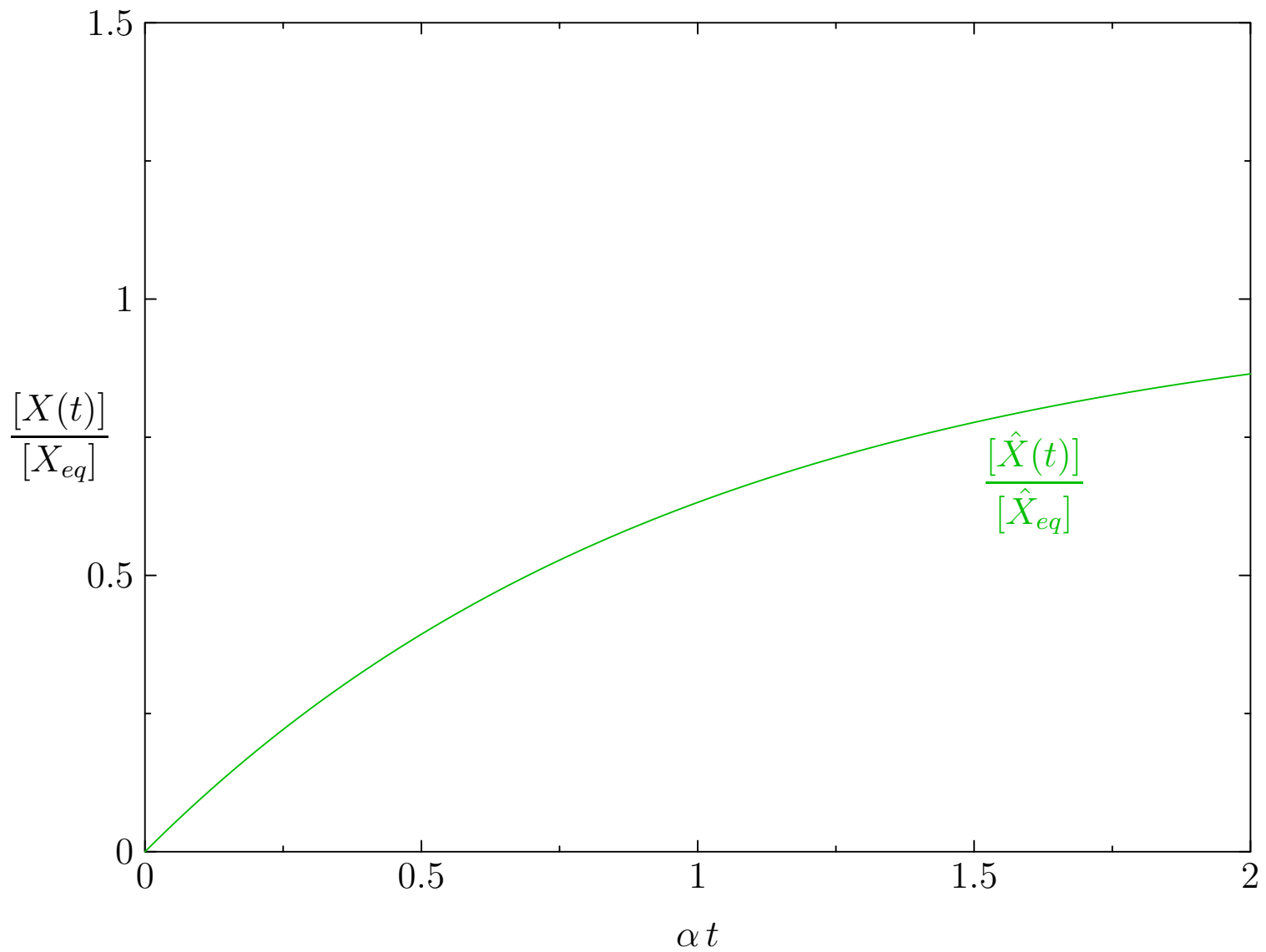
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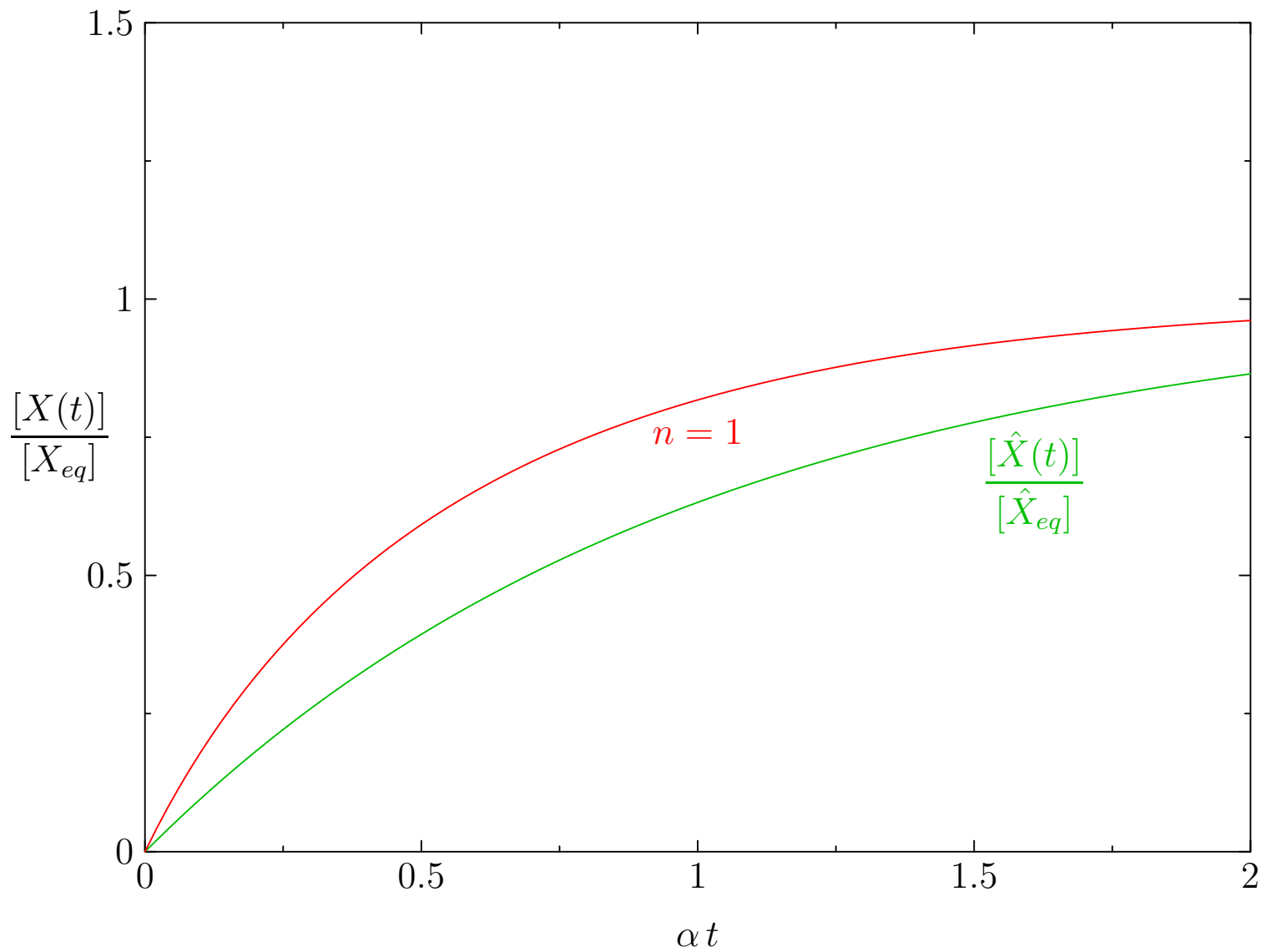
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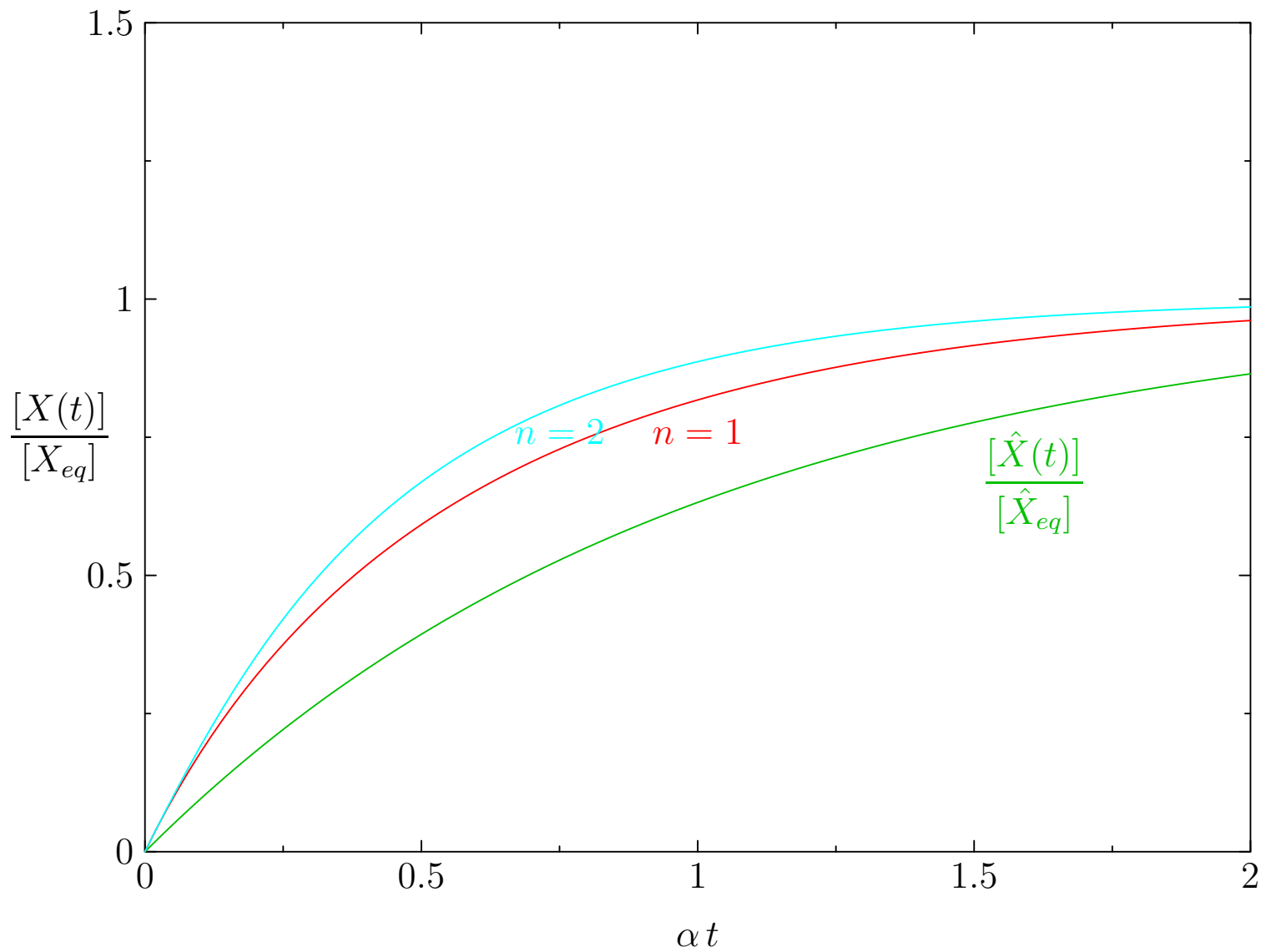
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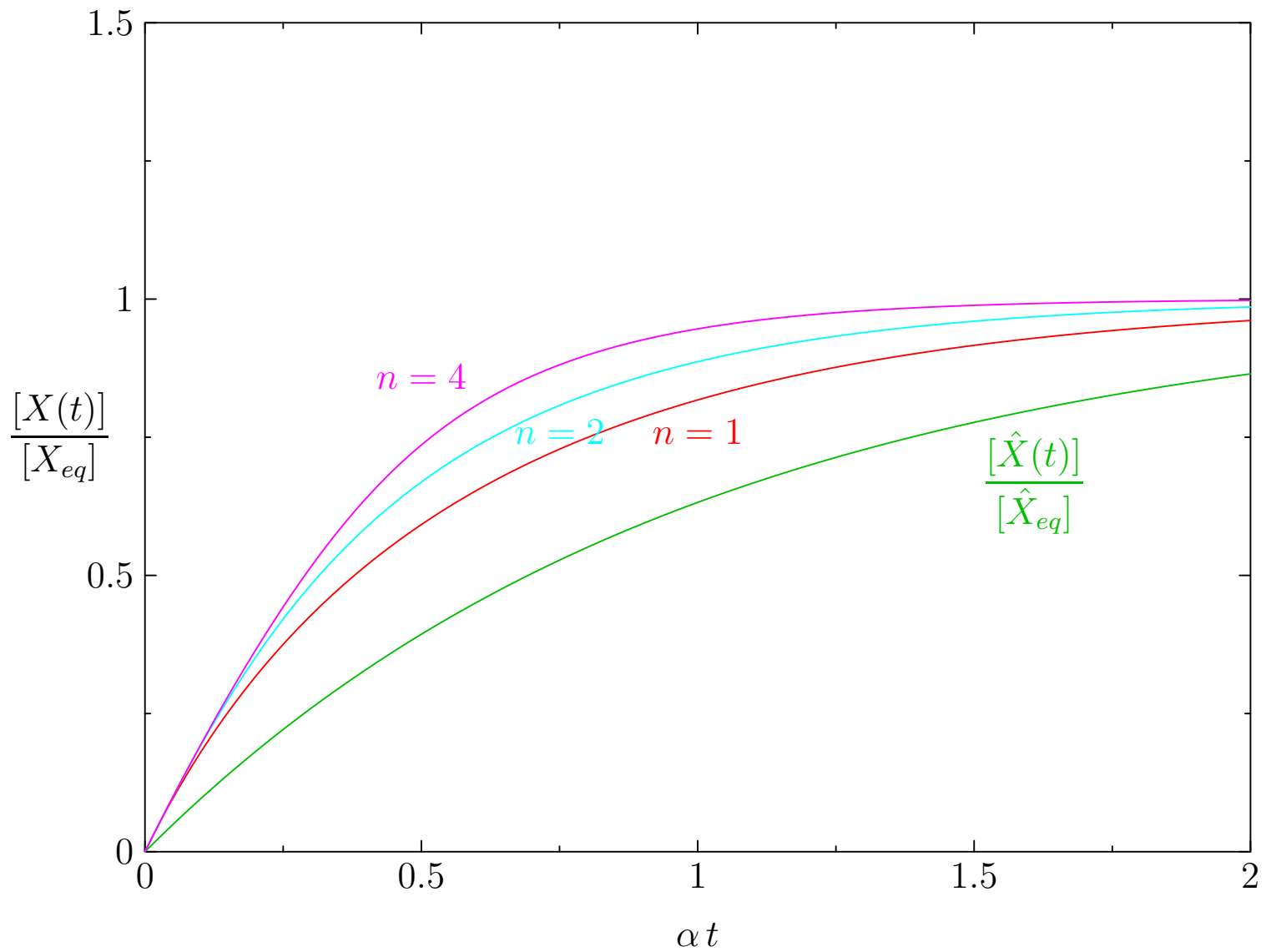
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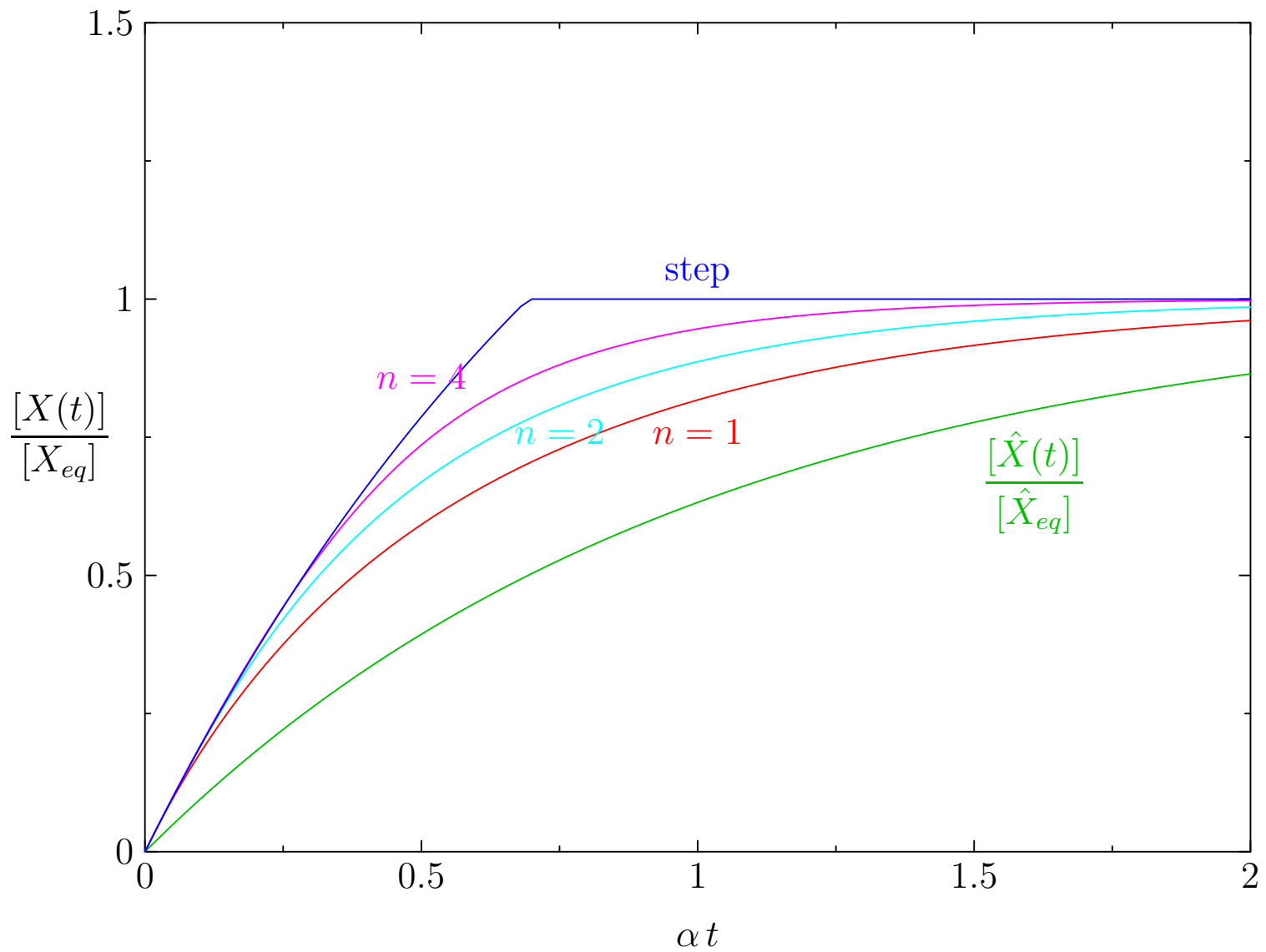
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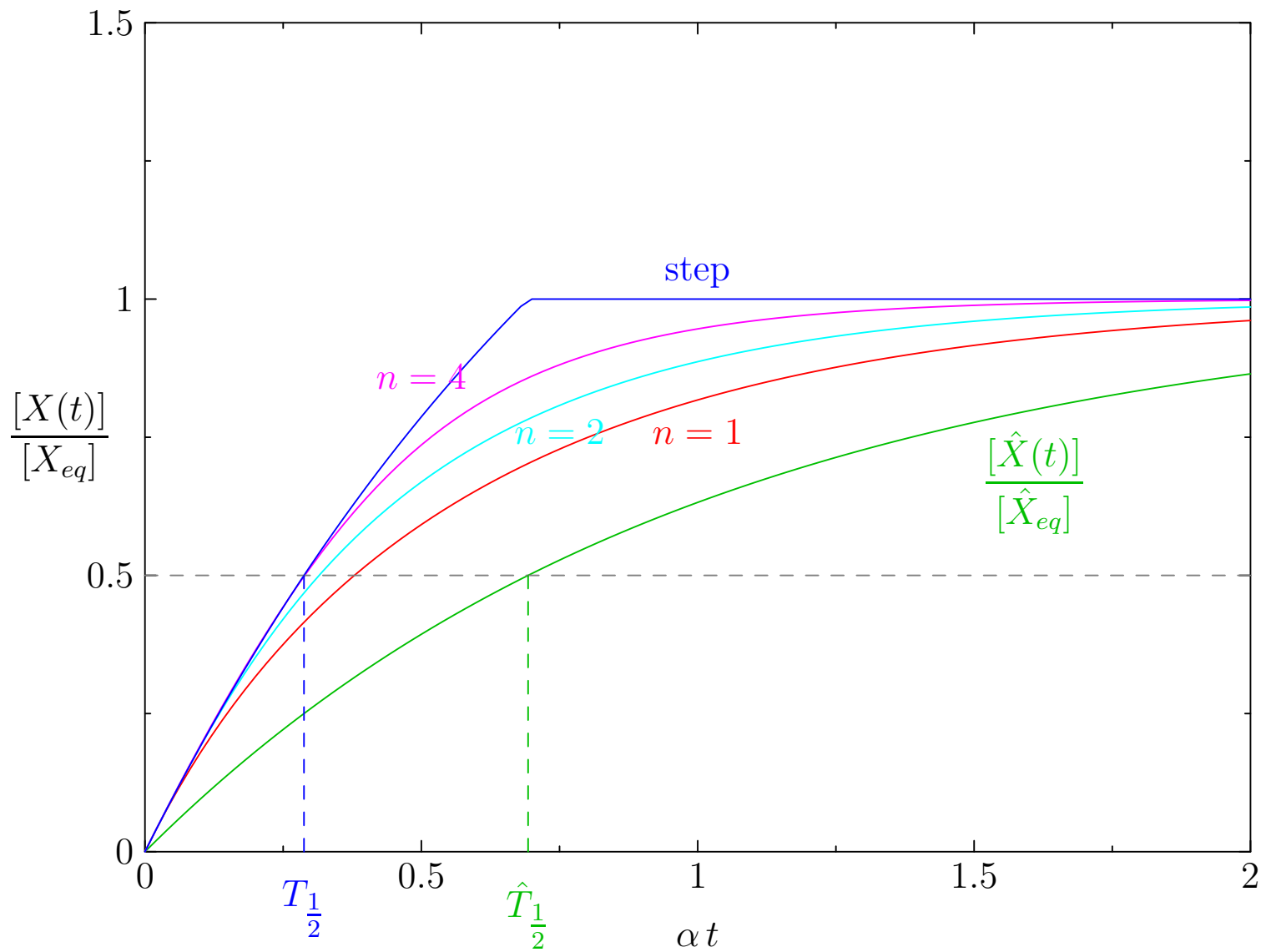
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- Auto-regulation allows us to use a large production rate β for the same equilibrium, speeding up the equilibration time
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